First TWSNMP FK

Most popular SNMP manager in Japan

Copyright (c) 2023-2026 Masayuki Yamai

At the beginning

TWSNMP is an SNMP manager that supports the most popular SNMPv3 in Japan for over 25 years.
It is TWSNMP FK that has been reprinted with the latest machine technology in 2023.
The TWSNMP FC that runs on the container is accessed from the web browser, but the FK is a desktop app and does not require a browser.

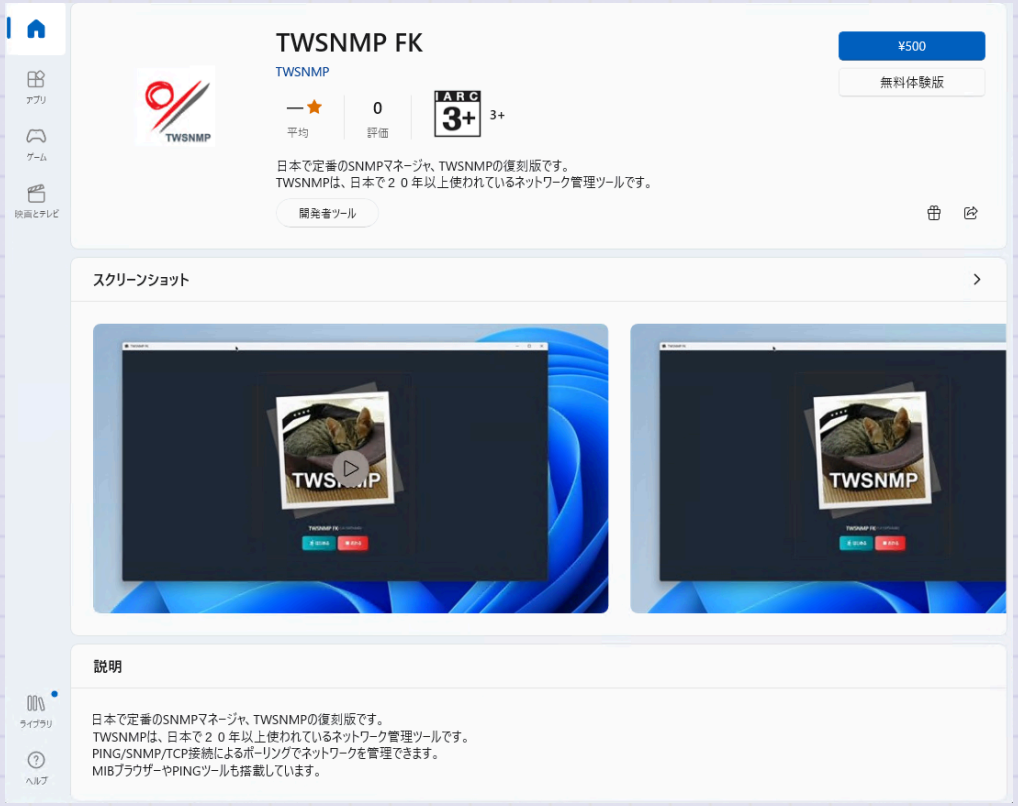
Copyright (c) 2023-2026 Masayuki Yamai

Microsoft Store

Windows version

https://www.microsoft.com/store/apps/9nsqn46p0mVL

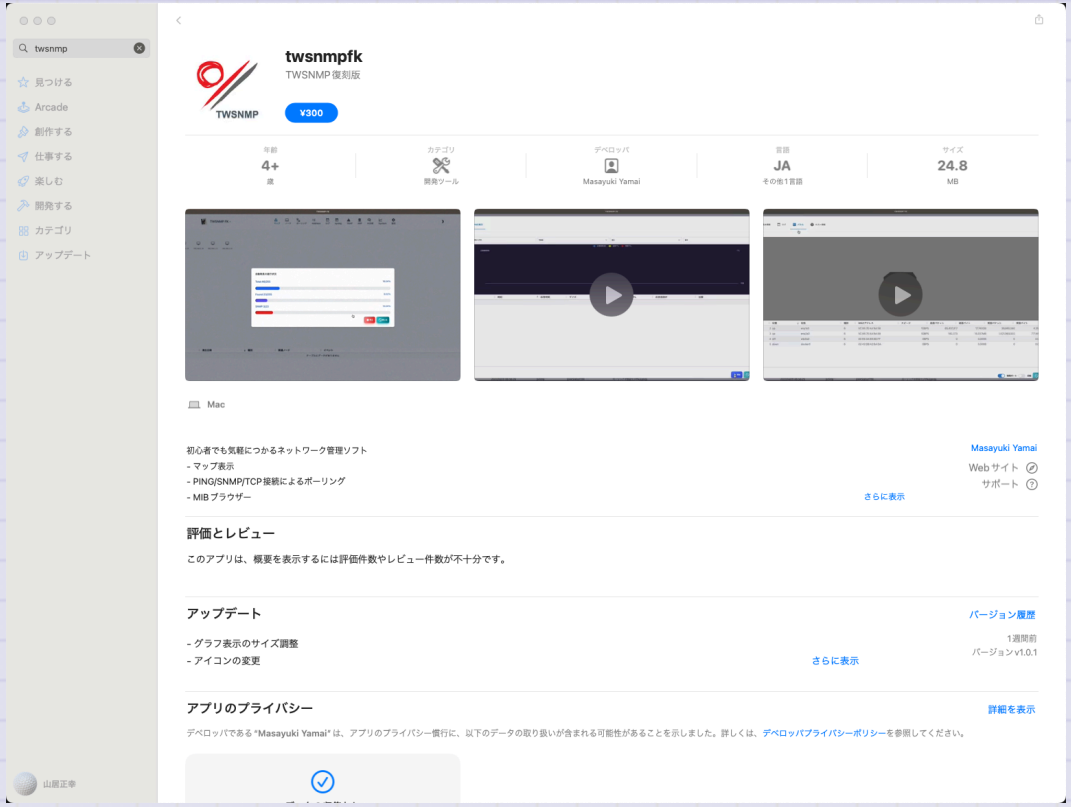
You can buy it.



Copyright (c) 2023-2026 Masayuki Yamai

App Store

The Mac version is
https://apps.apple.com/jp/app/twsnmpfk/id6468539128
You can buy it.



Copyright (c) 2023-2026 Masayuki Yamai

Notes for 24/7 Monitoring on macOS

Linux Version

You can download the package (`.tar.gz` format) from the GitHub Releases.

Notes for Running on Linux

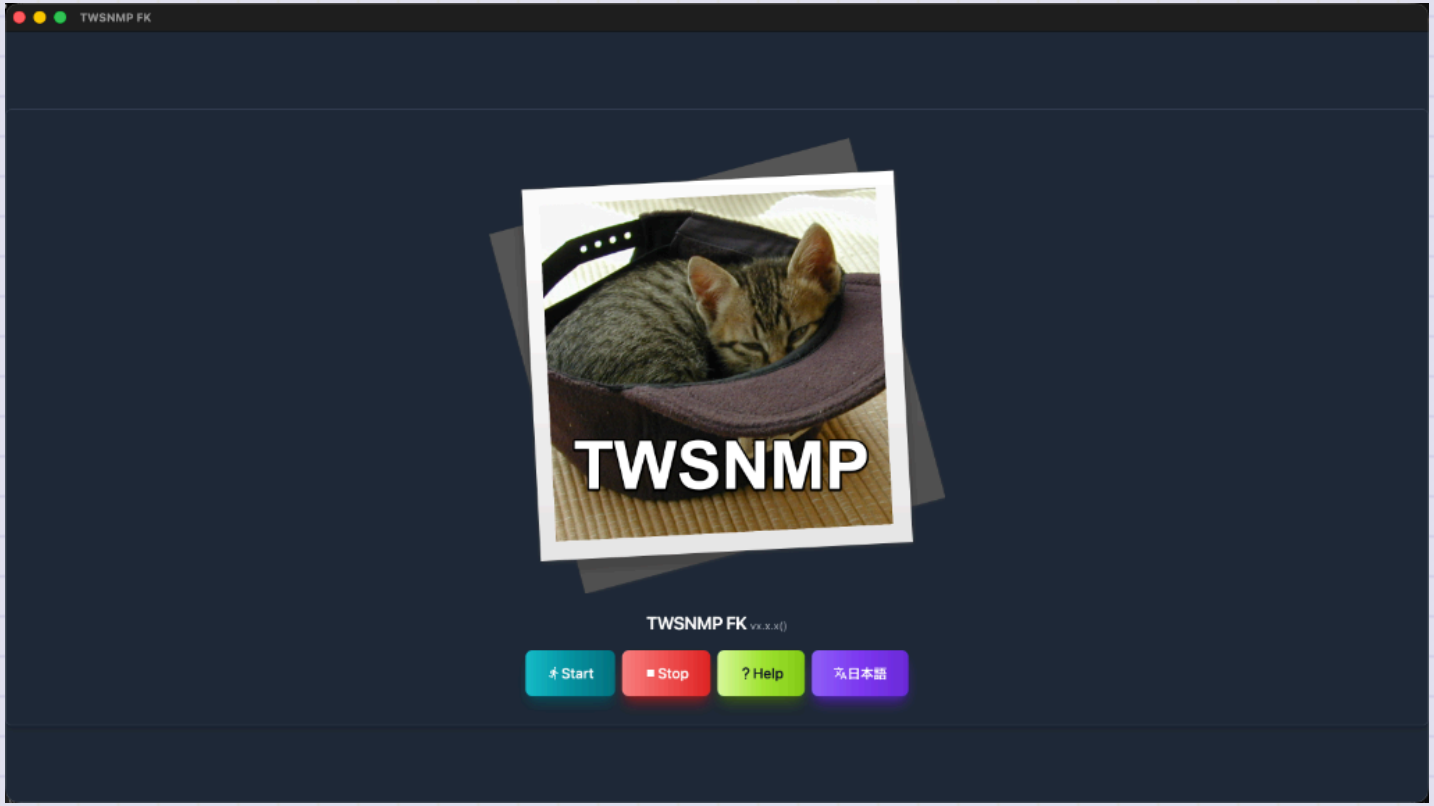
Running the application as a normal user will result in permission errors (no raw socket permission for ICMP Ping, no privileged port permission for 514/162).
Do not run the application directly with `sudo` (connection to the X11/Wayland display server will break).

1. **Grant Capabilities**:
   `sudo setcap 'cap_net_bind_service,cap_net_raw+ep' ./twsnmpfk`
2. **Install ARP Monitoring Tools** (adds `arp` command):
   `sudo apt-get update && sudo apt-get install -y net-tools`
3. **Run as a normal user**:
   `./twsnmpfk`

Copyright (c) 2023-2026 Masayuki Yamai

Starting TWSNMP FK

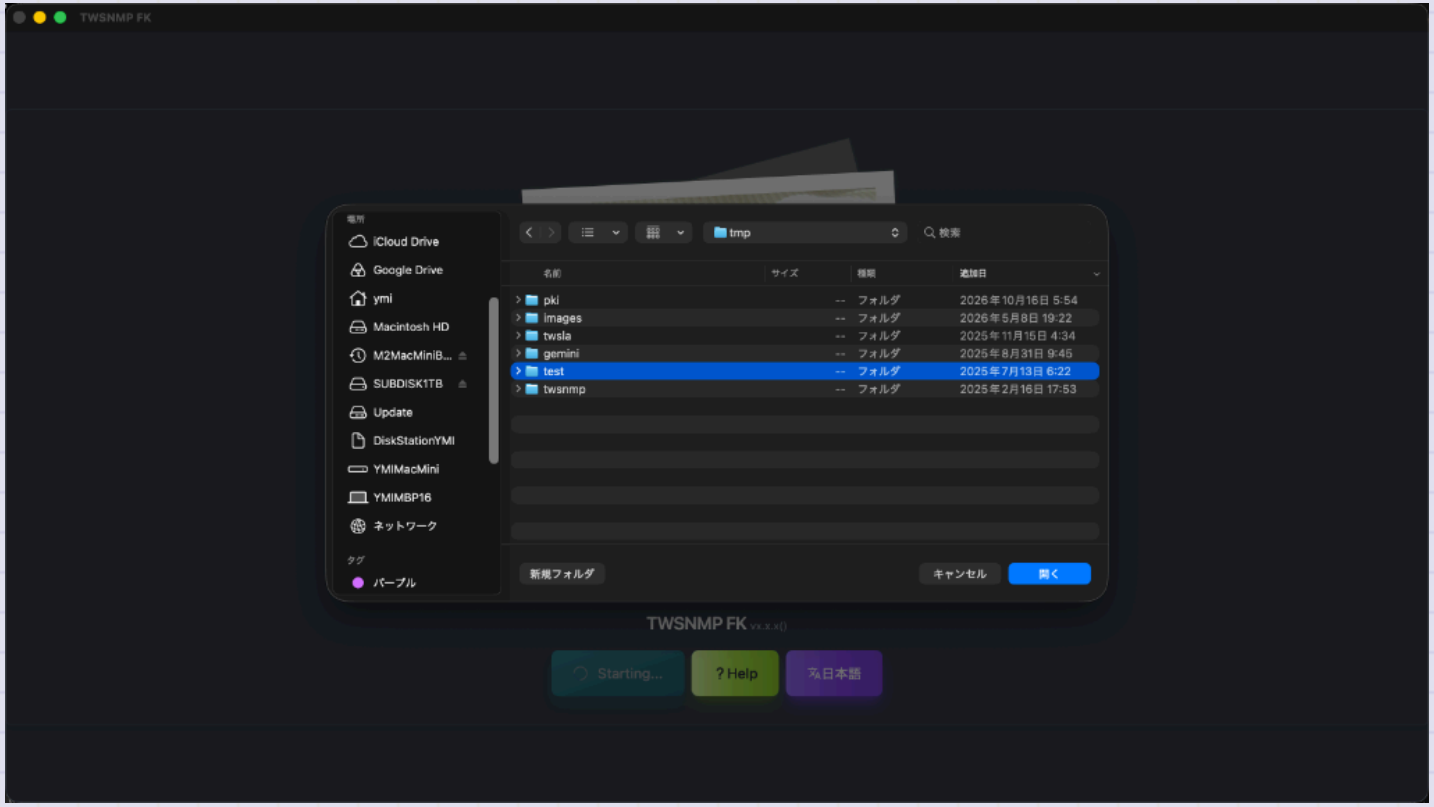
In the case of Windows, start from the start menu to the Mac OS in your favorite method, such as from the launcher.Welcome to the screen.Start with the <Start> button.Stop the program with the <Stop> button.The explanation screen of how to use it is displayed with the <Help> button.



Copyright (c) 2023-2026 Masayuki Yamai

Select a folder to save data

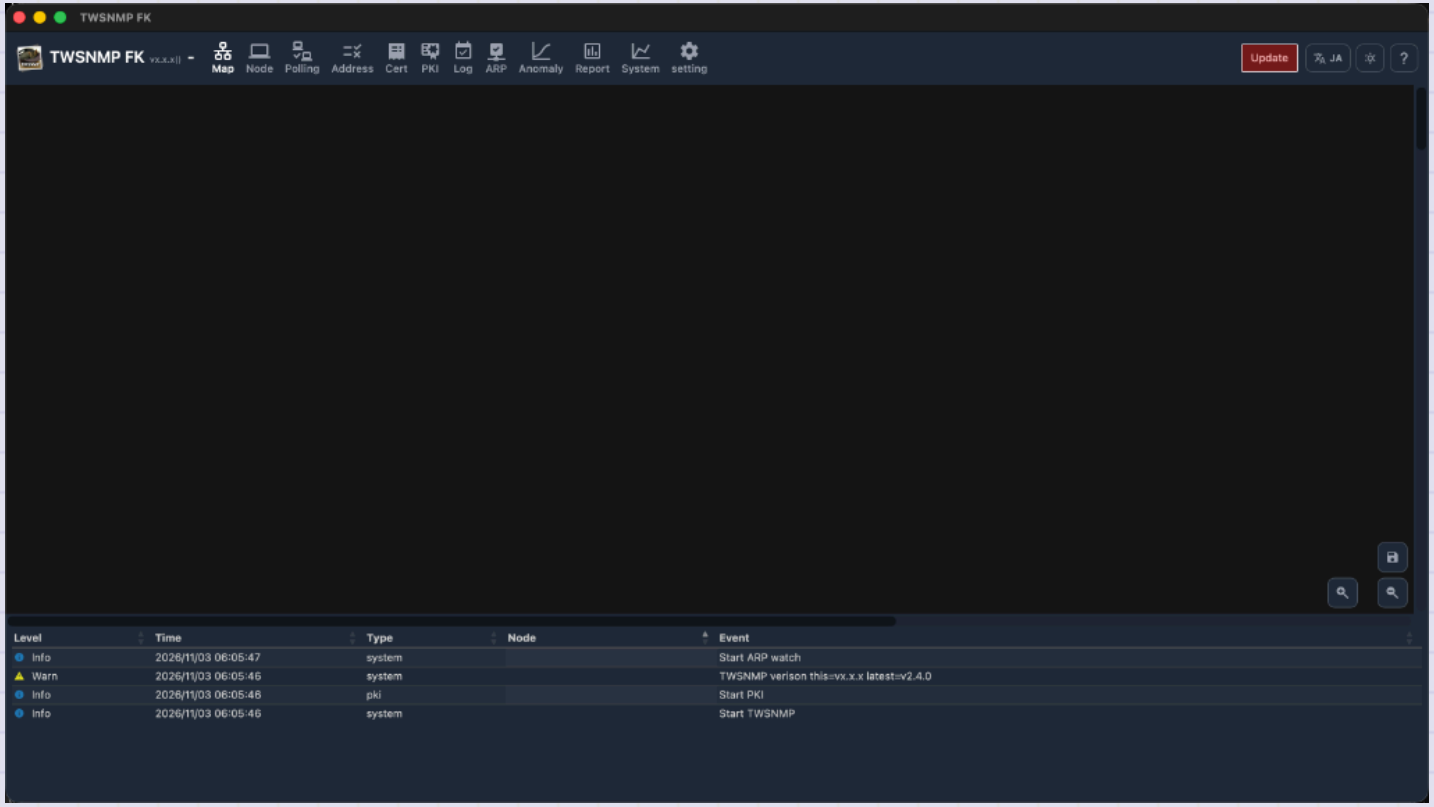
Click the <Start> button on the screen to display a dialog to select a folder to save the data.Please select a folder.You can also create a new one.



Copyright (c) 2023-2026 Masayuki Yamai

First map

Select a new folder and start a map without node.After a while, the log will be displayed.



Copyright (c) 2023-2026 Masayuki Yamai

Flow of the first map creation

The flow of creating a map is

- Click the appropriate position on the map
- Start "Automatic discovery" from the menu
- The IP address range to be searched
- Precrose automatic discovery
- Move node on map
- Line connection

You can now search for PCs, routers, servers, etc. connected to the managed network and register on the map.

Copyright (c) 2023-2026 Masayuki Yamai

New Features

TWSNMP FK is continuously enhanced. v1.30.0+ added several key features and improvements.

SNMPv3 Security (v1.33.0)

Added support for SHA256/AES128 and SHA512/AES256, providing stronger security for modern network monitoring requirements.

Map and UI Improvements (v1.33.0)

- **Node IP Display**: Option to show node IP addresses directly on the map.
- **Group Drawing Items**: New group (frame/background) items for better organization.
- **VPanel Enhancements**: Zoom and port wrap controls for the virtual panel.
- **Map Aesthetics**: Removed background rectangles for unselected nodes.

Copyright (c) 2023-2026 Masayuki Yamai

Drawing Item Enhancements (v1.33.0)

Added support for opacity (transparency) and improved UI for background images.

New Features in v2.5.0

- **PKI Enhancements (EST RFC 7030 & ACME)**: RFC 7030 compliant EST server (cacerts, simpleenroll, simplereenroll), mTLS/Basic auth, node matching, ACME DNS-01, IP-01, and device attestation (Apple / YubiKey PIV).
- **Local LLM Stability & Preset Models**: tensai v0.0.35 update, ARM64 NEON optimization, Qwen2.5/Qwen3/LLM-jp/Gemma/SmolLM/Phi presets, and GGUF tokenizer inference fixes.
- **Revamped IPAM Report (Hierarchical Drill-Down)**: Relative density heatmap (0–100%), /24 subnet block aggregation with host-level drill-down, subnet list, and inline host search.
- **Dynamic Polling Editor UI**: Dynamic mode dropdown by monitoring type, context-

Copyright (c) 2023-2026 Masayuki Yamai

PKI Enhancements: EST & Device Attestation (v2.5.0)

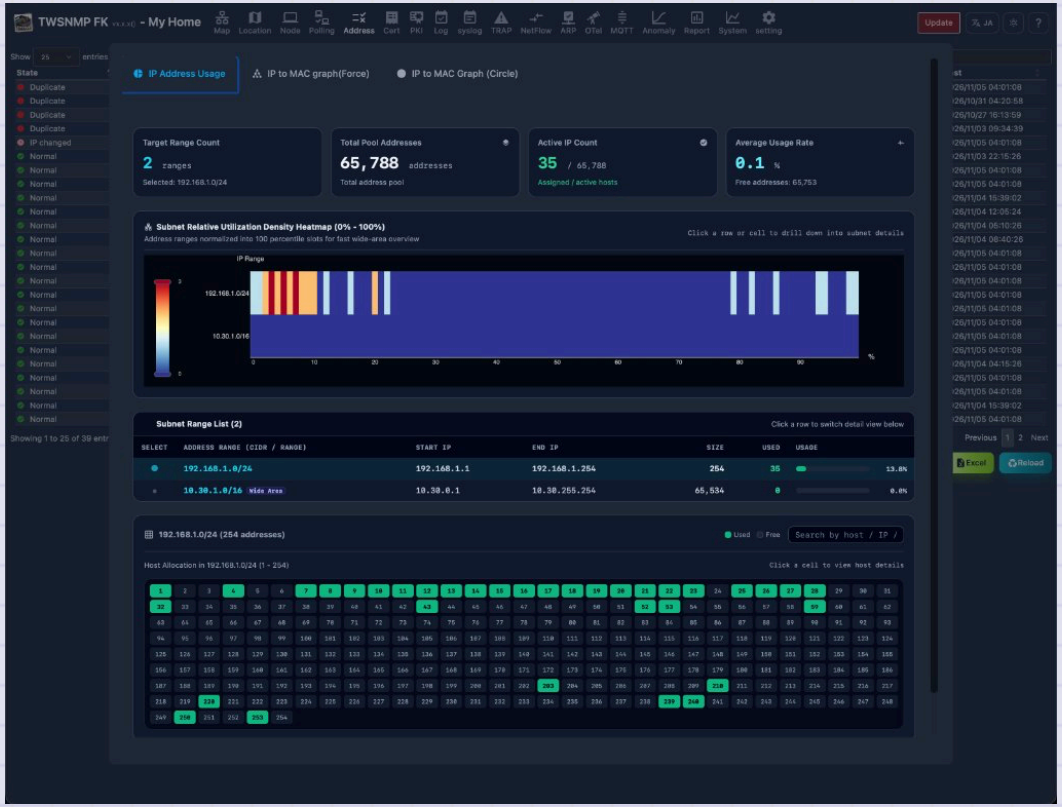
- **EST (RFC 7030) Server Integration**:
  - Automatically hosts EST services on the ACME HTTPS port
  - Serves `cacerts`, `simpleenroll`, and `simplereenroll` endpoints
  - Enforces mTLS or registered node credential HTTP Basic authentication
  - Strictly validates CSR SAN/Subject CN against registered map nodes
- **Extended ACME Challenges**:
  - `DNS-01` challenge (supports wildcard `*.example.com` certificates)
  - `IP-01` challenge (IPv4/IPv6 via `http-01` / `tls-alpn-01`)
- **Device Attestation**:
  - Apple Managed Device Attestation (iOS 16+/macOS 14+)
  - YubiKey PIV / Smallstep Attestation
  - Embeds RFC 4043 PermanentIdentifier in SAN extensions linked to nodes

Copyright (c) 2023-2026 Masayuki Yamai

IP Address Management (IPAM) Drill-Down (v2.5.0)

- **Subnet Relative Density Heatmap**:
  - Aggregates the entire address pool into 100 buckets for high-speed wide-area overview
- **Hierarchical Drill-Down**:
  - Aggregates wide address spaces into /24 subnet blocks
  - Click blocks to drill down into 1–254 individual host allocation statuses
- **Subnet Table & Inline Search**:
  - Summarizes address count, active count, and utilization percentage per subnet
  - Real-time search and highlighting by hostname, IP address, or MAC address

Copyright (c) 2023-2026 Masayuki Yamai



Copyright (c) 2023-2026 Masayuki Yamai

Usability, Polling Editor & Stability (v2.5.0)

- **Dynamic Polling Guidance**:
  - Dynamically populates available mode choices based on selected monitoring type
  - Displays contextual parameter examples, hints, and usable script variables
- **MIB Tree & gNMI Improvements**:
  - Single-click selection, preserved expansion state, and hover tooltips in MIB Tree
  - `Cmd/Ctrl + Enter` shortcut execution
- **macOS Tahoe WebKit Deadlock Fix & Performance**:
  - Content-layer reloading to eliminate WebKit/CVDisplayLink hangs; reinforced polling loop
  - Optimized log store format check to remove settings dialog opening latency
  - Fixed security vulnerabilities in dependencies

Copyright (c) 2023-2026 Masayuki Yamai

New Features in v2.4.0

- **Sigma Threat Detection & Compliance Auditing**: 75 embedded Sigma rule packs for Syslog threat detection (severity, tags, timeline, AI explanation), Sigma polling monitoring, and custom rule auto-loading.
- **Embedded Local LLM (tensai)**: Fully offline local LLM execution without external APIs, accompanied by a Local Model & WebGPU library management dialog.
- **Apache Parquet Log Storage & Migration**: Columnar Parquet format providing high compression, rapid querying, partition-based daily cleanup, and safe bbolt database migration.
- **twFathom-style KPI Cards & Redesigned Drawing Items**: Sparkline KPI cards, live editor previews, and one-click recommended size buttons.
- **Topology Discovery, Auto Line Connection & Layouts**: LLDP/CDP/STP/FDB/ARP topology discovery, auto line connection, right-click canvas layout optimization & Undo, and multi-credential SNMP discovery.

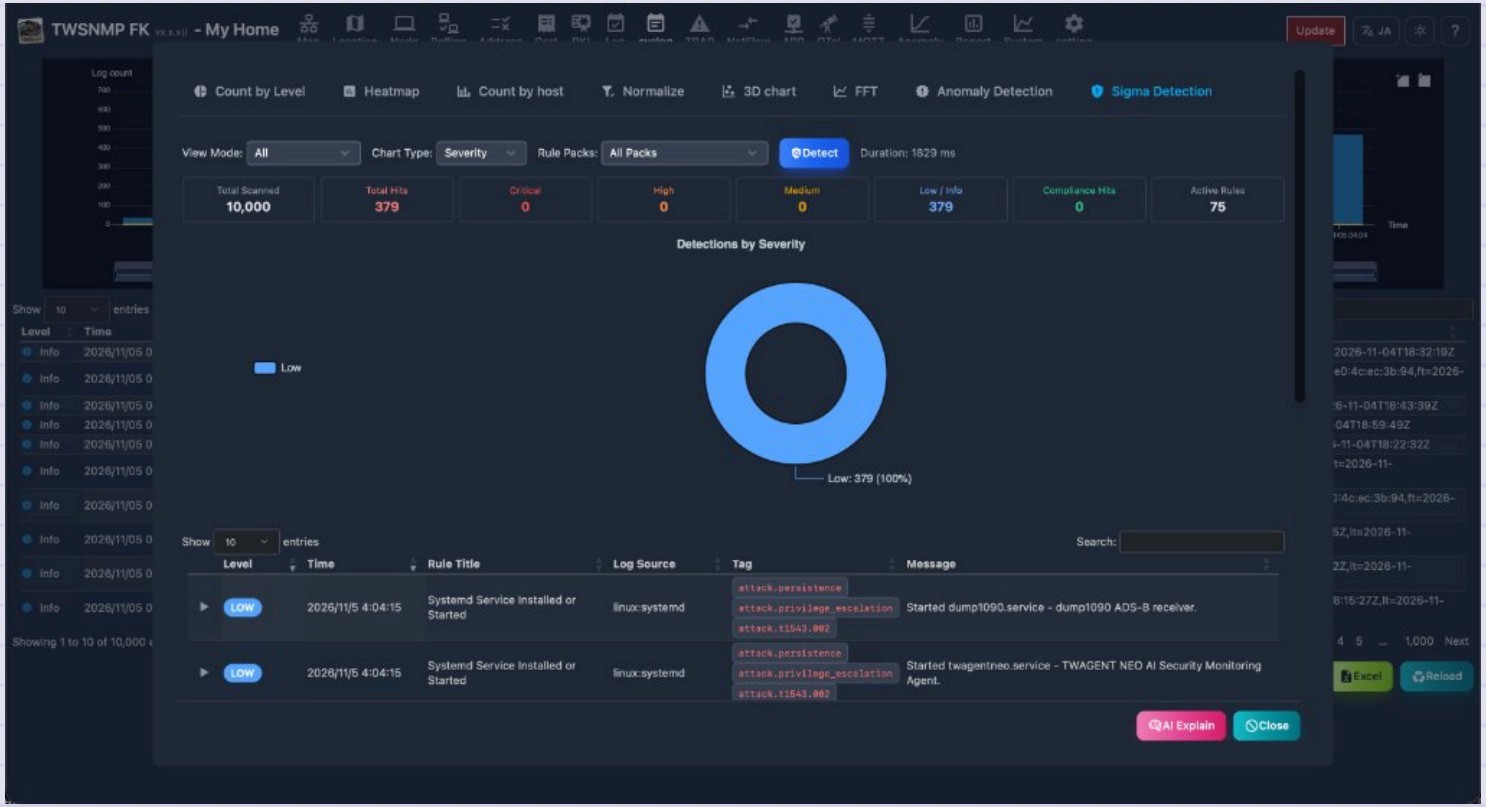
Copyright (c) 2023-2026 Masayuki Yamai

Threat Detection with Sigma Rules (v2.4.0)

Automatically detects cyber attacks and compliance violations within Syslog streams.

- **Syslog Report Sigma Detection Tab**:
  - Severity breakdown counts (Critical, High, Medium, Low)
  - Top detected rules and tags ranking
  - Detection timeline trend charts
  - Detailed per-rule detection log tables
  - AI (LLM) threat triage and remediation guidance
- **Sigma Polling Monitor**:
  - Monitors incoming log streams in real time
  - Predefined template "Syslog Threat Detection (Sigma)"
- **Custom Rule Deployment**:
  - Automatically loads custom Sigma YAML rules placed in `datastore/sigma`

Copyright (c) 2023-2026 Masayuki Yamai

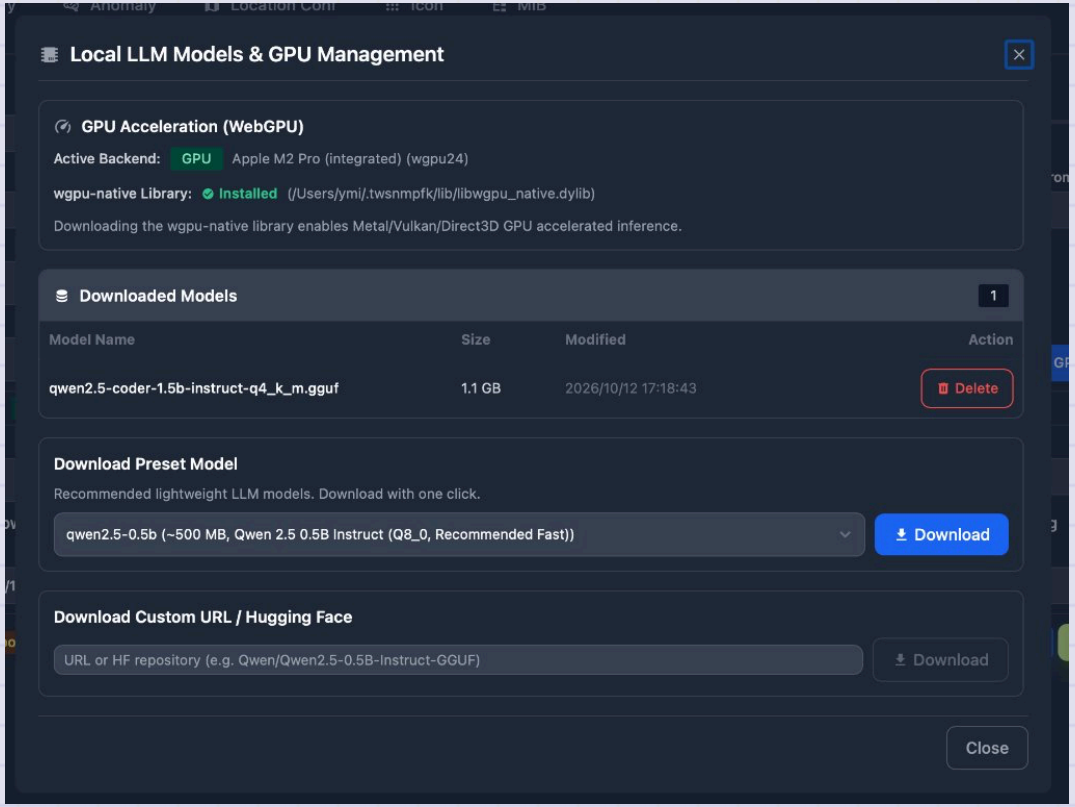


Copyright (c) 2023-2026 Masayuki Yamai

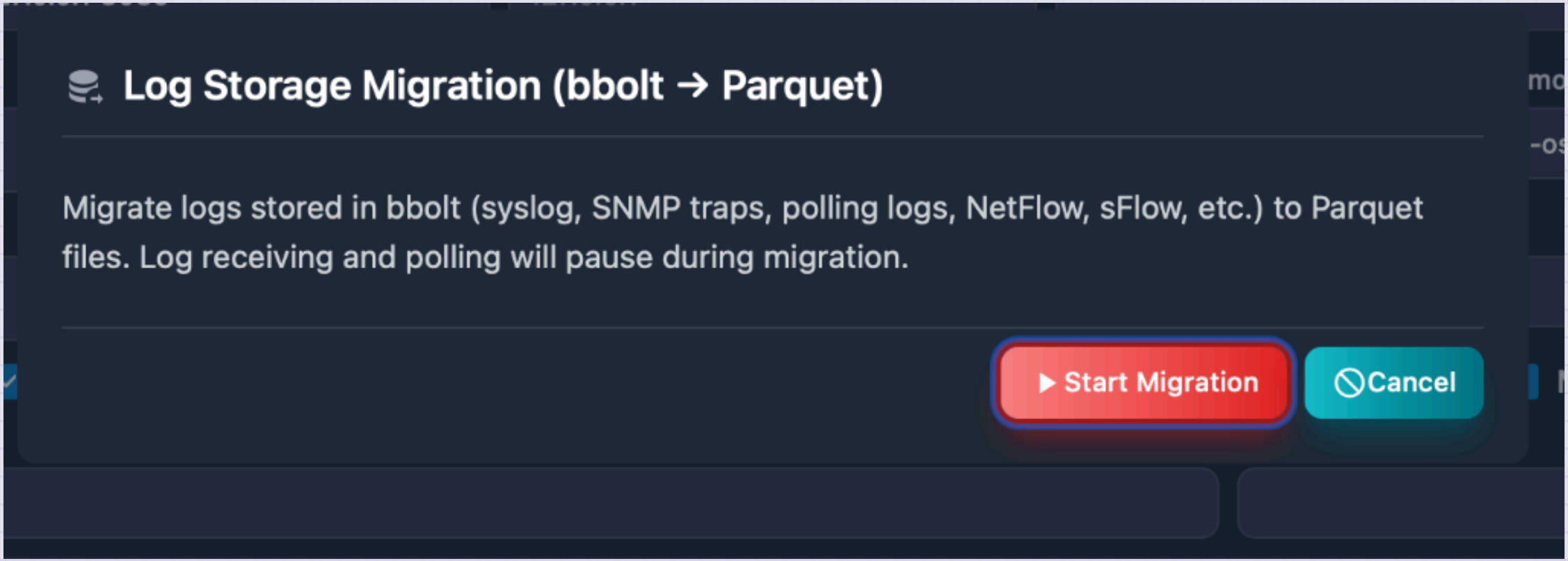
Local LLM & Apache Parquet Storage (v2.4.0)

- **Local LLM Engine (tensai)**:
  - Executes LLM models locally on your computer without external cloud APIs or daemons
  - "Local Model Manager" dialog for downloading and switching GGUF models
  - Automatic native WebGPU library management for hardware acceleration
- **Apache Parquet Log Storage**:
  - Stores Syslog, SNMP Trap, NetFlow, sFlow, ARP, and polling logs in columnar Parquet
  - Superior compression ratios and faster search performance compared to bbolt
  - High-speed folder deletion for logs past retention thresholds
  - Safe, automated batch migration tool from existing bbolt databases

Copyright (c) 2023-2026 Masayuki Yamai



Copyright (c) 2023-2026 Masayuki Yamai

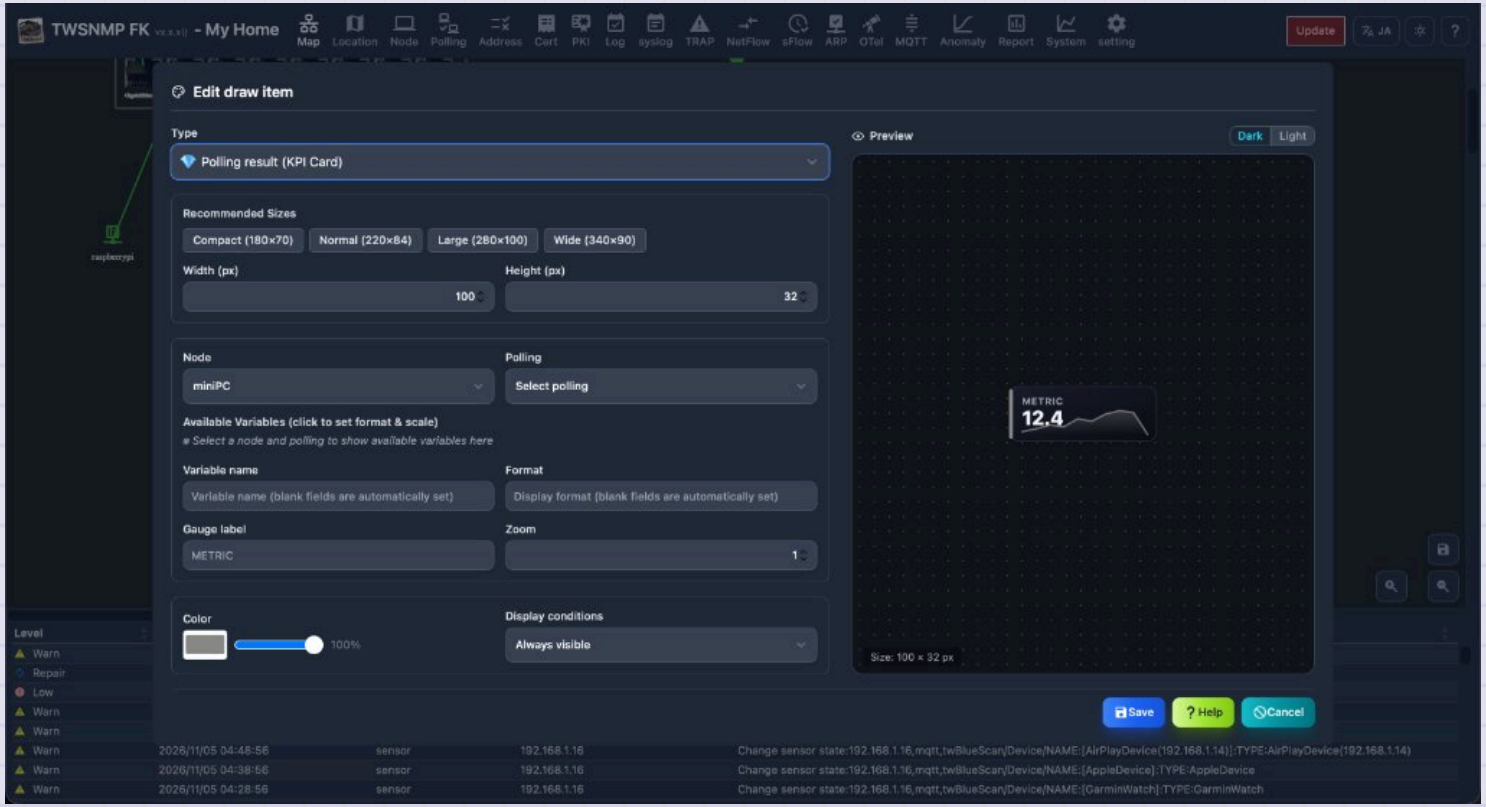


Copyright (c) 2023-2026 Masayuki Yamai

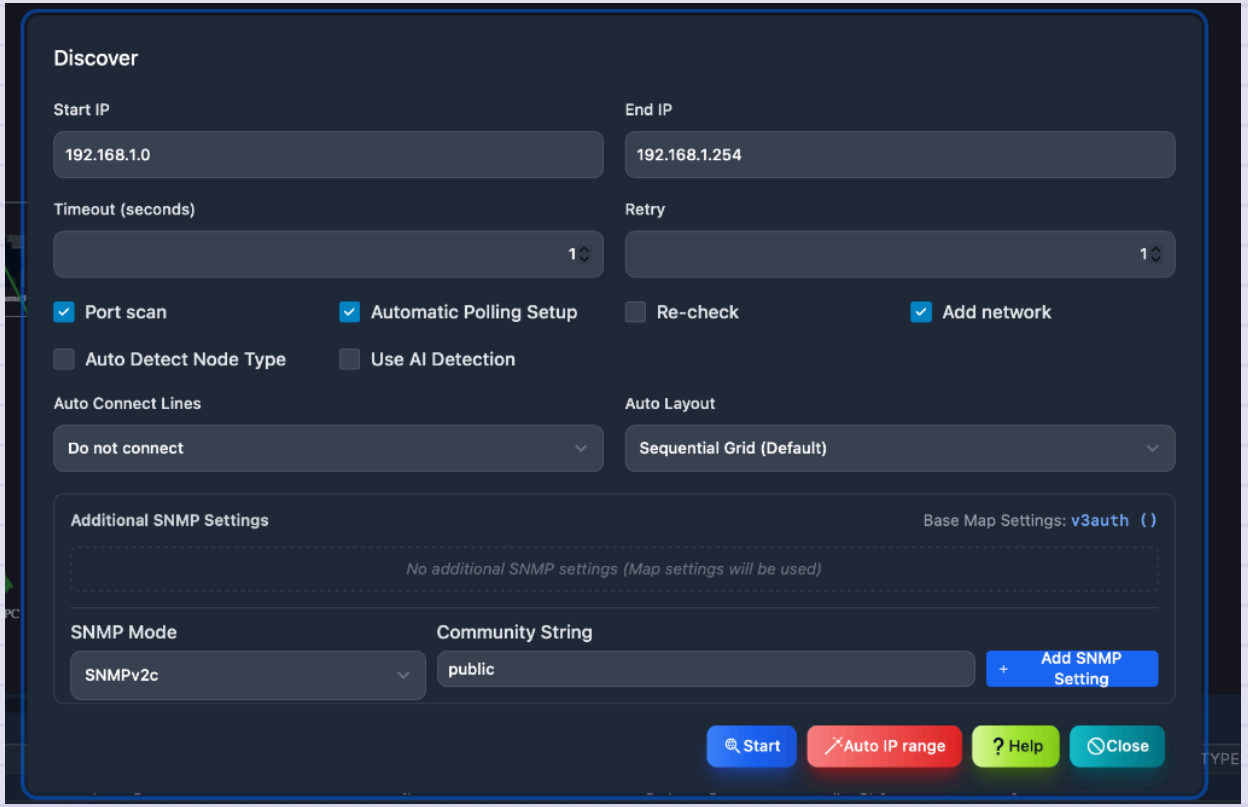
KPI Cards & Topology Enhancements (v2.4.0)

- **twFathom-style KPI Card Drawing Items**:
  - Real-time sparklines, status accent bars, and prominent metric values
  - Drawing item editor features live preview, recommended size buttons, and presets
- **Topology Discovery & Auto Line Connection**:
  - Multi-protocol topology engine (LLDP, CDP, STP, FDB, ARP)
  - Auto line connection options during discovery (strict or speculative)
  - "Find Neighbors" dialog displays confidence levels, reasoning, and AI inference
- **Map Layout Optimization & Undo**:
  - Automatic layout application during discovery (Tree, Cluster, Categorized)
  - Canvas right-click menu for layout optimization and Undo

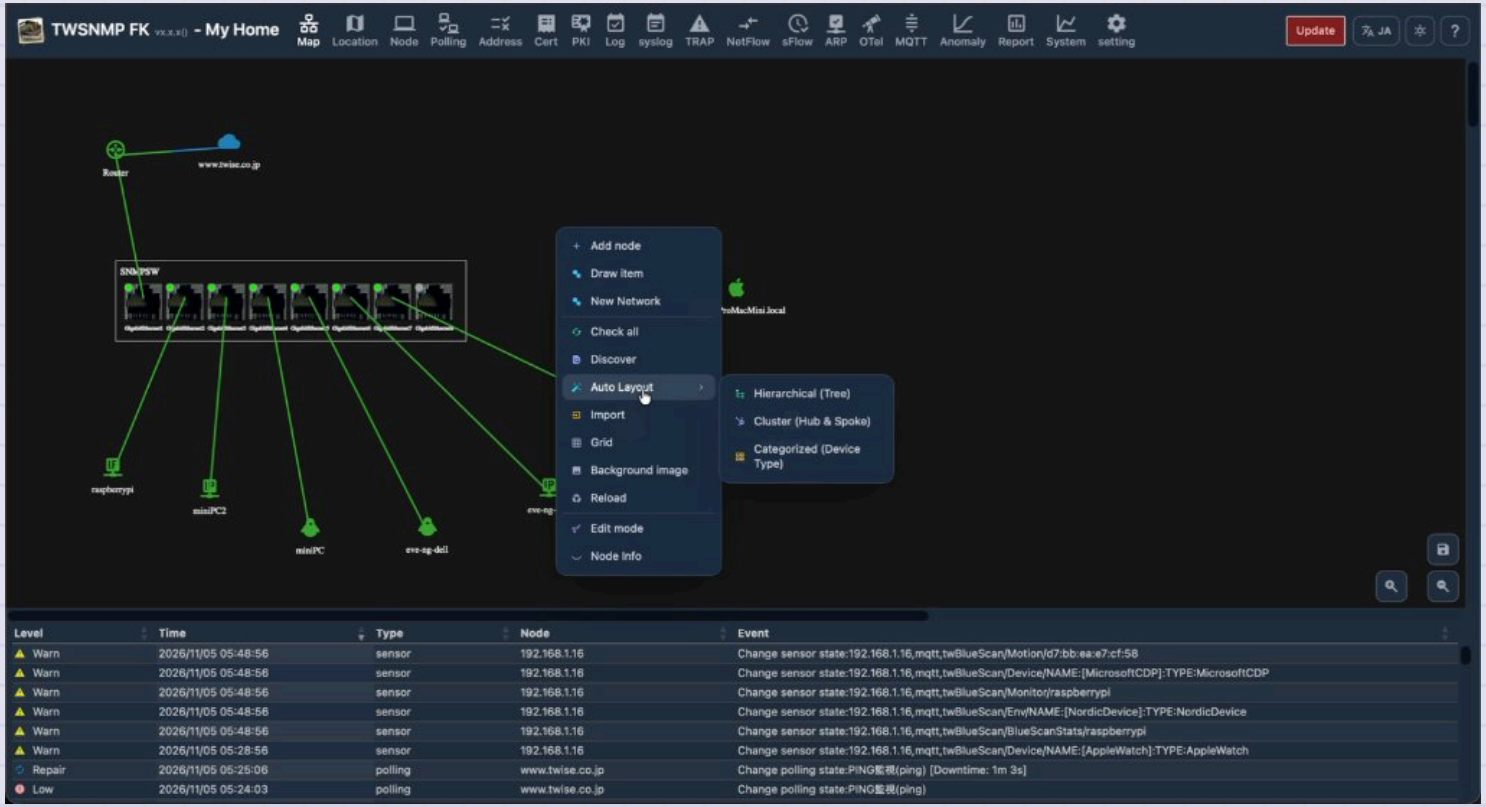
Copyright (c) 2023-2026 Masayuki Yamai



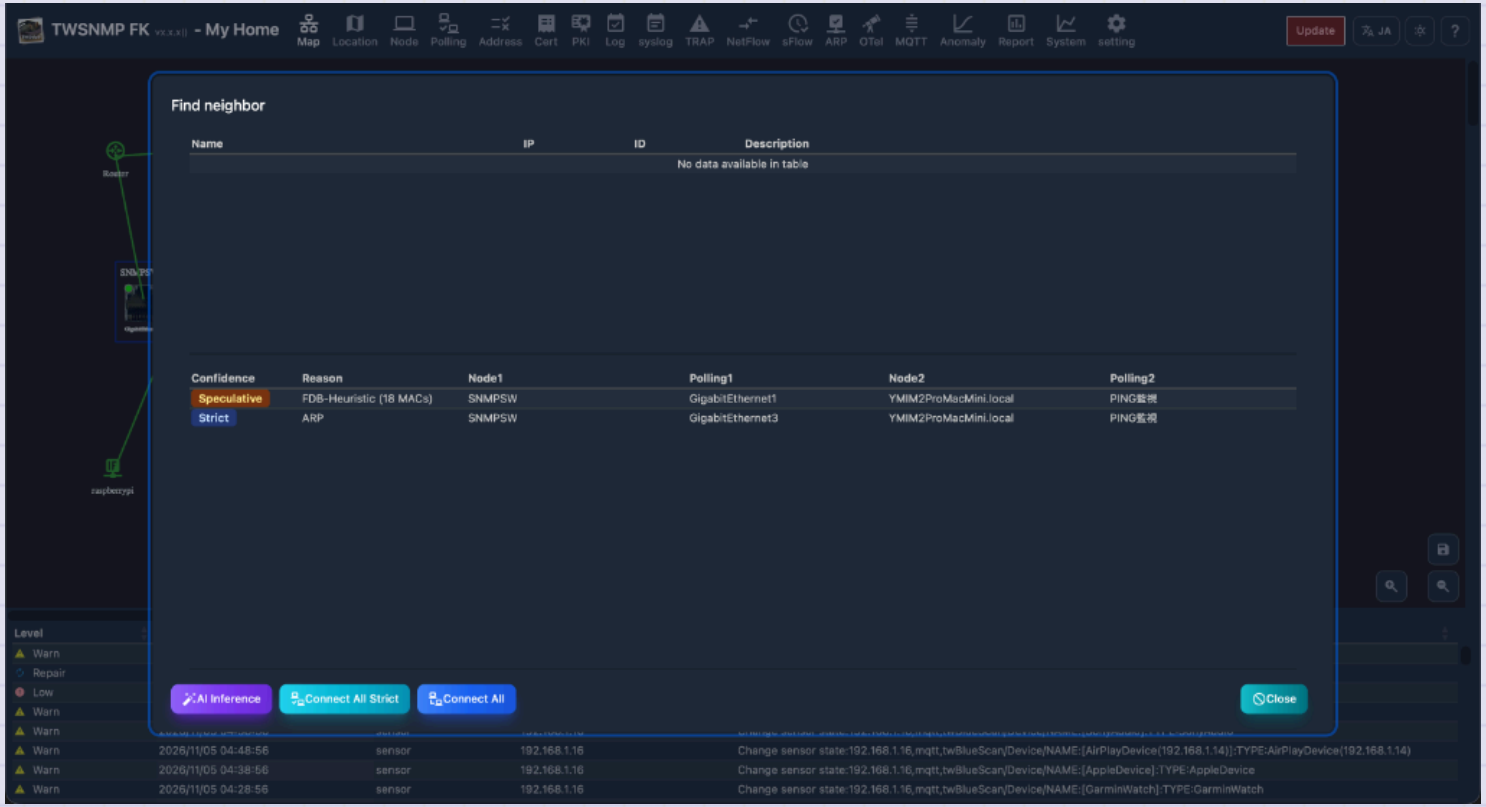
Copyright (c) 2023-2026 Masayuki Yamai



Copyright (c) 2023-2026 Masayuki Yamai



Copyright (c) 2023-2026 Masayuki Yamai



Copyright (c) 2023-2026 Masayuki Yamai

New Features in v2.3.0

- **STUN Global IP Inspection & Monitoring**: Added a `[STUN]` button to Address Management to inspect external global IP, mapped port, reverse DNS, local address, RTT, and GeoIP via STUN (IPv4/IPv6, copy/maps/VirusTotal links). Also added a native `stun` polling type with templates and AI assist support to monitor public IP modifications.
- **PING Diagnostics Expansion (Smokeping, MTR, AI Explanation)**: Added mode switching for Normal Ping, Smoke (high-frequency jitter & packet loss measurement), Traceroute, and MTR (My Traceroute with continuous per-hop sampling), with Smokeping gradation chart, MTR Hop Flow & stats, and AI explanation support.
- **Smokeping Polling Monitor**: Added `smoke` mode to PING polling with configurable continuous burst ping count, rich evaluation variables, templates, and AI assist support.

Copyright (c) 2023-2026 Masayuki Yamai

AI (LLM) Integration (v1.32.0)

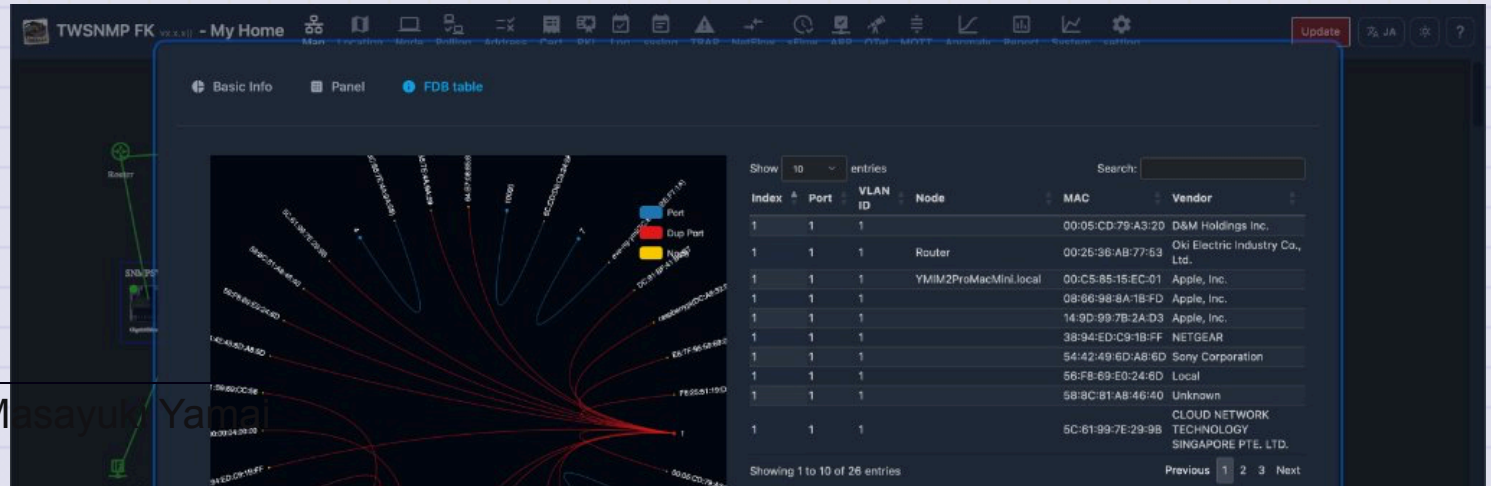
- **MIB Browser**: Natural language search and AI-powered MIB object explanations.
- **Log Analysis**: AI explanations for NetFlow, Syslog, and SNMP Trap logs.
- **Report Summarization**: AI-driven summaries for periodic reports.
- **Multi-Provider Support**: Supports Gemini, OpenAI, Claude, and Ollama.

Copyright (c) 2023-2026 Masayuki Yamai

TWSNMP FK provides a "Network Report" function that visualizes MAC addresses, IP addresses, hostnames, and other information of devices connected to each port from the FDB (Forwarding Database) table information of network switches, allowing you to grasp the overall connection status of the network.

Displaying the Report

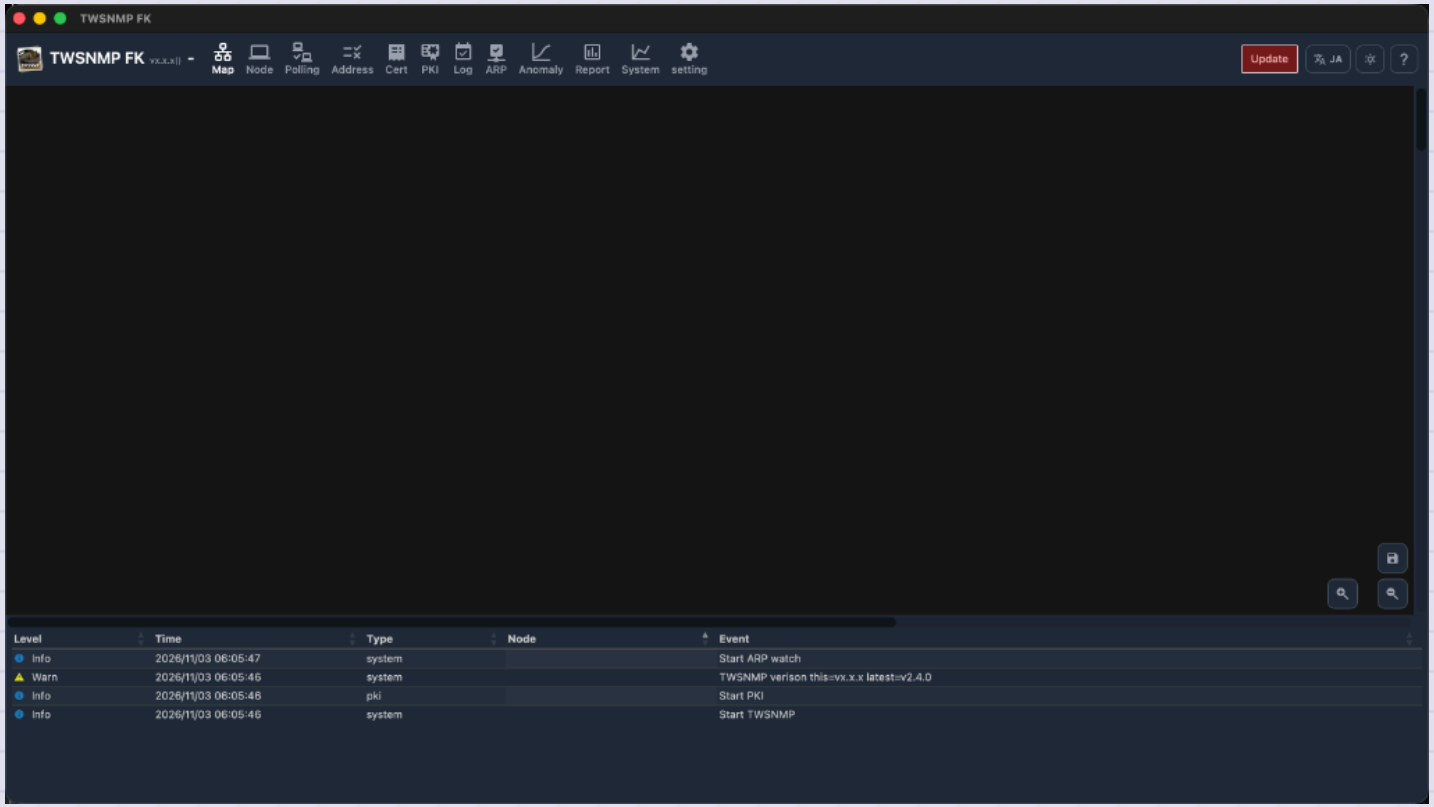
By clicking on the network icon on the map displayed on the "Network" screen, a detailed report of that network will be displayed. This report includes the FDB table, last change time of each port, and so on.



Copyright (c) 2023-2026 Masayuki Yamai

Map

The map screen has three large parts.



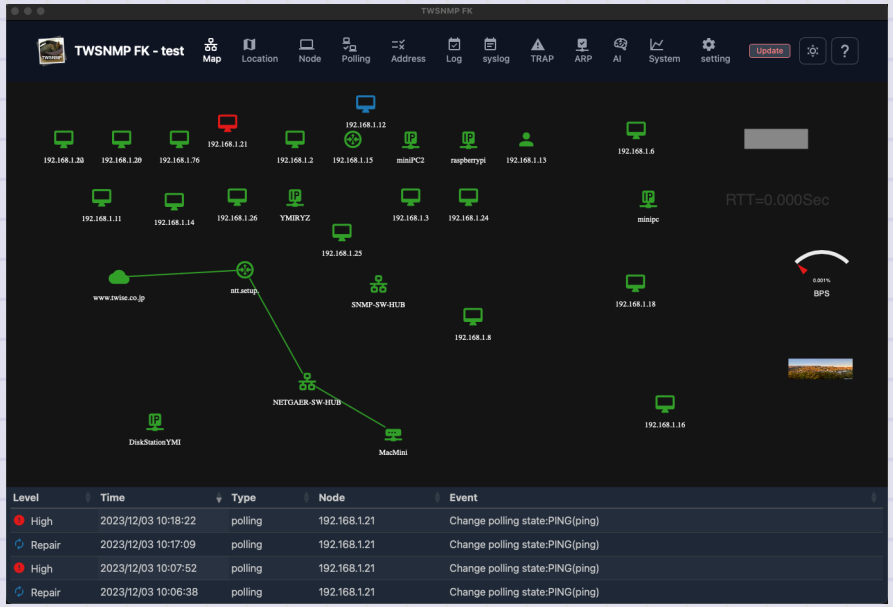
Copyright (c) 2023-2026 Masayuki Yamai

Screen	Contents
Toolbar	Switch the screen.
Map	This is the part that displays the composition of the network.
Event Log	Displays the latest 100 event logs.

Copyright (c) 2023-2026 Masayuki Yamai

Light/dark mode switching

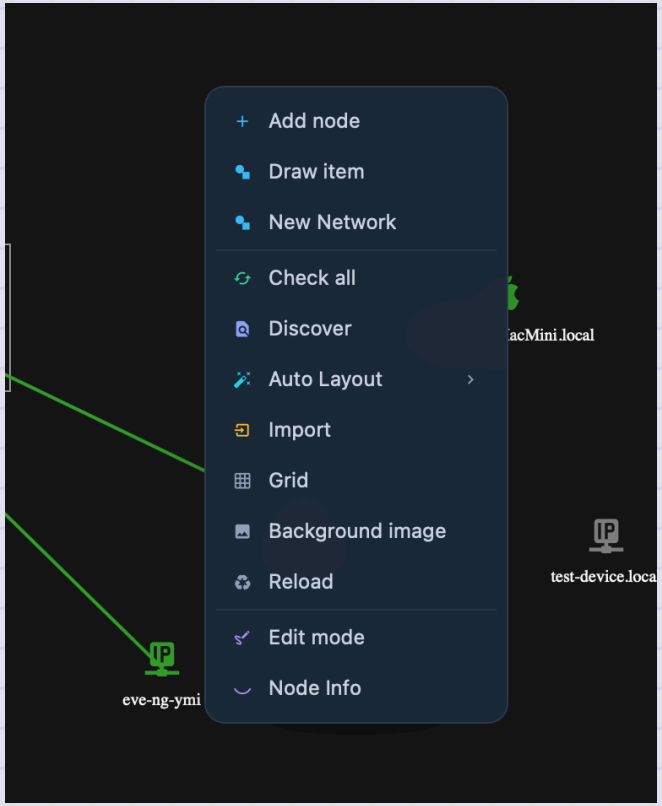
Click the 🌙 mark on the upper right to dark mode.I like dark mode.Probably the person who aims for a white hacker likes dark mode.There are only white hackers in the cat world.By Cat of the predecessor assistant.The current assistant cat seems to like both because the pattern is black and white.



Copyright (c) 2023-2026 Masayuki Yamai

Map menu

Right -click the location other than the node and drawing items on the map to display.



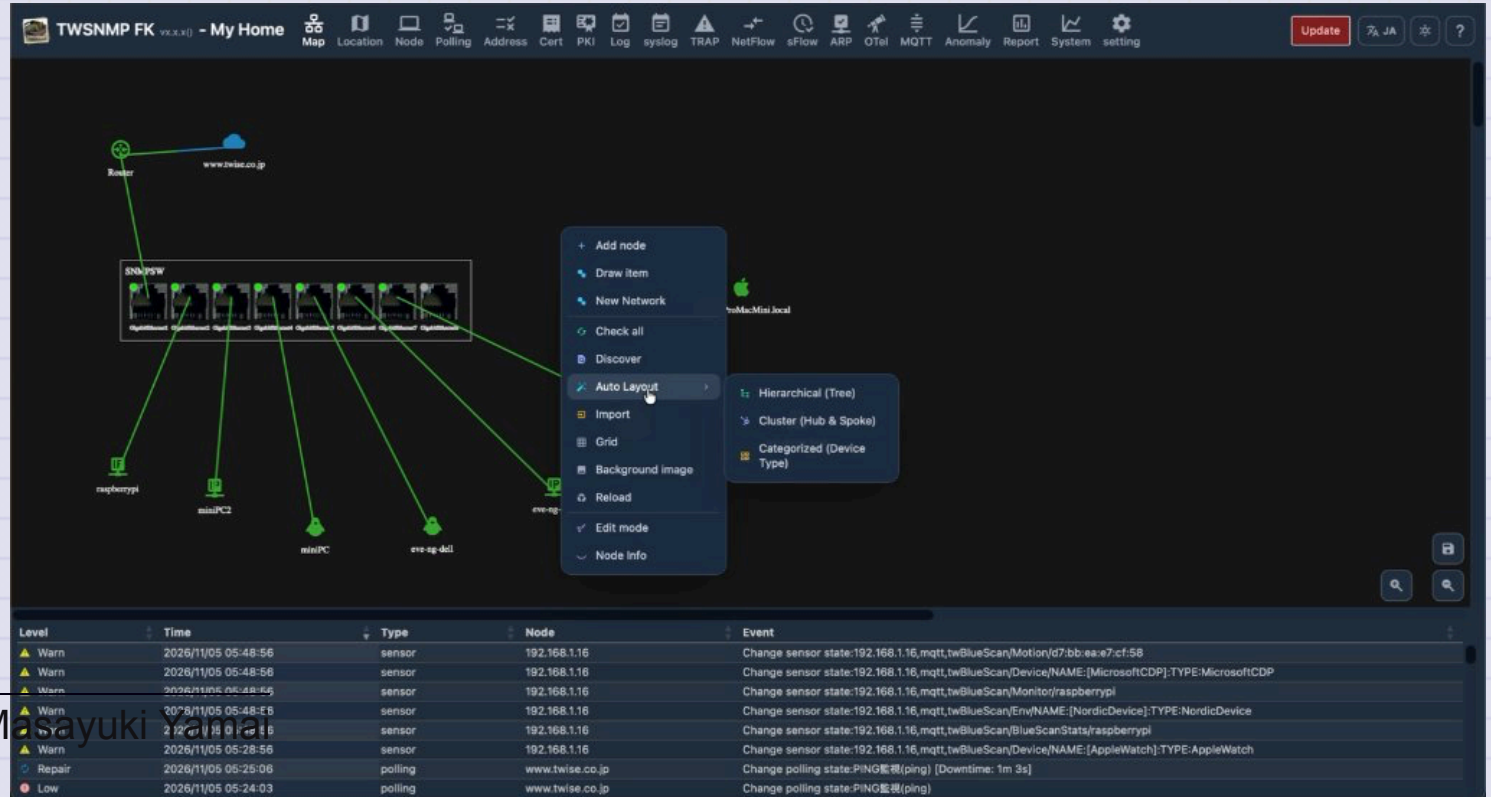
Copyright (c) 2023-2026 Masayuki Yamai

Menu	Operation
Add node	Add the node to the map manually.
Draw item	Add drawing items to the map.
New Network	Add network elements to the map.
Check all	Reconfirm the node that has occurred.
Discover	Displays the automatic discovery screen.
Import	Import map file from TWSNMP v4.x.
Export	Export map to Image (PNG, SVG, PDF), Draw.io, or Data (JSON, CSV, Excel) formats.
Grid	Align the position of the node at the specified interval.
Background image	Set background image for the map.

Copyright (c) 2023-2026 Masayuki Yamai

Auto Layout

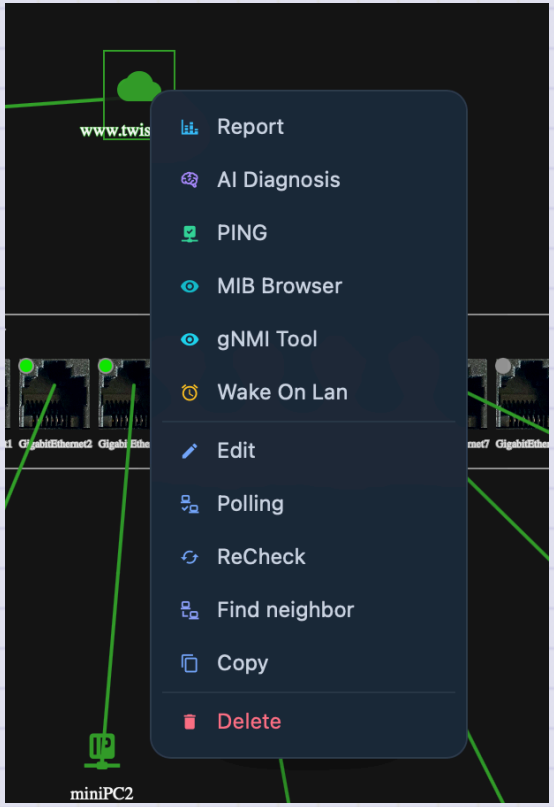
Right-click empty space on the map and select "Auto Layout" to instantly re-organize and optimize node positions based on network topology hierarchy (Hierarchical Tree), subnet clustering (Cluster / Hub & Spoke), or device categories (Categorized). Use "Undo Layout" to instantly revert to the previous layout state.



Copyright (c) 2023-2026 Masayuki Yamai

Node menu

Right -click the node on the map to display it.



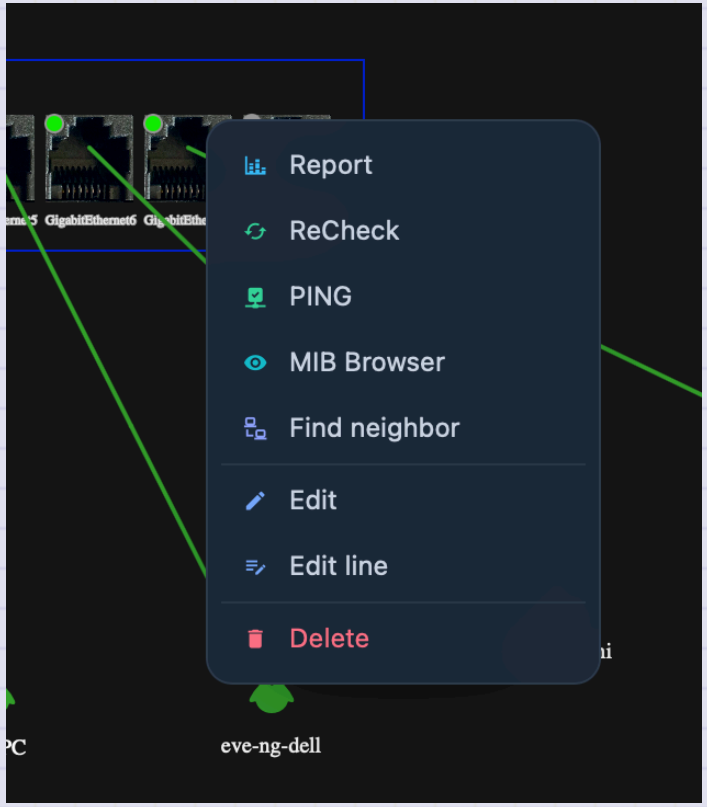
Copyright (c) 2023-2026 Masayuki Yamai

Menu	Operation
Report	Displays the report screen related to the node.
AI Diagnosis	Diagnose node status, related logs, and responses using AI (LLM).
PING	Displays the ping screen.
MIB Browser	Displays MIB browser.
gNMI Tool	Displays gNMI Tool.
Wake On Lan	Send Wake On Lan packet.
Edit	Displays the screen to edit the node settings.
Polling	Displays a polling list related to nodes.
ReCheck	Reconfirm the condition of the node by executing polling.
Copy	Create a node duplication.

Copyright (c) 2023-2026 Masayuki Yamai

Network node menu

Right-click a network node on the map to display this menu.



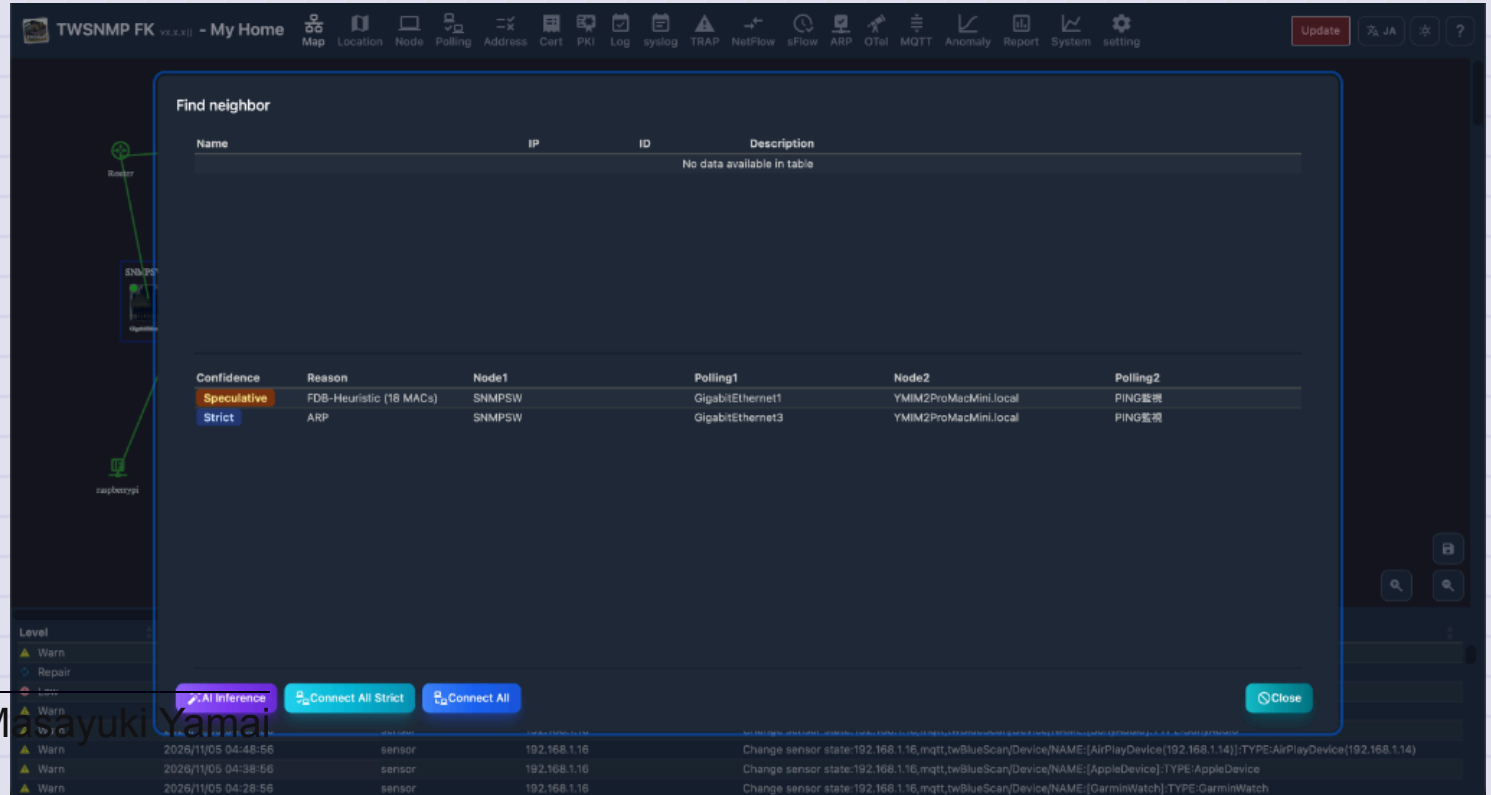
Copyright (c) 2023-2026 Masayuki Yamai

Menu	Operation
Report	Displays the network report screen.
ReCheck	Reconfirm polling for the network node.
PING	Displays the ping screen.
MIB Browser	Displays MIB browser.
Find neighbor	Search connection targets via switch FDB table.
Edit	Edit network node settings.
Edit line	Edit connection lines.
Delete	Delete network node.

Copyright (c) 2023-2026 Masayuki Yamai

Find Neighbors

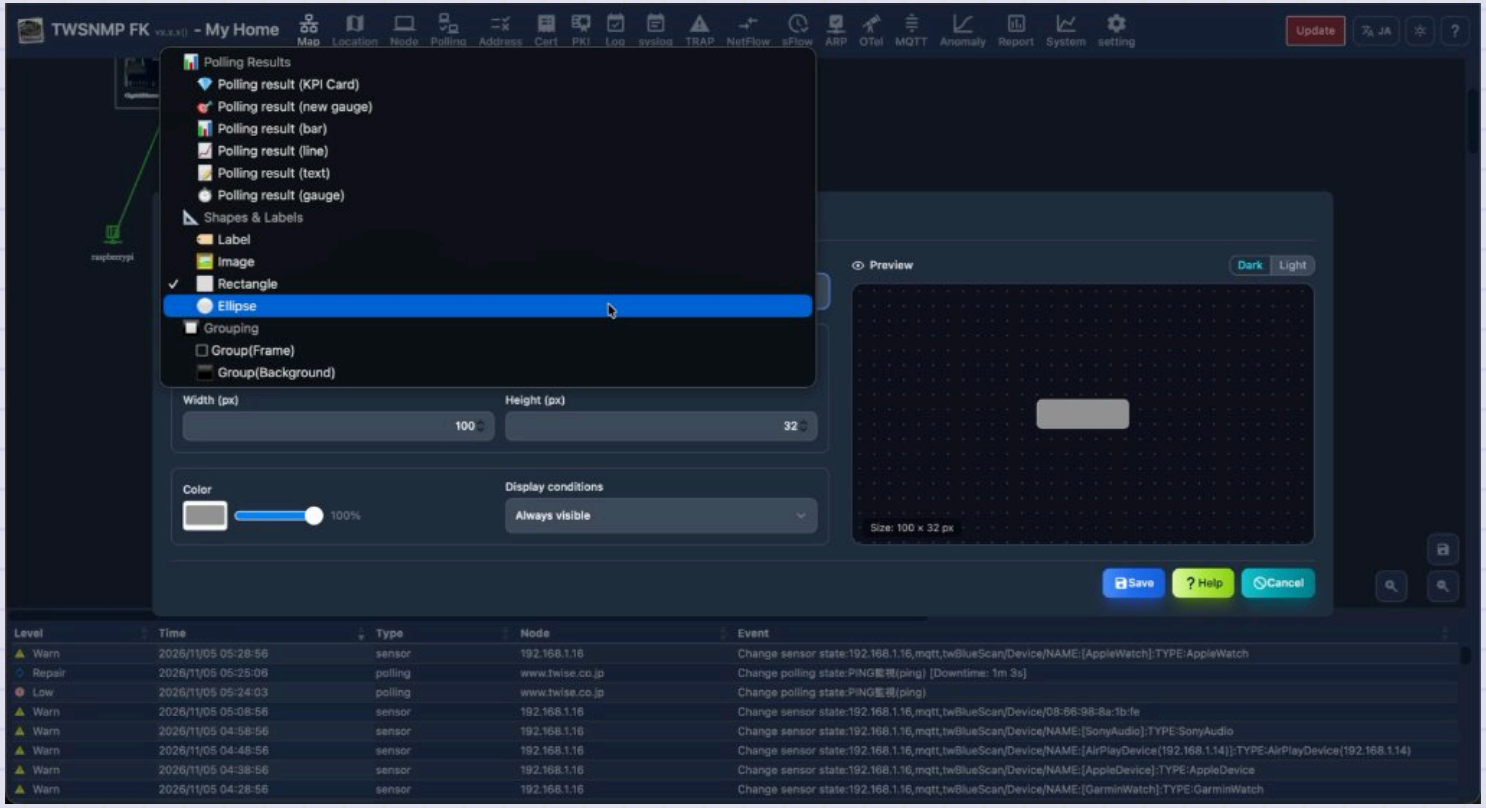
Explores topology data from LLDP, CDP, STP, FDB, and ARP tables to display candidate connections between network devices and nodes. Users can inspect confidence levels (High, Medium, Speculative) and reasoning, run AI connection inferences, and batch-connect all detected links using the "Connect All" button.



Copyright (c) 2023-2026 Masayuki Yamai

Draw item menu

Right -click the drawing item on the map to display it.



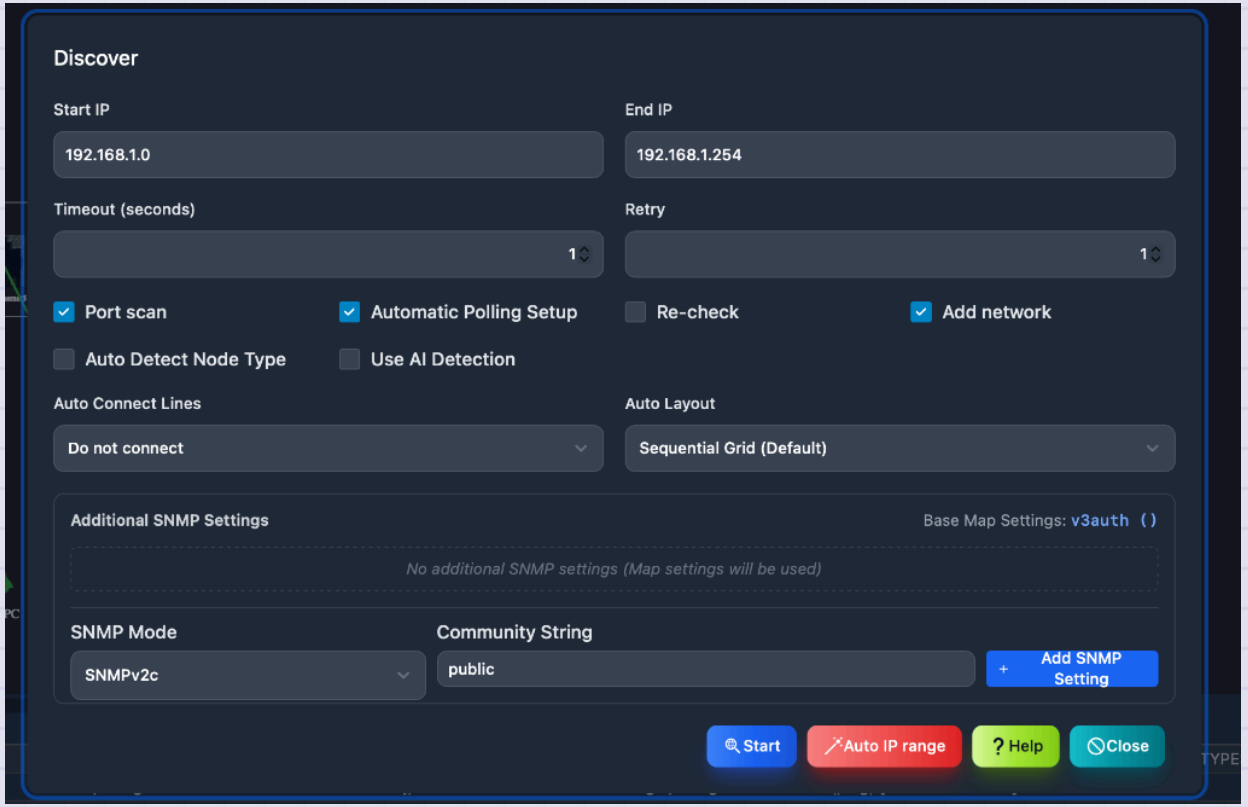
Copyright (c) 2023-2026 Masayuki Yamai

Menu	Operation
Edit	Displays the screen to edit the drawing item settings.
Copy	Create drawing items.
Delete	Delete drawing items.

Copyright (c) 2023-2026 Masayuki Yamai

Discover

Automatic discovery screen.



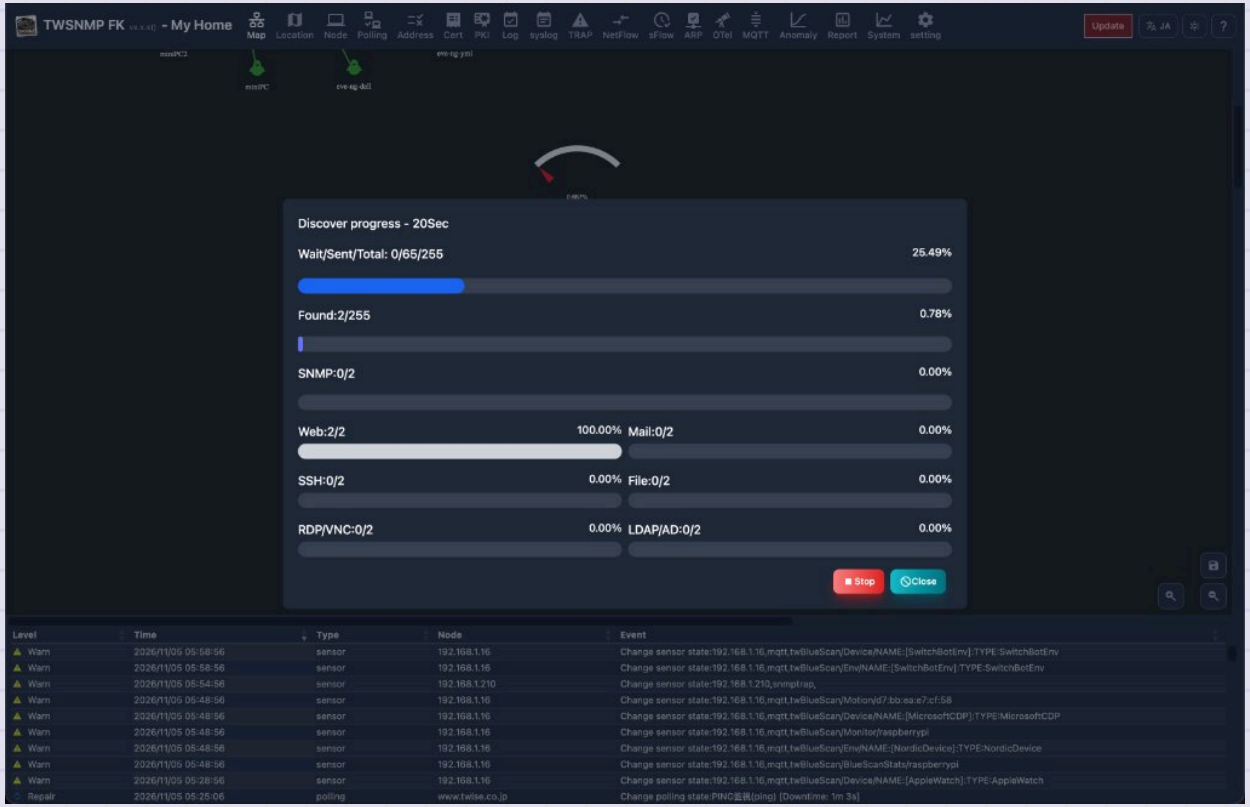
Copyright (c) 2023-2026 Masayuki Yamai

Items	Contents
Start IP	The first IP address range to search.
End IP	The end of the IP address range to search.
Timeout	This is the timeout of ping when searching.
Retry	This is the number of retries of ping when searching.
Port scan	Perform a port scan on the found node.
Automatic Polling Setup	Polling is automatically set on the found node.
Re-check	Re-scans including already registered nodes.
Add network	Automatically generate network elements from discovered subnets.
Auto Detect Node	Automatically detect node types (Server, Router, etc.)

Copyright (c) 2023-2026 Masayuki Yamai

Automatic discovery is being performed

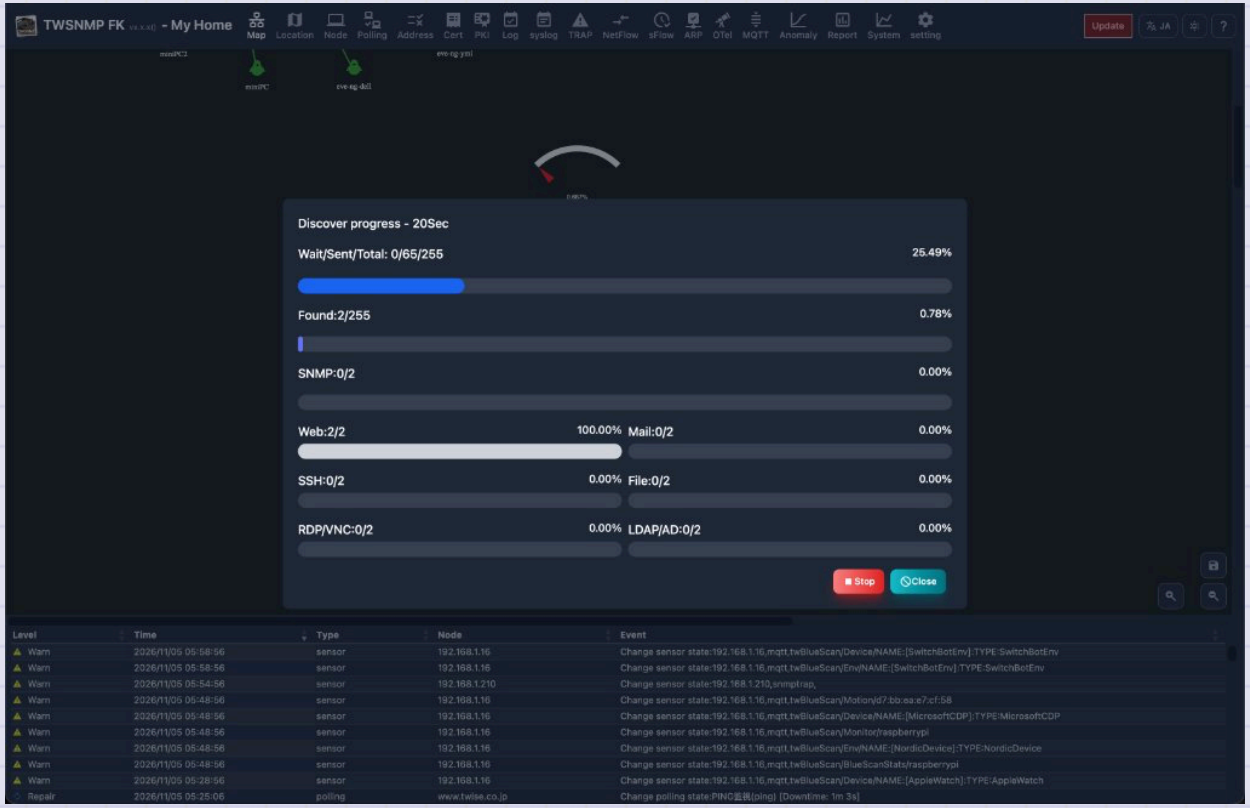
The number of nodes you have executed or discovered is displayed.



Copyright (c) 2023-2026 Masayuki Yamai

Automatic discovery is being executed (with port scanning)

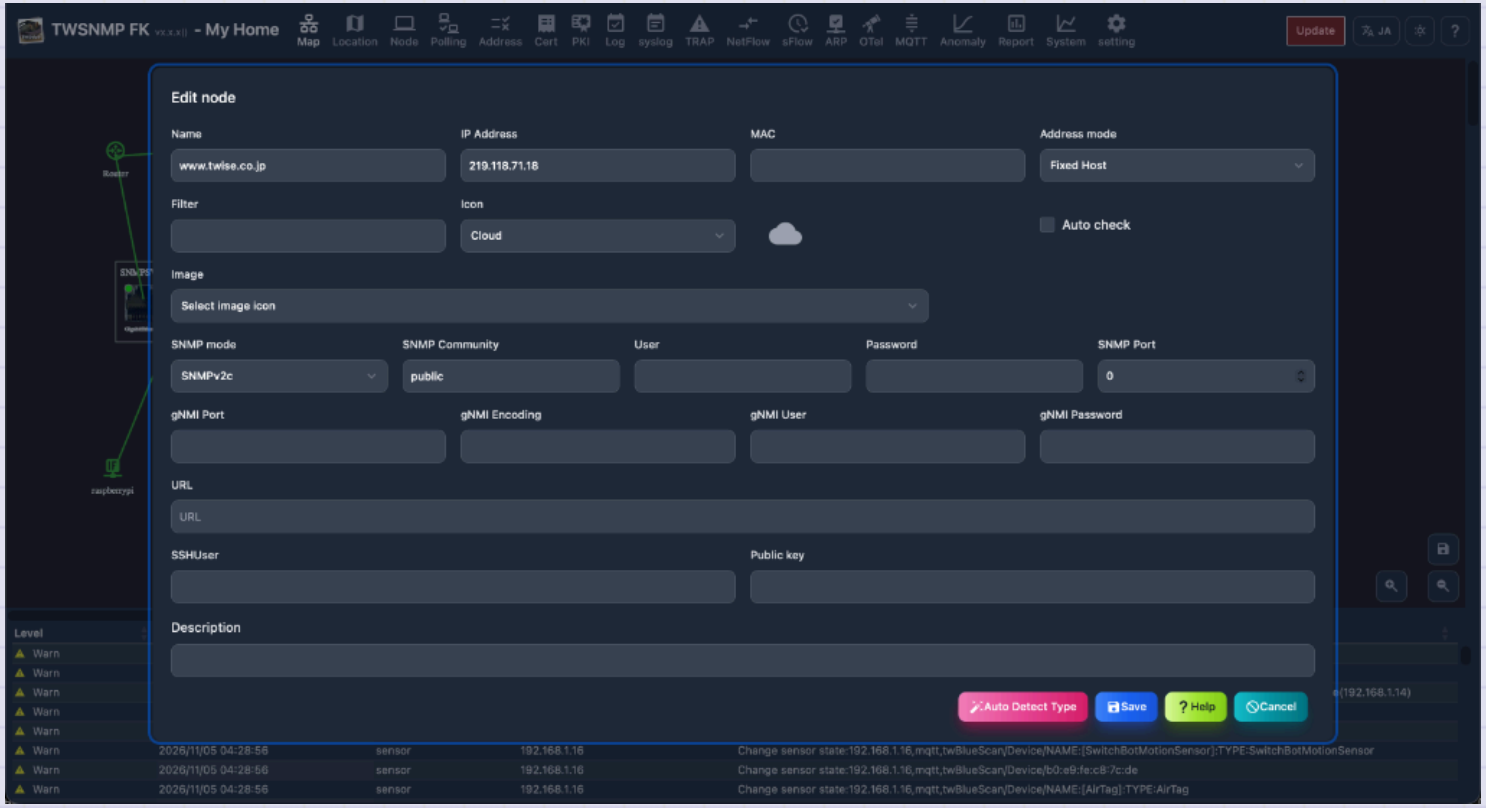
The number of nodes you have executed or discovered is displayed.When performing a port scan, the discovered server function is also displayed.



Copyright (c) 2023-2026 Masayuki Yamai

Node editing

You can edit the node from the menu or button by selecting a node on the map screen or node list.



Copyright (c) 2023-2026 Masayuki Yamai

Items	Contents
Name	Node name.
IP address	Node IP address.
Address mode	IP address fixation (default), MAC address fixing, host name fixed.
Icon	It is an icon to be displayed.
Auto recheck	When it is returned, it will be automatically normal.
SNMP mode	SNMP mode.There are SNMPv1, V2C, V3 (authentication and encryption).

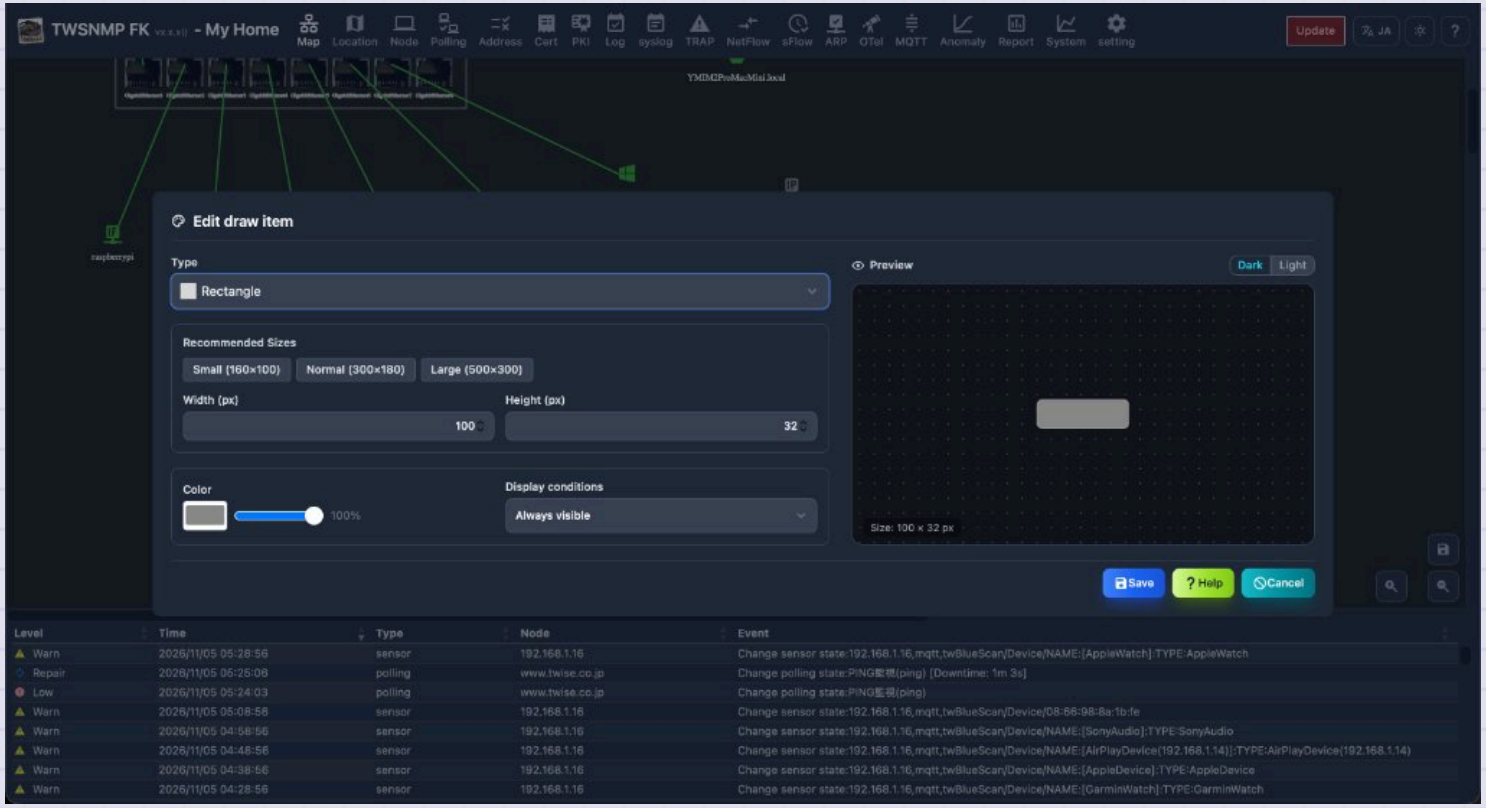
Copyright (c) 2023-2026 Masayuki Yamai

Items	Contents
SNMP Community	Community name for SNMPV1, V2C.
User	User ID when accessing with SNMPv3.
Password	Password when accessing with SNMPv3.
Public key	This is the public key of the node when polling with SSH. In the case of blank, automatically set at the first connection.
URL	URL when accessing with browser etc. It will be displayed on the right -click menu. You can specify multiple by separation of comma.
Description	Supplementary information is described.

Copyright (c) 2023-2026 Masayuki Yamai

Drawing item (rectangle, elliptical)

It is an edit screen of drawing item (rectangle, elliptical).



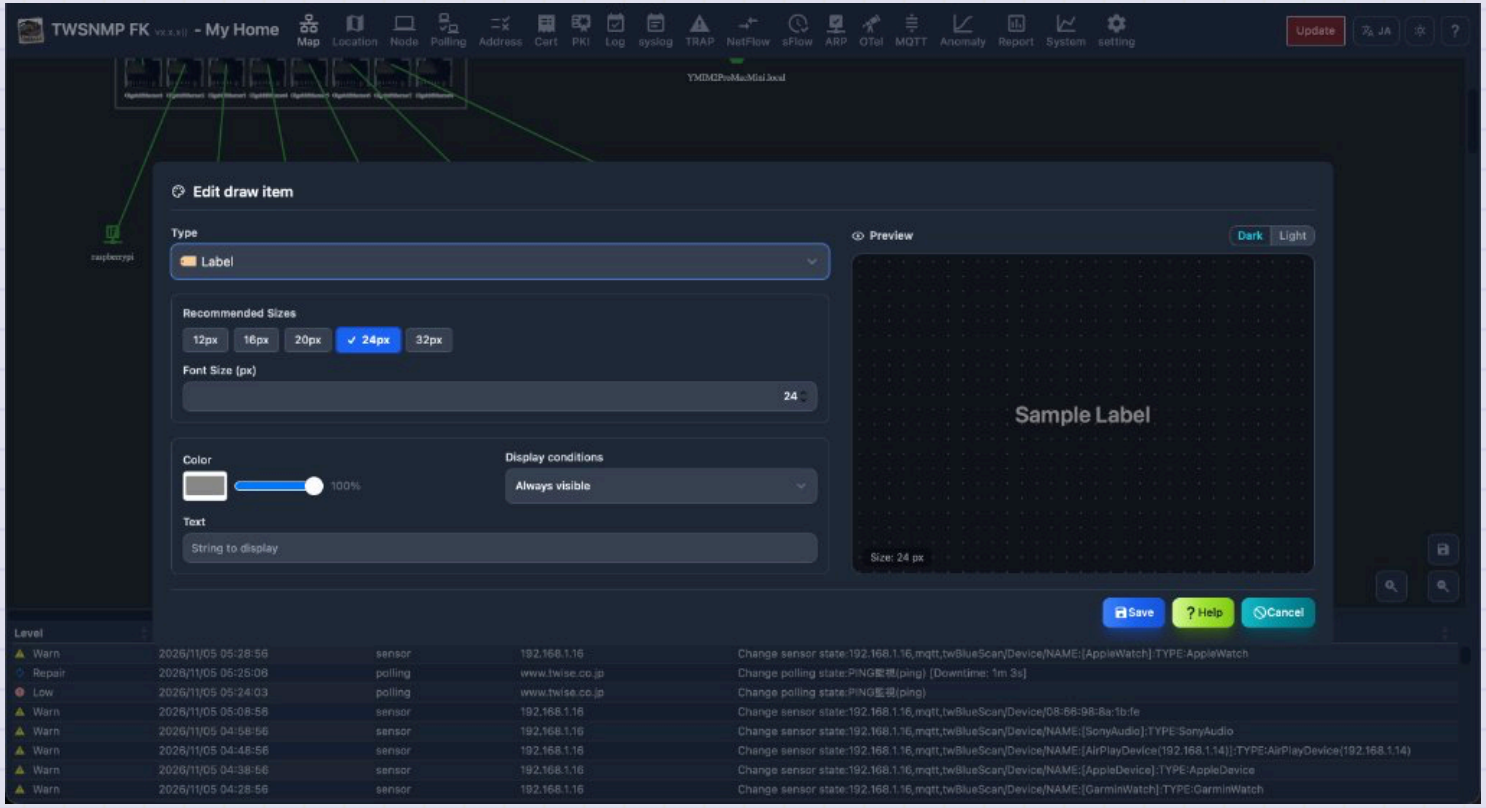
Copyright (c) 2023-2026 Masayuki Yamai

Items	Contents
Type	It is a type of drawing item.You can only change it when you add it.
Width	The width of the drawing item.
Height	It is the height of the drawing item.
Color	It is the color of the drawing item.
Display condition	It is a state of the map that displays drawing items.
Magnification	The display rate of drawing items.

Copyright (c) 2023-2026 Masayuki Yamai

Drawing item (label)

It is the editing screen of the drawing item (label).



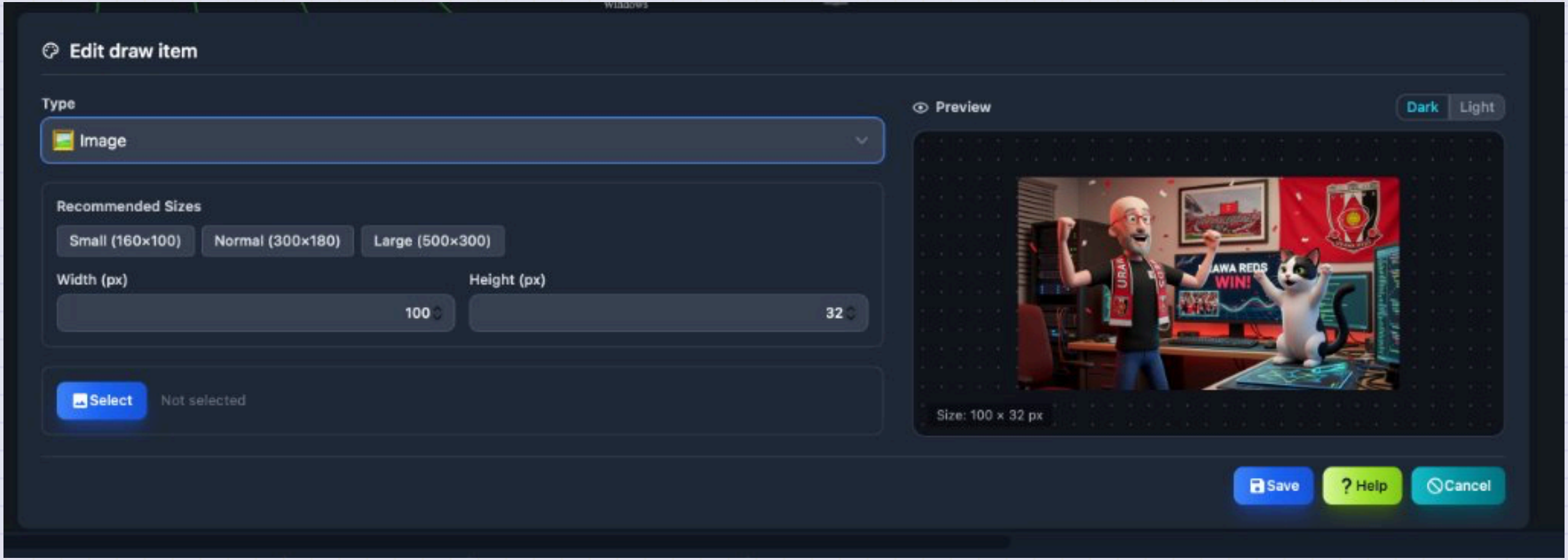
Copyright (c) 2023-2026 Masayuki Yamai

Items	Contents
Type	It is a type of drawing item.You can only change it when you add it.
Character size	Label character size.
Color	It is the color of the drawing item.
Display condition	It is a state of the map that displays drawing items.
Character string	It is a string to be displayed.
Magnification	The display rate of drawing items.

Copyright (c) 2023-2026 Masayuki Yamai

Drawing item (image)

It is the editing screen of drawing item (image).



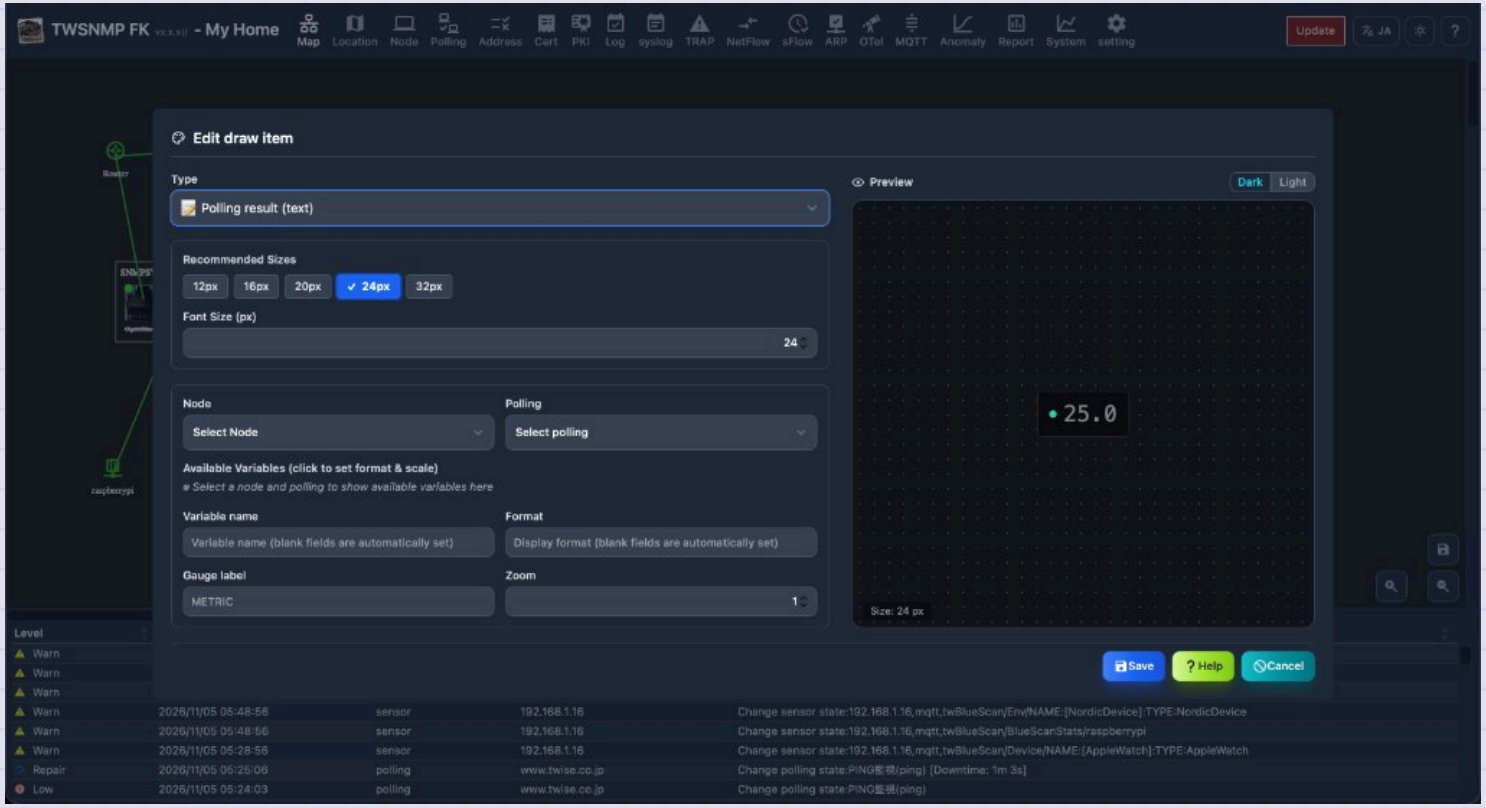
Copyright (c) 2023-2026 Masayuki Yamai

Items	Contents
Type	It is a type of drawing item.You can only change it when you add it.
Width	It is the width of the image.
Height	It is the height of the image.
Display condition	It is a state of the map that displays drawing items.
Image	It is an image to be displayed.Select an image file with the <Select> button.
Magnification	The display rate of drawing items.

Copyright (c) 2023-2026 Masayuki Yamai

Drawing item (polling result)

It is the editing screen of drawing item (polling result: text).



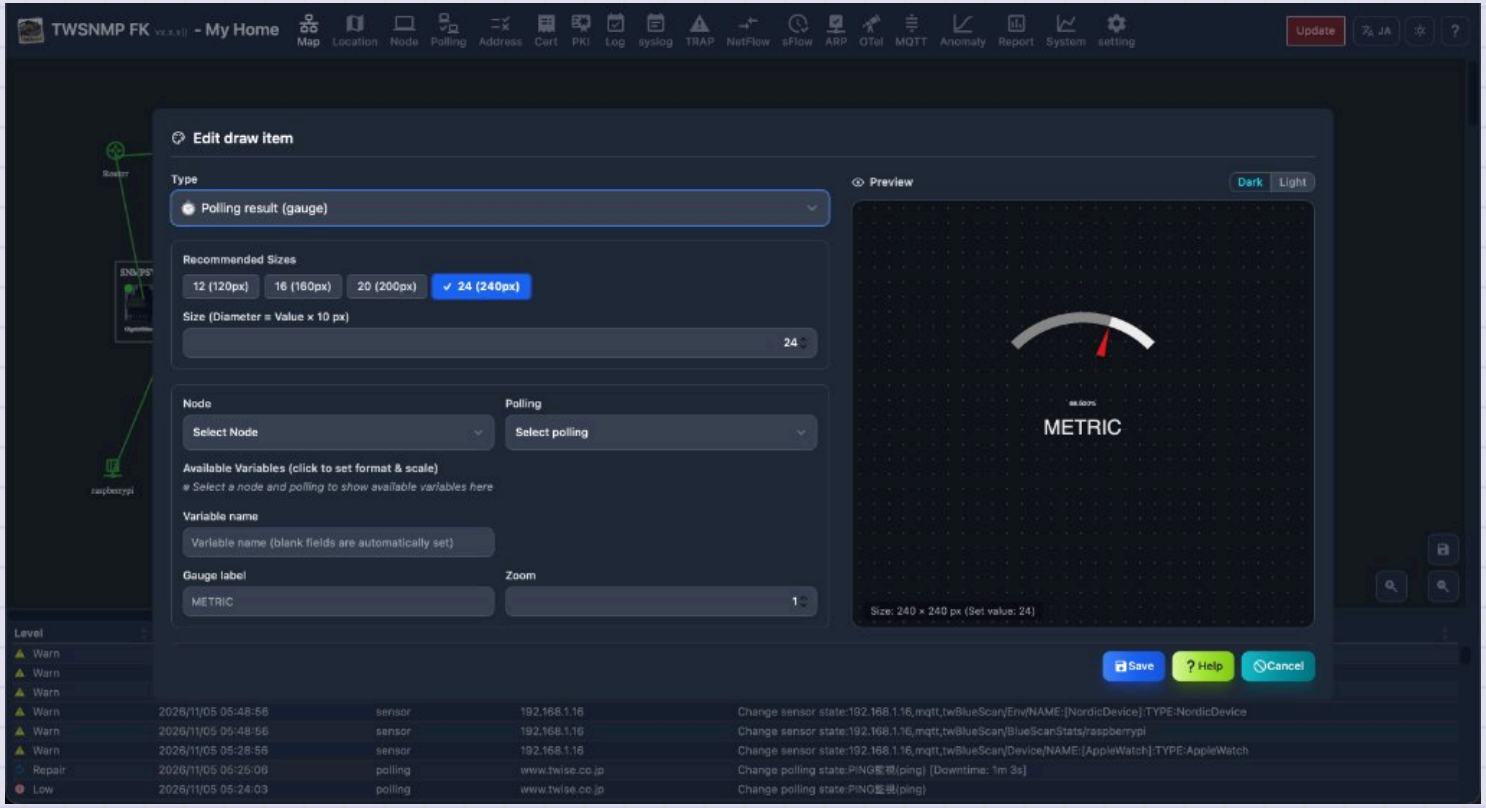
Copyright (c) 2023-2026 Masayuki Yamai

Items	Contents
Type	It is a type of drawing item.You can only change it when you add it.
Size	Character size.
Node	This is a node list for selecting polling.
Polling	Polling that displays results.
Variable name	The name of the variable displayed from the polling results.
Display format	Format when displaying.
Magnification	The display rate of drawing items.

Copyright (c) 2023-2026 Masayuki Yamai

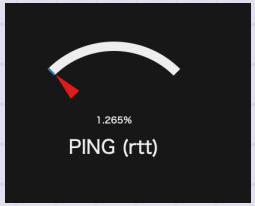
Drawing item (polling result: gauge)

It is the editing screen of drawing item (polling result: gauge).It can be used to display % data.



Copyright (c) 2023-2026 Masayuki Yamai

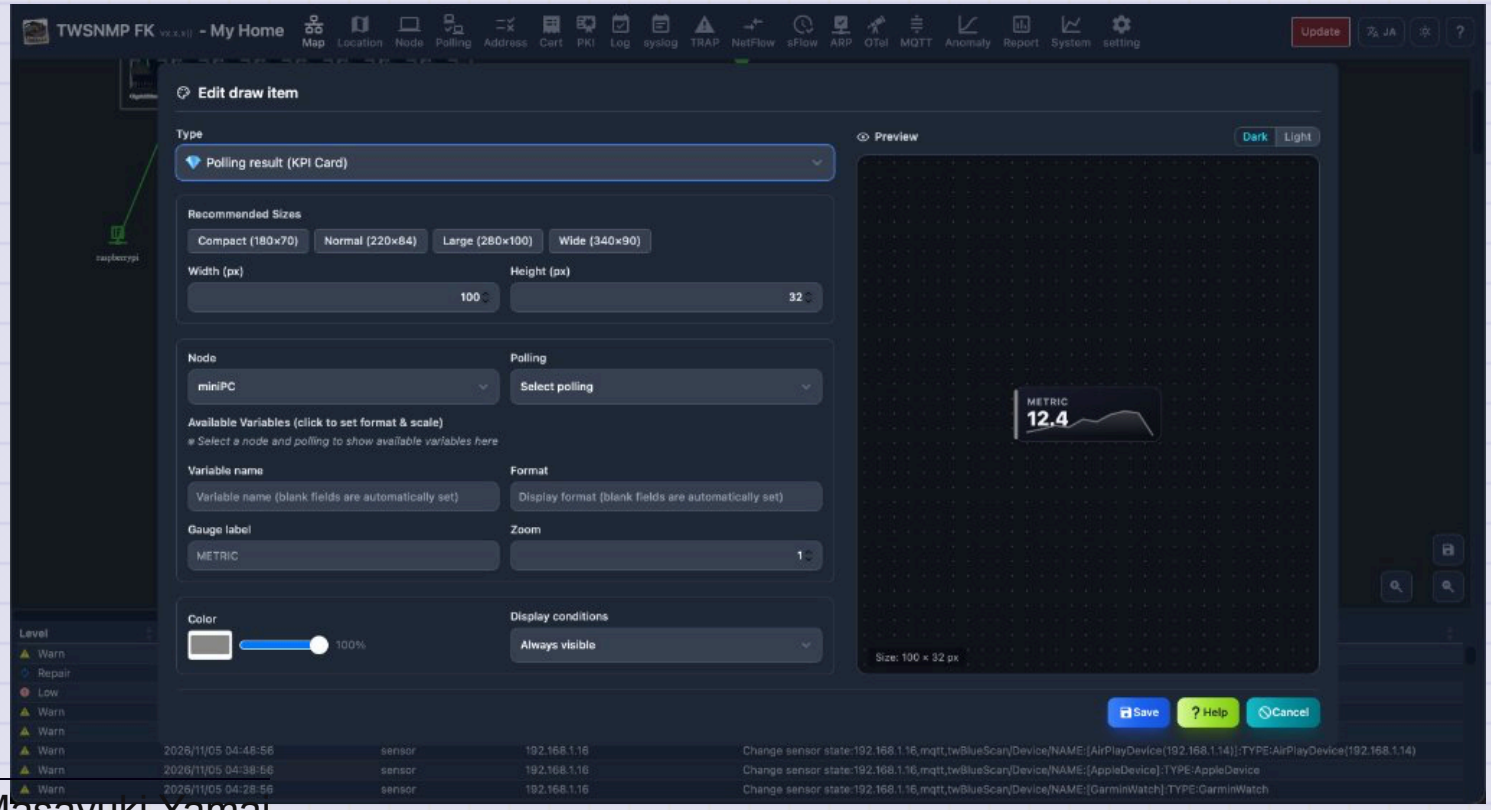
Items	Contents
Type	It is a type of drawing item.You can only change it when you add it.
Size	Gauge size.
Node	This is a node list for selecting polling.
Polling	Polling that displays results.
Variable name	The name of the variable displayed from the polling results.
Gauge label	This is a character string displayed under the gauge.
Magnification	The display rate of drawing items.



Copyright (c) 2023-2026 Masayuki Yamai

Drawing item (Polling result: KPI card)

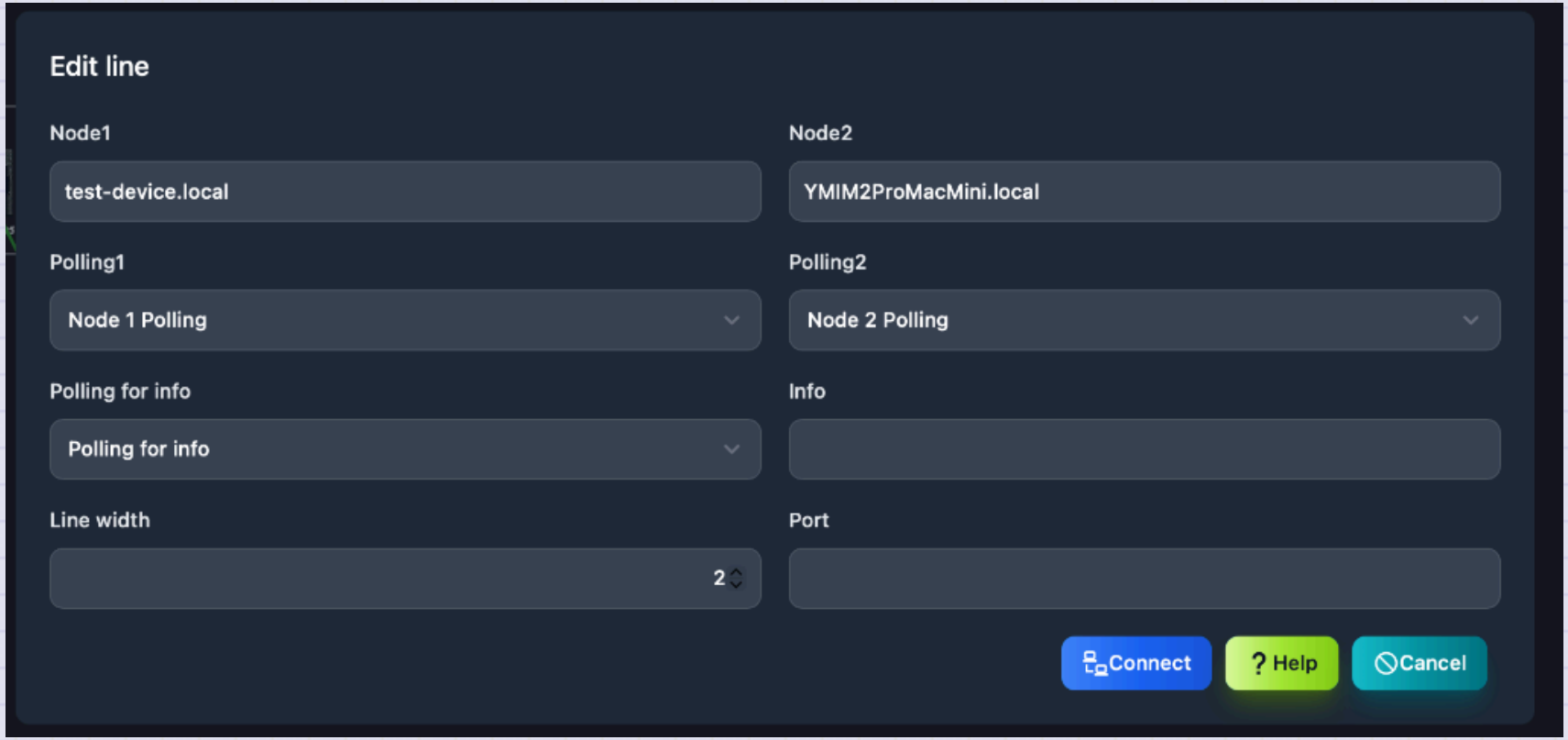
Displays polling results as modern twFathom-style KPI cards. Features sparklines, status accent bars, and prominent metric values, with live preview and recommended size buttons in the editor.



Copyright (c) 2023-2026 Masayuki Yamai

Line editing

To edit the line, press the two nodes while pressing the shift key on the map screen.



Copyright (c) 2023-2026 Masayuki Yamai

Items	Contents
Node1	This is the first node to connect the line.
Polling1	This is the first node polling that determines the color on one side of the line.
Node2	This is the second node to connect the line.
Polling2	This is the second node polling that determines the color on one side of the line.

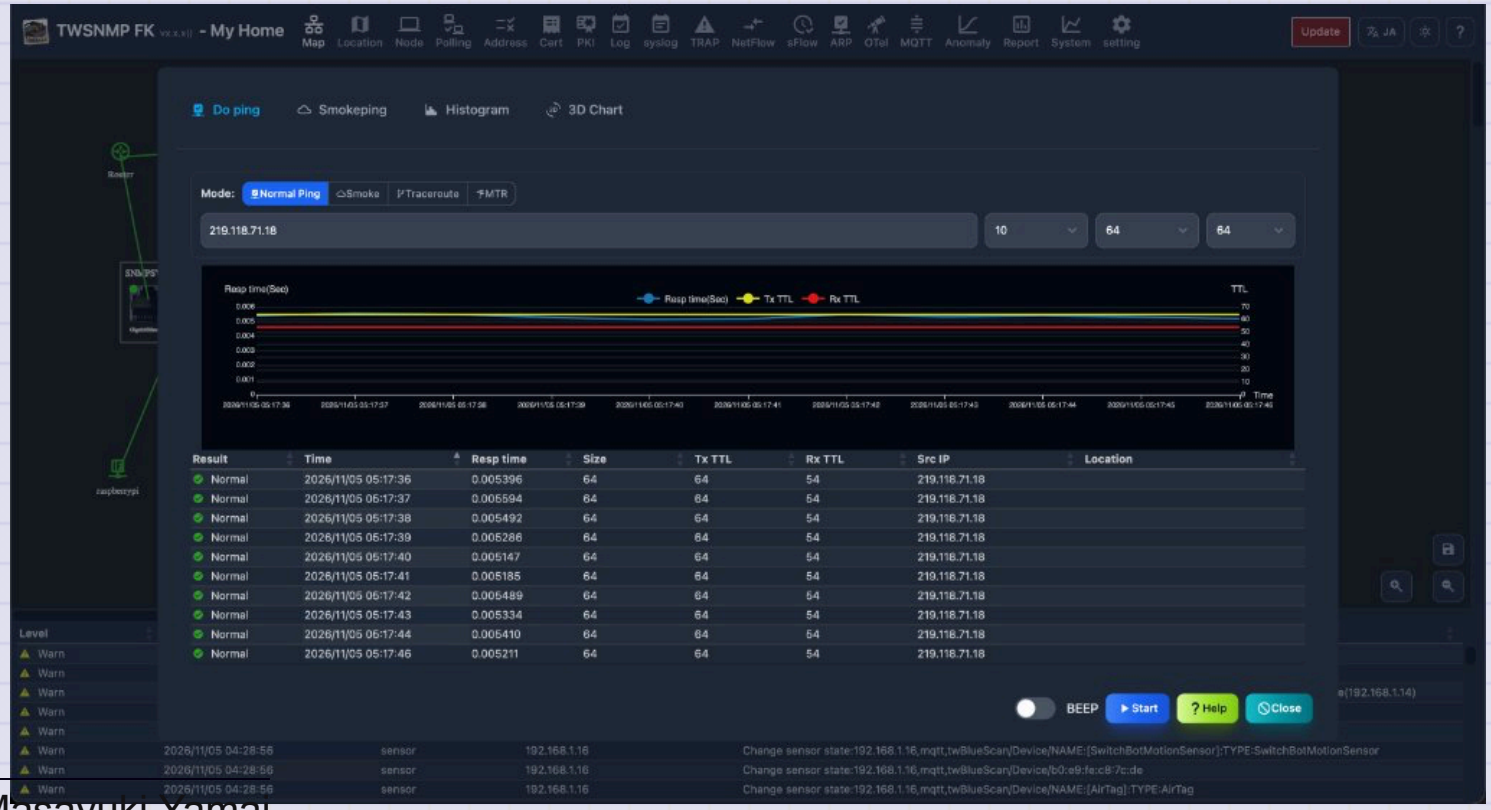
Copyright (c) 2023-2026 Masayuki Yamai

Items	Contents
Polling for information	Polling for information displayed next to the line. Specify the traffic monitor polling.
Information	Set the character string to be displayed next to the line. It will be overwritten by setting a polling for information.
Thickness of the line	It is the thickness of the line.
Port	Specify the port number used when displaying the panel.

Copyright (c) 2023-2026 Masayuki Yamai

PING

This is the screen to execute ping diagnostics. Four operation modes are available: Normal Ping, Smoke (jitter analysis), Traceroute, and MTR (real-time per-hop metrics). To get location information, a GeoIP database file is required.



Copyright (c) 2023-2026 Masayuki Yamai

Items	Contents
Operation Mode	Select the execution mode (Normal Ping, Smoke, Traceroute, MTR).
IP address	The target IP address or hostname to run ping.
Count / Duration	Number of execution times for Normal Ping, or duration (1 min, 3 min, 5 min, 10 min, Continuous) for Smoke and MTR modes.
Size	Ping packet payload size. Select "Inc Size" in Normal Ping to run pings with increasing packet size.
TTL	TTL value of ping packets (configurable in Normal Ping mode).
Result Graph	Time-series graph of ping response time and TTL values.

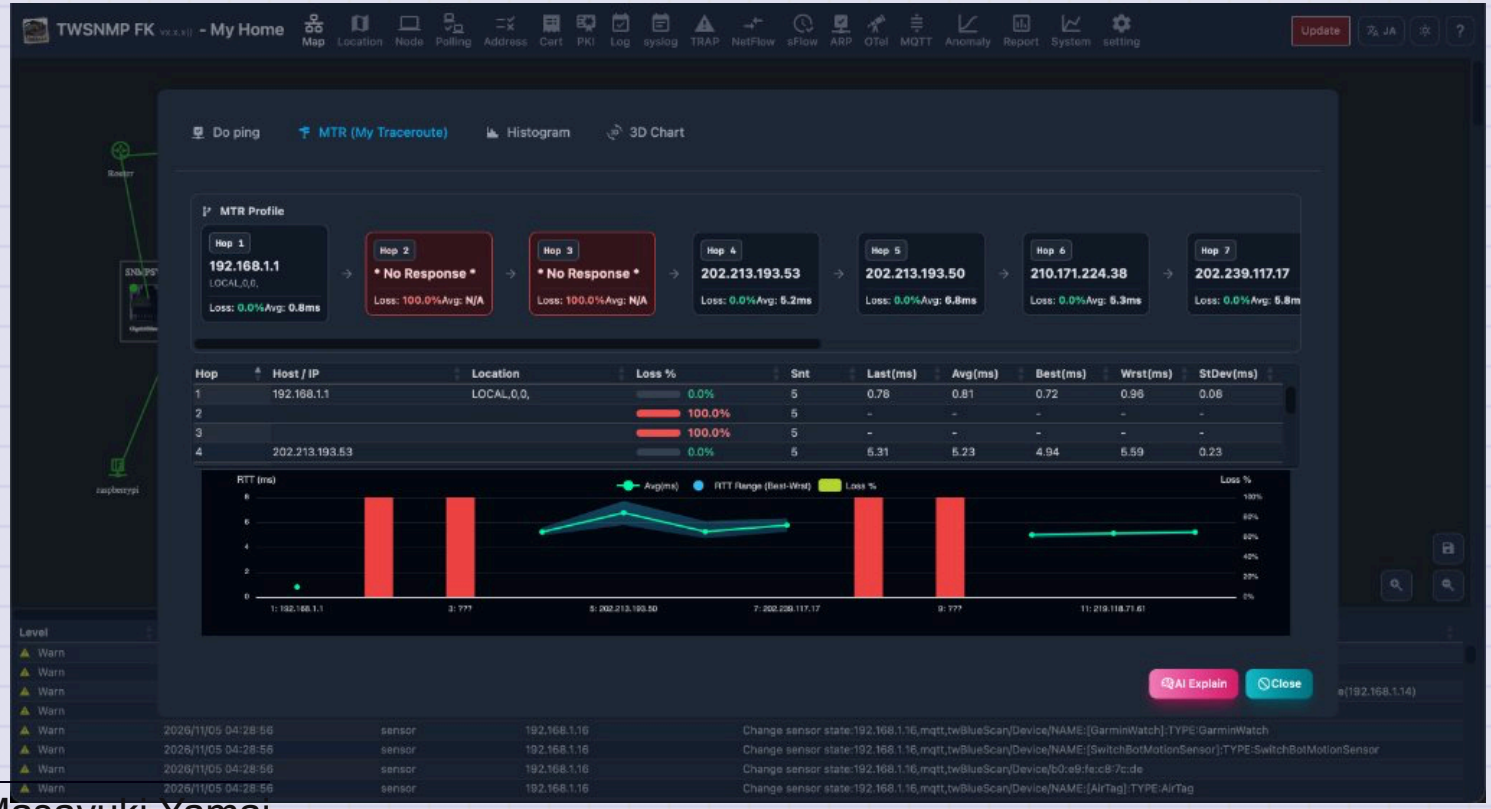
Copyright (c) 2023-2026 Masayuki Yamai

Items	Contents
Results	Ping execution log results. Timestamp, response time, size, Tx/Rx TTL, source IP, location.
Beep	Notify ping execution results with sound.
Start	Start ping in the selected mode.
Stop	Stop ping.
AI Explain	Send displayed charts and report data to AI (LLM) for automated diagnostic analysis.
Close	Close the ping window.

Copyright (c) 2023-2026 Masayuki Yamai

MTR (My Traceroute)

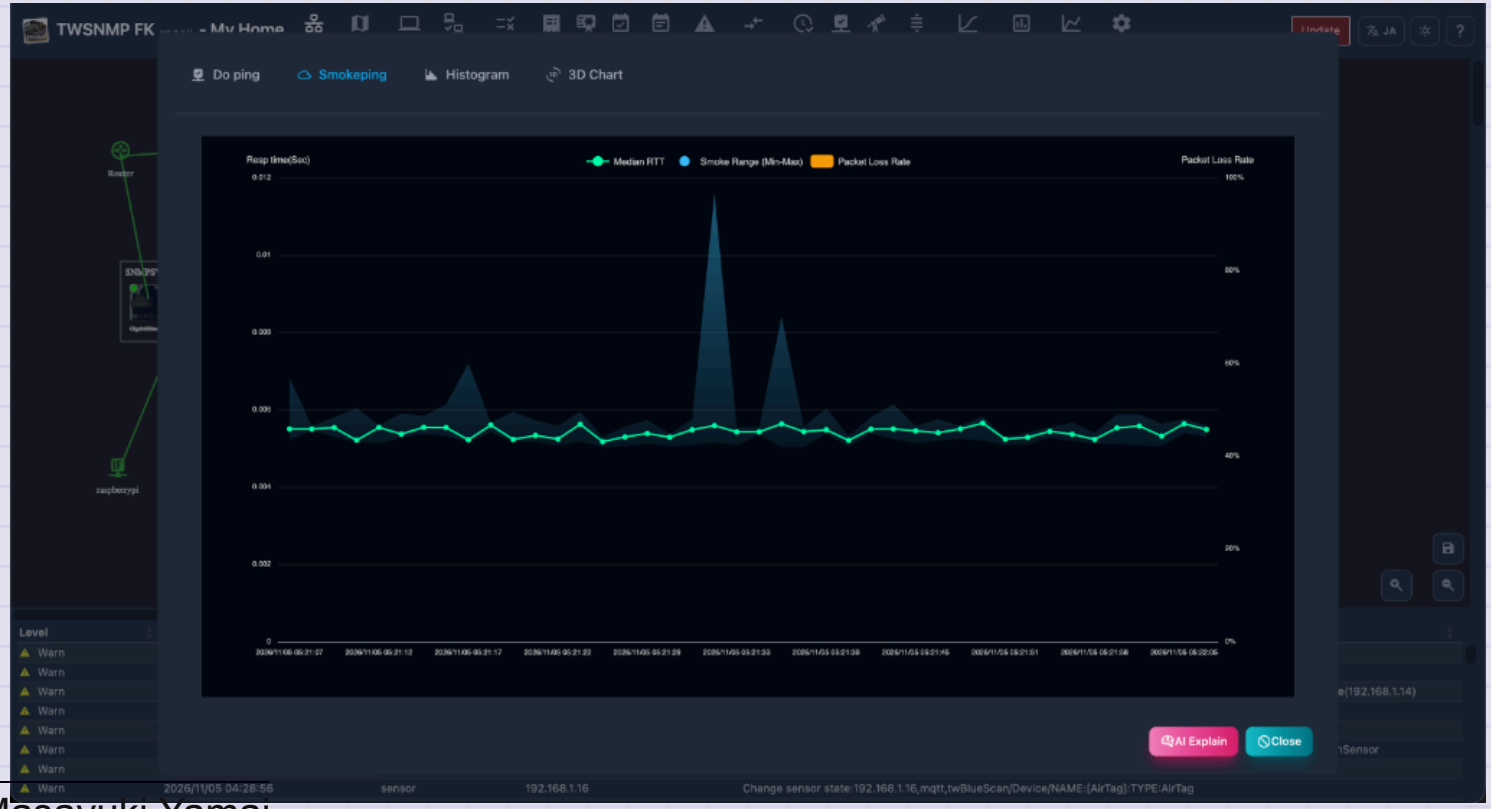
Automatically discovers all router hops along the path to the target, and performs continuous ping sampling to each hop. Displays per-hop metrics (Sent, Avg, Best, Worst, StDev, Loss %) and a hop latency profile chart.



Copyright (c) 2023-2026 Masayuki Yamai

Smokeping

Measures and visualizes latency variation (jitter), min/max/median response times, and packet loss intensity using high-frequency pings (200ms interval) with color gradient charts.



Copyright (c) 2023-2026 Masayuki Yamai

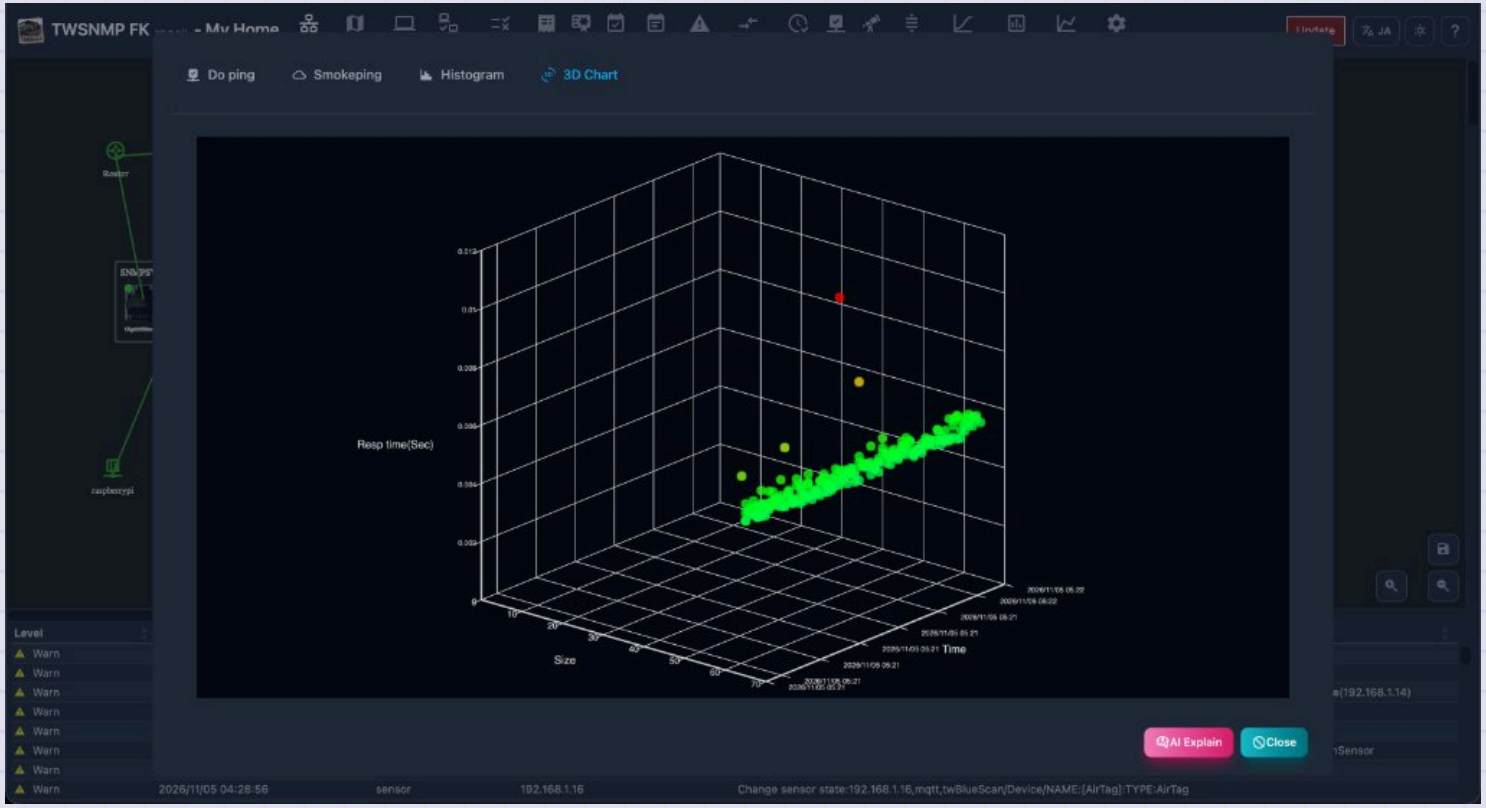
PING Histogram

It is a histogram of response time.

Copyright (c) 2023-2026 Masayuki Yamai

PING 3D analysis

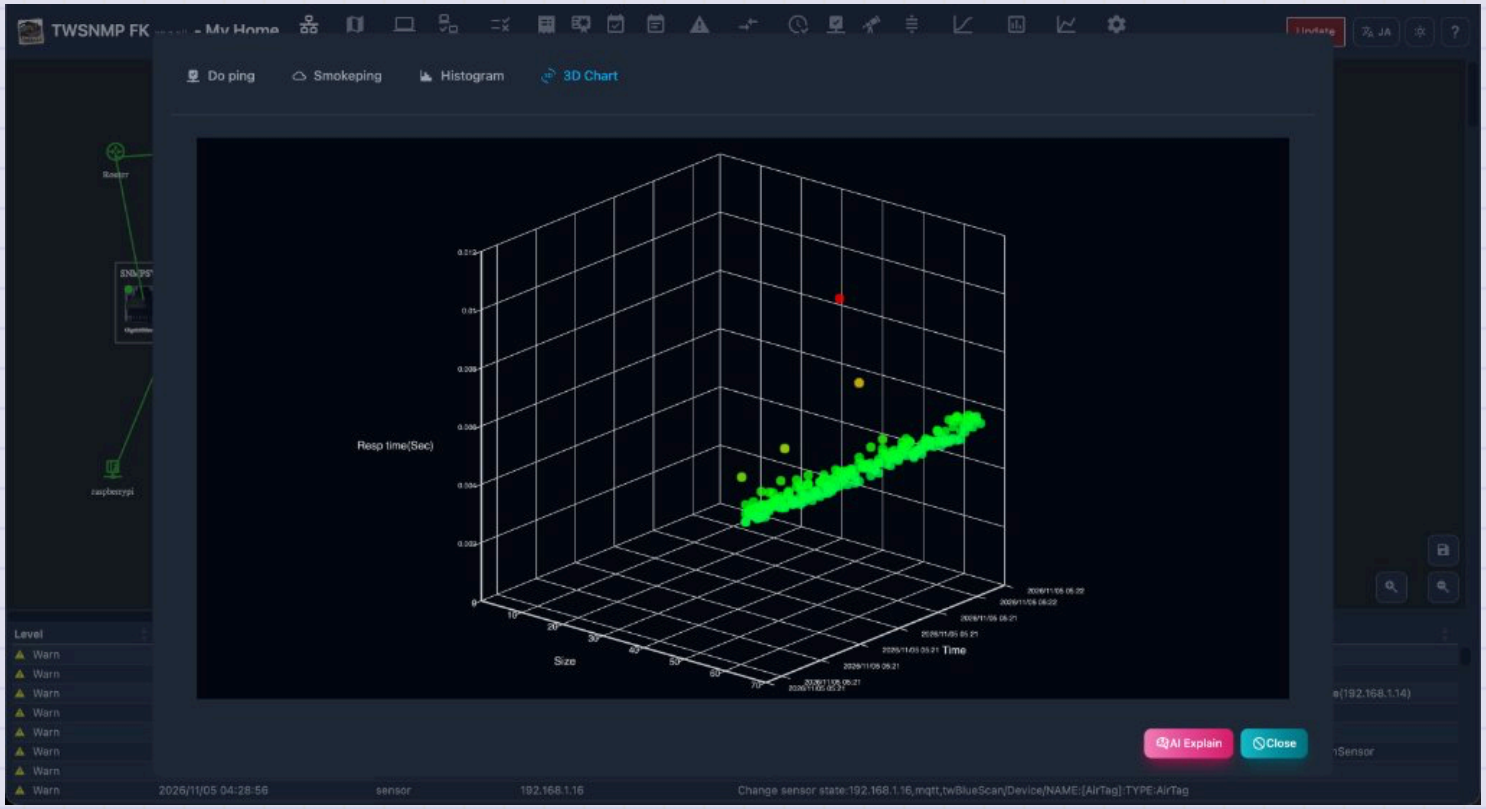
The response time, size, and implementation date and time are displayed in 3D graphs.



Copyright (c) 2023-2026 Masayuki Yamai

PING Line speed prediction

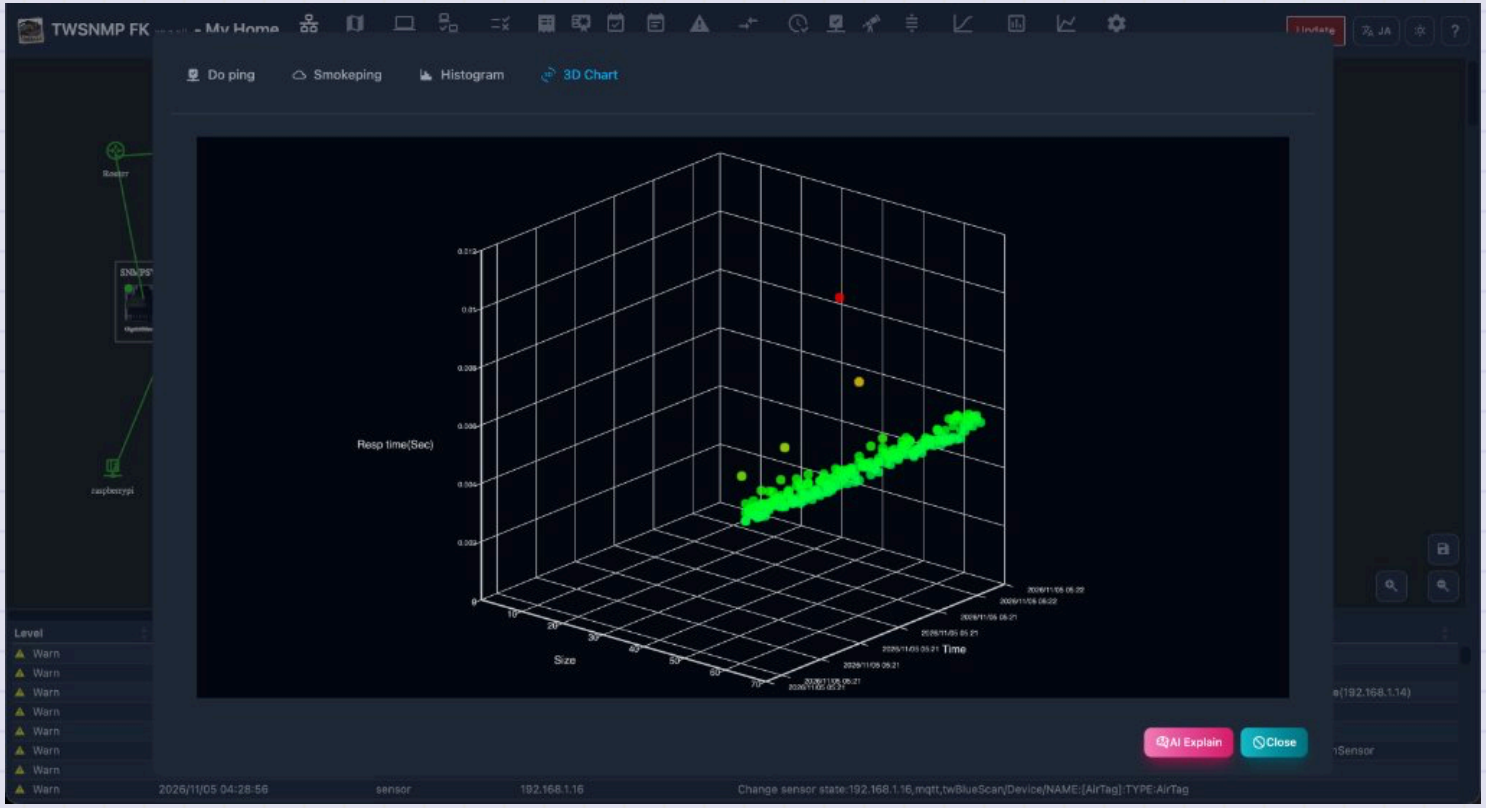
From the change in response time if the size is changed
This is a report that predicts the line speed.



Copyright (c) 2023-2026 Masayuki Yamai

PING Route analysis

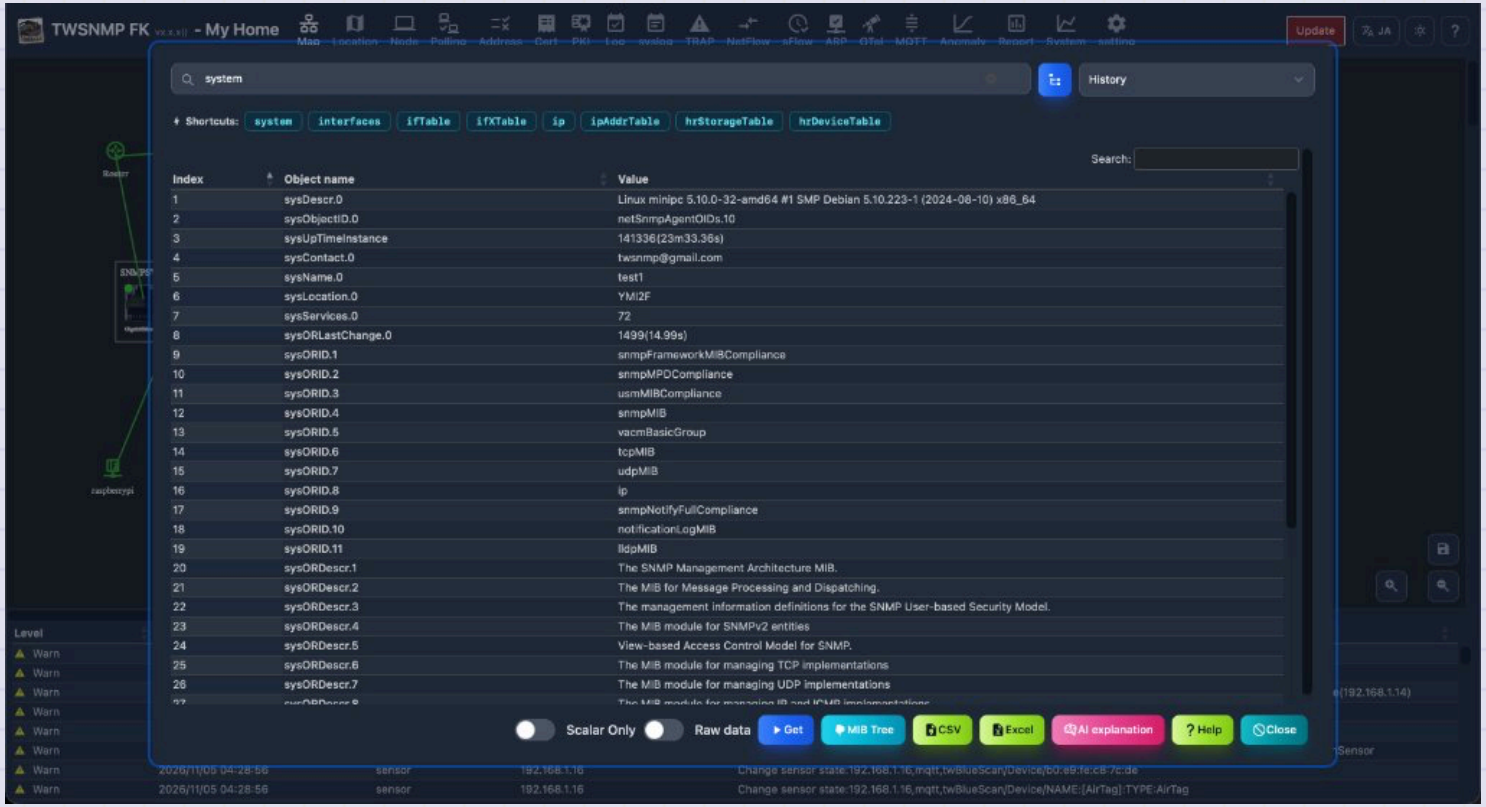
Display location information. It cannot be displayed without a GEOIP database.



Copyright (c) 2023-2026 Masayuki Yamai

MIB browser

This is a screen to get MIB information of SNMP from the node.
It is necessary to set SNMP access information in the node setting.
If you want to use MIB other than built -in, save the MIB file to the extmibs of the data folder.



Copyright (c) 2023-2026 Masayuki Yamai

Items	Contents
Object name	Specify the object name of the MIB you want to get. You can choose from the MIB tree.Example: System
<MIB Tree> Button	Display MIB tree.
History	It is the history of the object name obtained so far.You can select and get it again.
Results	Acquired MIB information.In the case of MIB in a table format, it is automatically displayed in a table format.

Copyright (c) 2023-2026 Masayuki Yamai

Items	Contents
Raw data	Displays the acquired MIB information without converting it. In the case of off, convert the time data to an easy -to -understand display.
Acquisition	Get MIB information.
CSV	Export the obtained MIB information of the CSV file.
Excel	Export the acquired MIB information of the Excel file.

Copyright (c) 2023-2026 Masayuki Yamai

MIB tree

This is a screen for selecting the obtained MIB object name.

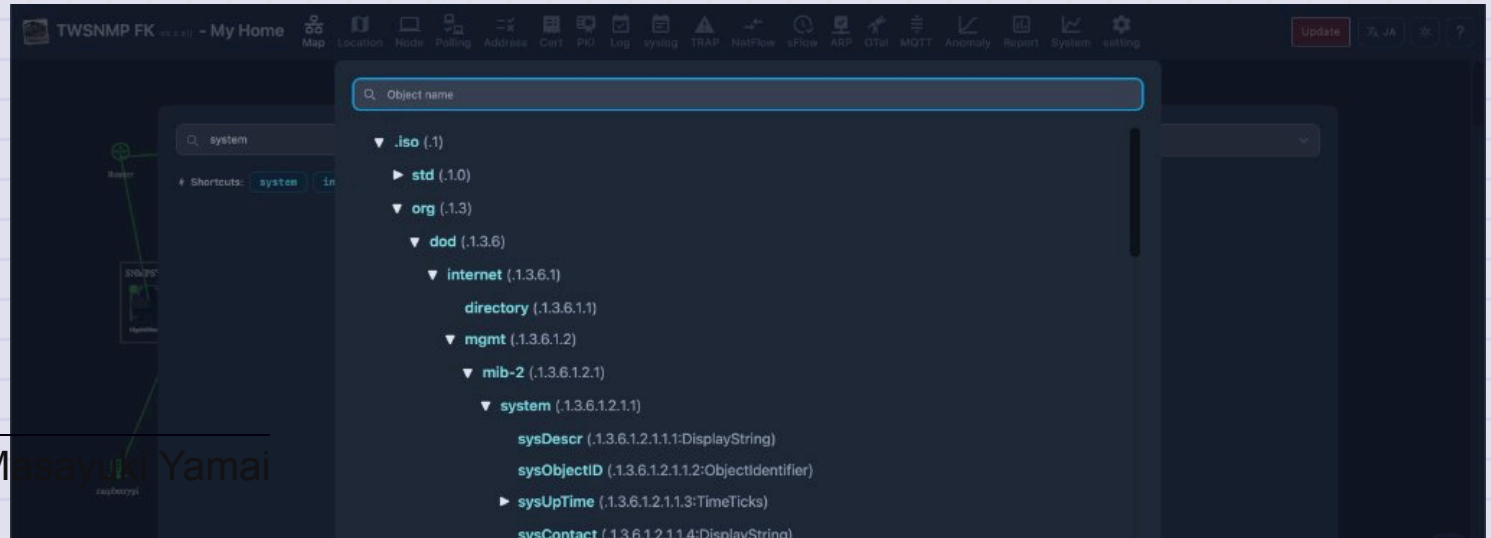
You can narrow down the displayed MIBs by entering keywords in the search box at the top of the MIB tree.

OIDs, MIB names, descriptions, etc., are subject to search.

Open the tree and click the object name to see the explanation.

Double click to select.

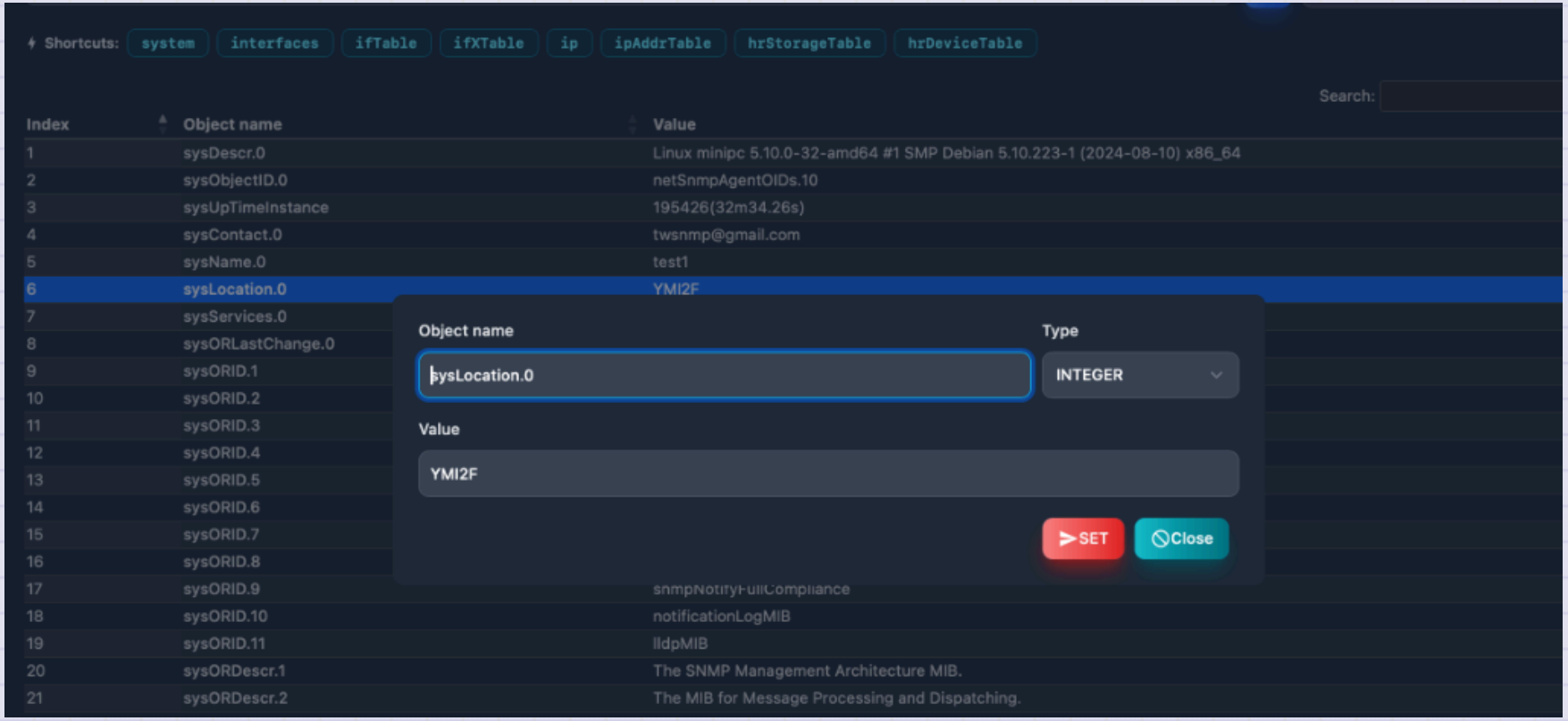
Also, AI-powered object explanation is available.



Copyright (c) 2023-2026 Masayuki Yamai

Set dialog

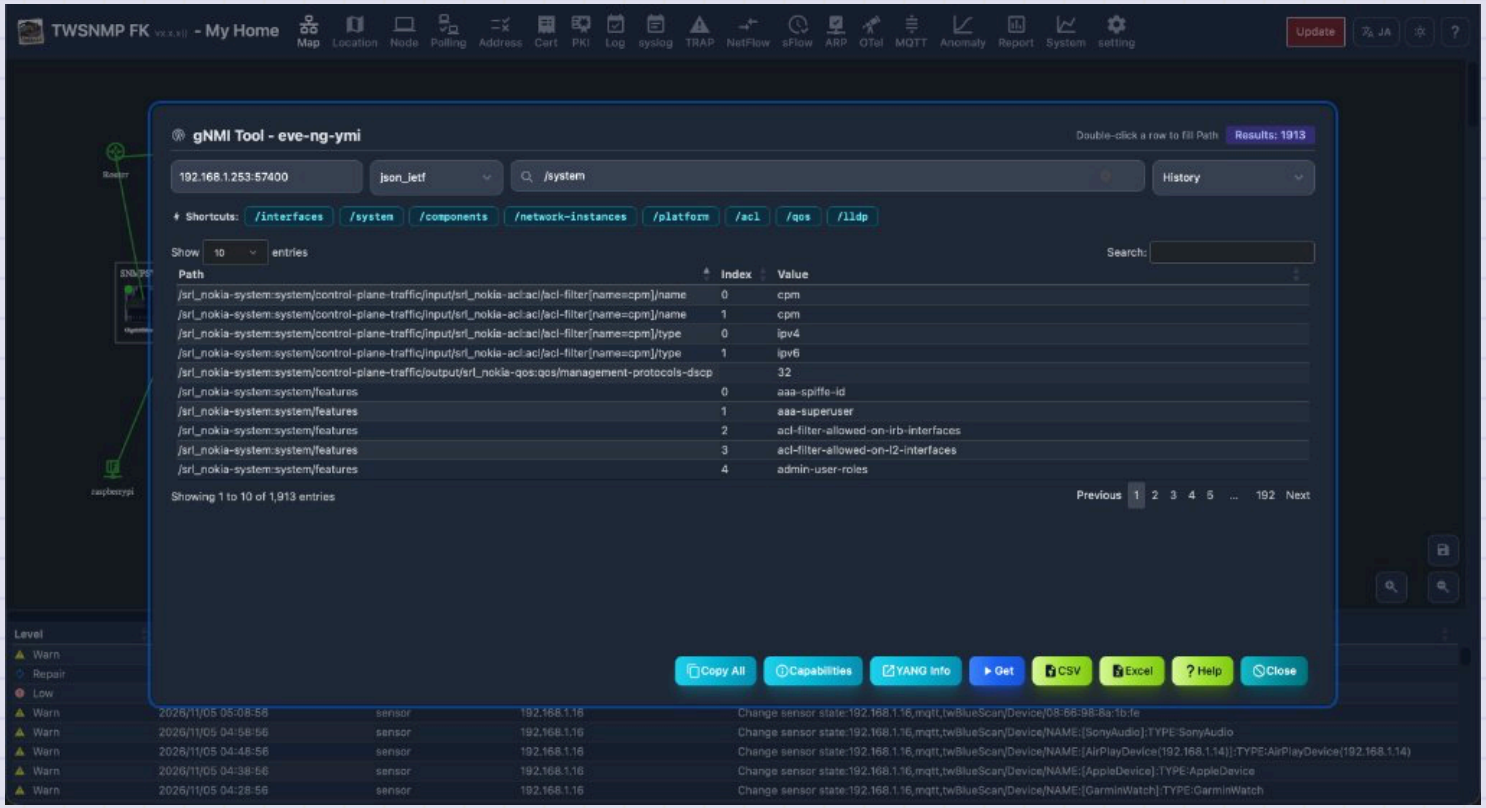
This is the screen for executing SNMP Set. Specify the object name, type, and value and press the <SET> button to send a Set request.



Copyright (c) 2023-2026 Masayuki Yamai

gNMI Tool

This is a screen to acquire management information from a node via gNMI.
You need to configure gNMI access in the node settings beforehand.



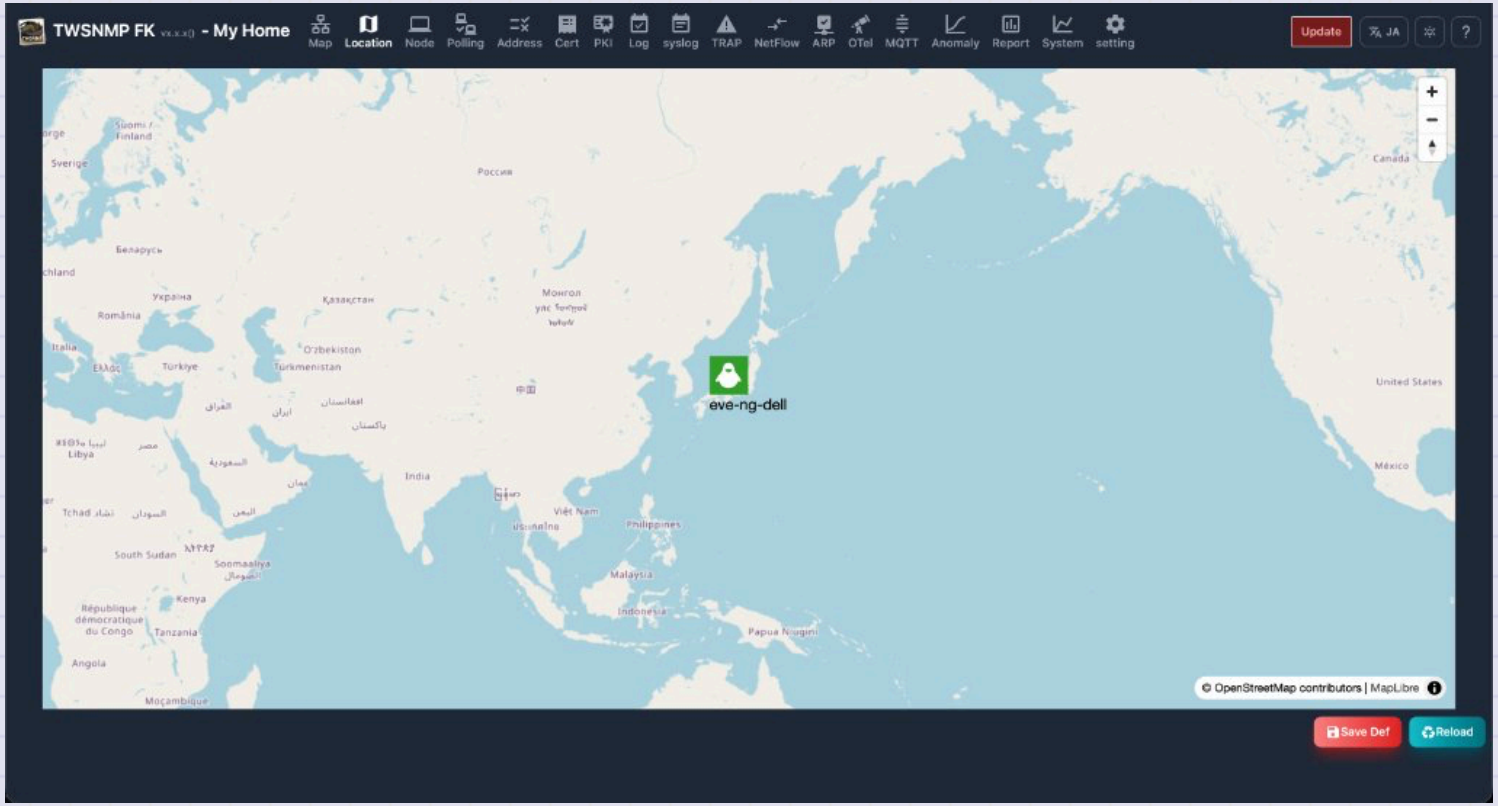
Copyright (c) 2023-2026 Masayuki Yamai

Items	Contents
Target	Specify the IP:port to access with gNMI.
Encoding	Specify gNMI encoding (json
Path	Specify the sensor path to retrieve.
History	History of paths acquired so far. Select to reload.
Result	Table list of acquired gNMI values.
Copy All	Copies the acquired results to the clipboard.
Polling	Create a polling monitor from the selected result.
Capabilities	Get gNMI capabilities and supported models of the target node.
YANG Info	Opens the YangModels GitHub repository in a browser.
Get	Execute GET under the specified path (can also be executed with Cmd/Ctrl + Enter)

Copyright (c) 2023-2026 Masayuki Yamai

Location Map screen

This is a screen that displays the node on the map.
Map data can be used in OpenStreetMap, which is used in location information services.
You can select by clicking the node.You can move by dragging.Multiple choices cannot be selected.



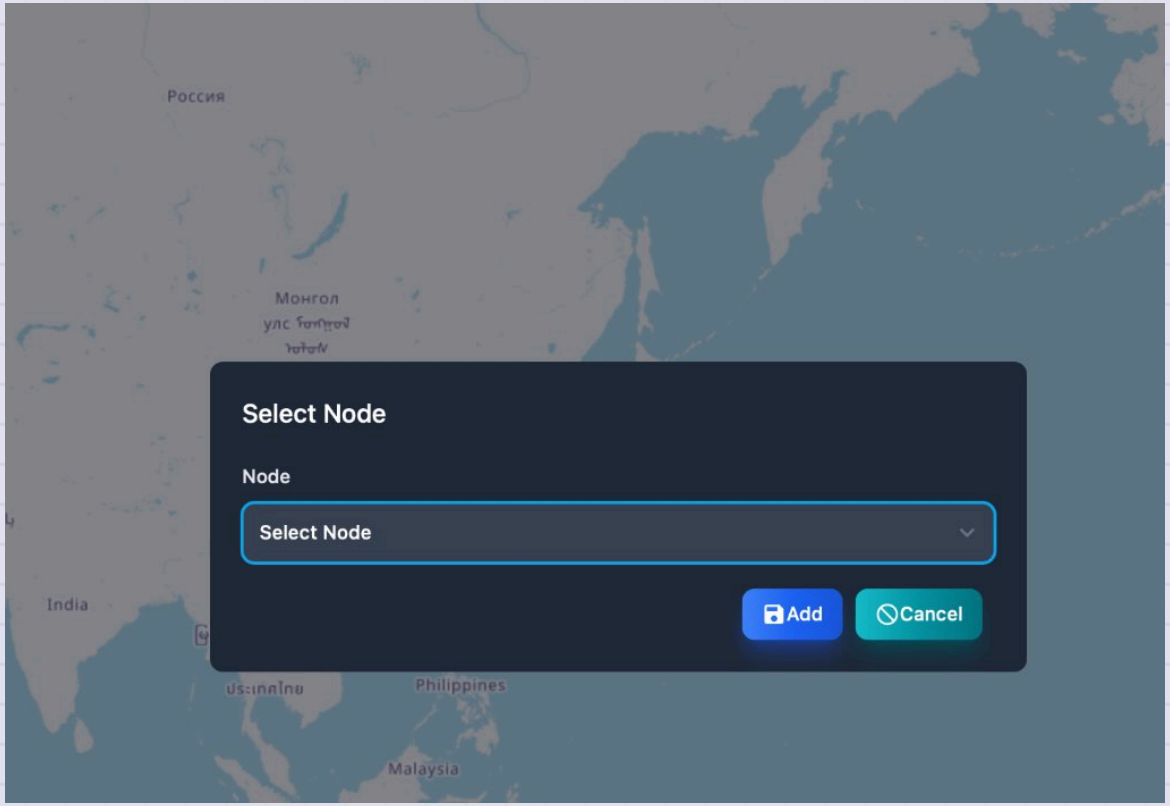
Copyright (c) 2023-2026 Masayuki Yamai

Items	Contents
Edit	Displays the screen of the selected node.
Polling	Displays the selected node polling.
Delete	Delete the selected node from the map screen.
Report	Displays the selected node report screen.
Initial display	Save the center and zoom level of the map.The next time you open the map screen, it will be in the same state.
Reload	Update the list of event logs to the latest state.

Copyright (c) 2023-2026 Masayuki Yamai

Add node to location map

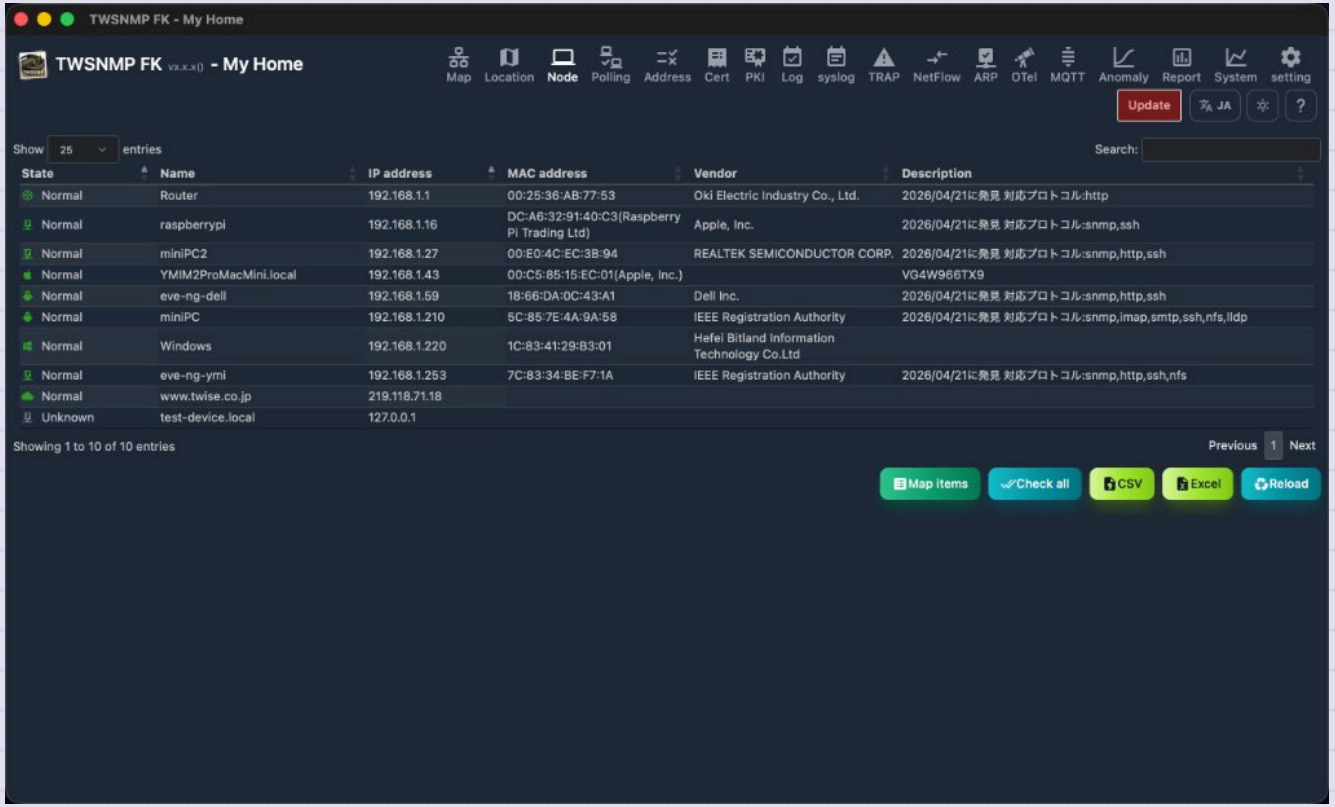
Right -click where you want to place the node on the map and the dialog to add is displayed.You can add it by selecting a node.



Copyright (c) 2023-2026 Masayuki Yamai

Node list

A list of nodes to be managed.



Copyright (c) 2023-2026 Masayuki Yamai

Items	Contents
State	Node condition. Severe, mild, precautions, return, normal, unknown.
Name	Node name.
IP address	Node IP address.
MAC address	Node MAC address.
Vendor	The name of the vendor corresponding to the MAC address.
Description	Supplementary information about nodes.

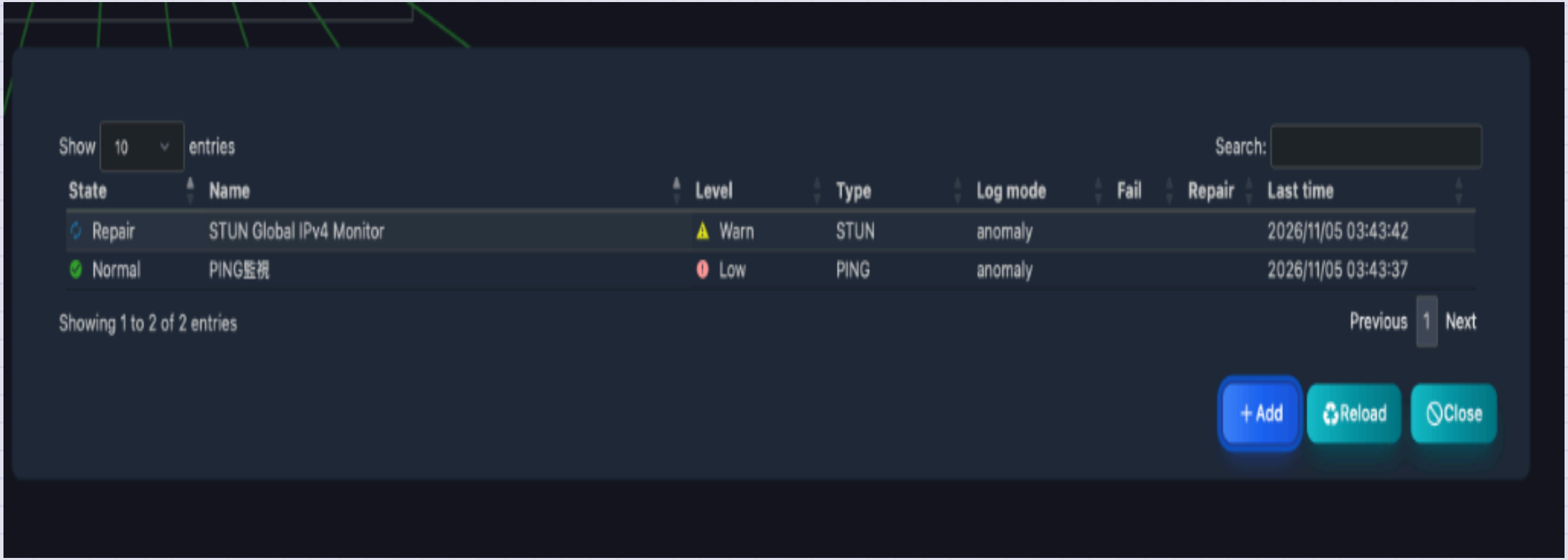
Copyright (c) 2023-2026 Masayuki Yamai

Items	Contents
Edit	Edit node settings.
Polling	Displays a list of polling related to the selected node.
AI Diagnose	Comprehensive AI-powered (LLM) diagnosis of node status, related logs, and responses.
Report	Displays the selected node analysis report.
Delete	Delete the selected node.
Reconfirm	Reconfirm the polling of the selected node.
Remost confirmation	Reconfirm all nodes polling.
CSV	Export the node list to the CSV file.
Excel	Export the node list to the Excel file.

Copyright (c) 2023-2026 Masayuki Yamai

Node polling list

A list of polling related to nodes.



Copyright (c) 2023-2026 Masayuki Yamai

Items	Contents
State	Polling state. Severe, mild, precautions, return, normal, unknown.
Name	Polling name.
Level	Pauling level.
Type	Polling type. Ping, SNMP, TCP, etc.
Log	Log mode.
Last time	This is the last date and time when polling was implemented.

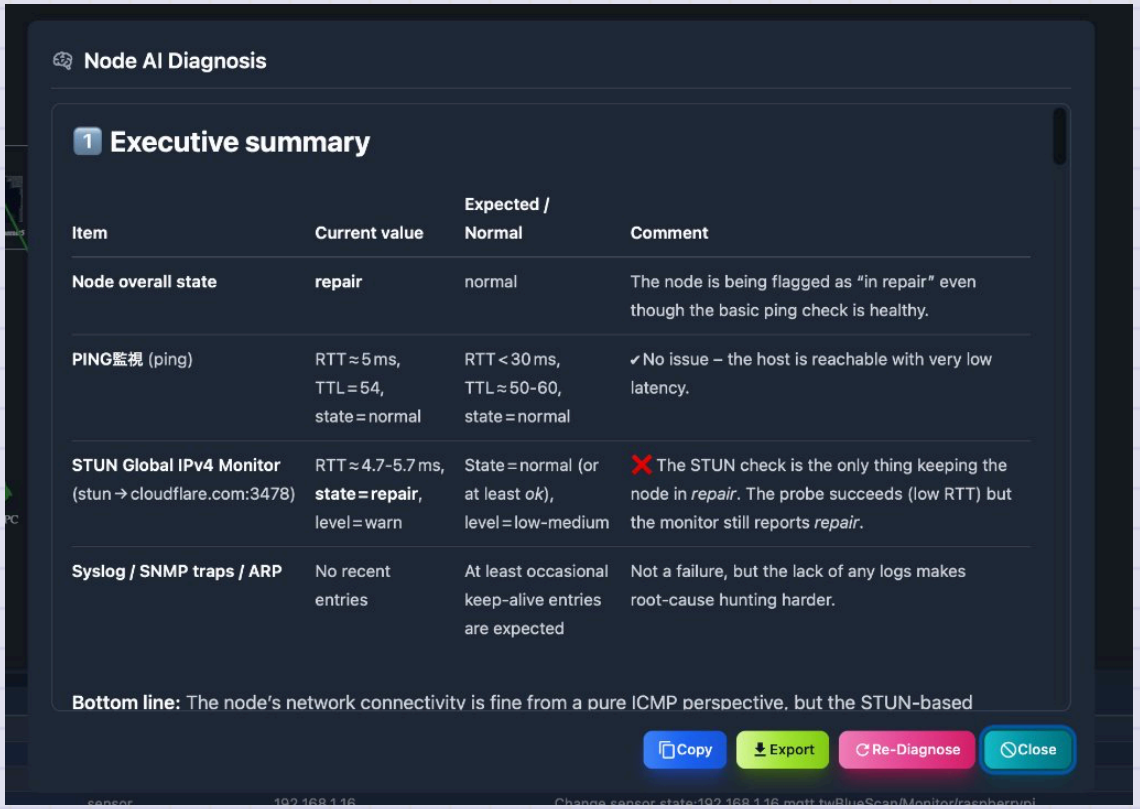
Copyright (c) 2023-2026 Masayuki Yamai

Items	Contents
Add	Add polling to nodes.
Edit	Edit the selected polling.
Copy	Create a selected polling copy.
Report	Displays the selected polling analysis report.
Delete	Delete the selected polling.
Reload	Update the polling list to the latest state.
Close	Close the list of polling.

Copyright (c) 2023-2026 Masayuki Yamai

AI Diagnosis

Uses AI (LLM) to comprehensively analyze the selected node's current status, polling responses, and related logs.



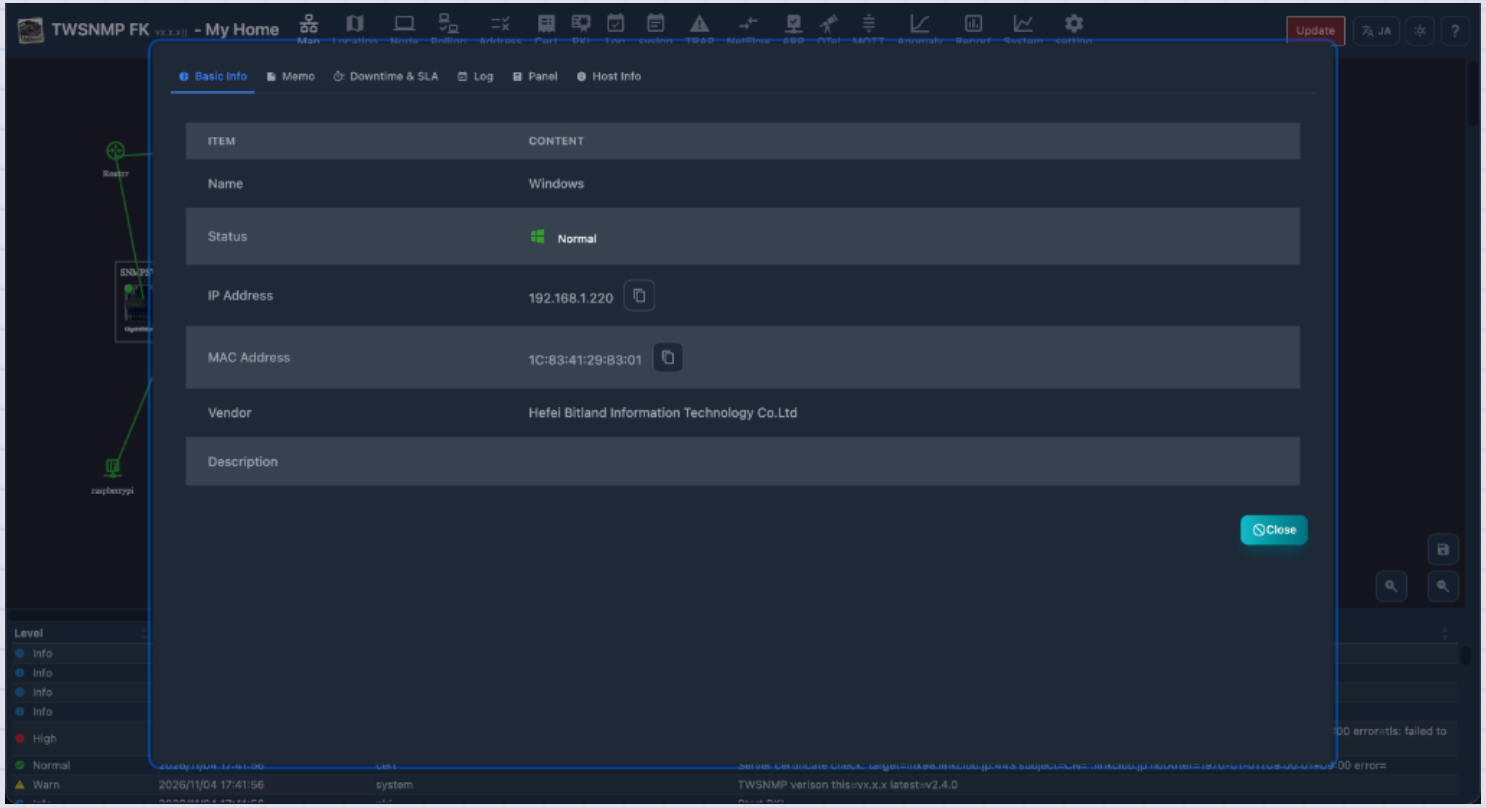
Copyright (c) 2023-2026 Masayuki Yamai

Items	Contents
Copy	Copy the AI diagnosis report content to the clipboard.
Export	Export the AI diagnosis report content to a text file.
Re-Diagnose	Re-run the AI diagnosis using the latest node status and log data.
Close	Close the AI diagnosis dialog.

Copyright (c) 2023-2026 Masayuki Yamai

Basic information report

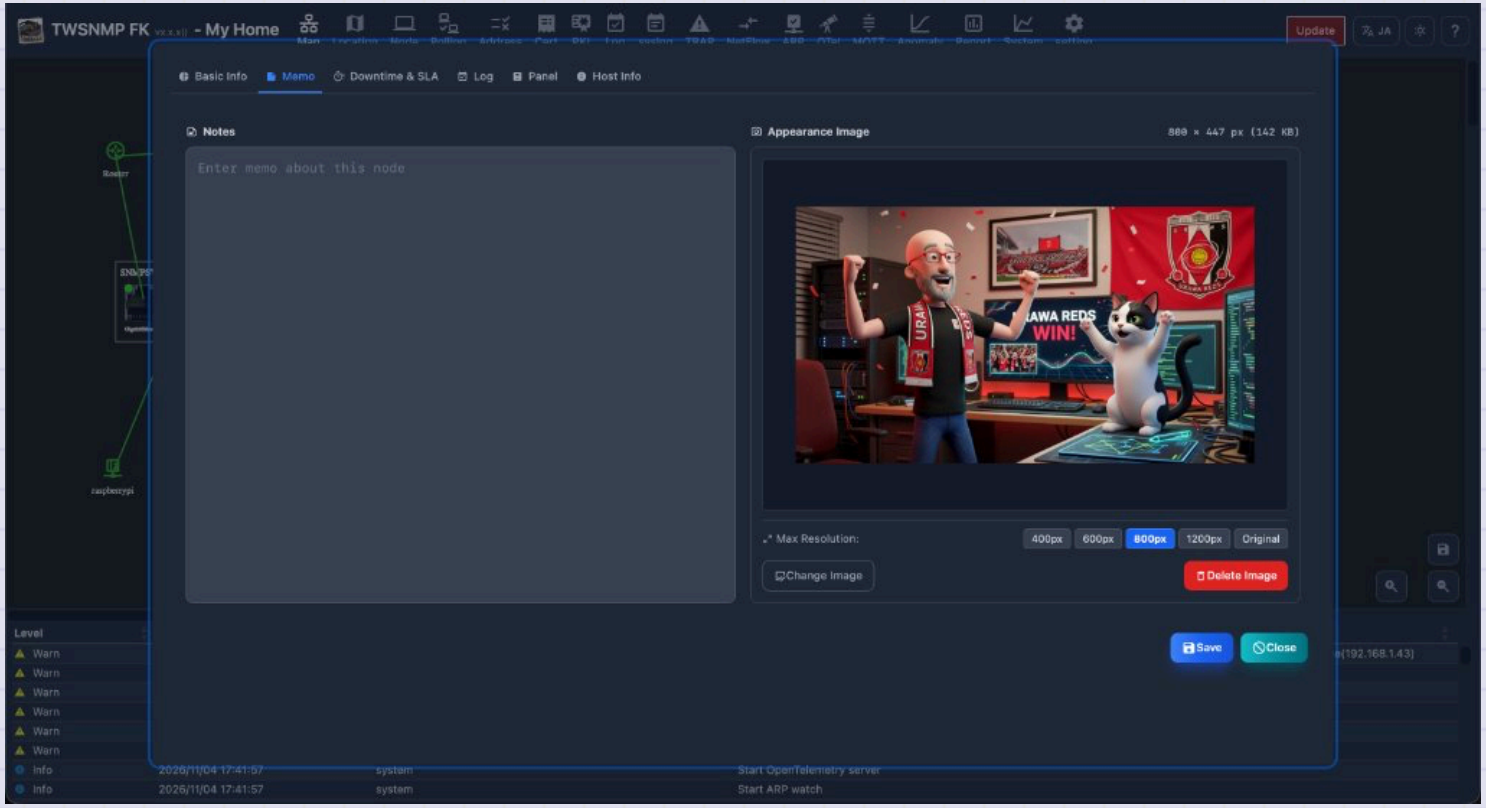
Basic information about nodes.



Copyright (c) 2023-2026 Masayuki Yamai

Memo

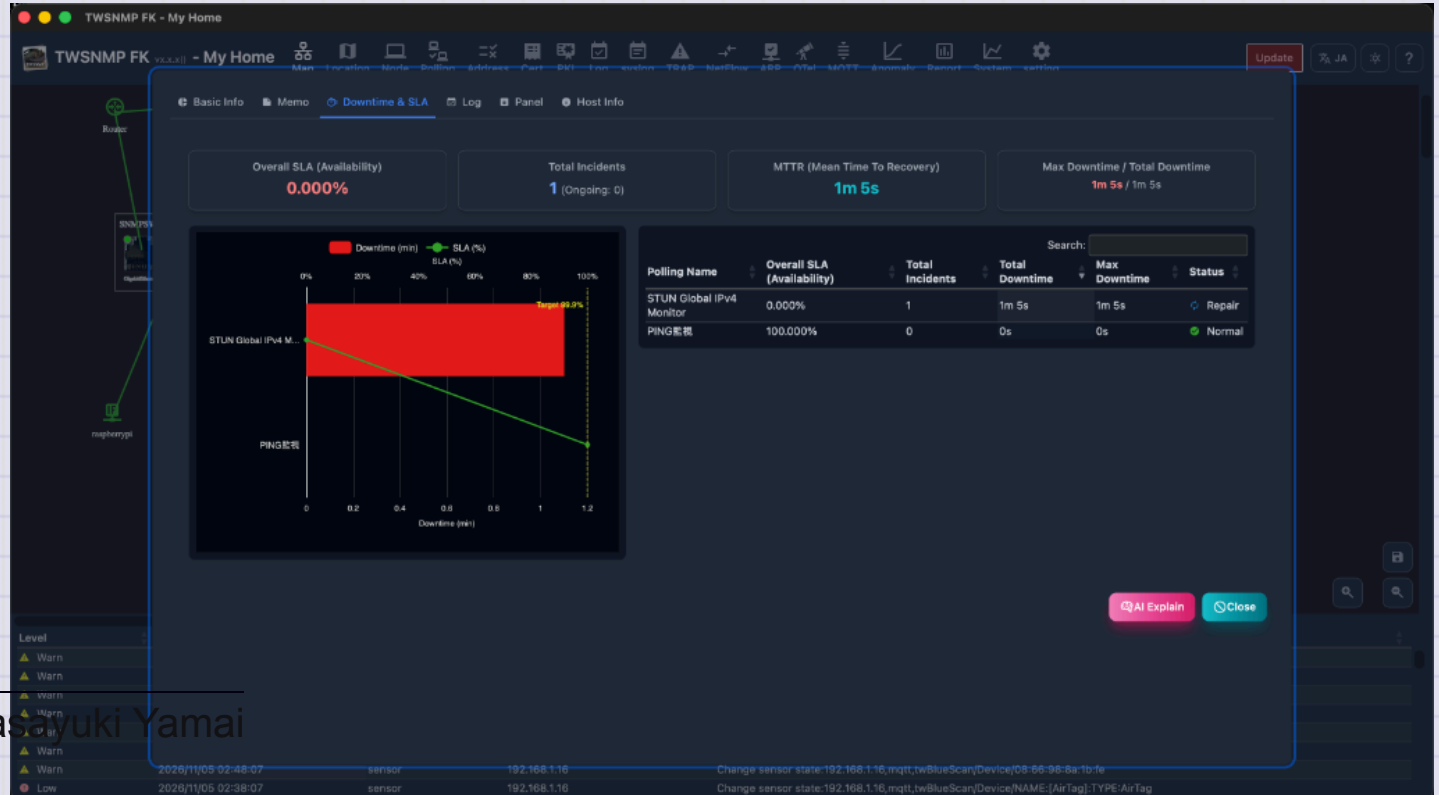
Memo about the node.



Copyright (c) 2023-2026 Masayuki Yamai

Downtime & SLA

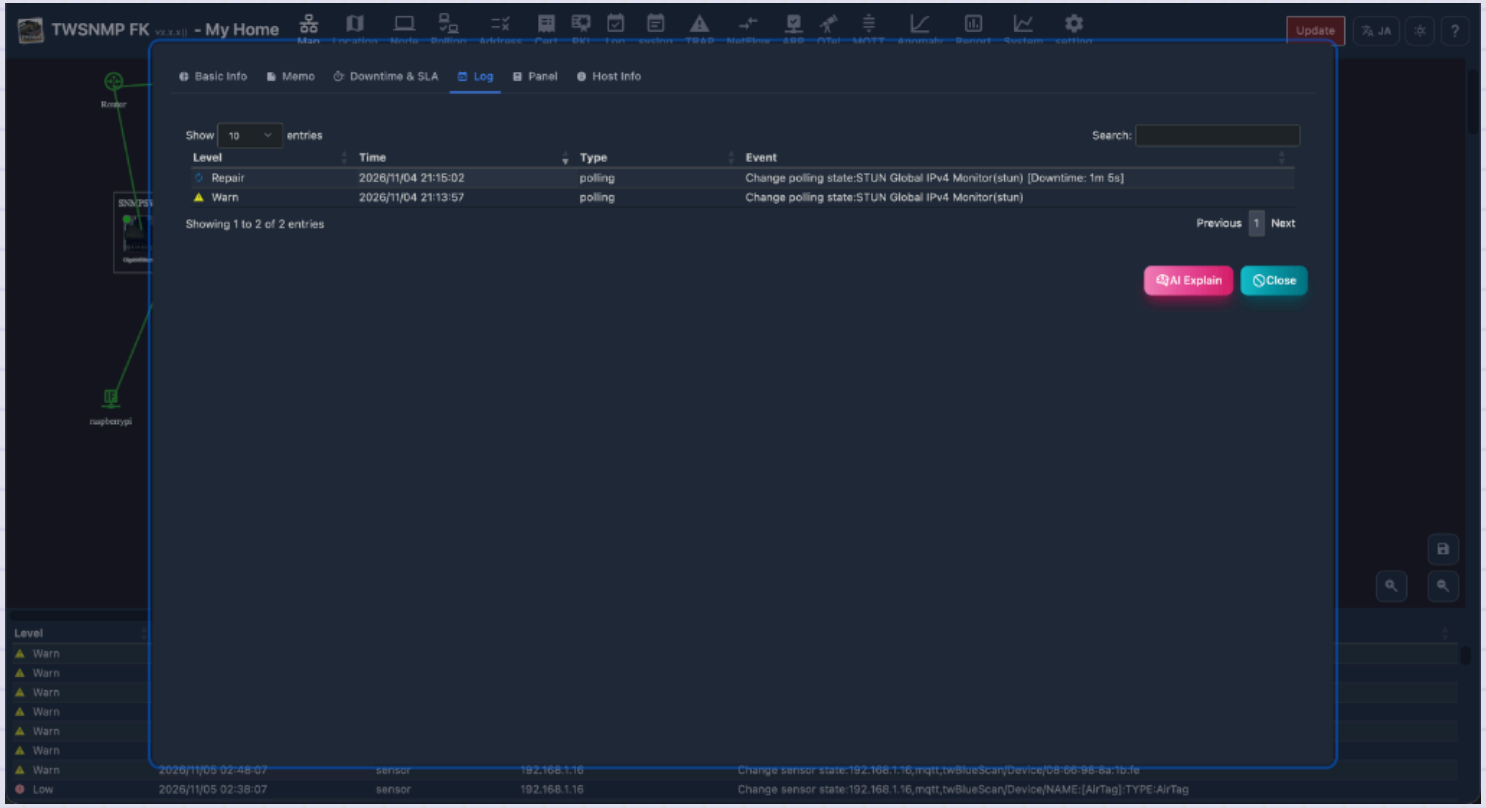
An aggregate report analyzing overall node SLA (Availability %), total incidents count, MTTR, max/total downtime, and per-polling downtime & SLA ranking chart and detail table based on polling failure and recovery logs. Supports AI explanation via the `[AI Explain]` button.



Copyright (c) 2023-2026 Masayuki Yamai

node event log

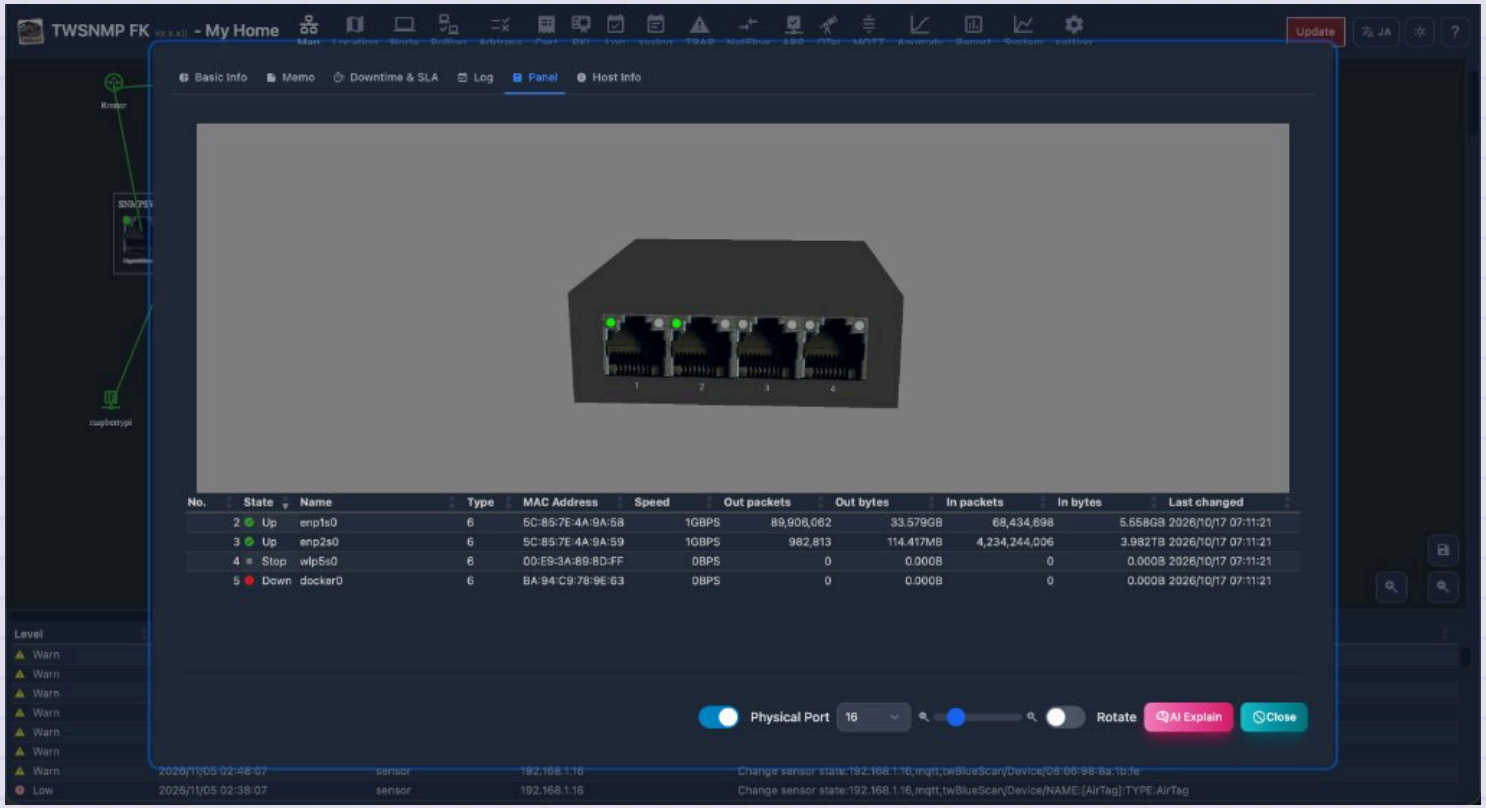
This is an event log related to the node.



Copyright (c) 2023-2026 Masayuki Yamai

Panel

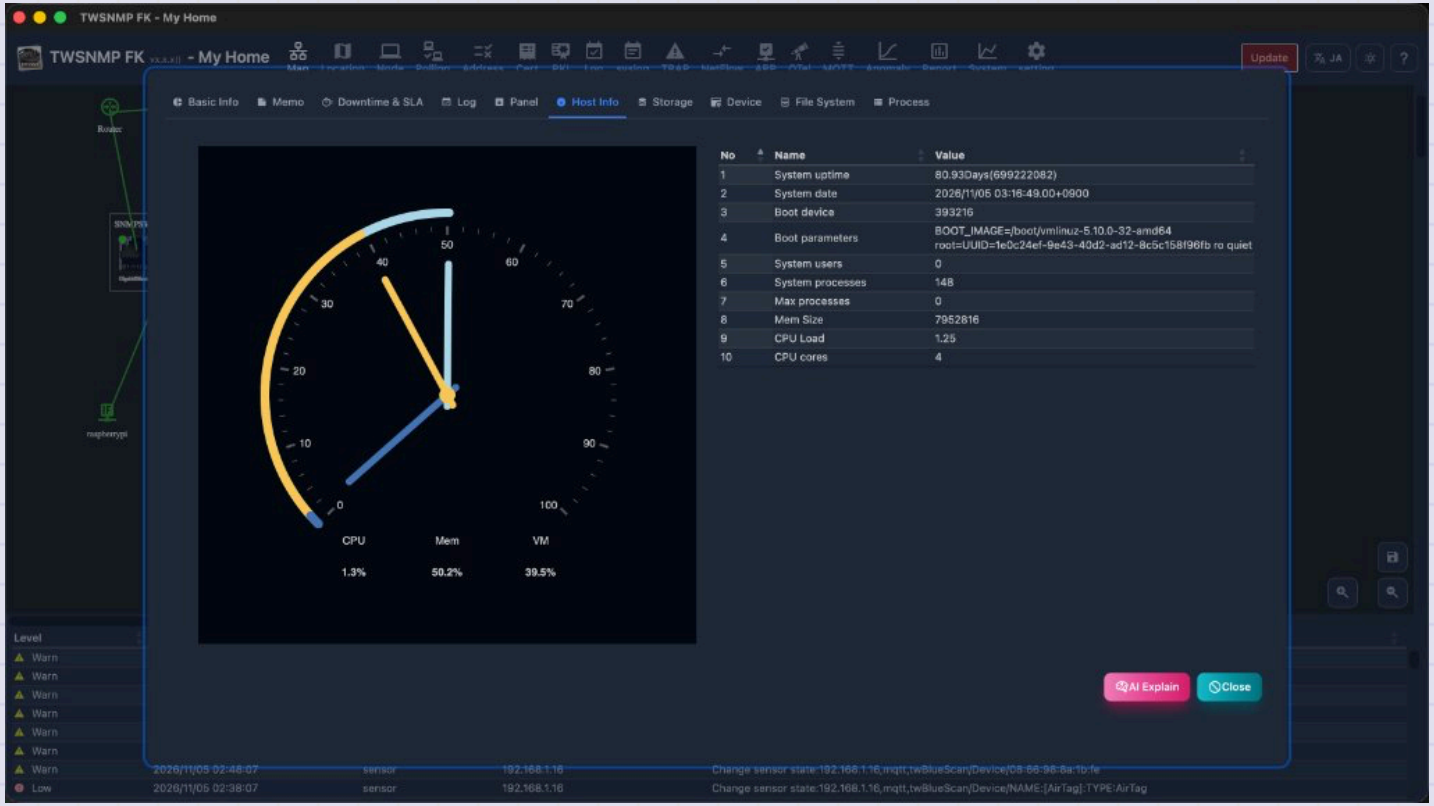
Displays the appearance of the node.Displays the port from the acquisition of the interface mib by SNMP or the line connection information.The <physical port> switch can only be displayed on the physical port.Rotate the panel display with the <rotation> switch.



Copyright (c) 2023-2026 Masayuki Yamai

Host information

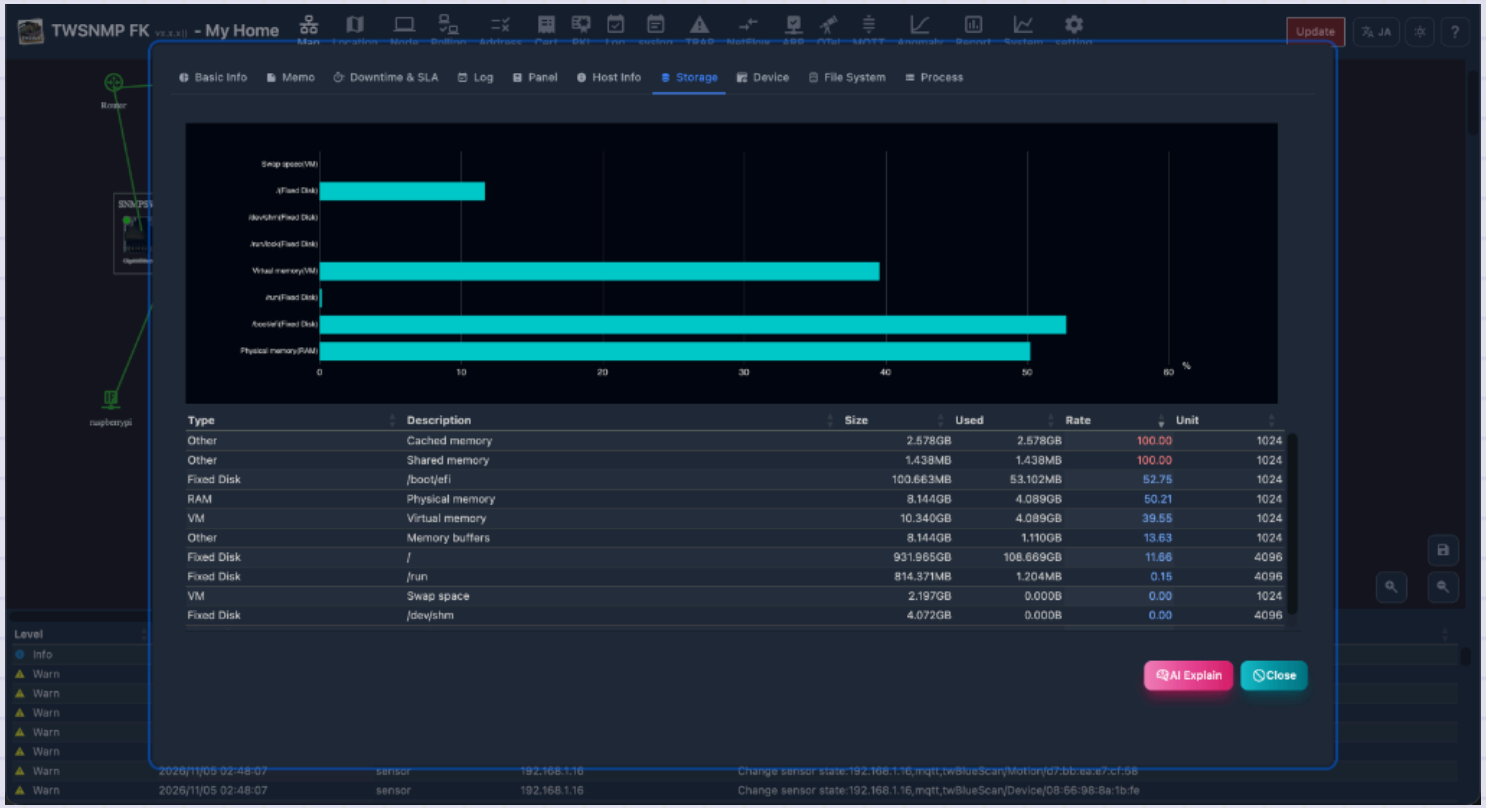
Displays the information of the host resource mib of SNMP.If it is not compatible with the host resource MIB, it cannot be displayed.



Copyright (c) 2023-2026 Masayuki Yamai

Storage

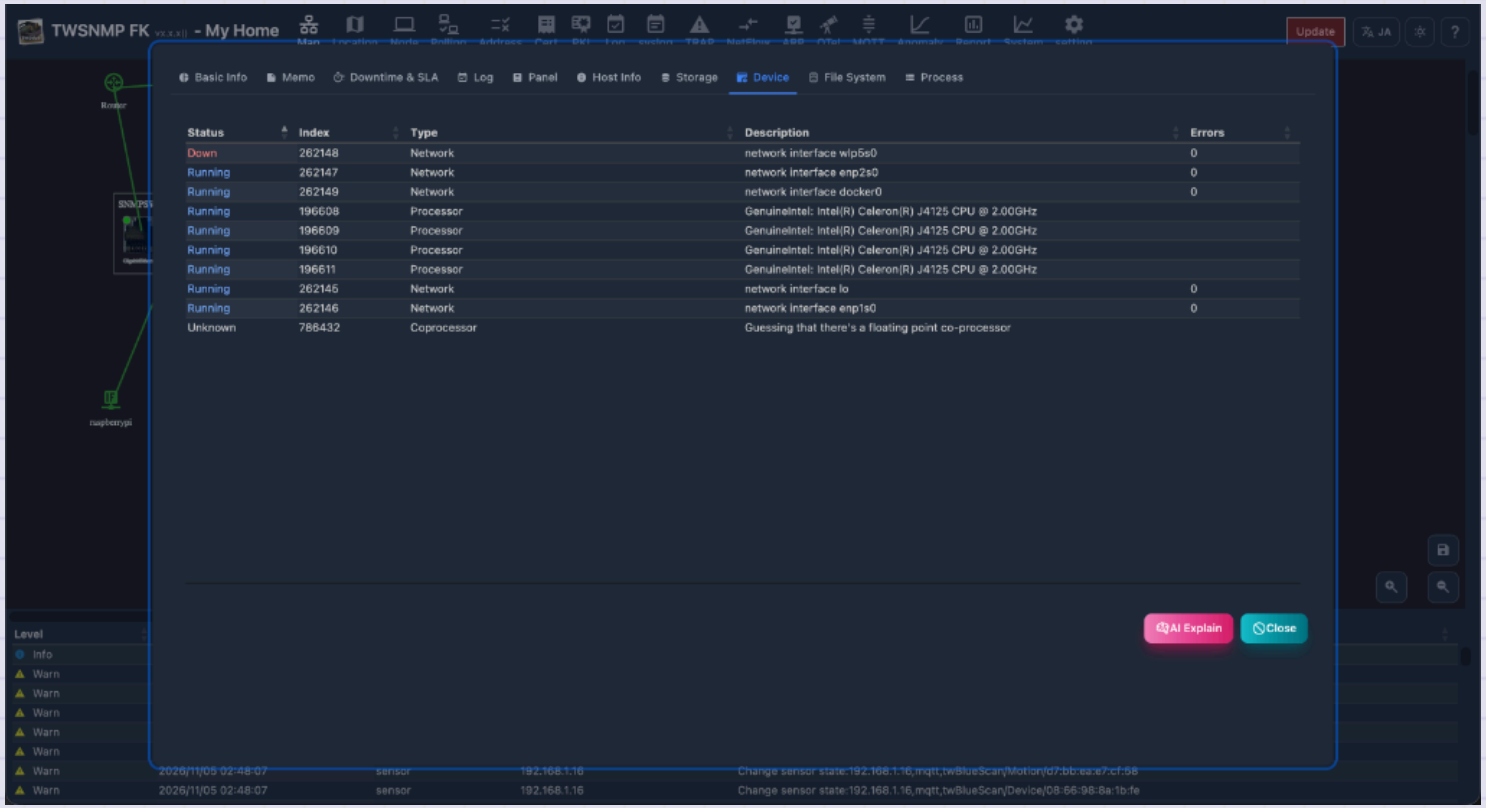
Displays the storage information of SNMP host resource mib.When you select, the addition button of the polling will be displayed.If it is not compatible with the host resource MIB, it cannot be displayed.



Copyright (c) 2023-2026 Masayuki Yamai

Device

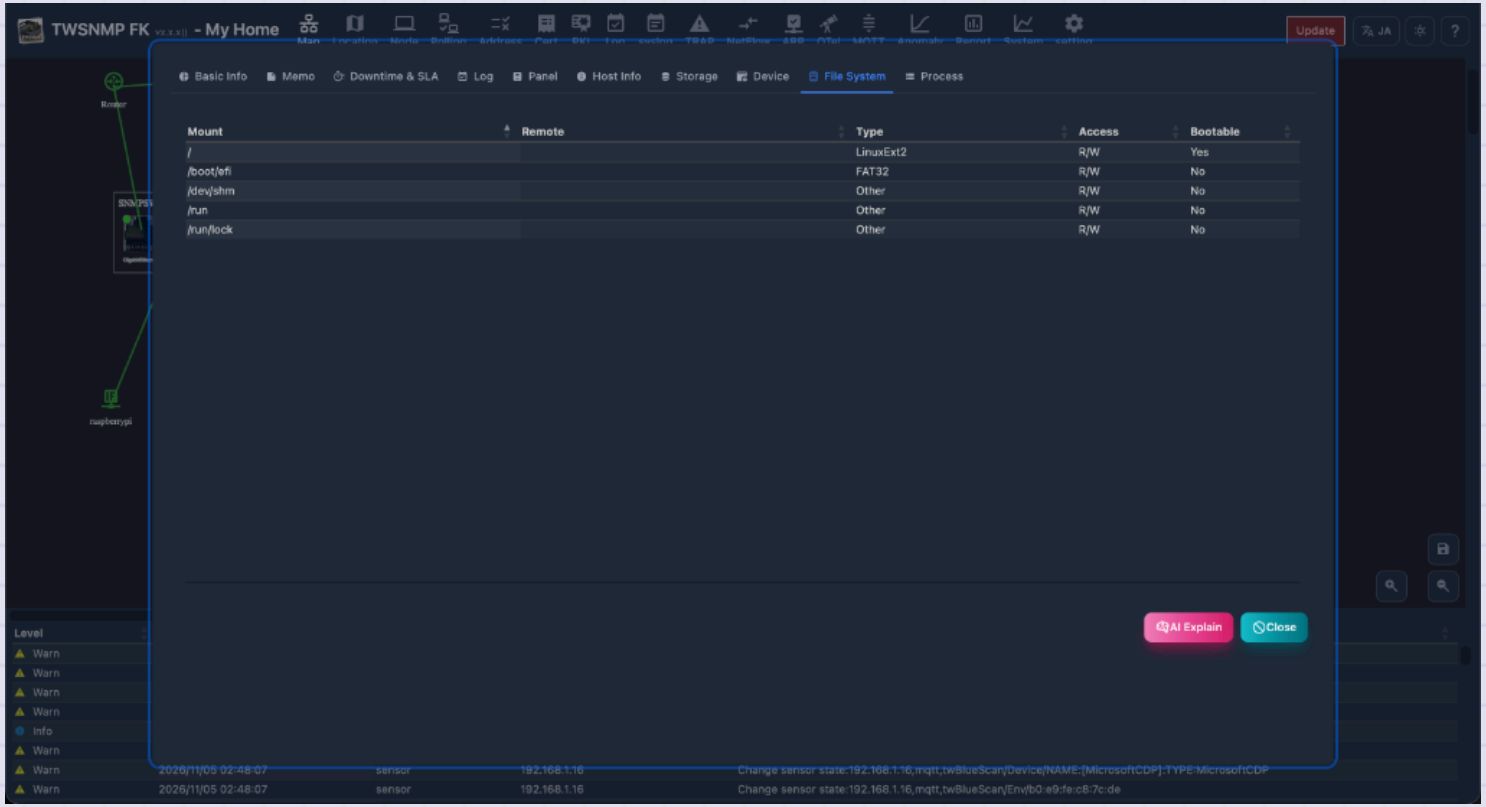
Displays the device information of the SNMP host resource MIB.If it is not compatible with the host resource MIB, it cannot be displayed.



Copyright (c) 2023-2026 Masayuki Yamai

File System

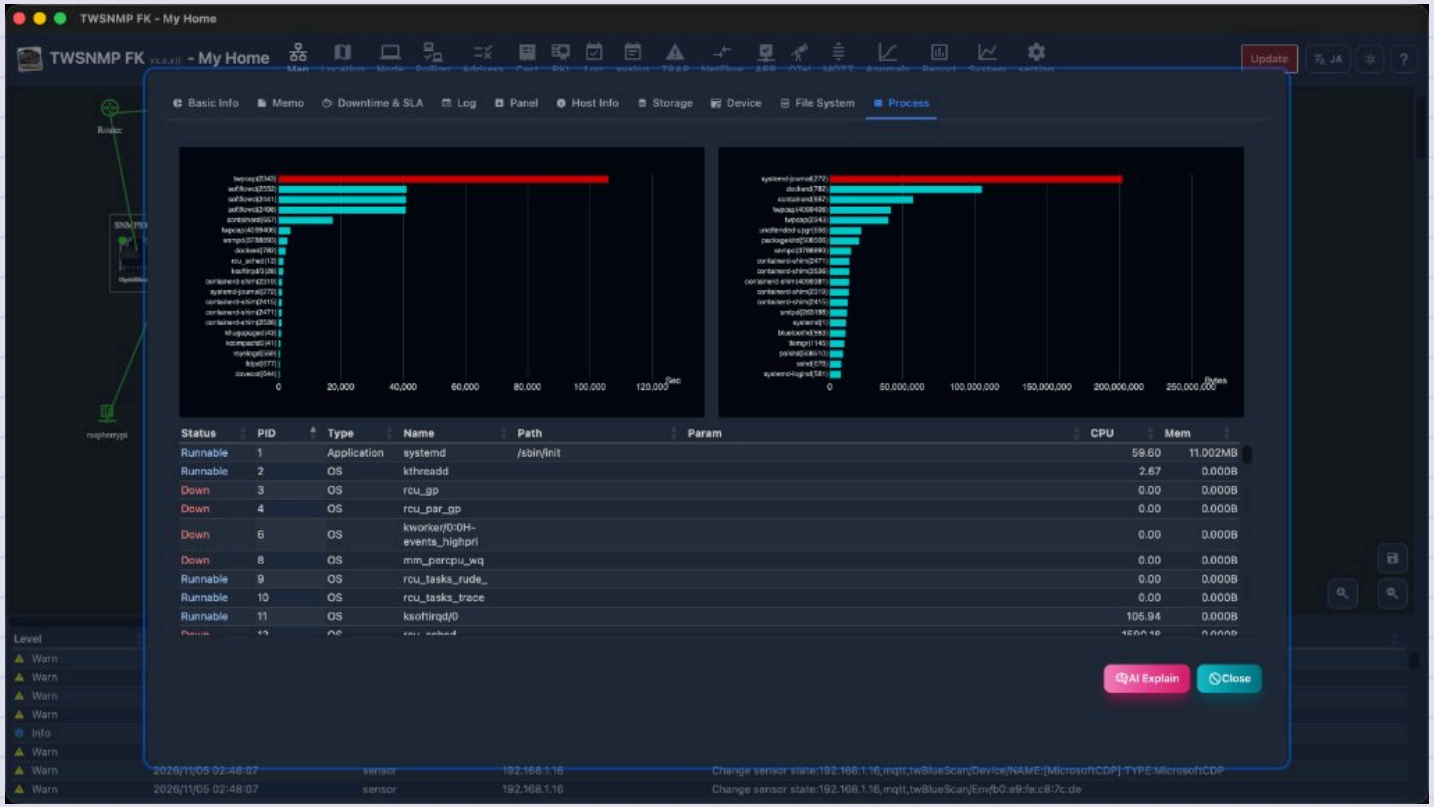
Displays File System, information on SNMP host sources MIB.If it is not compatible with the host resource MIB, it cannot be displayed.



Copyright (c) 2023-2026 Masayuki Yamai

Process

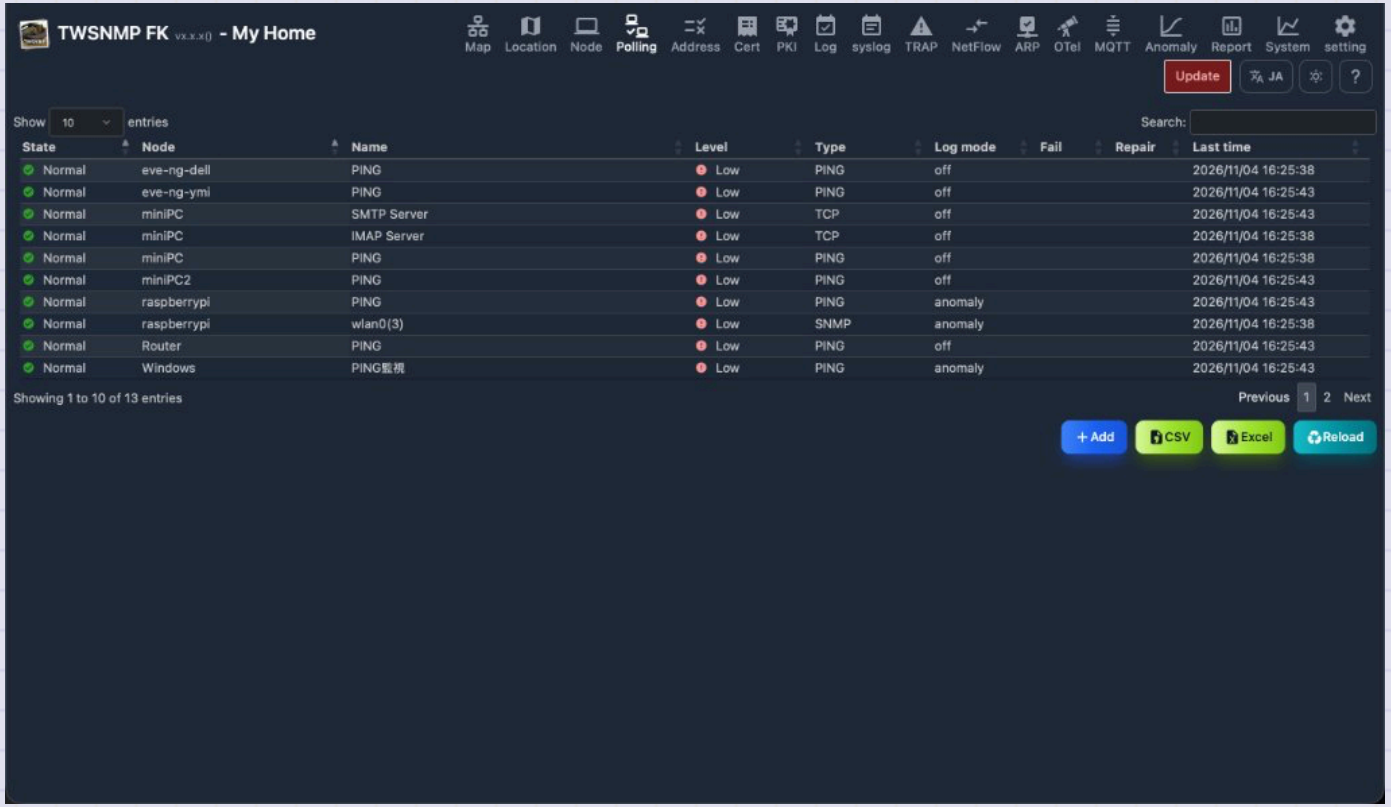
Displays the process information of SNMP host resource mib.When you select, the addition button of the polling will be displayed.If it is not compatible with the host resource MIB, it cannot be displayed.



Copyright (c) 2023-2026 Masayuki Yamai

Polling list

A list of polling to be managed.



Copyright (c) 2023-2026 Masayuki Yamai

Items	Contents
State	Polling state. Severe, mild, precautions, return, normal, unknown.
Node name	Node related to polling.
Name	Polling name.
Level	Pauling disability level.
Type	Polling type.(Ping, SNMP, TCP, HTTP, DNS, TWSNMP, Syslog, Mail, etc.)
Log	Polling log mode.
Final confirmation	Polling final confirmation date and time.

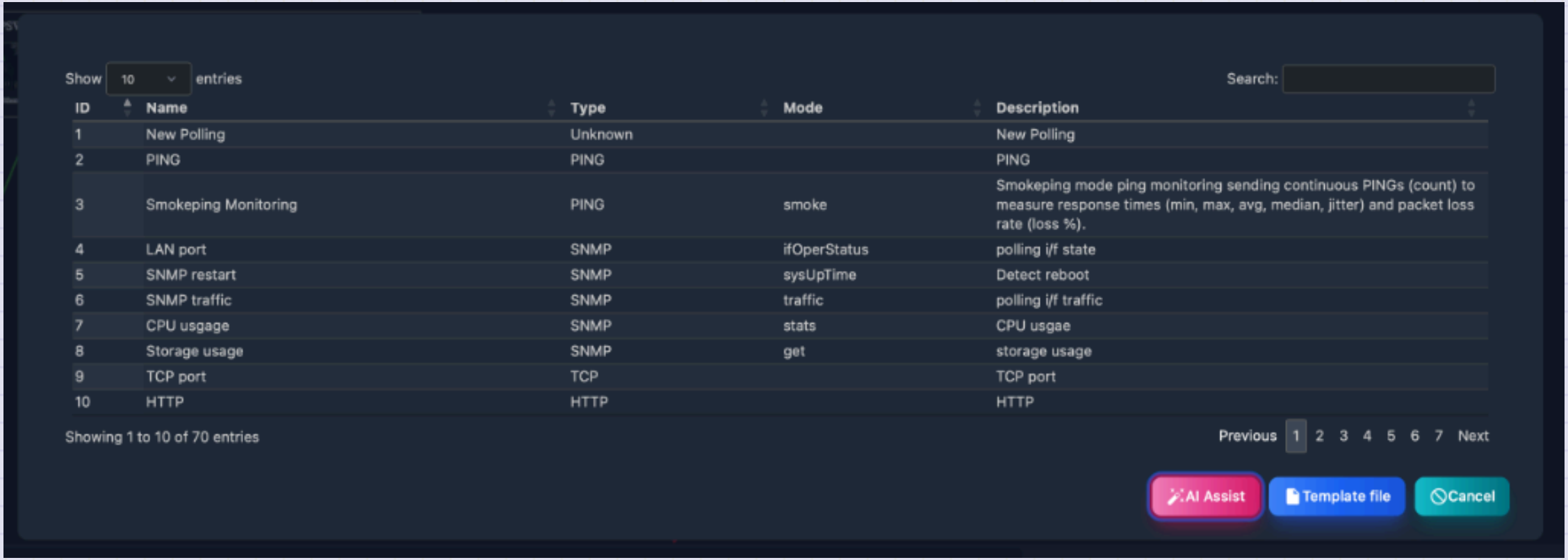
Copyright (c) 2023-2026 Masayuki Yamai

Items	Contents
Add	Add polling.
Edit	Edit the selected polling.
Copy	Copy the selected polling.
Export	Export the selected polling settings.
Report	Displays the selected polling analysis report.
Delete log	Delete the selected polling logs.
Delete	Delete the selected polling.
CSV	Export the polling list to the CSV file.
Excel	Export the polling list to the Excel file.
Reload	Update the polling list to the latest state.

Copyright (c) 2023-2026 Masayuki Yamai

Polling template selection

This is the selection screen of the template displayed when adding polling.



Copyright (c) 2023-2026 Masayuki Yamai

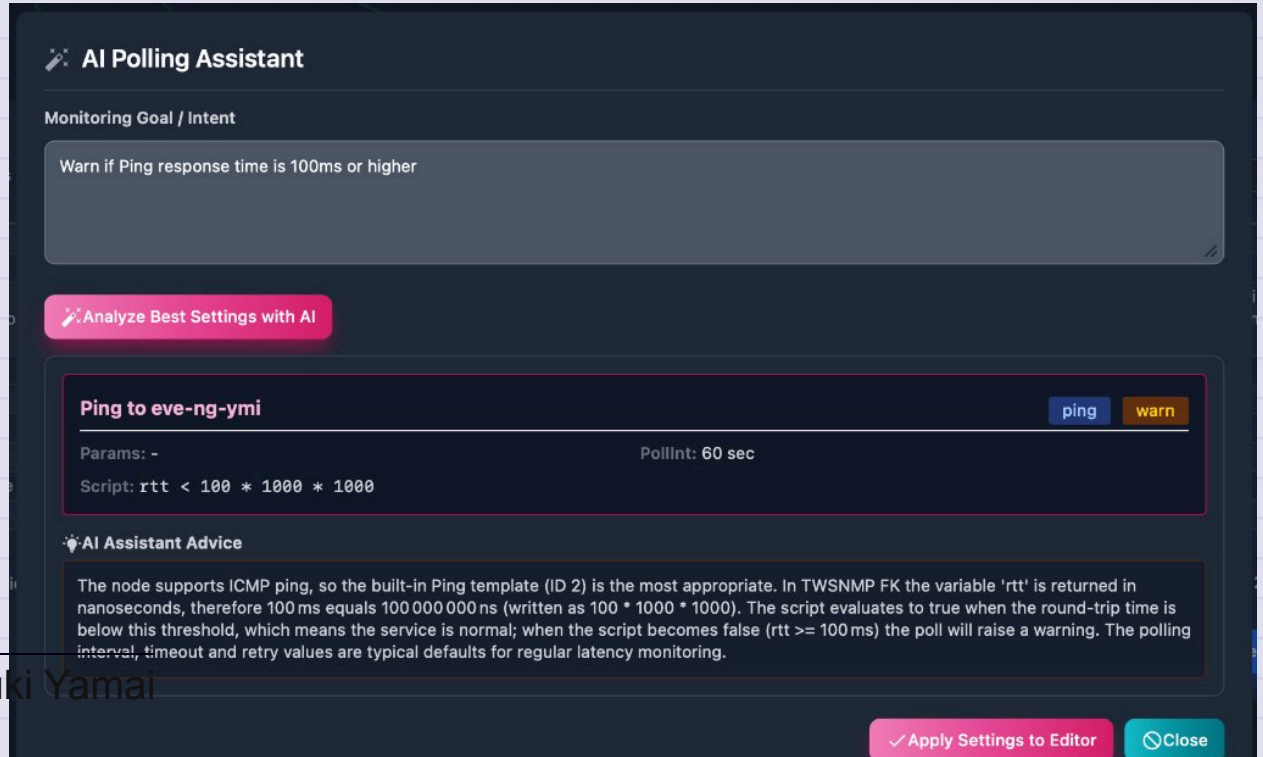
Items	Contents
ID	Template number.
Name	Polling name.
Type	Polling type. Ping, SNMP, TCP, etc.
Mode	Polling mode.
Description	Polling explanation.
Template file	Import polling from template file.
Add	Select polling.
Cancel	Polling Closes.

Copyright (c) 2023-2026 Masayuki Yamai

AI Polling Setting Assistant

An assistant feature that leverages AI (LLM) to automatically generate optimal polling types, threshold parameters, evaluation JavaScript scripts, and setup advice from natural language prompts or received log messages.

Natural Language Prompting



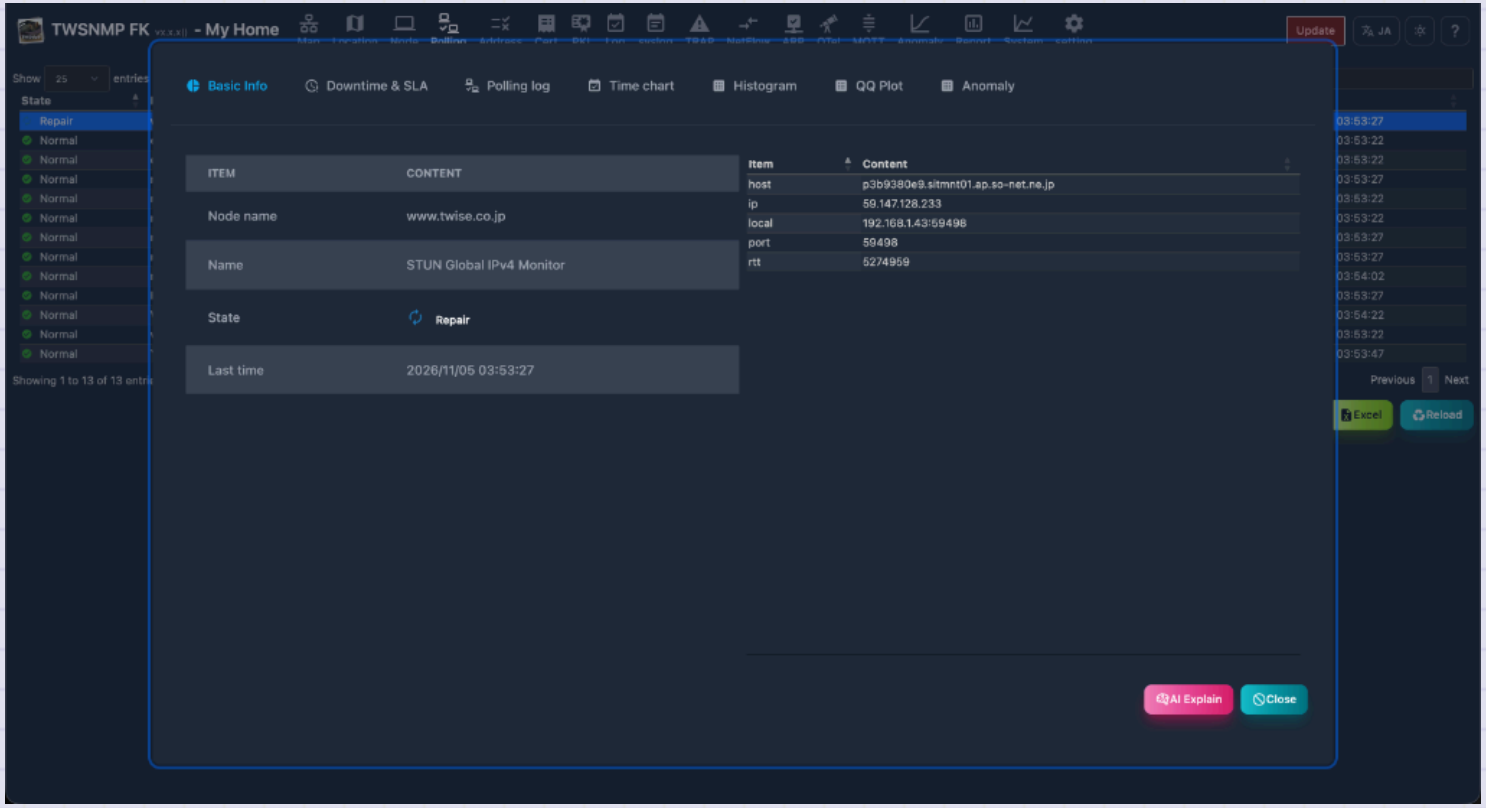
Copyright (c) 2023-2026 Masayuki Yamai

Items	Contents
Monitoring Goal / Intent	Enter desired monitoring details or alert conditions in natural language.
Quick Examples	One-tap input for common monitoring scenarios (Ping Latency, HTTP/HTTPS, Syslog Error, etc.).
Analyze Best Settings with AI	Send request to AI (LLM) based on the instructions to analyze and generate optimal settings.
Setting preview	Preview of the generated polling type, evaluation script, polling interval, and status level.
AI Assistant Advice	Detailed explanation and guidance from AI regarding the generated configuration logic.
Apply Settings to Editor	Apply the AI-generated settings directly to the polling configuration/edit screen.

Copyright (c) 2023-2026 Masayuki Yamai

Basic information

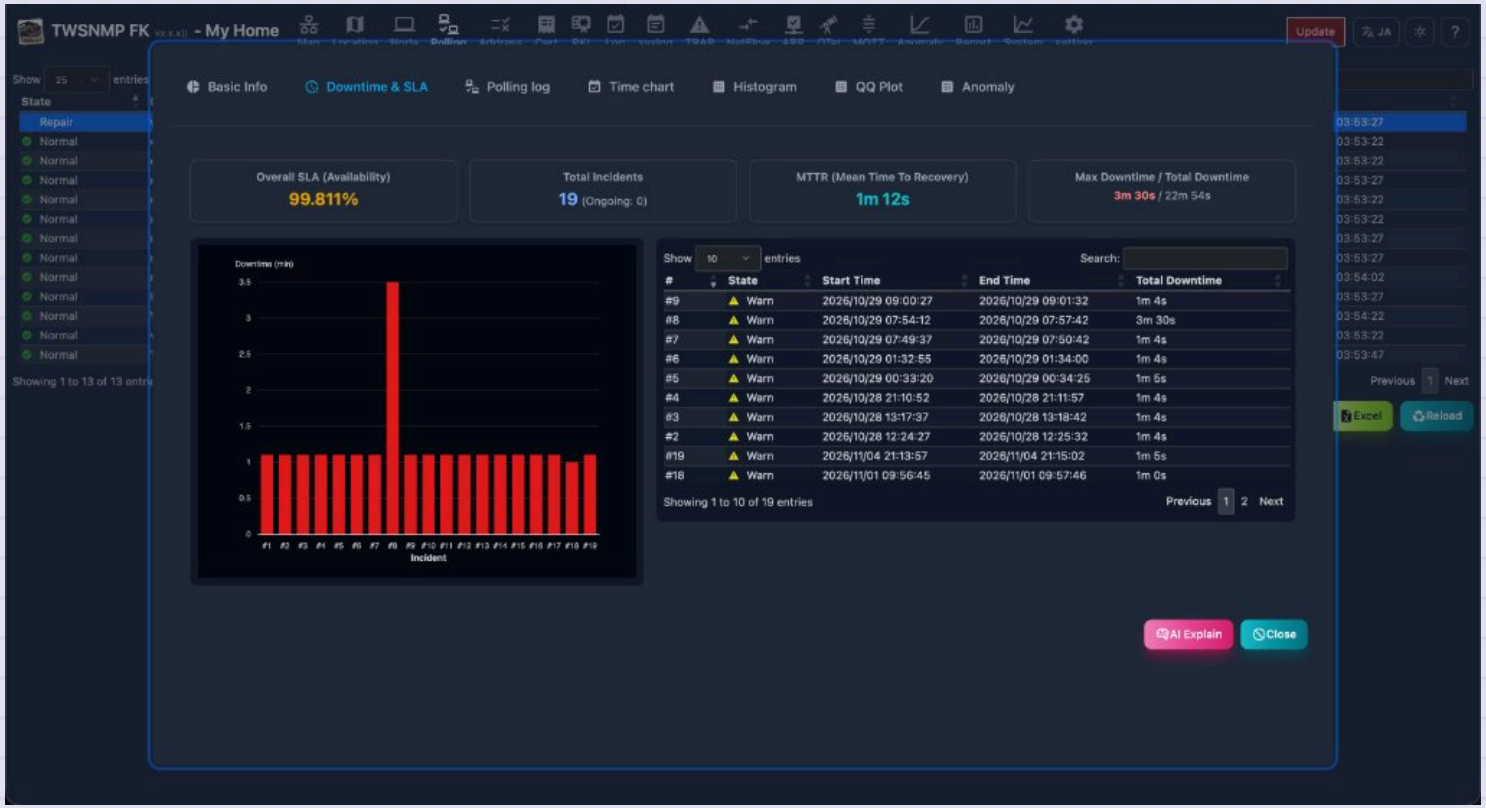
Basic information about polling.



Copyright (c) 2023-2026 Masayuki Yamai

Downtime & SLA

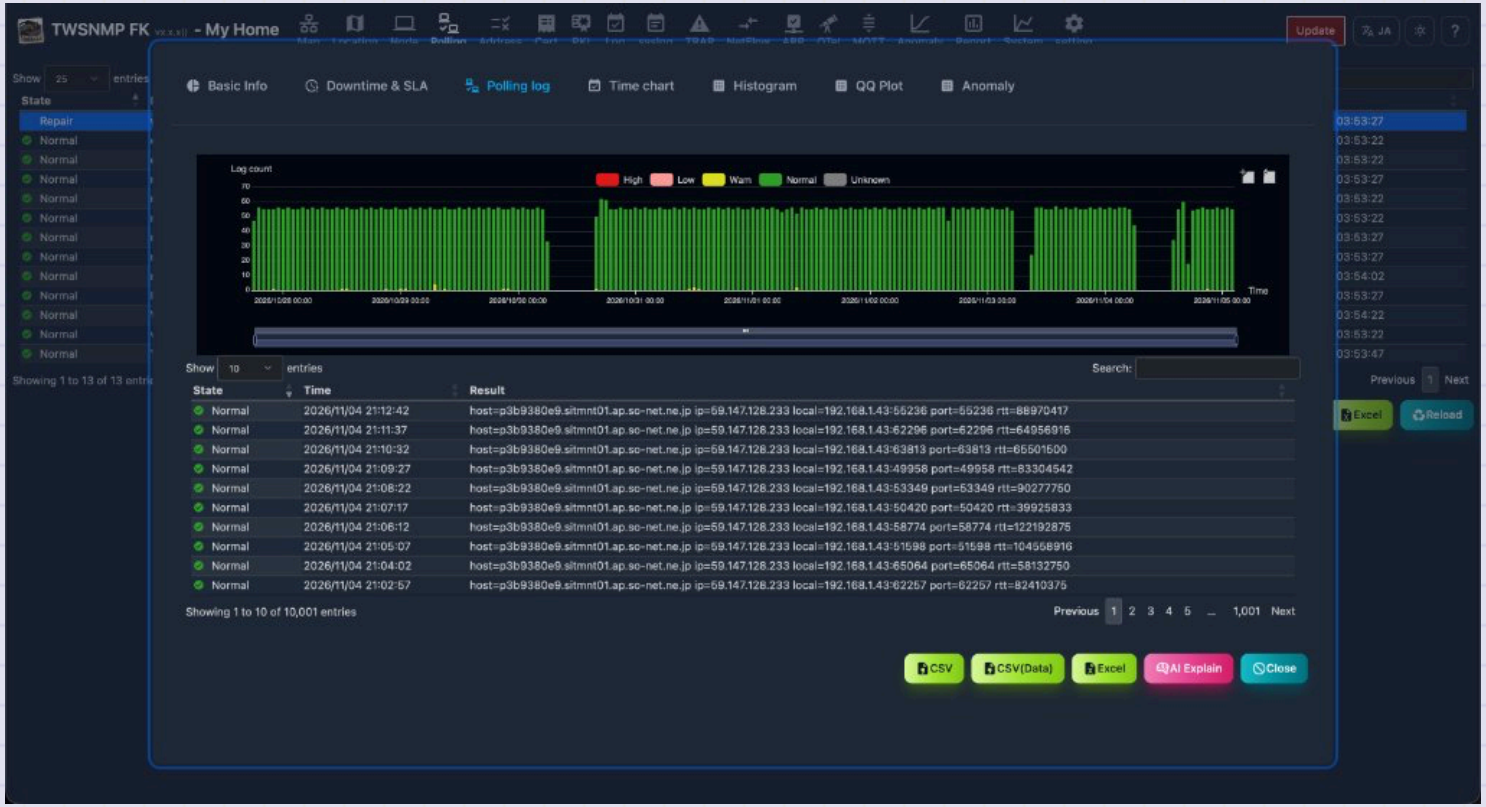
Calculates and displays overall SLA (Availability %), total incidents count (with ongoing status), MTTR (Mean Time To Recovery), and max/total downtime based on polling logs. Only displayed when log mode is enabled.



Copyright (c) 2023-2026 Masayuki Yamai

Polling log

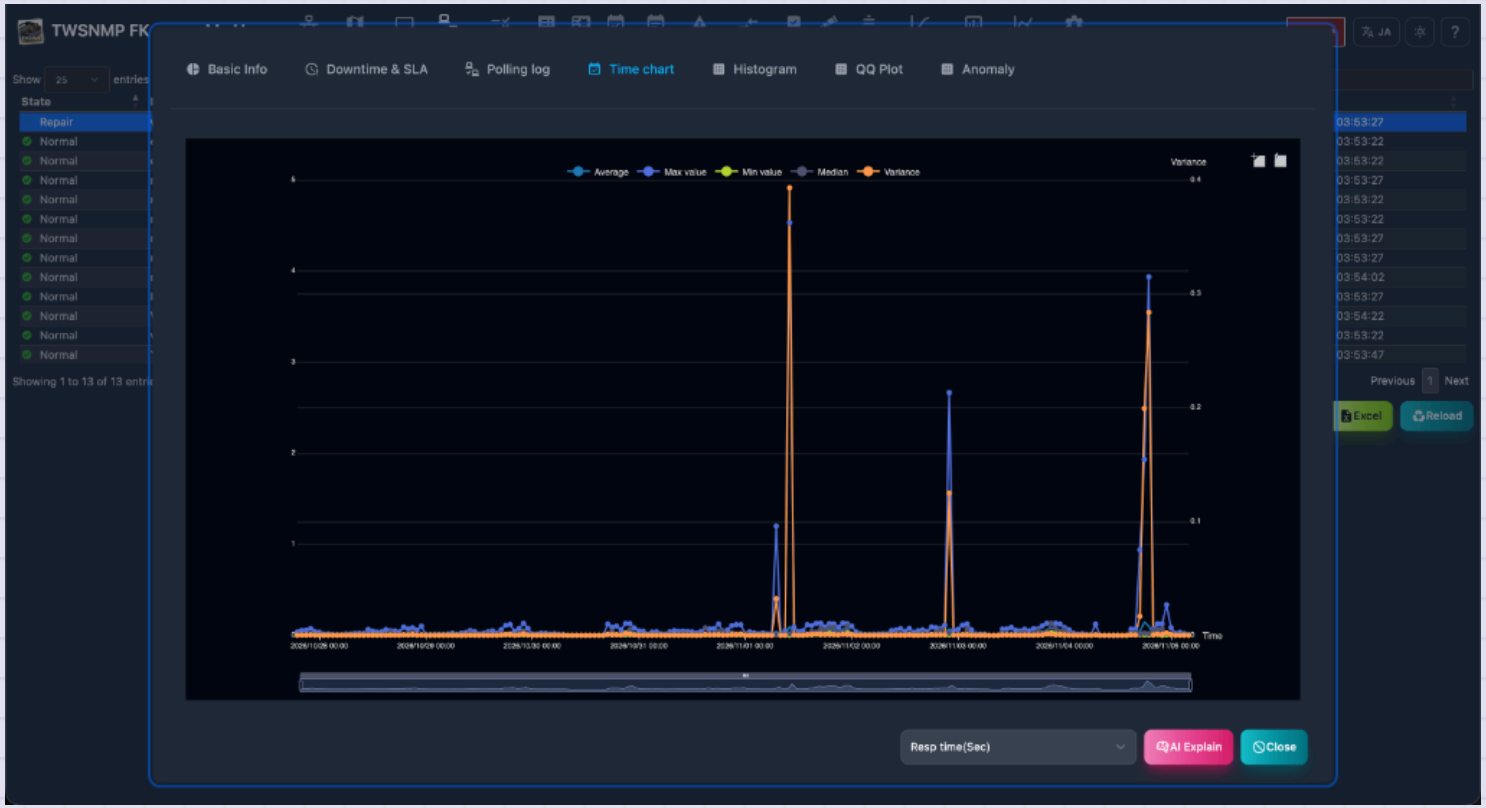
This is a log of the polling result.It is displayed only when the log mode is not output.



Copyright (c) 2023-2026 Masayuki Yamai

Time chart

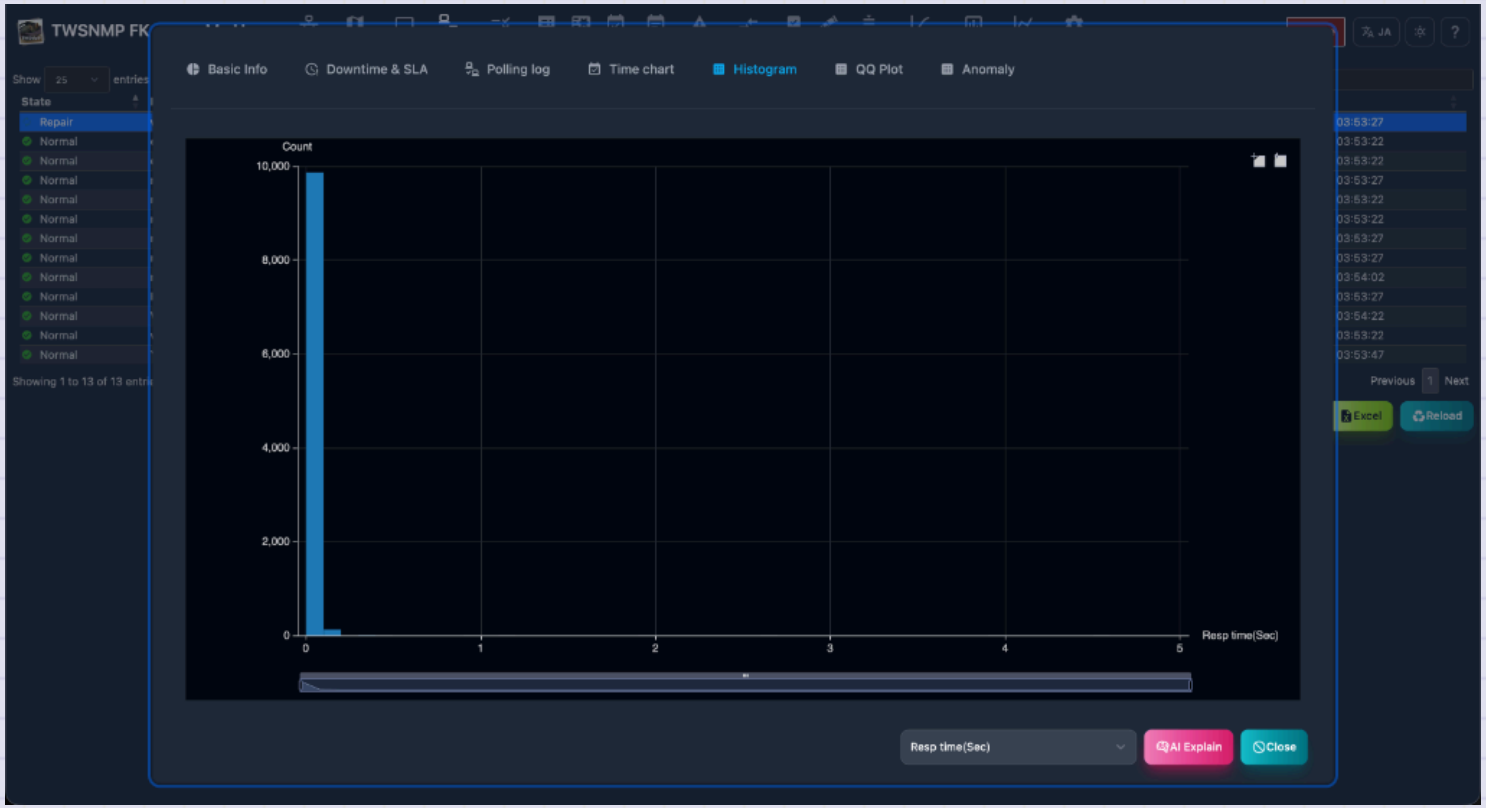
In the log of the polling result, the numerical data is displayed in a chronological graph.The displayed items can be selected at the top of the graph.It is displayed only when the log mode is not output.



Copyright (c) 2023-2026 Masayuki Yamai

Histogram

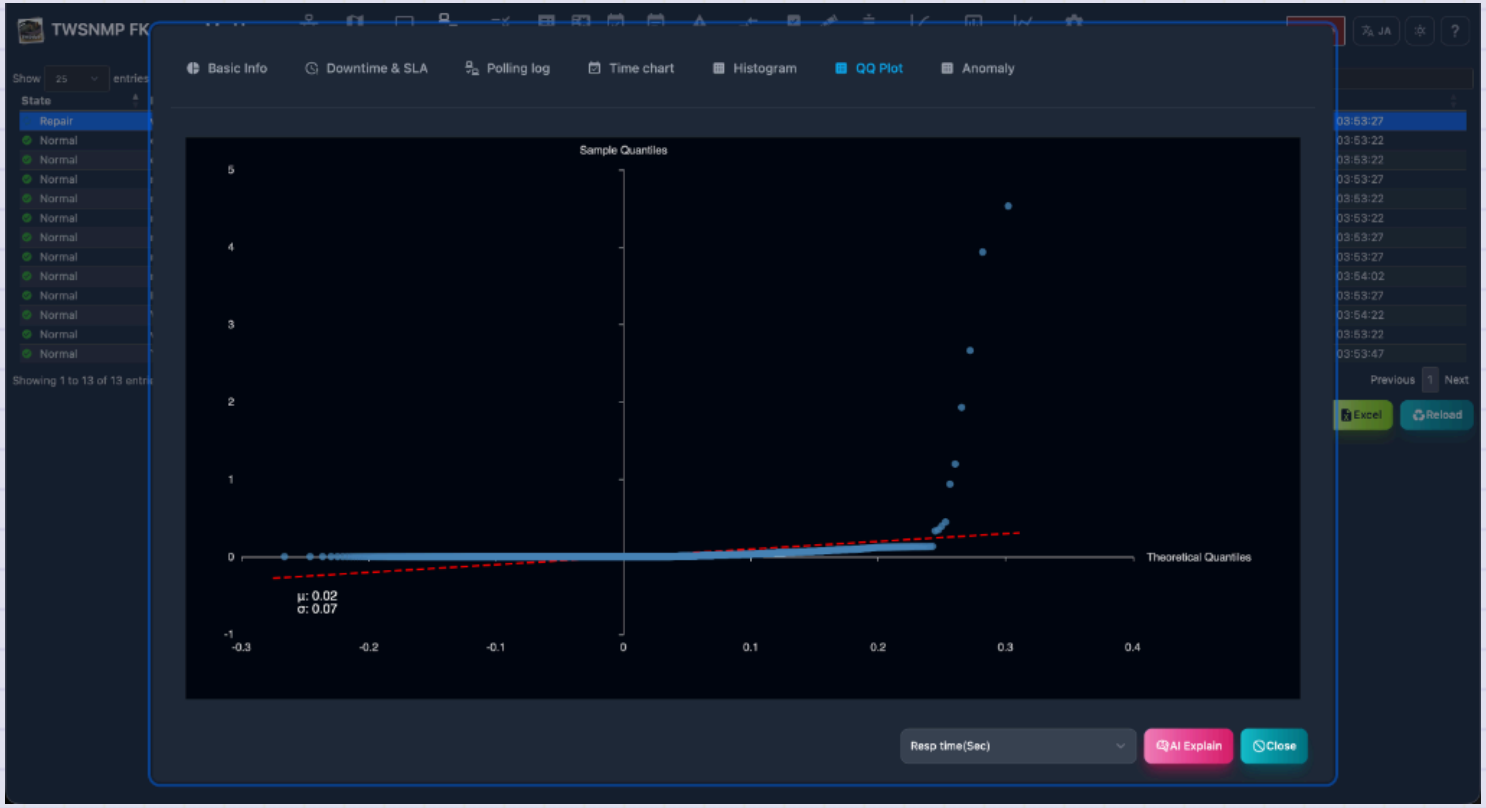
The numerical data in the log of the polling result is displayed on the histogram.The displayed items can be selected at the top of the graph.It is displayed only when the log mode is not output.



Copyright (c) 2023-2026 Masayuki Yamai

QQ Plot

Displays a Quantile-Quantile (Q-Q) plot to evaluate whether the numerical data in the polling logs follows a theoretical normal distribution, plotting sample quantiles against theoretical quantiles along with the fitted line and estimated parameters. It is displayed only when the log mode is not output.



Copyright (c) 2023-2026 Masayuki Yamai

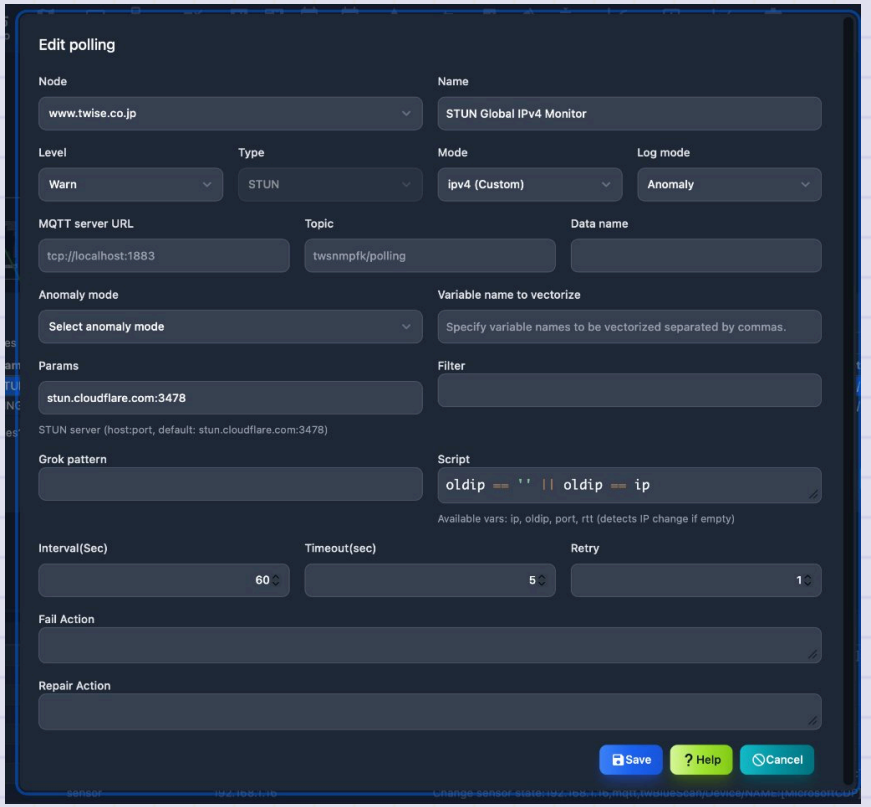
AI analysis

This is the result of AI analysis of numerical data in the log of the polling results.It is displayed only when the log mode is set to AI analysis and sufficient data is obtained.

Copyright (c) 2023-2026 Masayuki Yamai

Polling editing

Polling edit can be displayed by clicking the button on the polling list. The AI Assist feature is available when adding or editing a polling monitor.



Copyright (c) 2023-2026 Masayuki Yamai

Items	Contents
Name	Polling name.
Level	Pauling disability level.
Type	Polling type. Ping, SNMP, TCP, STUN, etc.
Mode	Operation mode dynamically populated based on the selected polling type.
Log mode	How to save the polling result log ("None", "Always", "On change", "Anomaly").

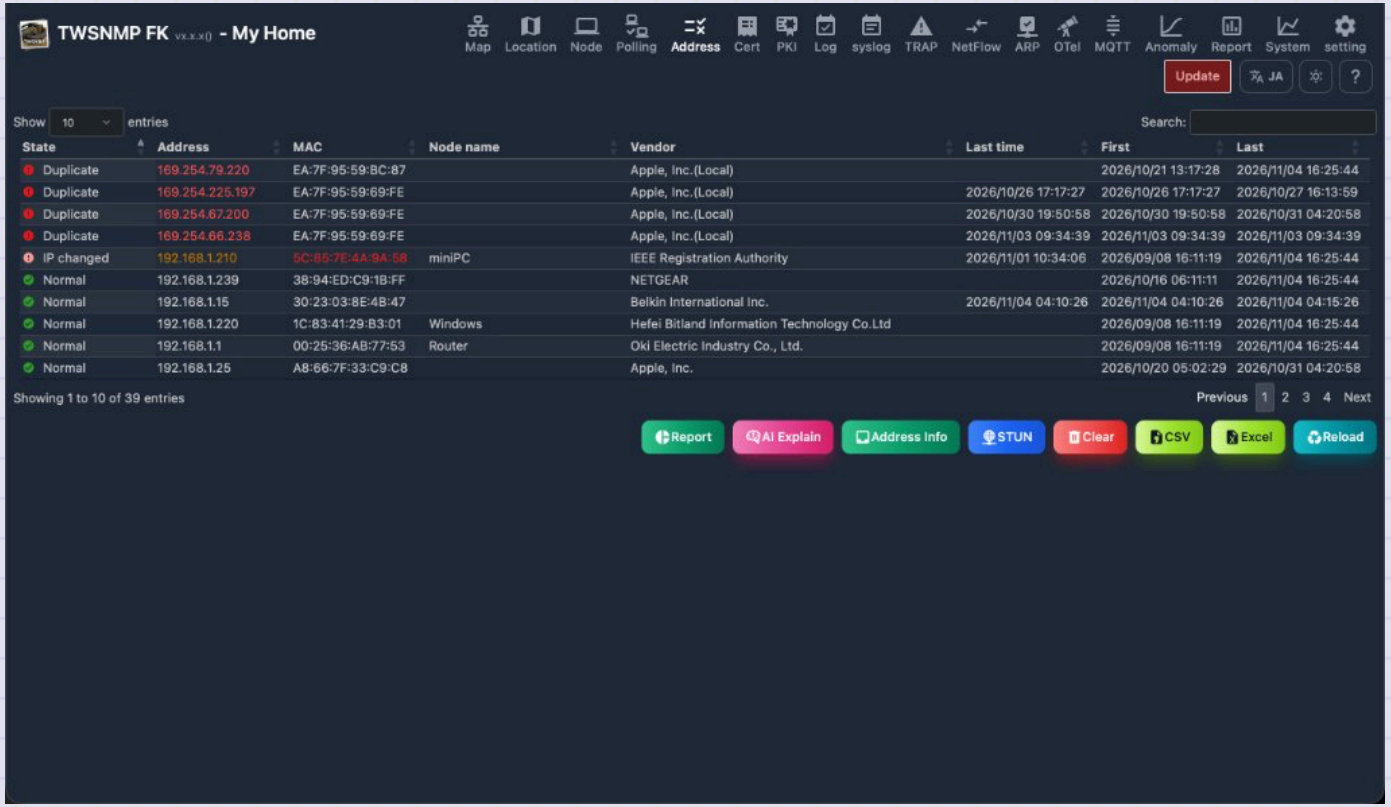
Copyright (c) 2023-2026 Masayuki Yamai

Items	Contents
Parameter	Polling type and mode -dependent parameters. For email polling, set the following: • **Mail Server**: Hostname or IP address of the IMAP/POP3 server • **Port**: Port number of the IMAP/POP3 server (993 for IMAP, 995 for POP3 are common) • **User Name**: Username for the mailbox • **Password**: Password for the mailbox • **Protocol**: IMAP or POP3 • **Secure Connection**: Whether to use SSL/TLS • **Keyword**: Check if the mail subject or body contains specific keywords (optional)
Filter	Polling type and filter condition that depends on mode.

Copyright (c) 2023-2026 Masayuki Yamai

Address list

This is a list of IP address found by TWSNMP.Only the IP address in the same segment found in the ARP watch function is displayed.You can detect duplicate and the change in the address.



Copyright (c) 2023-2026 Masayuki Yamai

Items	Contents
State	It is the state of the address.(Normal, duplicate, IP change, Mac change.)
Address	IP address.
Domain	Domain information obtained from reverse IP lookup, etc.
MAC address	MAC address.
Node name	The name of the node registered on the map as a management target.
Vendor	The name of the vendor corresponding to the MAC address.
Risk	Risk level determined from the IP address.
Final change	This is the last change date and time.

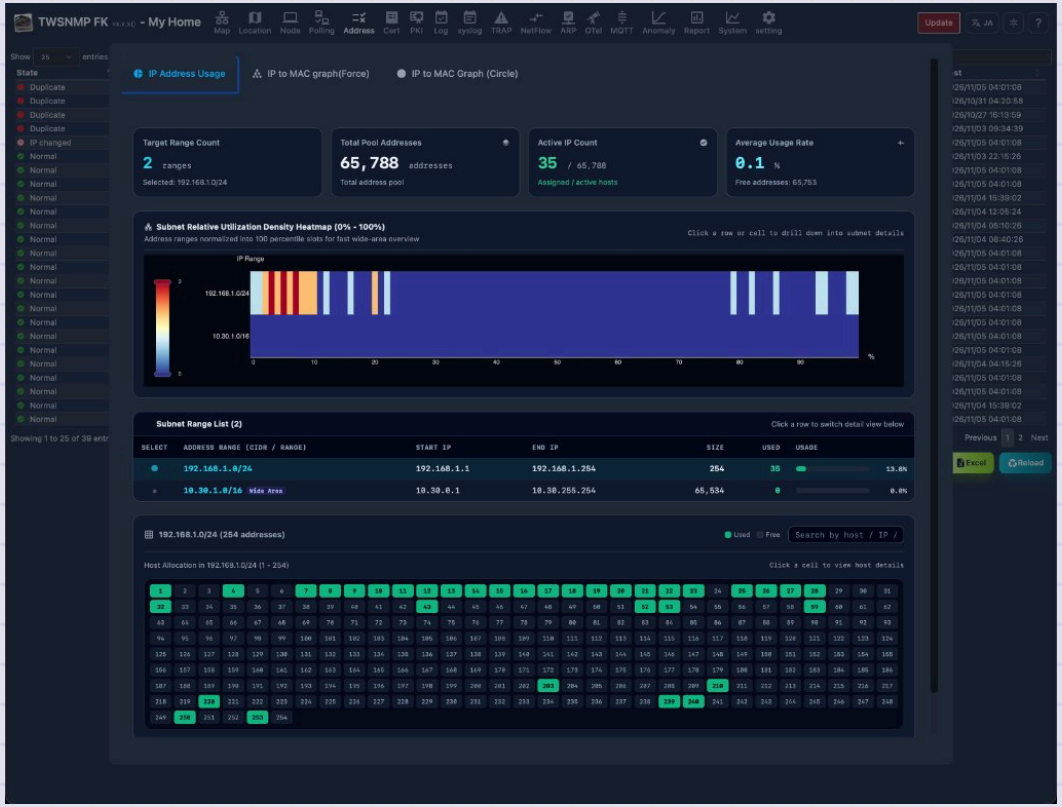
Copyright (c) 2023-2026 Masayuki Yamai

Items	Contents
Add node	Add the selected IP address to the map.It is displayed only when it is not registered.
Delete	Delete the selected IP address.
Report	Display the address list report.
AI Explanation	Sends current tab data to AI (LLM) for subnet allocation efficiency, IP conflict risk analysis, and recommendations (only visible when AI integration is enabled).
clear	Clear all address lists.
CSV	Export the address list to the CSV file.
Excel	Export the address list to the Excel file.
Reload	Update the address list to the latest state.

Copyright (c) 2023-2026 Masayuki Yamai

IP address usage status

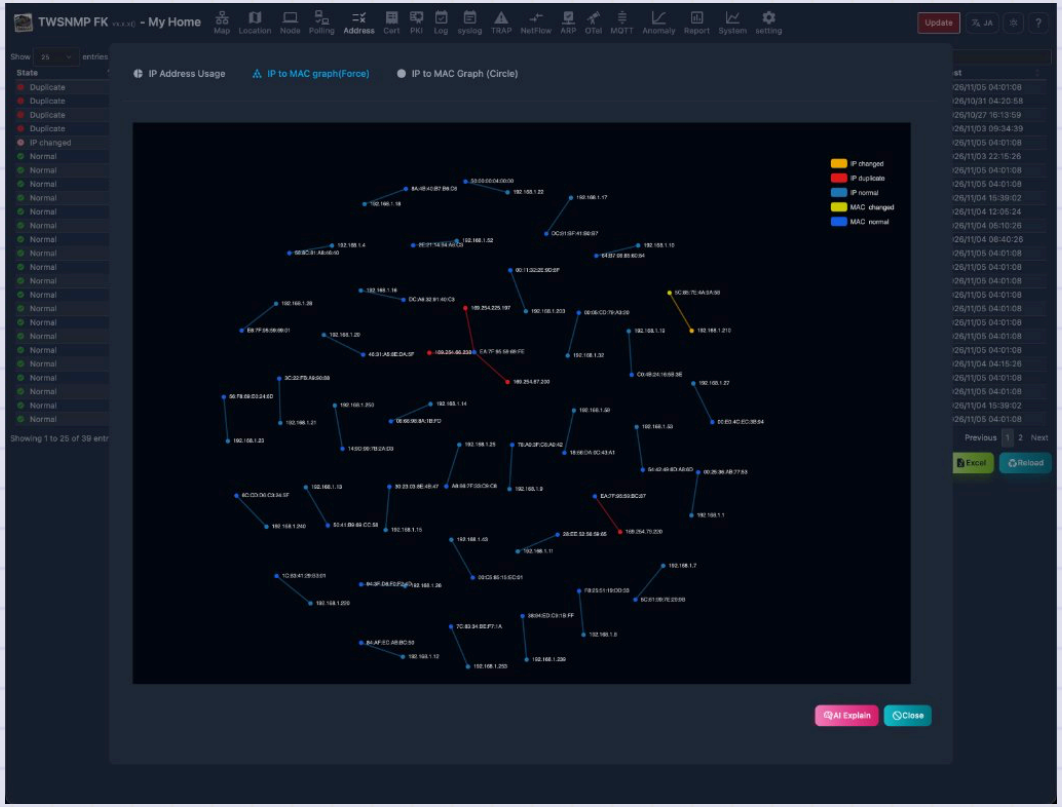
This is a report on the status of the set IP address.



Copyright (c) 2023-2026 Masayuki Yamai

Relationship between IP and MAC address (force model)

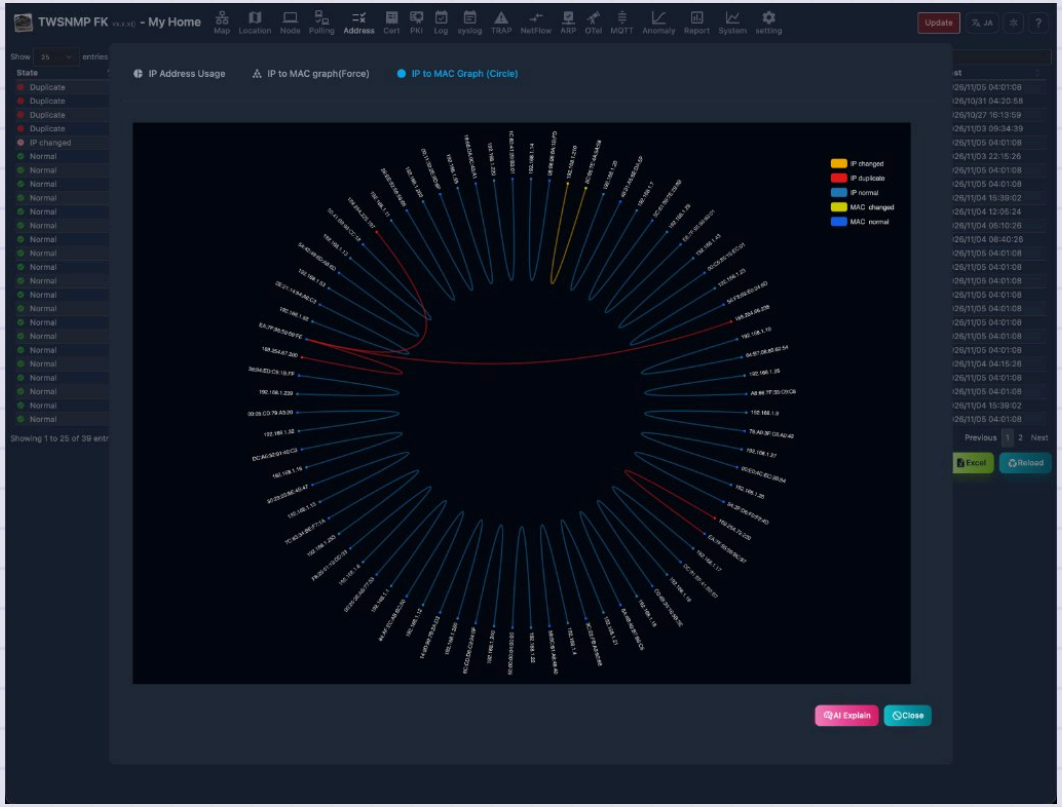
This is a report that shows the relationship between IP address and MAC address with an force model.The normal address is one -on - one for the IP address and the MAC address.You can detect MAC addresses using the same IP address on multiple Macs or having multiple IP addresses.



Copyright (c) 2023-2026 Masayuki Yamai

Relationship between IP and MAC address (circular model)

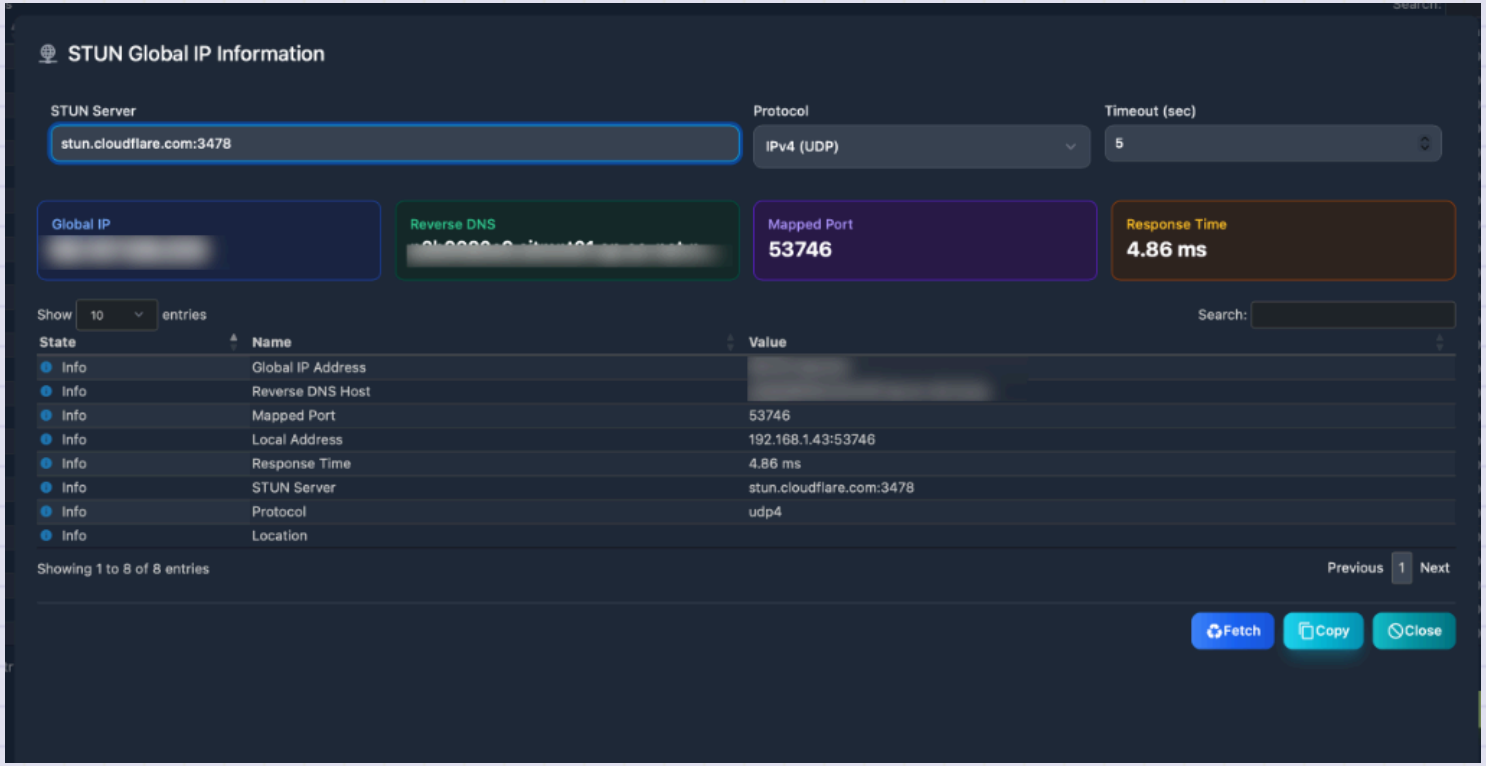
This is a report that shows the relationship between IP address and MAC address with a circular model.The normal address is one -on -one for the IP address and the MAC address.You can detect MAC addresses with the same IP address on multiple Macs or have multiple IP addresses.



Copyright (c) 2023-2026 Masayuki Yamai

STUN Global IP Information

Inspects external public global IP address, mapped port, reverse DNS, local address, RTT, and GeoIP location using STUN servers.



Copyright (c) 2023-2026 Masayuki Yamai

Address Analysis

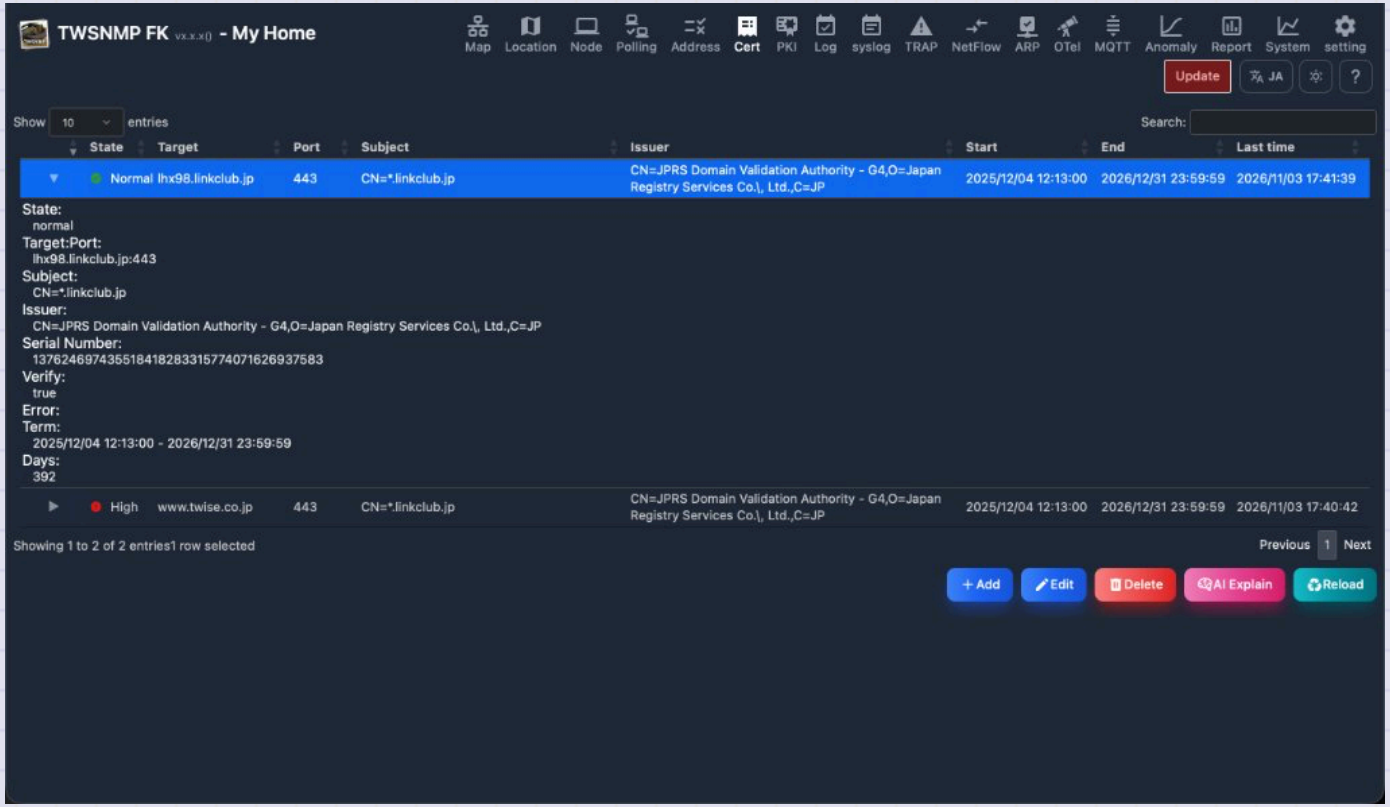
Displays detailed info on IP addresses, MAC addresses, and domain names.
Available from the address list or log lists (Syslog, NetFlow, etc.).

Item	Contents
State	Information level (severity).
Name	Information item name.
Value	Information value.

Copyright (c) 2023-2026 Masayuki Yamai

Server certificate list

Monitor server certificates from TWSNMP. Add targets via the edit screen.

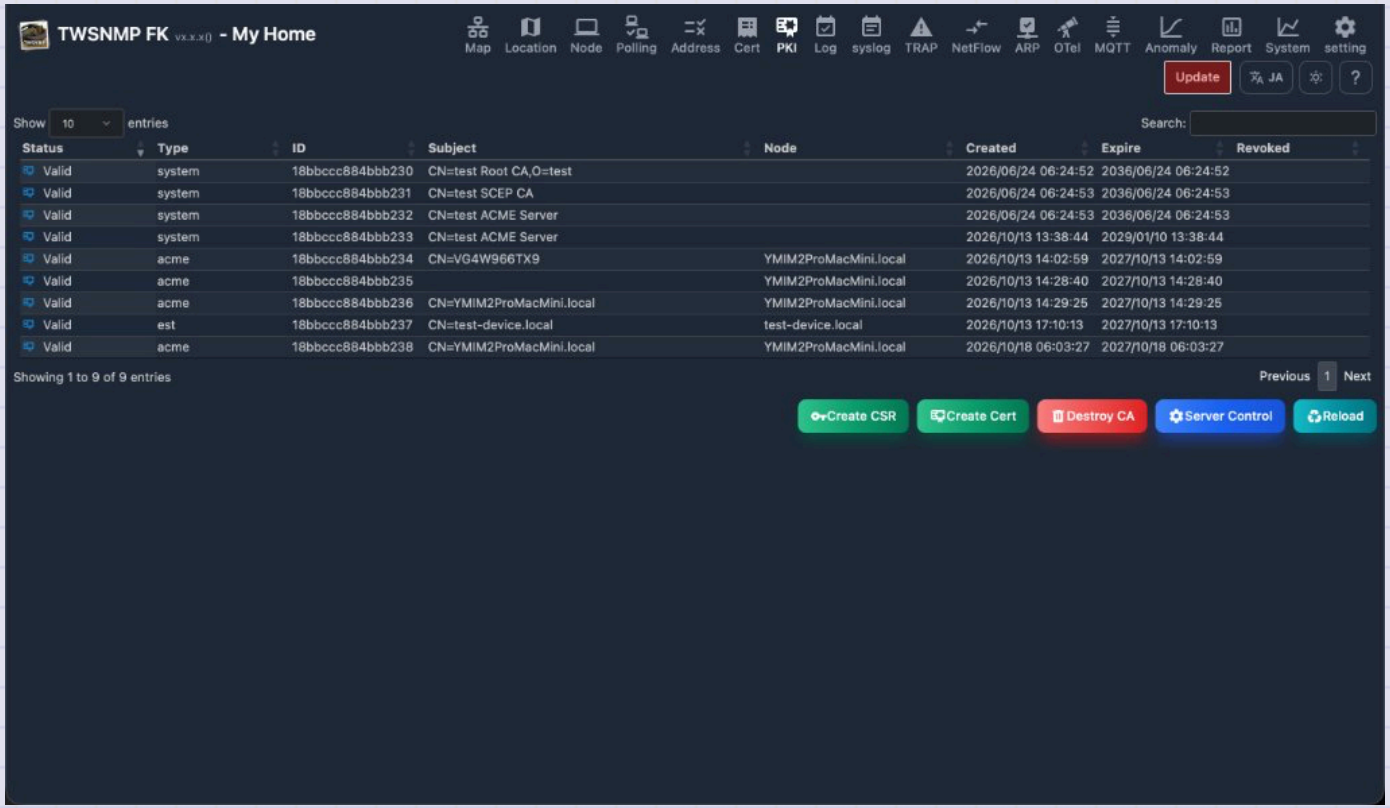


Item	Content
>	Expand the server certificate details.

Copyright (c) 2023-2026 Masayuki Yamai

PKI CA construction

This is the screen before building a CA for the PKI function.



Copyright (c) 2023-2026 Masayuki Yamai

Item	Content
Name	This is the name of the CA.I'll try to use the Subject of the CA certificate.
DNS name	Specify the CDP of the certificate to be issued, the OCSP address, the host name and IP address to be used for SANs for the certificate of the ACME server, separated by commas.
ACME URL	This is the basic URL for the ACME server.Blanks will be automatically set from the host name.
OCSP/SCEP Server URL	This is the basic URL for the CRL/OCSP/SCEP Server.Blanks will be automatically set from the host name.
CA key type	Specify the CA key type.

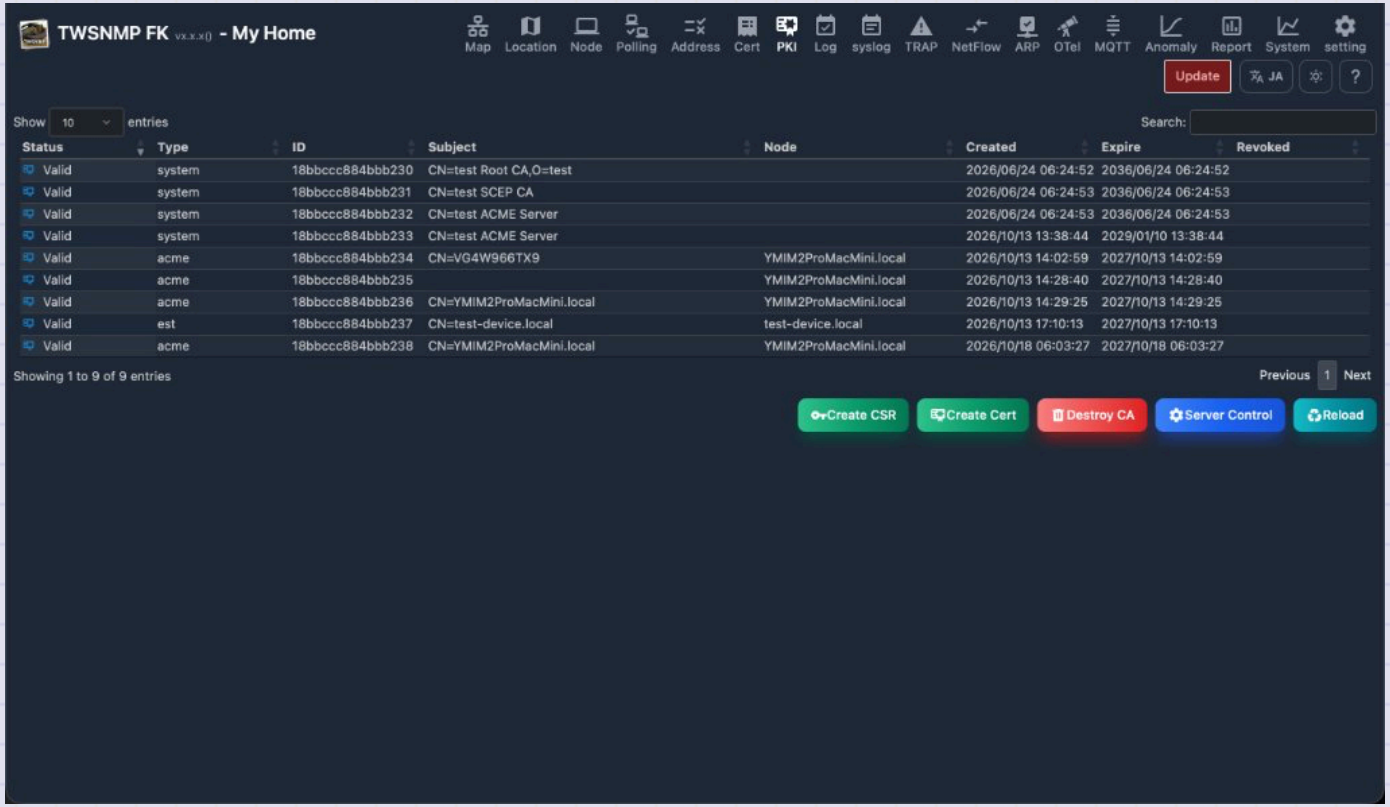
Copyright (c) 2023-2026 Masayuki Yamai

Item	Content
CA certificate duration	Specify the number of years the certificate is valid.
CRL Update Interval	Specify the CRL update interval in hours.
Certificate Period	Specify the period of the certificate to be issued in hours.
CRL/OCSP/SCEP server port number	Specify the HTTP server port number.Cannot be changed later.
ACME Server Port Number	Specify the ACME Server Port Number.Cannot be changed later.

Copyright (c) 2023-2026 Masayuki Yamai

Certificate list

After the CA is built, the certificate list screen will be displayed.You can check the issued certificate.



Copyright (c) 2023-2026 Masayuki Yamai

Item	Content
Status	Certificate status.
Type	Certificate type.
ID	Certificate serial number.
Subject	A Subject for the certificate.
Node	The node where the certificate was obtained.
Created	The start date and time of the certificate period.
Expire	The end date and time of the certificate period.
Revoked	The date and time the certificate was revoked.

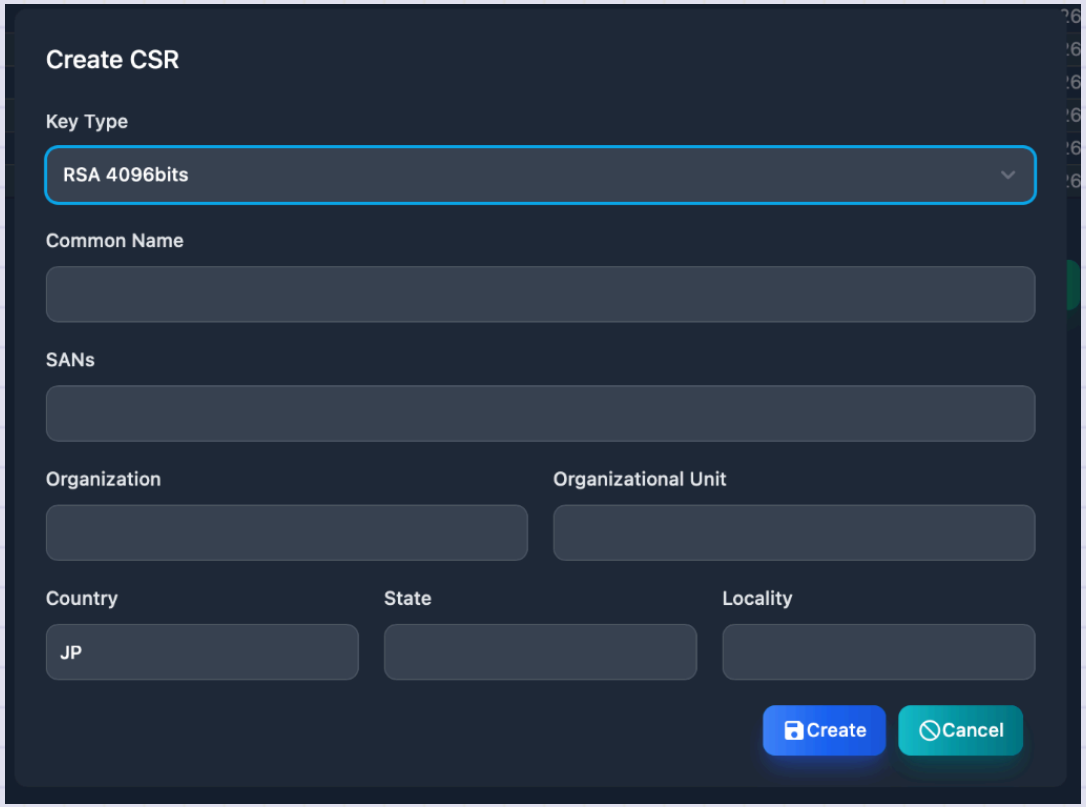
Copyright (c) 2023-2026 Masayuki Yamai

Item	Content
Create CSR	Displays the screen for creating a certificate request (CSR).
Certificate creation	Read the CSR and issue the certificate.
CA Initialization	Destroy CA.
Server Control	Displays the server control screen.
Renew	Update the certificate list.
Revokes	Revokes the selected certificate.
Export	Saves the selected certificate to a file.

Copyright (c) 2023-2026 Masayuki Yamai

Create CSR

This is the screen for creating a certificate request (CSR).



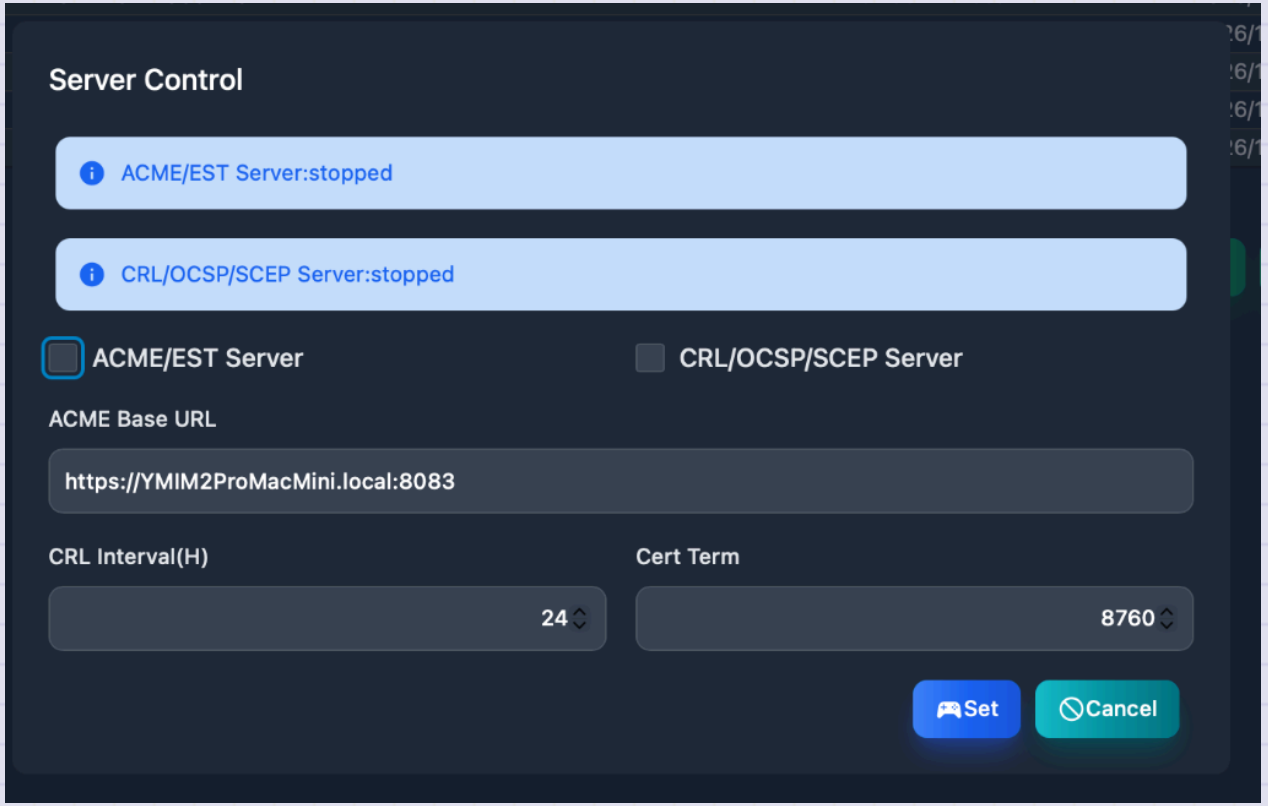
Copyright (c) 2023-2026 Masayuki Yamai

Item	Content
Key type	Specifies the key type for CSR.
Name	Specifies the value for CN.
SANs	Subject Alt Names are specified, separated by commas.
Organization name	Specify the organization name.It's OK to leave blank.
Organization Unit	Specify an organizational unit.It's OK to leave blank.
Country code	Specify the country code.It's OK to leave blank.
State/Province name	Specify the state or prefecture name.It's OK to leave blank.
City name	Specify the city name.It's OK to leave blank.

Copyright (c) 2023-2026 Masayuki Yamai

Server Control

This is a screen that controls the operation of the PKI server.



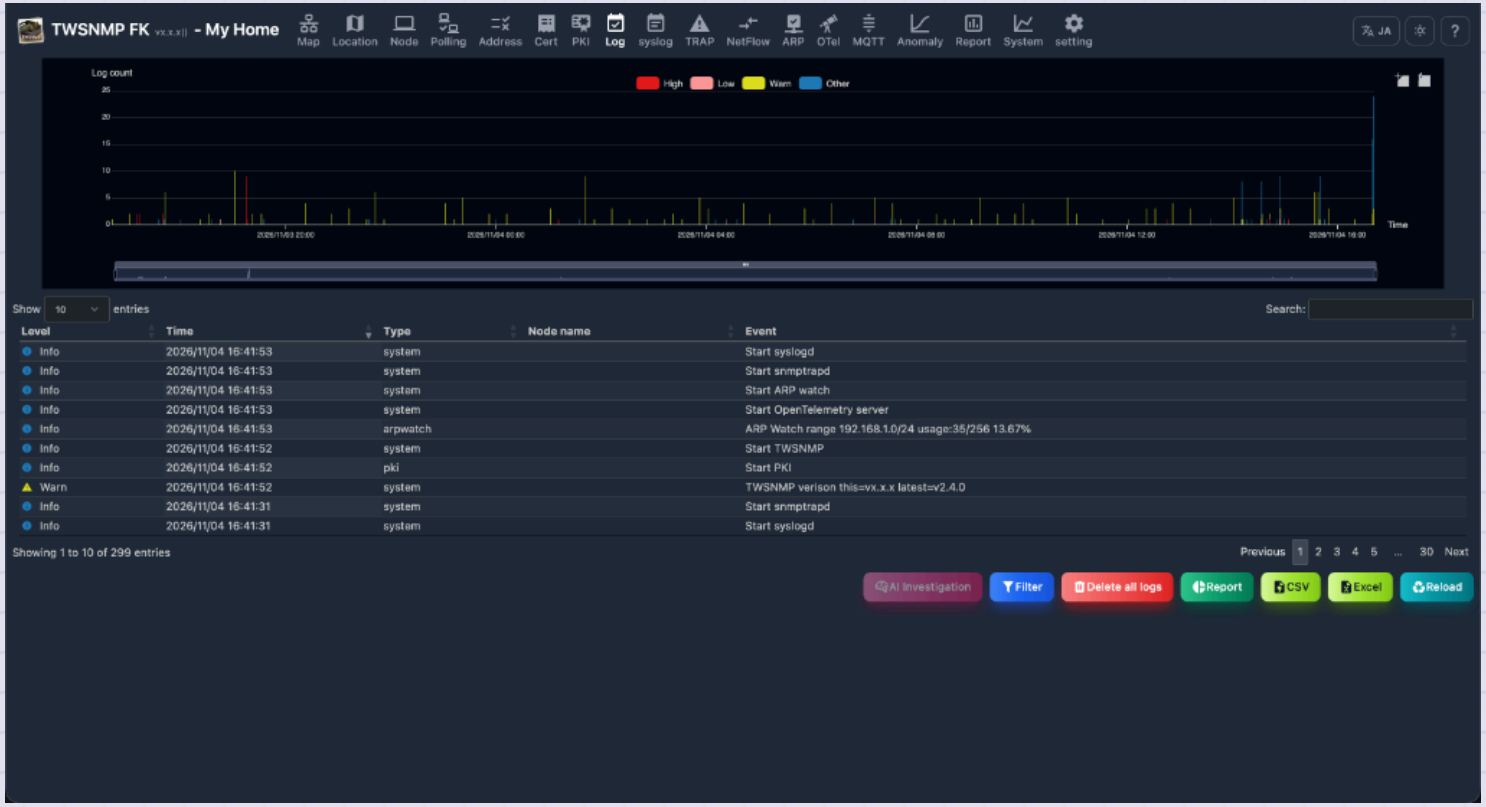
Copyright (c) 2023-2026 Masayuki Yamai

Item	Content
ACME/EST Server	Start the ACME and EST (RFC 7030) servers.
CRL/OCSP/SCEP Server	Start the CRL/OCSP/SCEP server.
ACME Server Basic URL	Specifies the basic URL that the ACME server responds to.
CRL Update Interval	Specify the CRL update interval in hours.
Certificate Period	Specify the period of the certificate to be issued in hours.

Copyright (c) 2023-2026 Masayuki Yamai

Event Log

This is the event log screen.At the top, there is a graph showing the number of logs in chronological order.



Copyright (c) 2023-2026 Masayuki Yamai

Items	Contents
Level	Log level.There is severe, mild, attention, return, and information.
Date and time	The date and time of the log is recorded.
Type	Log type. Polling, System, Oprate, User, ArpWatch,
Related node	Name of node related to logs. The blank means that there is no related node.
Event	This is an event that occurred.

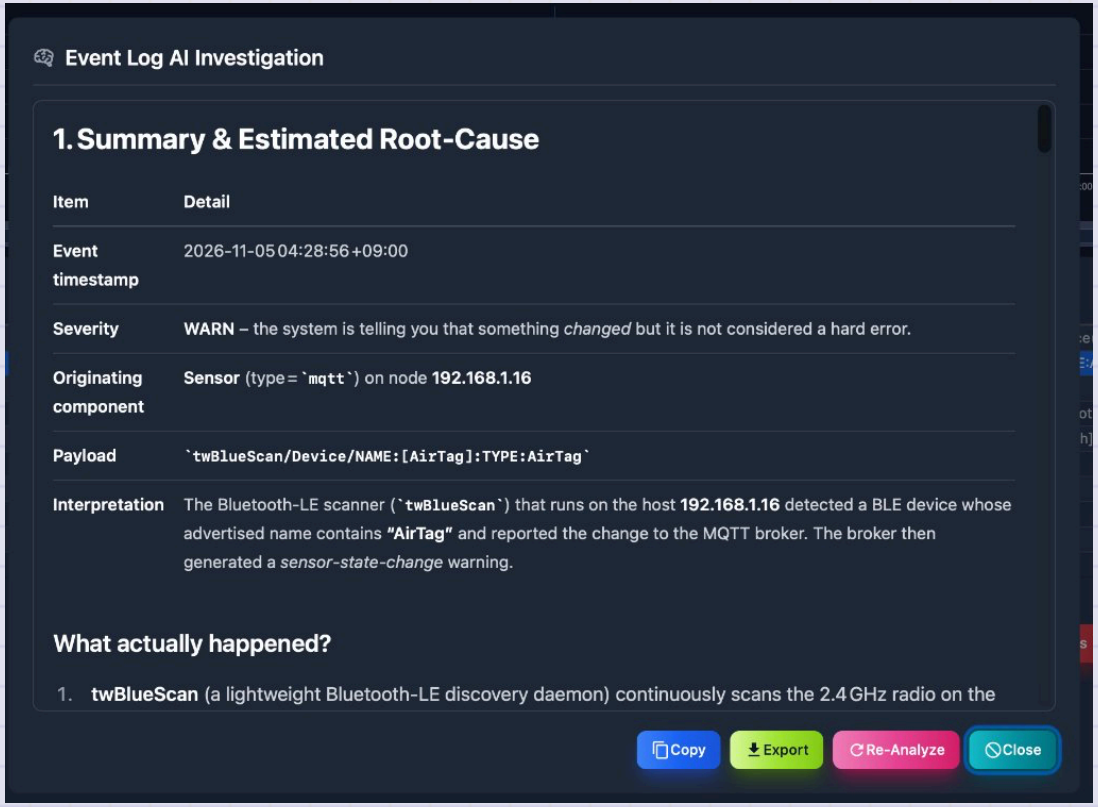
Copyright (c) 2023-2026 Masayuki Yamai

Items	Contents
AI Investigation	Automatically analyzes and investigates the root cause and suggested measures for selected event logs using AI (LLM).
Filter	Specify the search conditions and display the log.
Delete all logs	Delete all event logs.
Report	Displays the event log analysis report.
CSV	Export the event log to the CSV file.
Excel	Export the event log to the Excel file.
Reload	Update the list of event logs to the latest state.

Copyright (c) 2023-2026 Masayuki Yamai

AI Investigation

Uses AI (LLM) to automatically analyze the root cause of recorded system events and warning logs, timeline event transitions, and recommended troubleshooting steps.



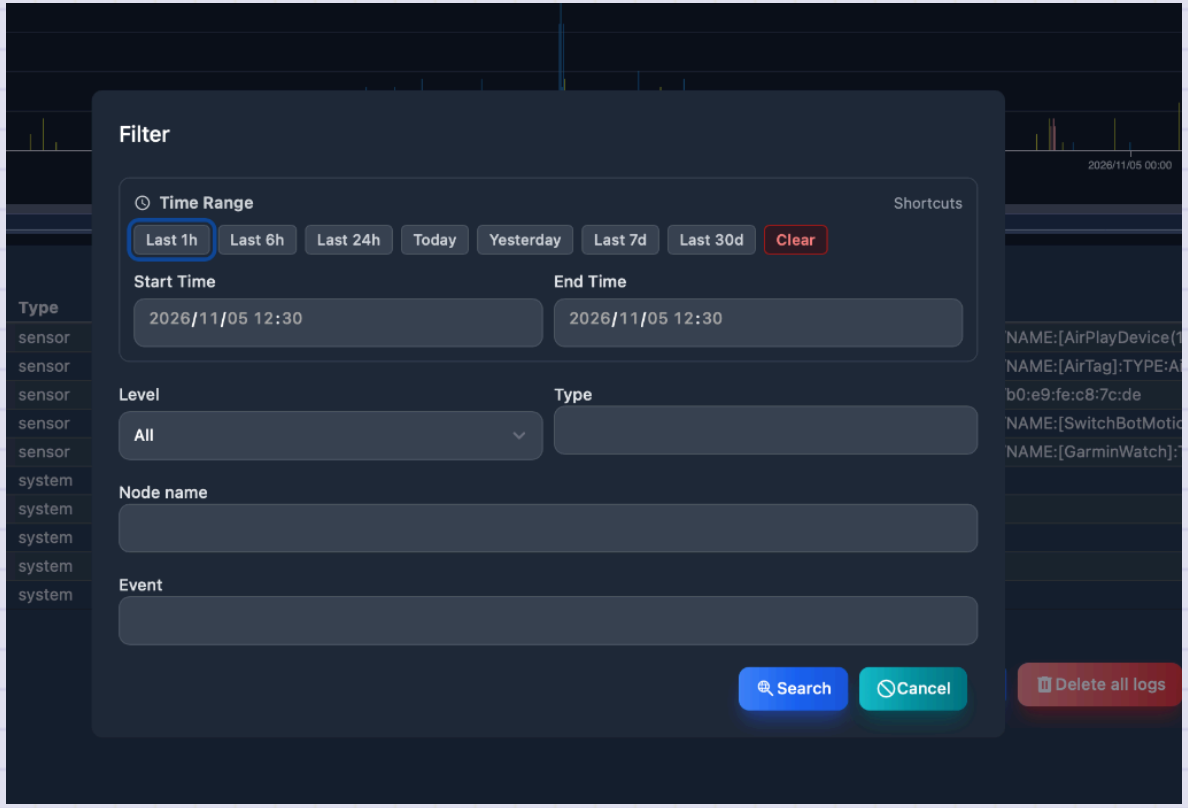
Copyright (c) 2023-2026 Masayuki Yamai

Items	Contents
Copy	Copy the investigation report content to the clipboard.
Export	Export the investigation report content to a text file.
Re-Analyze	Re-run the AI cause investigation using the latest event log data.
Close	Close the AI investigation dialog.

Copyright (c) 2023-2026 Masayuki Yamai

Event log filter

This is a dialog that specifies the search conditions for the event log.



Copyright (c) 2023-2026 Masayuki Yamai

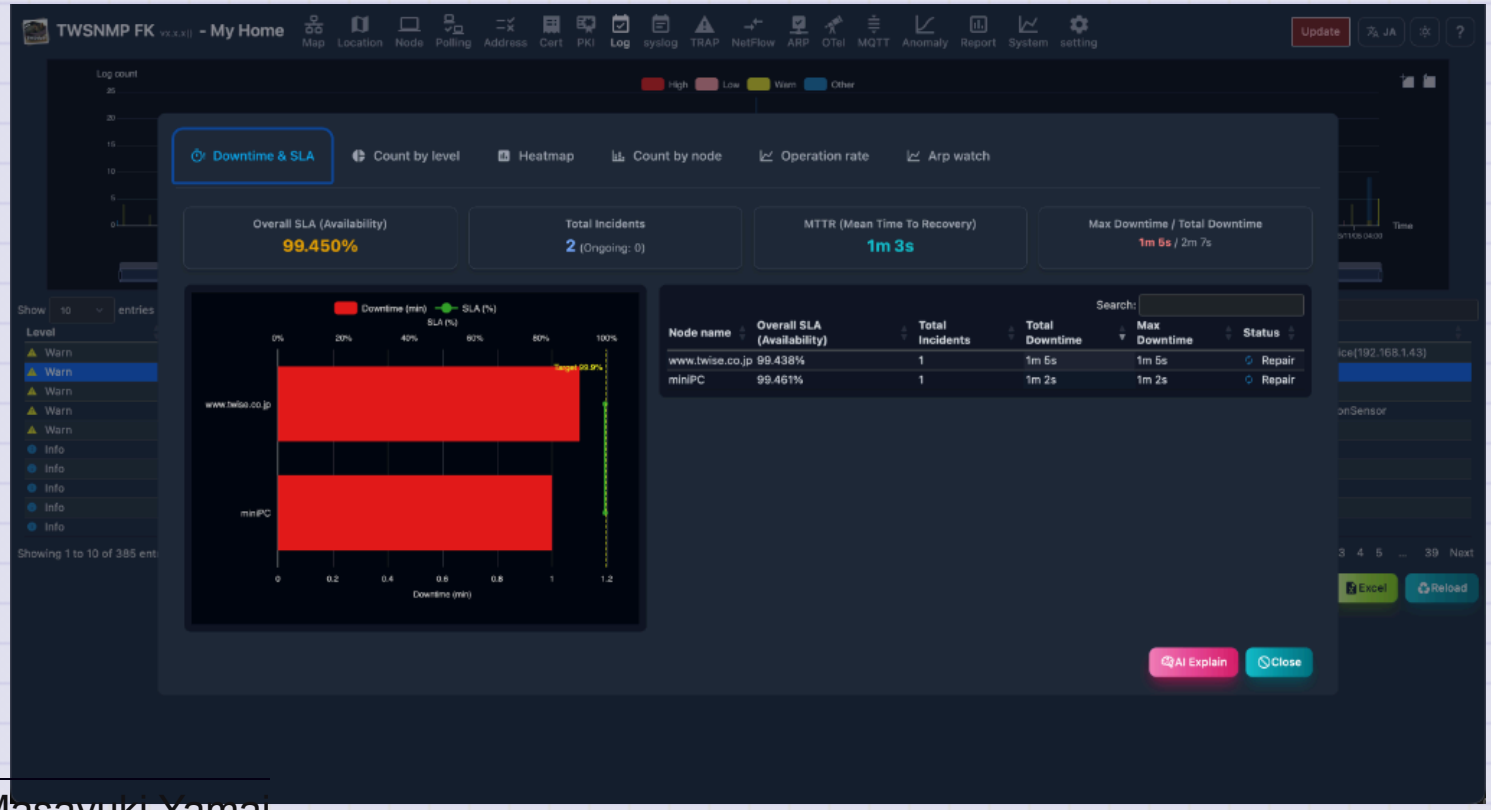
Items	Contents
Level	Log level.All, there are more attention, more than severe, mild.
Type	Log type. Polling, System, Oprate, User, ArpWatch,
Related node	Search by node name related to the log.
Event	Search by the string of the event that occurred.

The string can be searched by regular expression.

Copyright (c) 2023-2026 Masayuki Yamai

Event log downtime & SLA

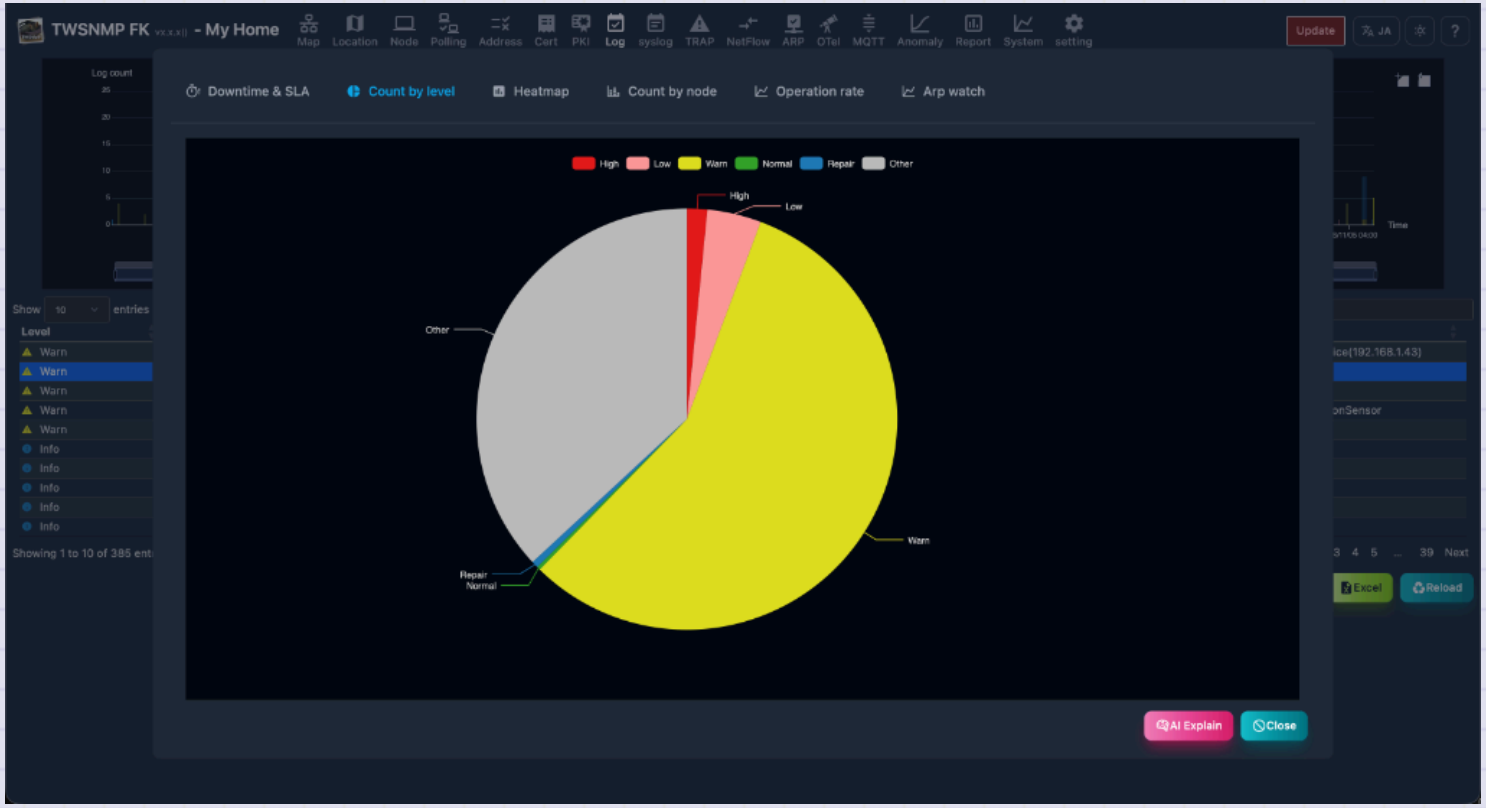
An aggregate report analyzing overall SLA (Availability %), total incidents count, MTTR (Mean Time To Recovery), max/total downtime, and node-by-node SLA/downtime ranking chart and detail table based on polling failure and repair events.



Copyright (c) 2023-2026 Masayuki Yamai

Event log count by state

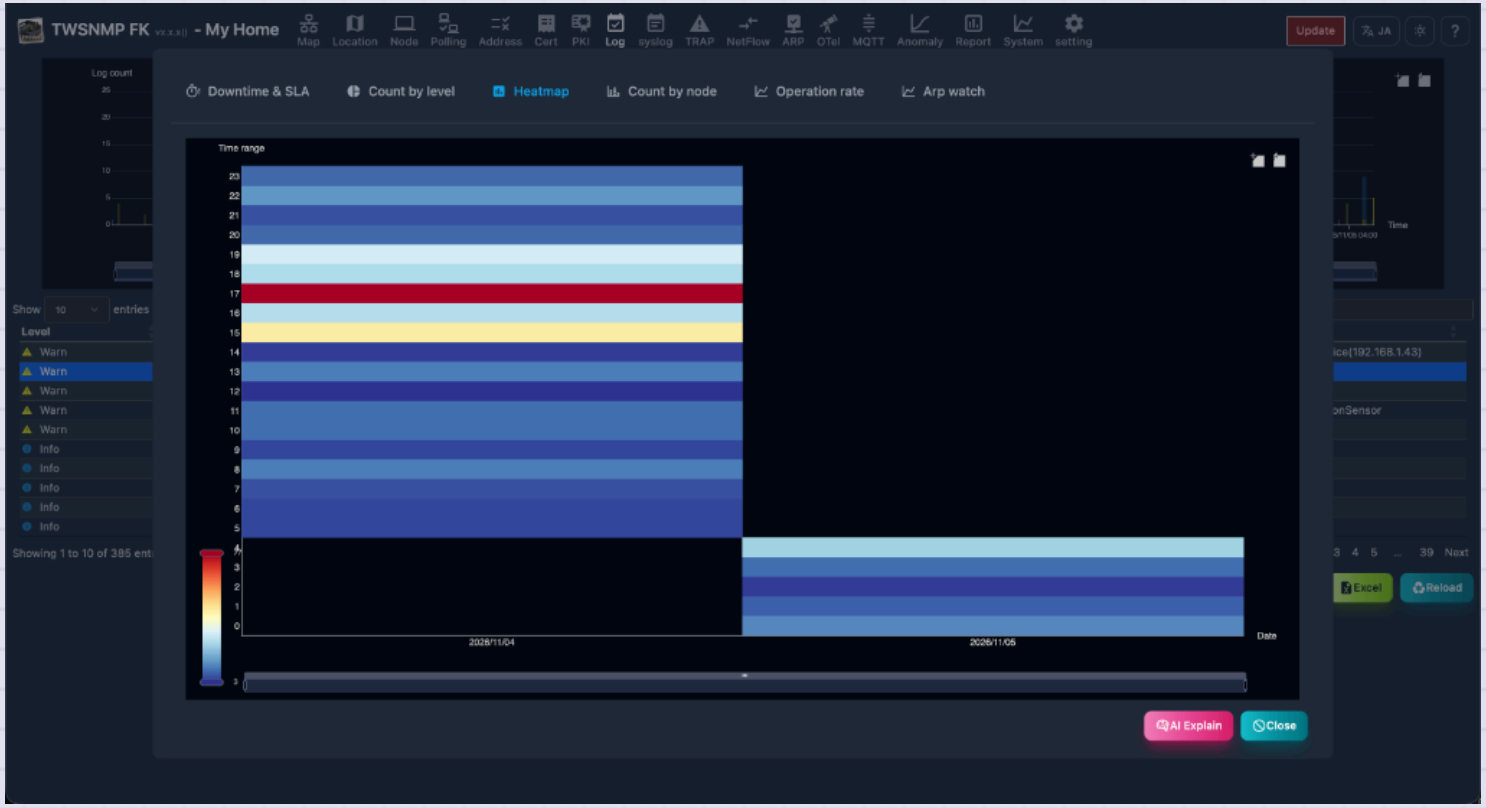
This is a report of the number of event logs by state (level).



Copyright (c) 2023-2026 Masayuki Yamai

Event log Heatmap

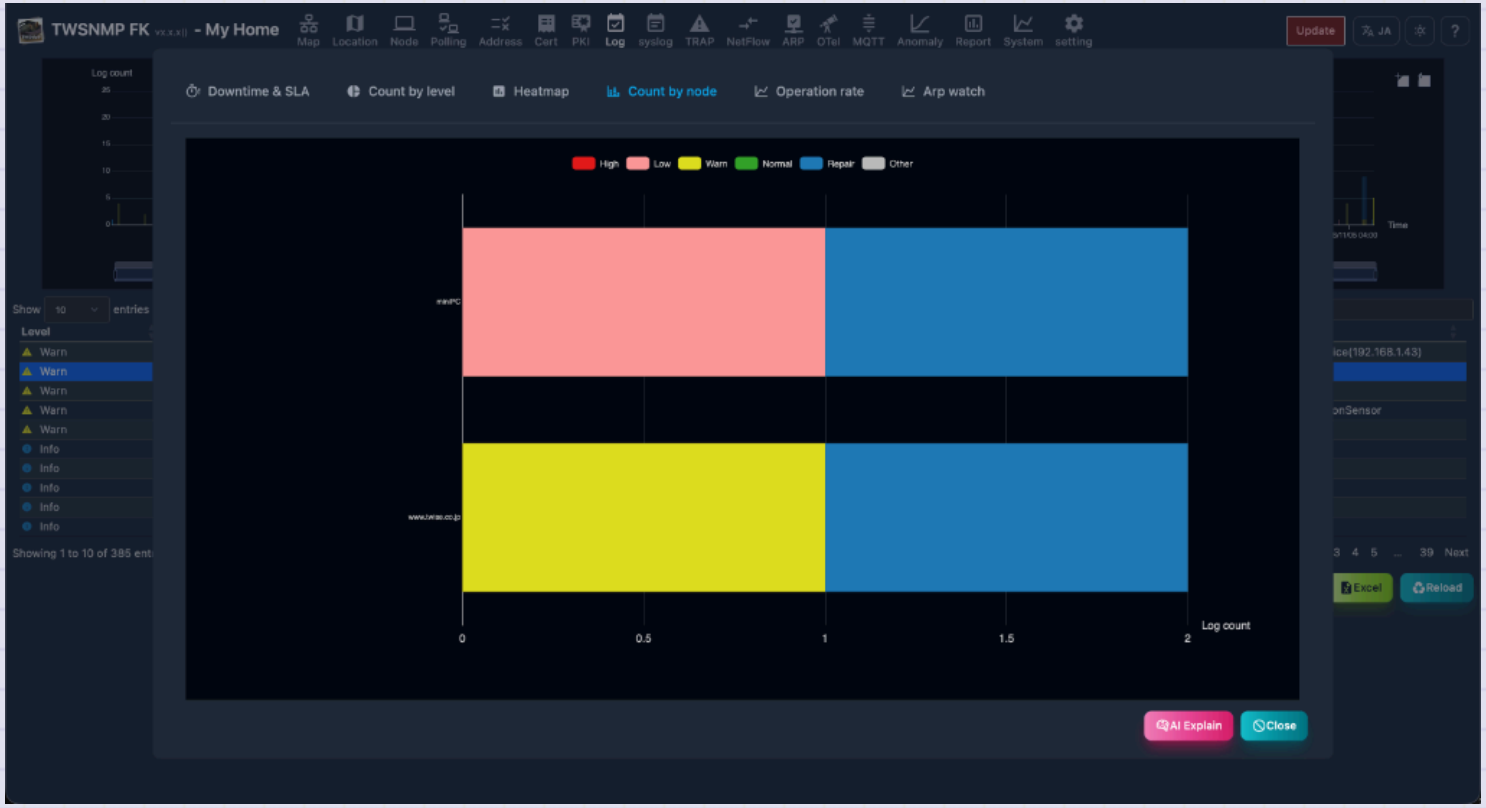
This is a report of the number of cases of each event log on the heat map.



Copyright (c) 2023-2026 Masayuki Yamai

Event log count by node

This is a report of the number of event logs by node.



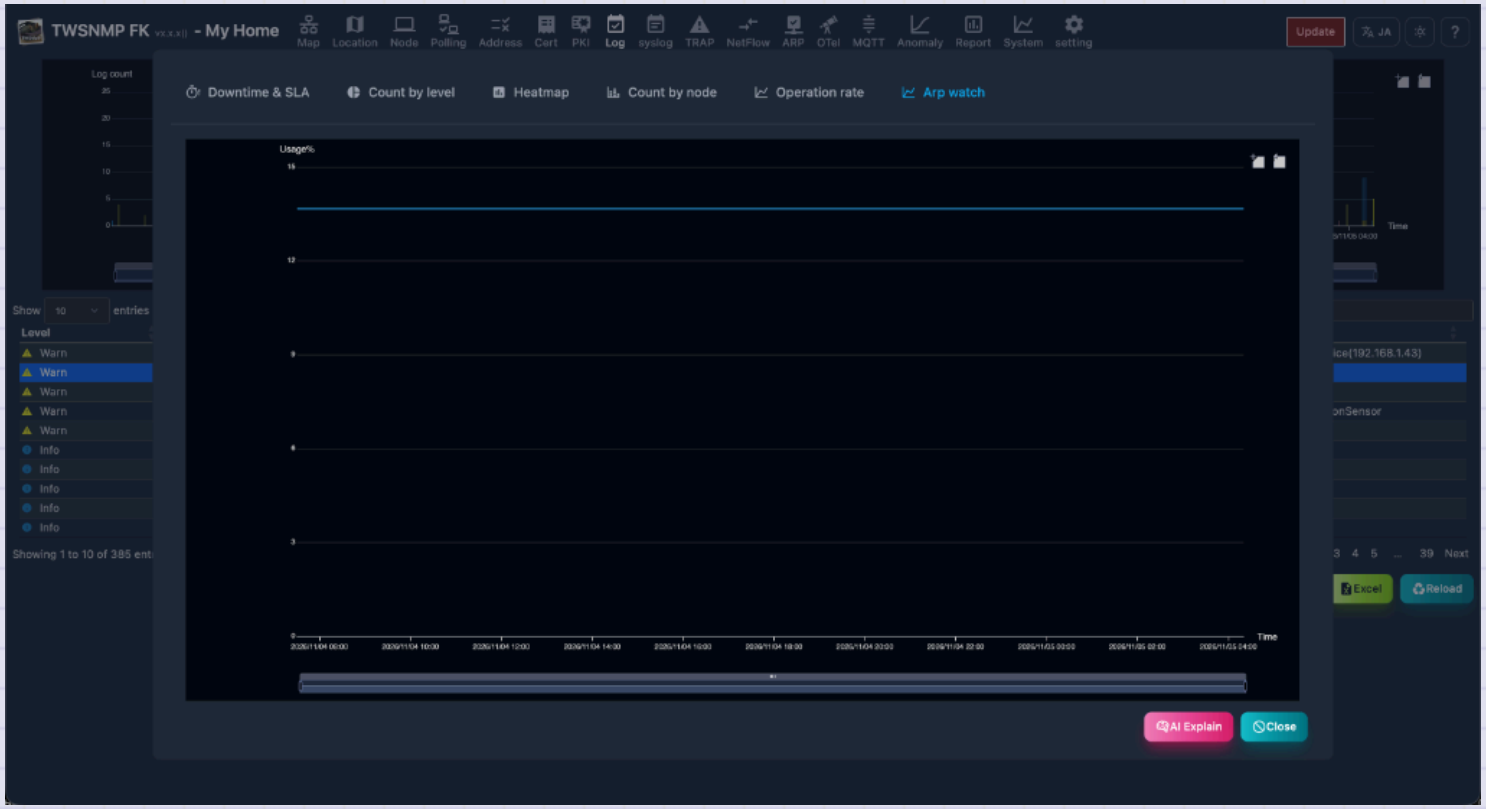
Operating rate

This is a report that uses a chronological graph of the value of the operating rate (OPRATE) in the event log.

Copyright (c) 2023-2026 Masayuki Yamai

ARP watch

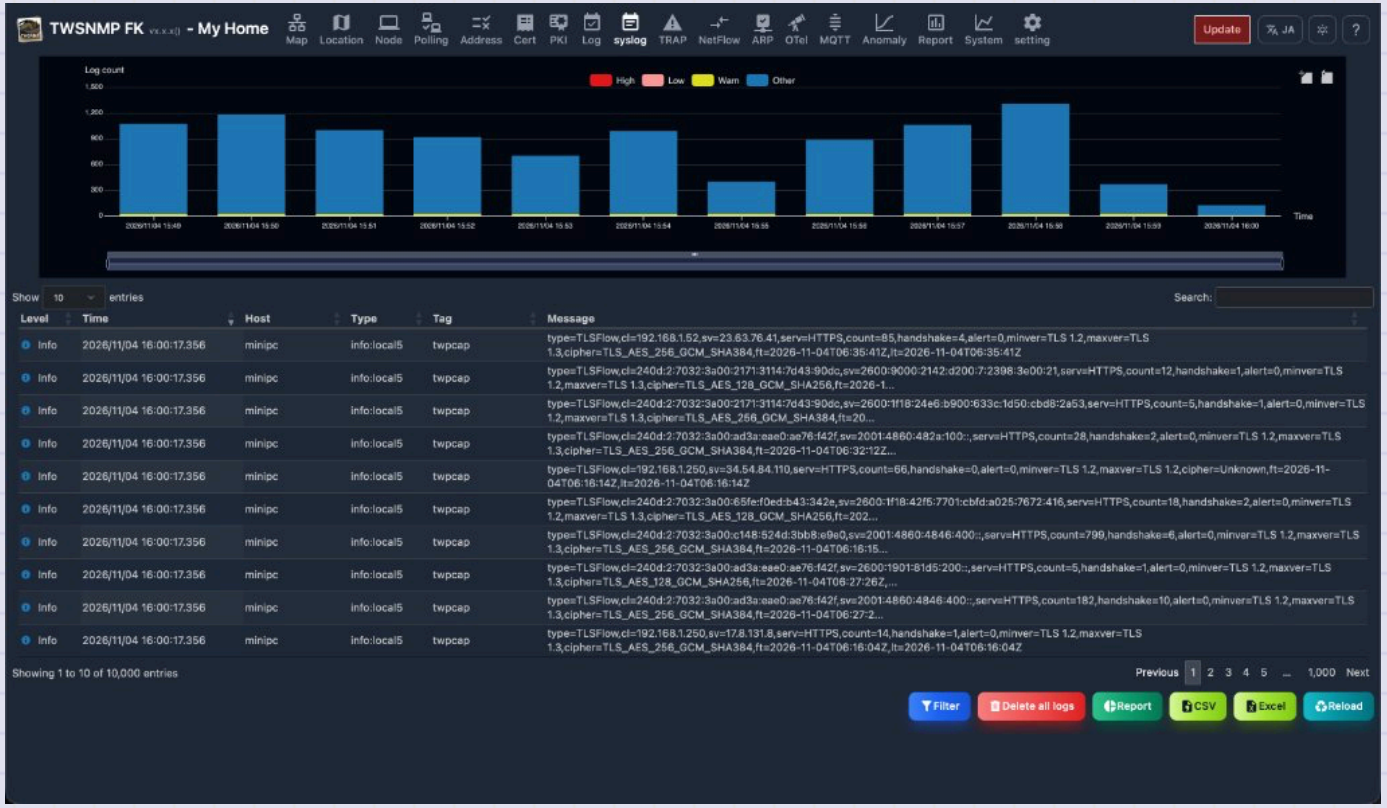
This is a report of the value of the address usage rate (ARPWATCH) in the event log as a chronological graph.



Copyright (c) 2023-2026 Masayuki Yamai

Syslog

Syslog screen.At the top, there is a graph showing the number of logs in chronological order.



Copyright (c) 2023-2026 Masayuki Yamai

Items	Contents
Level	Syslog level.There is severe, High,Low, Warn, and information.
Date and time	It is the date and time when I received syslog.
Host	SYSLOG source host.
Type	Syslog Facility and priority string.
Tags	Syslog tag.Process and process ID.
Message	Syslog message.

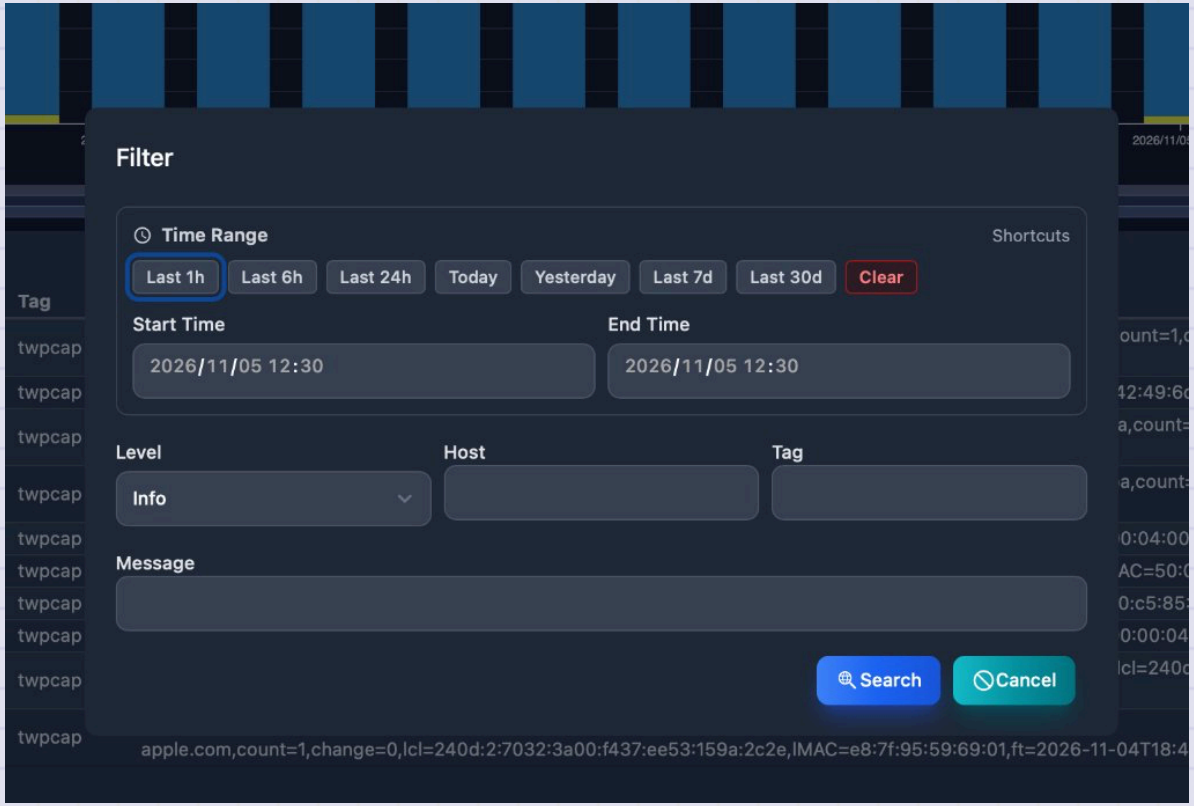
Copyright (c) 2023-2026 Masayuki Yamai

Items	Contents
Polling	Register the polling from the selected syslog.
Filter	Specify the search conditions and display syslog.
Delete all logs	Delete all syslogs.
Report	Displays Syslog analysis reports.
Export CSV	syslog to CSV file.
Excel	EXCEL file is exported to syslog.
Reload	Update the list of syslog to the latest state.

Copyright (c) 2023-2026 Masayuki Yamai

Syslog Filter

This is a dialog that specifies the search conditions for syslog.



Copyright (c) 2023-2026 Masayuki Yamai

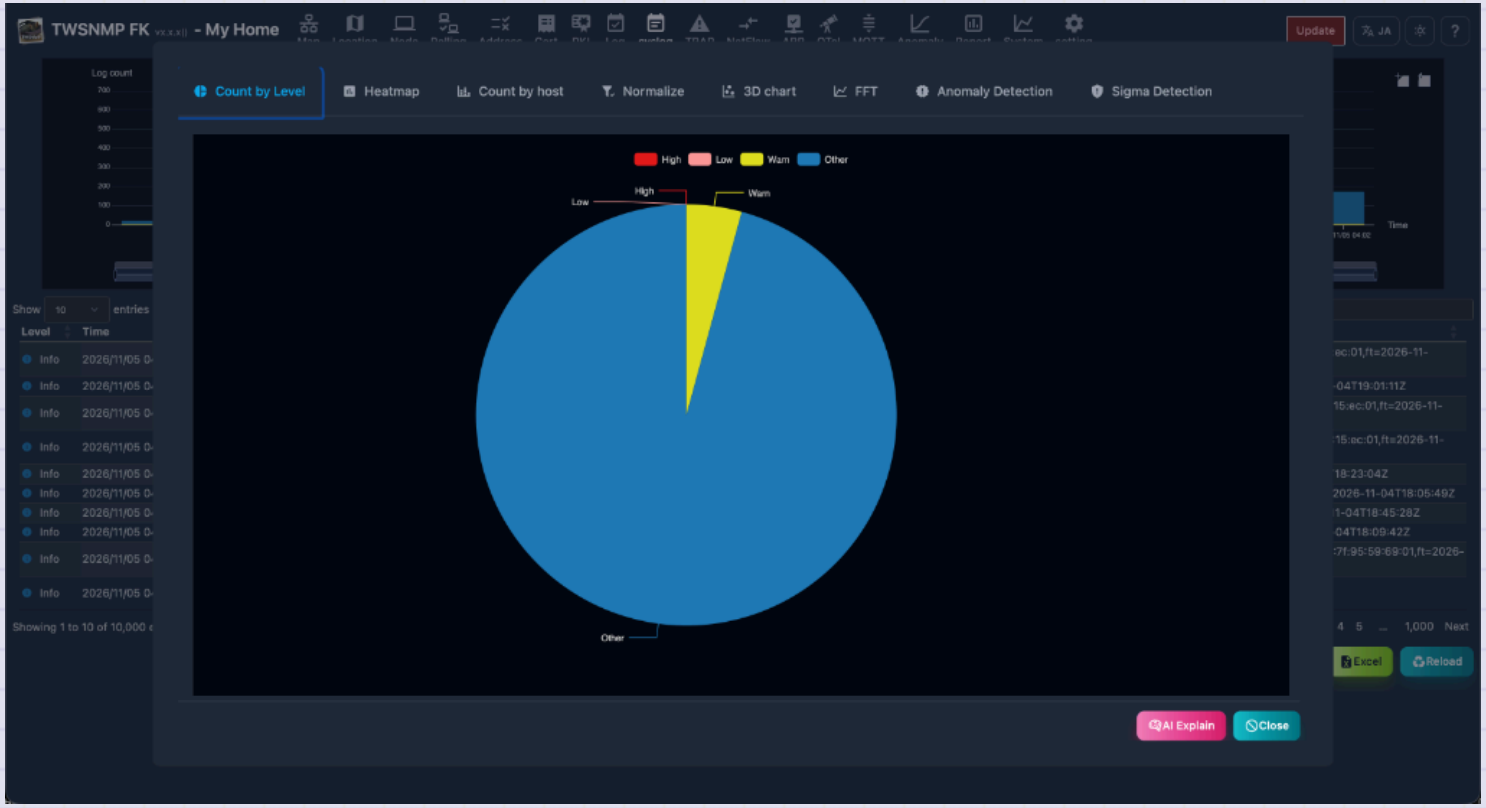
Items	Contents
Level	Syslog level. All, more than information, more than caution, mild or higher, more severe.
Host	It is the source host.
Tags	The value of the syslog tag.
Message	Syslog message.

*Host,Tag,Message can be searched in regular expressions.

Copyright (c) 2023-2026 Masayuki Yamai

Syslog count by state

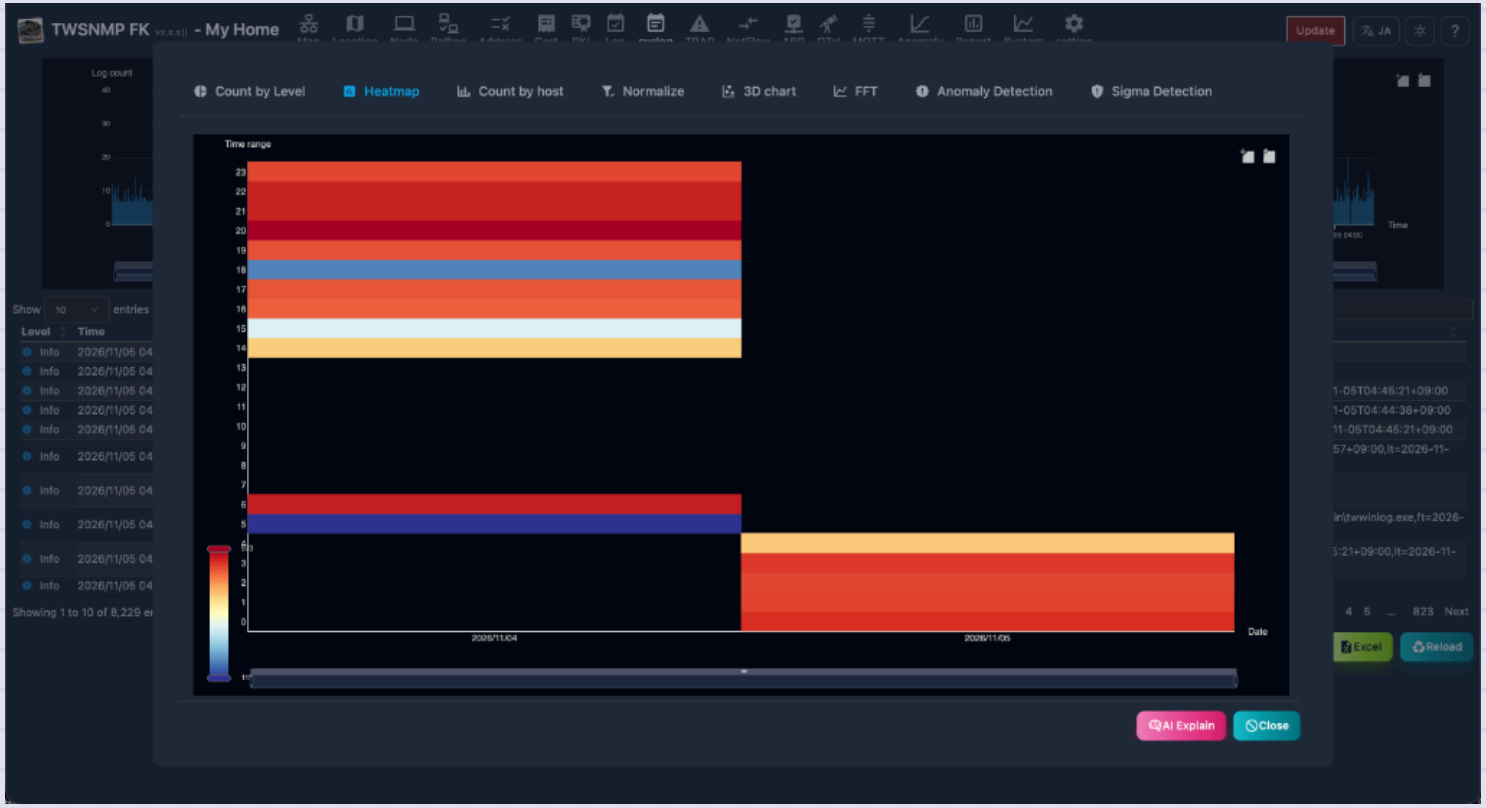
This is a report of the number of syslogs by state.



Copyright (c) 2023-2026 Masayuki Yamai

Syslog Heatmap

This is a report of the number of cases of syslog on the heat map.

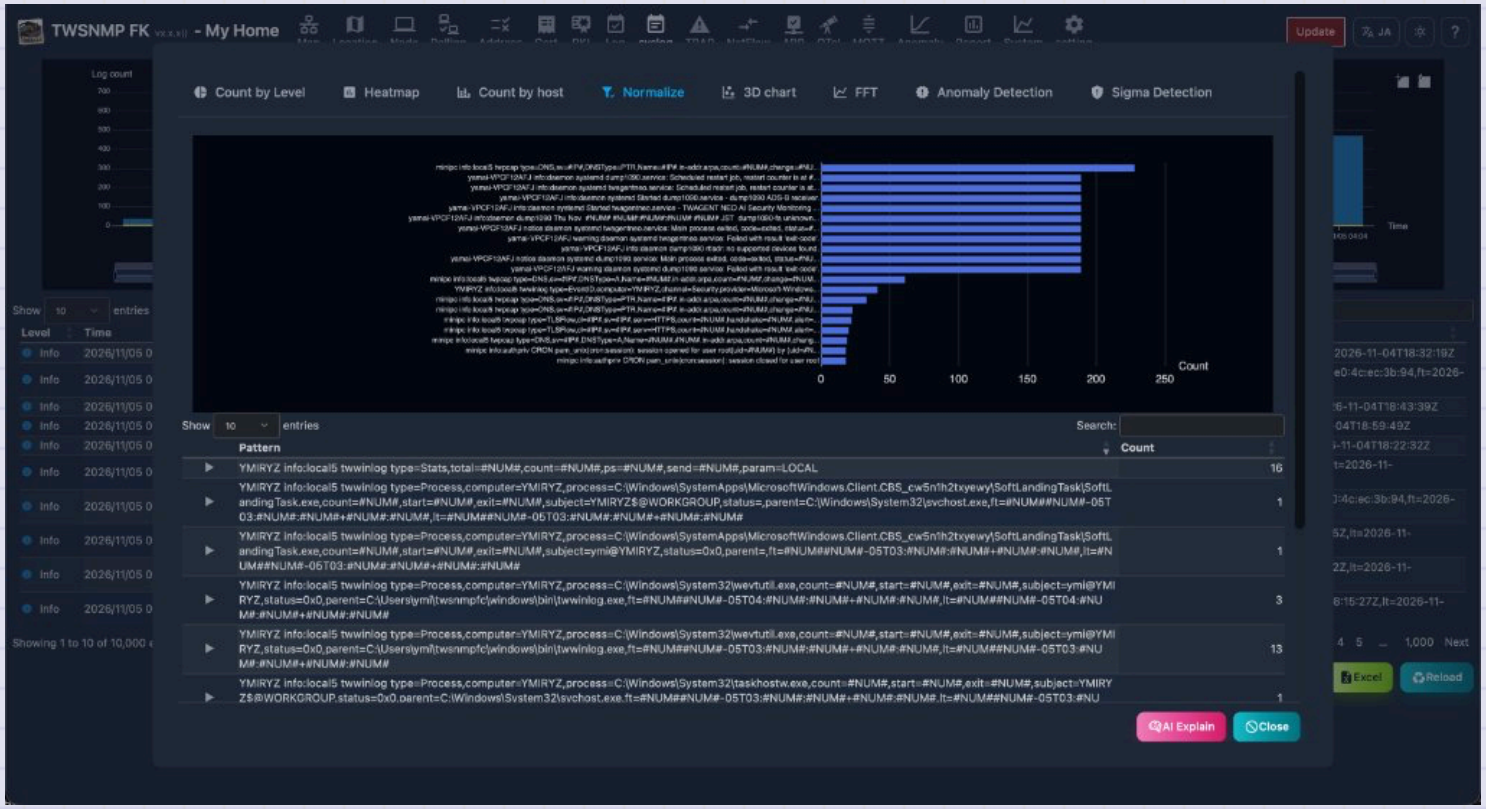


Copyright (c) 2023-2026 Masayuki Yamai

Normalization Analysis

Report that normalizes syslog and aggregates it.

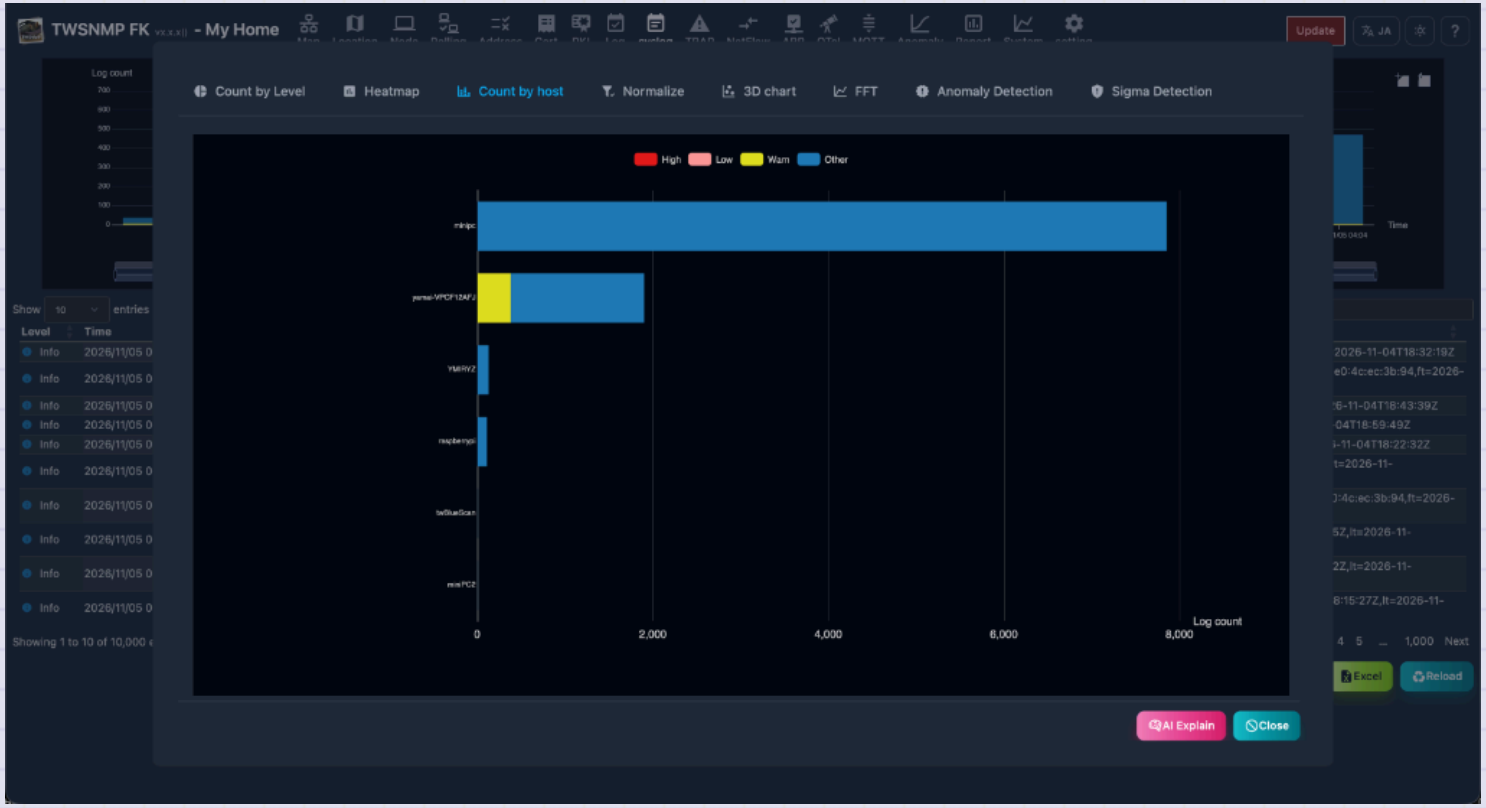
Counts numbers, IP addresses, email addresses, etc. with common characters.



Copyright (c) 2023-2026 Masayuki Yamai

Syslog count by host

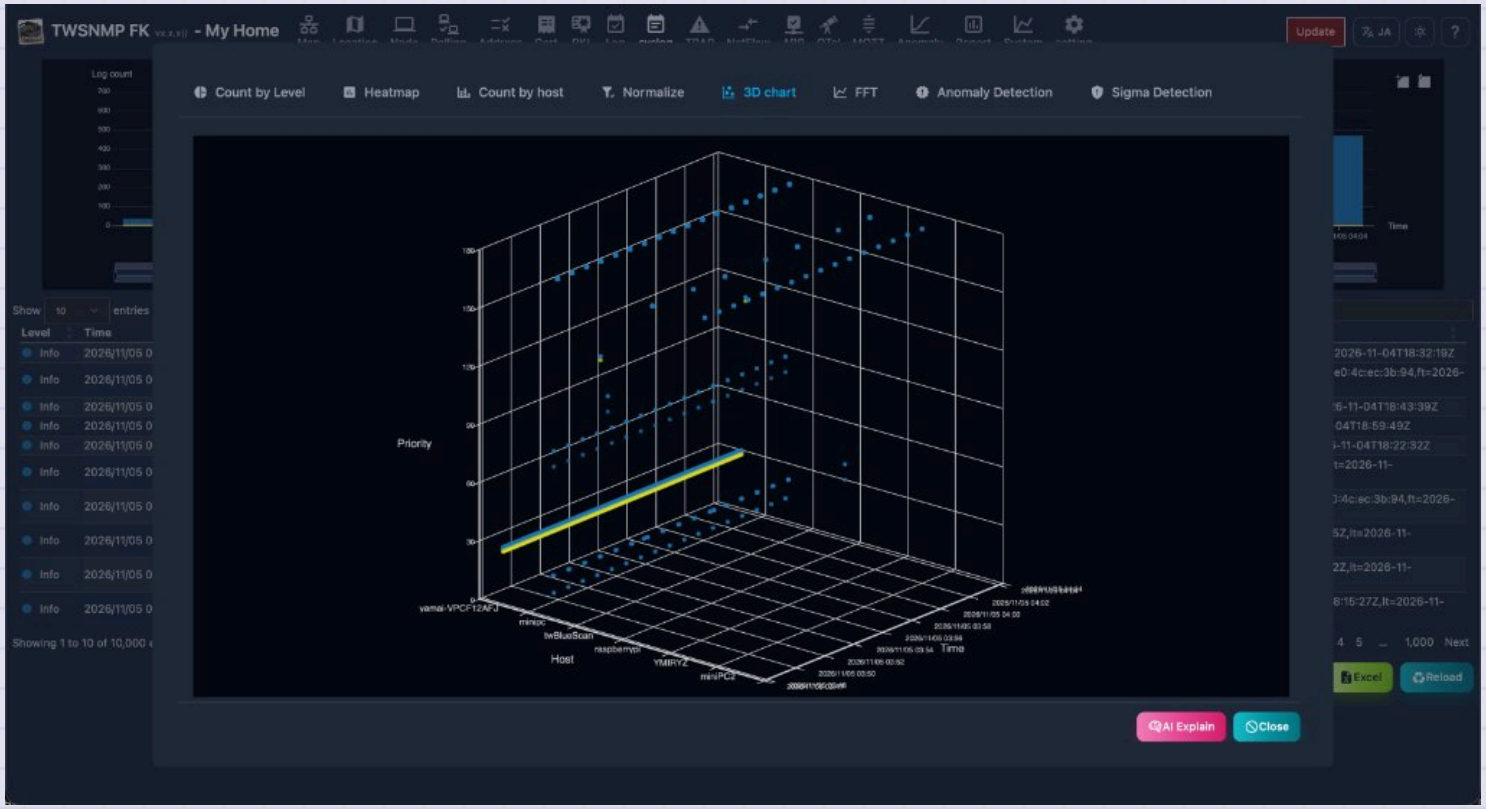
This is a report of the number of syslogs by the source host.



Copyright (c) 2023-2026 Masayuki Yamai

Syslog count by host (3D)

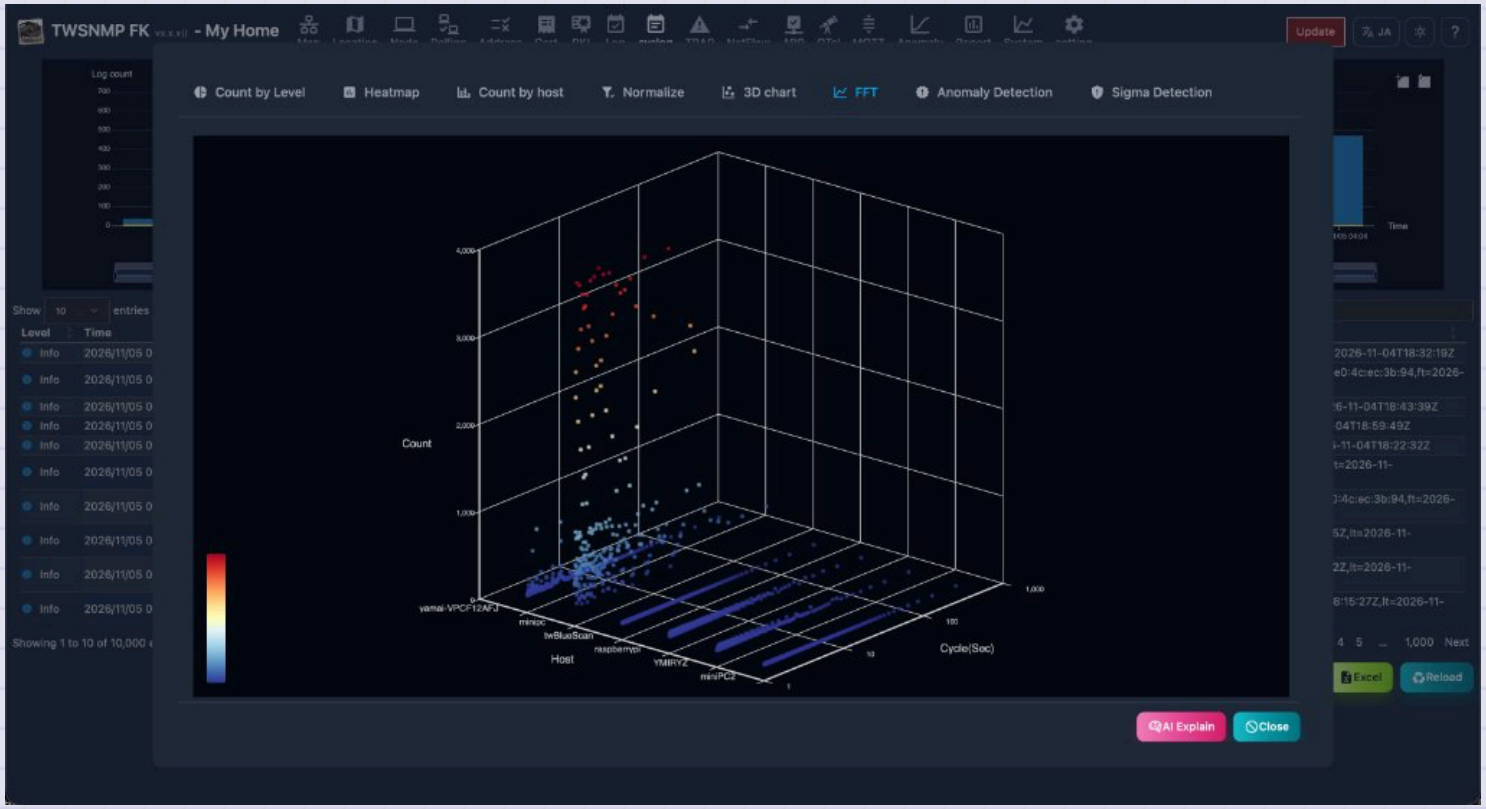
This is a report displayed in three -dimensional graphs of Syslog, source host, priority, and time.



Copyright (c) 2023-2026 Masayuki Yamai

Syslog FFT

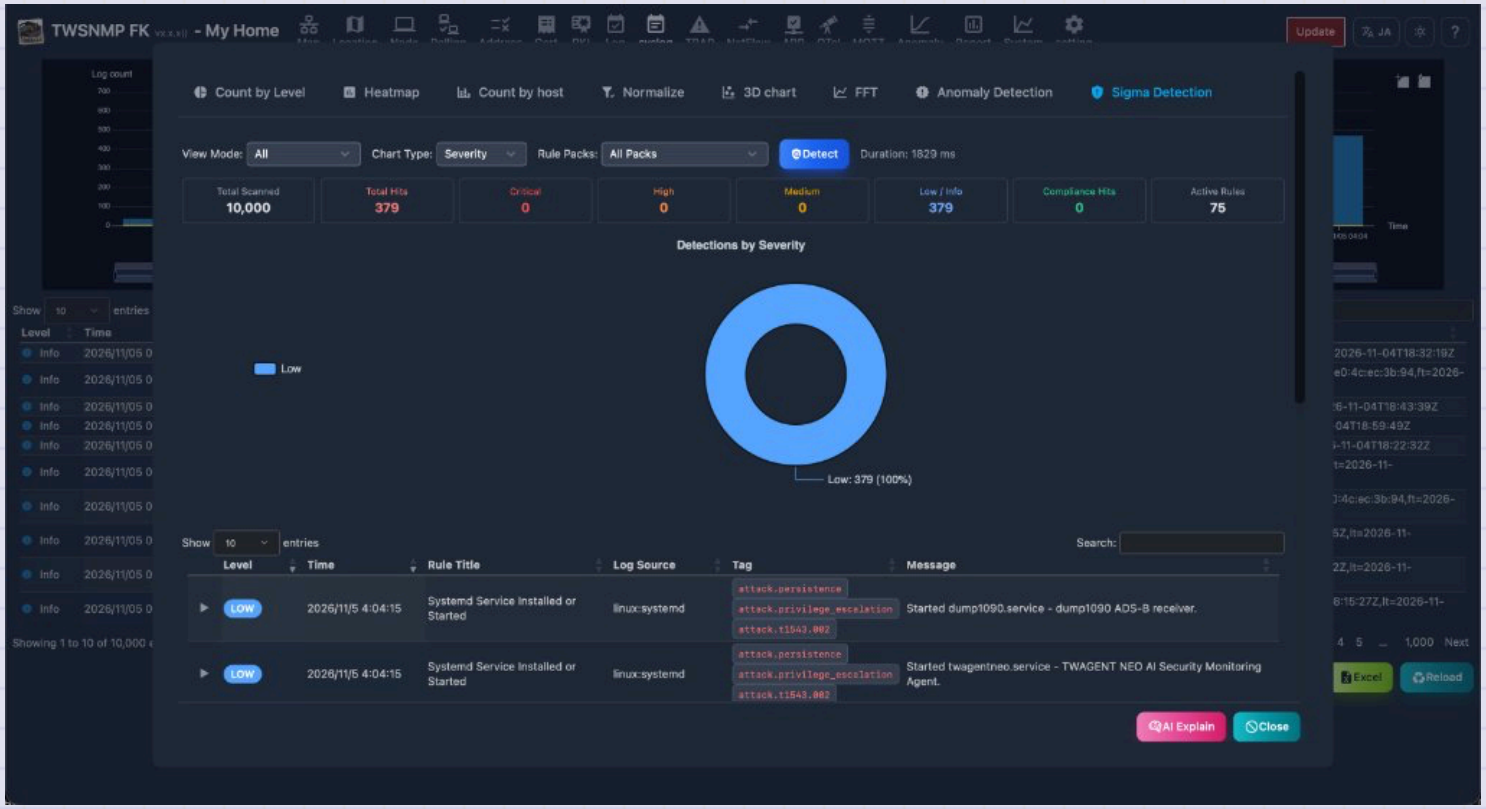
This is a report that analyzes Syslog for each host and analyzes the number of receiving cases.



Copyright (c) 2023-2026 Masayuki Yamai

Syslog Sigma Detection

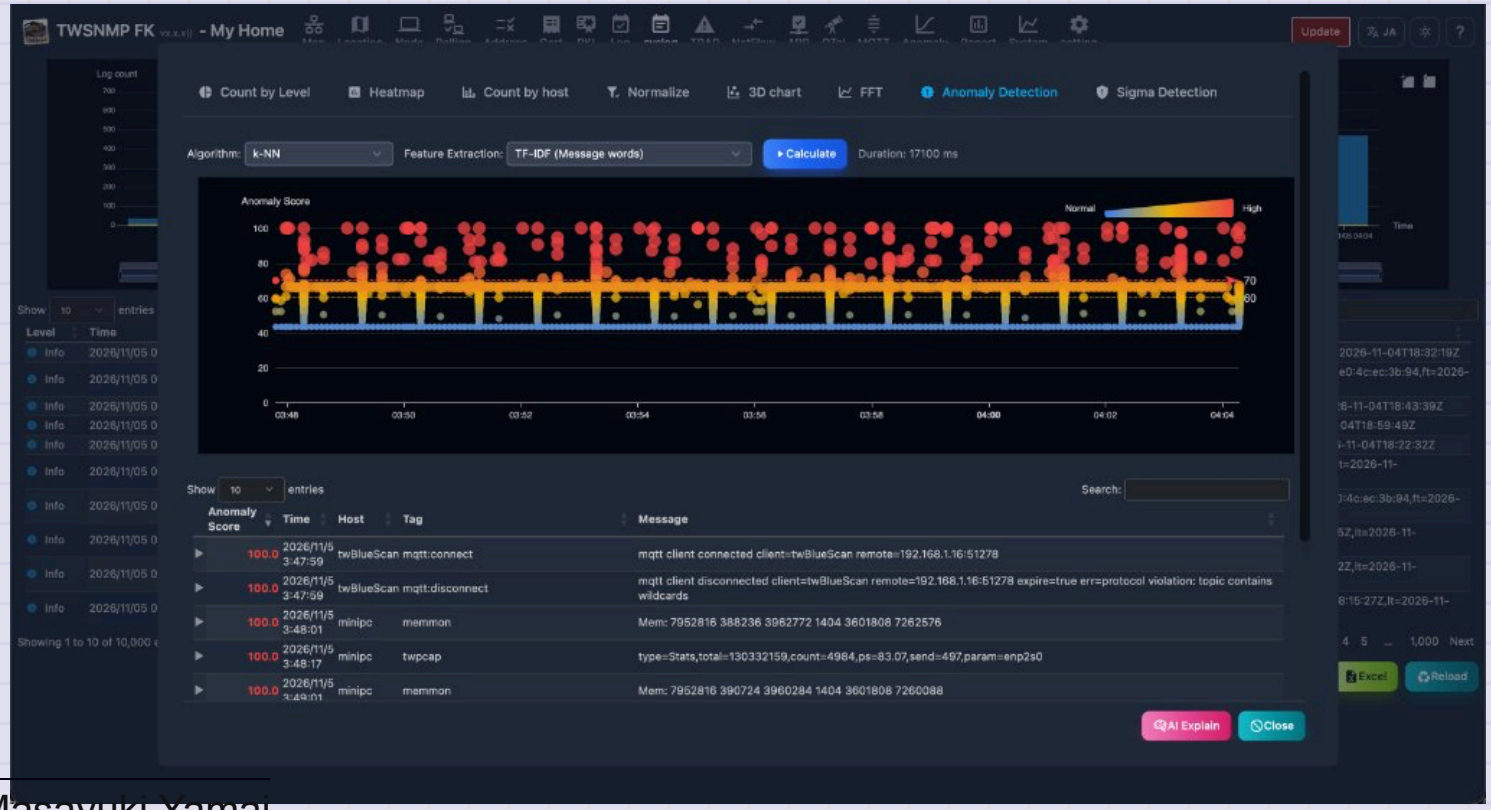
Reports cyber attacks, unauthorized access, and compliance violations detected from Syslog messages based on Sigma rules.



Copyright (c) 2023-2026 Masayuki Yamai

Syslog Anomaly Detection

Analyzes anomaly scores of Syslog messages using machine learning algorithms (Mahalanobis distance, Isolation Forest, etc.) and feature extraction techniques (TF-IDF, Threat keywords, etc.).



Copyright (c) 2023-2026 Masayuki Yamai

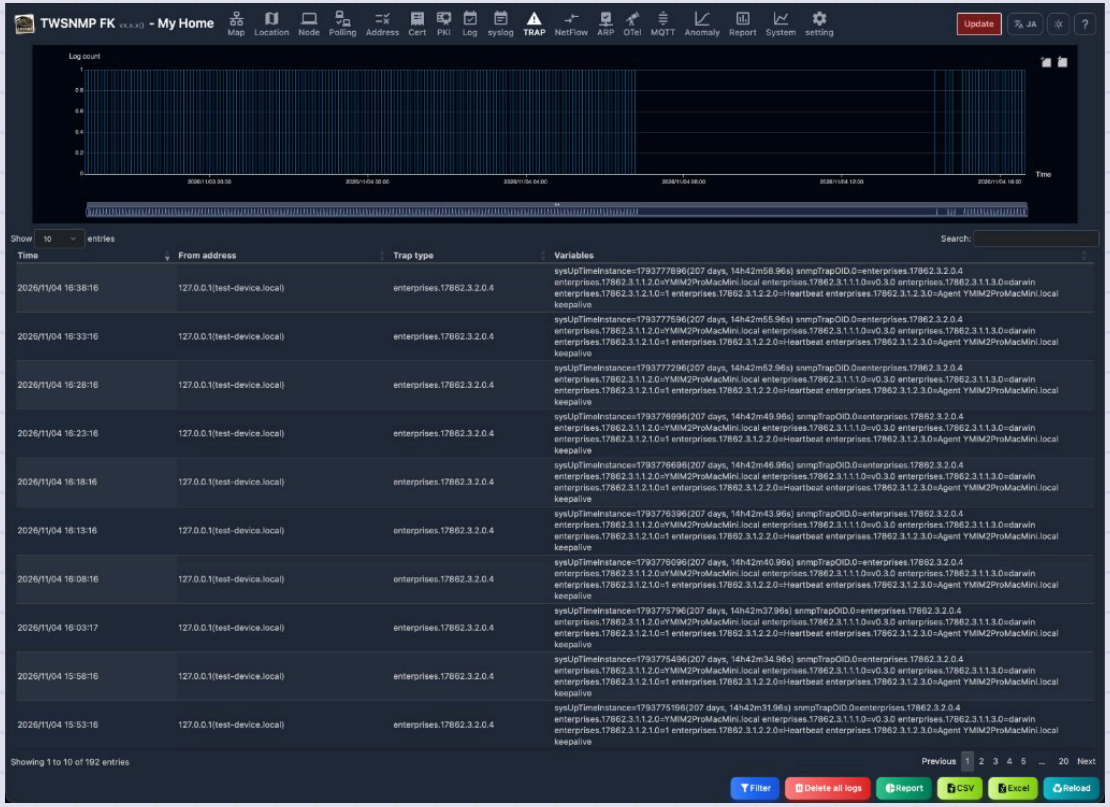
AI Explanation

Sends current Syslog reports, severity distributions, and top sender host metrics to AI (LLM) for log spike analysis, anomaly explanation, and actionable remediation steps (only visible when AI integration is enabled).

Copyright (c) 2023-2026 Masayuki Yamai

SNMP TRAP

SNMP Trap log screen.At the top, there is a graph showing the number of logs in chronological order.



Copyright (c) 2023-2026 Masayuki Yamai

Items	Contents
Date and time	This is the date and time of receiving SNMP Trap.
Sending source	SNMP Trap's source host.
Type	SNMP Trap type.
Variables	Variables attached to SNMP Trap.

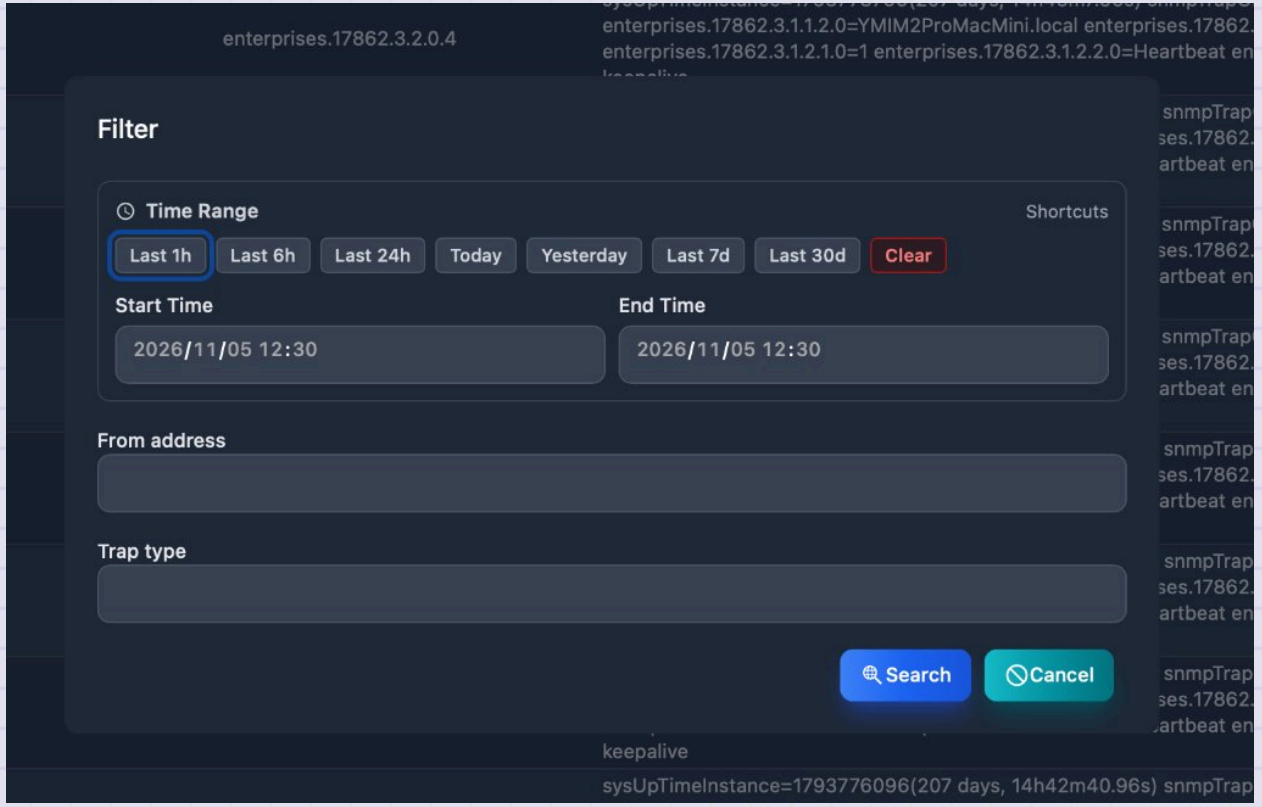
Copyright (c) 2023-2026 Masayuki Yamai

Items	Contents
Polling	Register the polling from the selected SNMP Trap.
Filter	Specify the search conditions and display SNMP Trap.
Delete all logs	Delete all syslogs.
Report	Displays the analysis report of SNMP Trap.
CSV	Sport the SNMP Trap to the CSV file.
Excel	Export SNMP Trap to Excel file.
Reload	Update the SNMP Trap list to the latest state.

Copyright (c) 2023-2026 Masayuki Yamai

SNMP TRAP Filter

This is a dialog that specifies the search conditions for SNMP Trap.



Copyright (c) 2023-2026 Masayuki Yamai

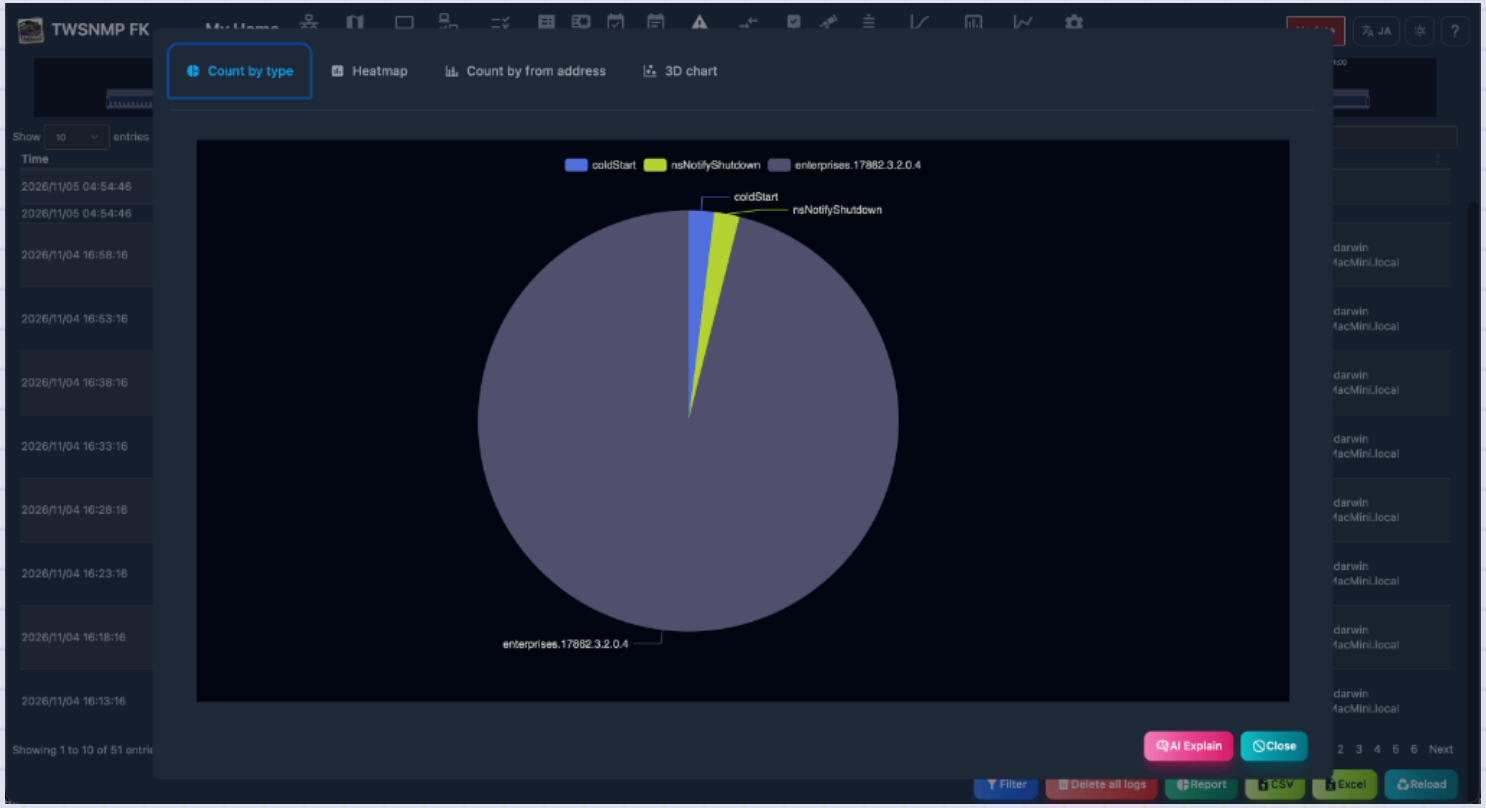
Items	Contents
Sending source	It is the source host.
Type	SNMP Trap type.

*Character strings can be searched in regular expressions.

Copyright (c) 2023-2026 Masayuki Yamai

SNMP TRAP count by TRAP type

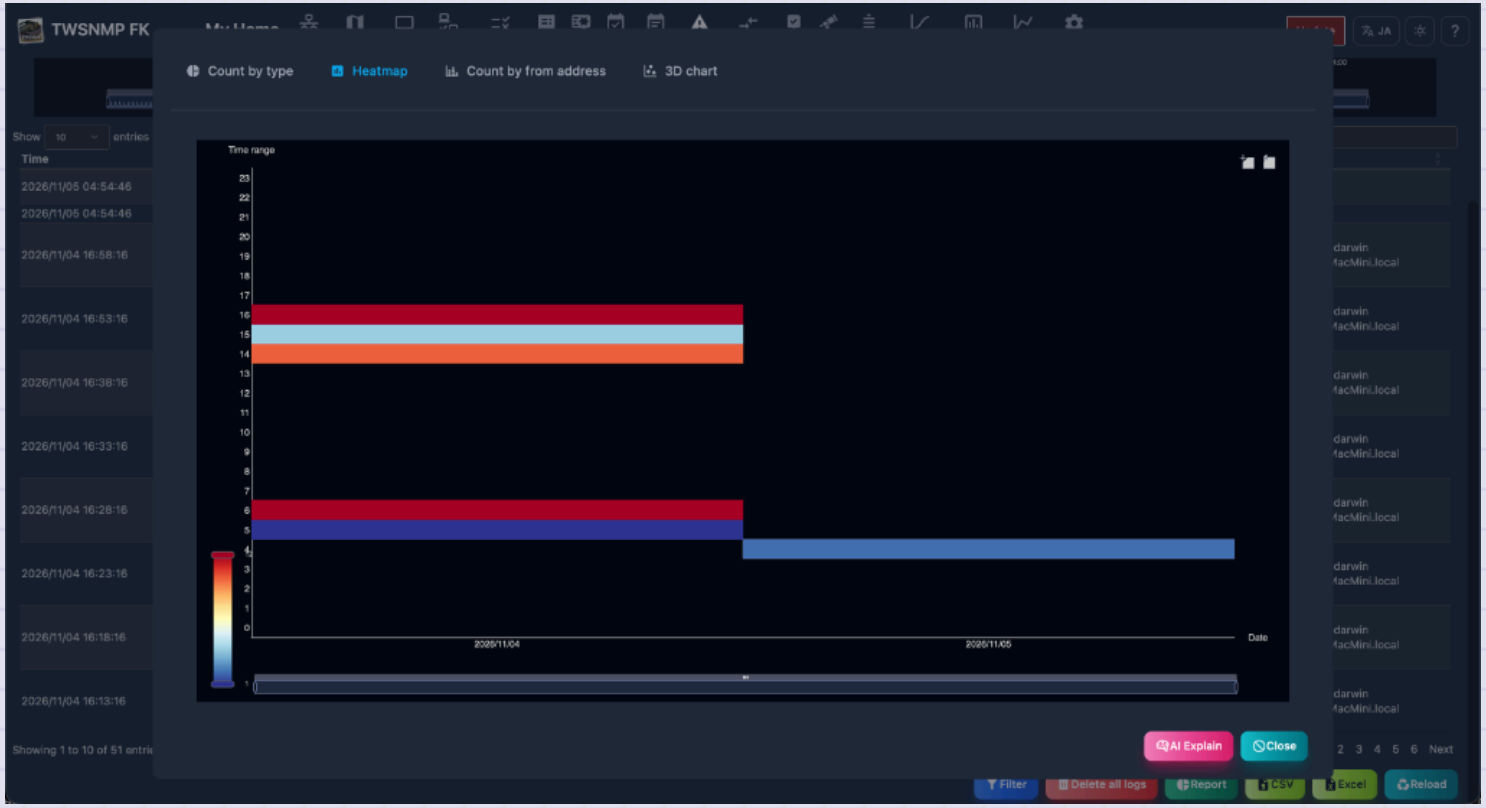
This is a report of the number of SNMP traps by type.



Copyright (c) 2023-2026 Masayuki Yamai

SNMP TRAP Heatmap

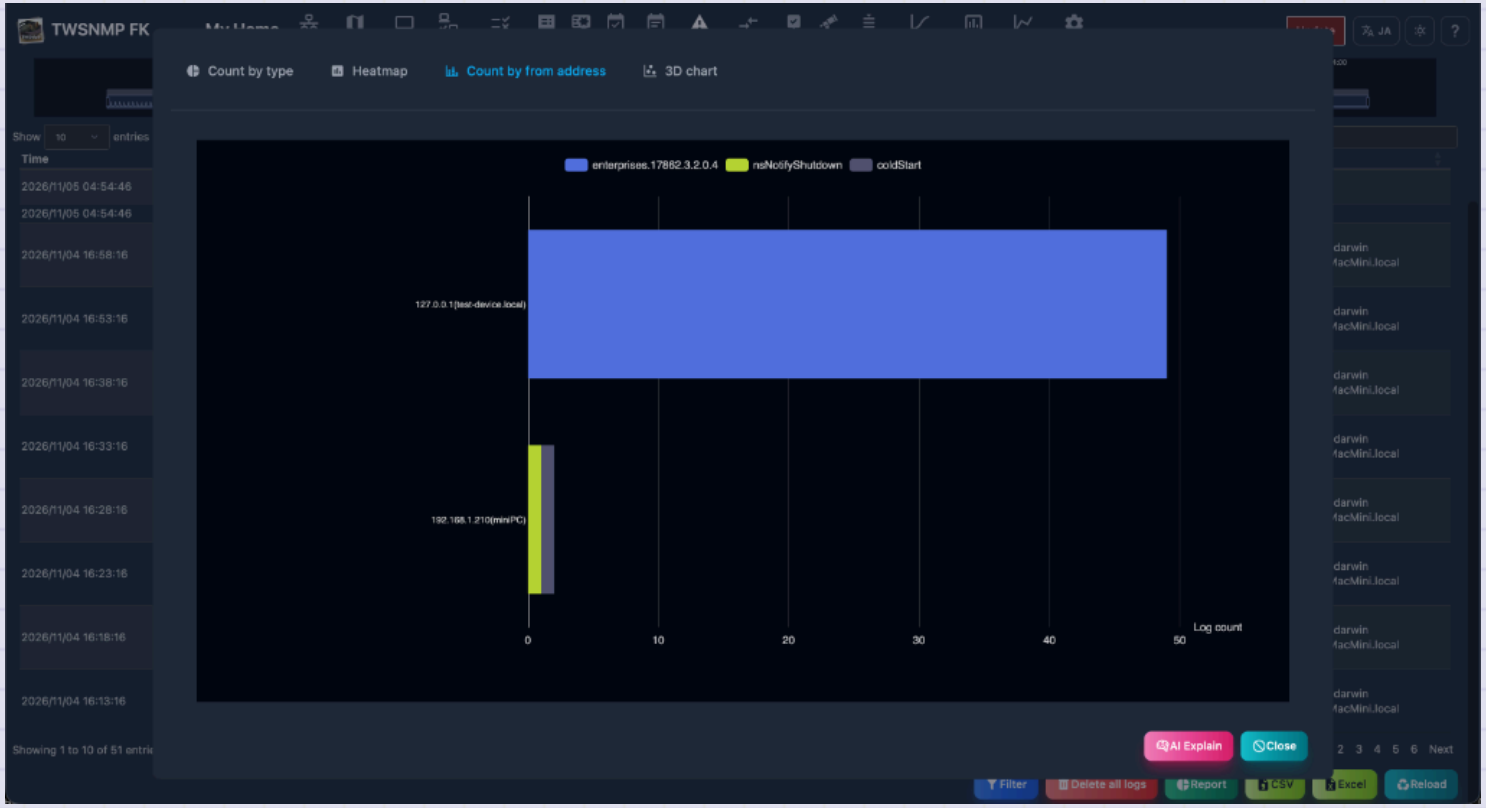
This is a report of the number of cases of SNMP TRAP on the heat map.



Copyright (c) 2023-2026 Masayuki Yamai

SNMP TRAP count by host

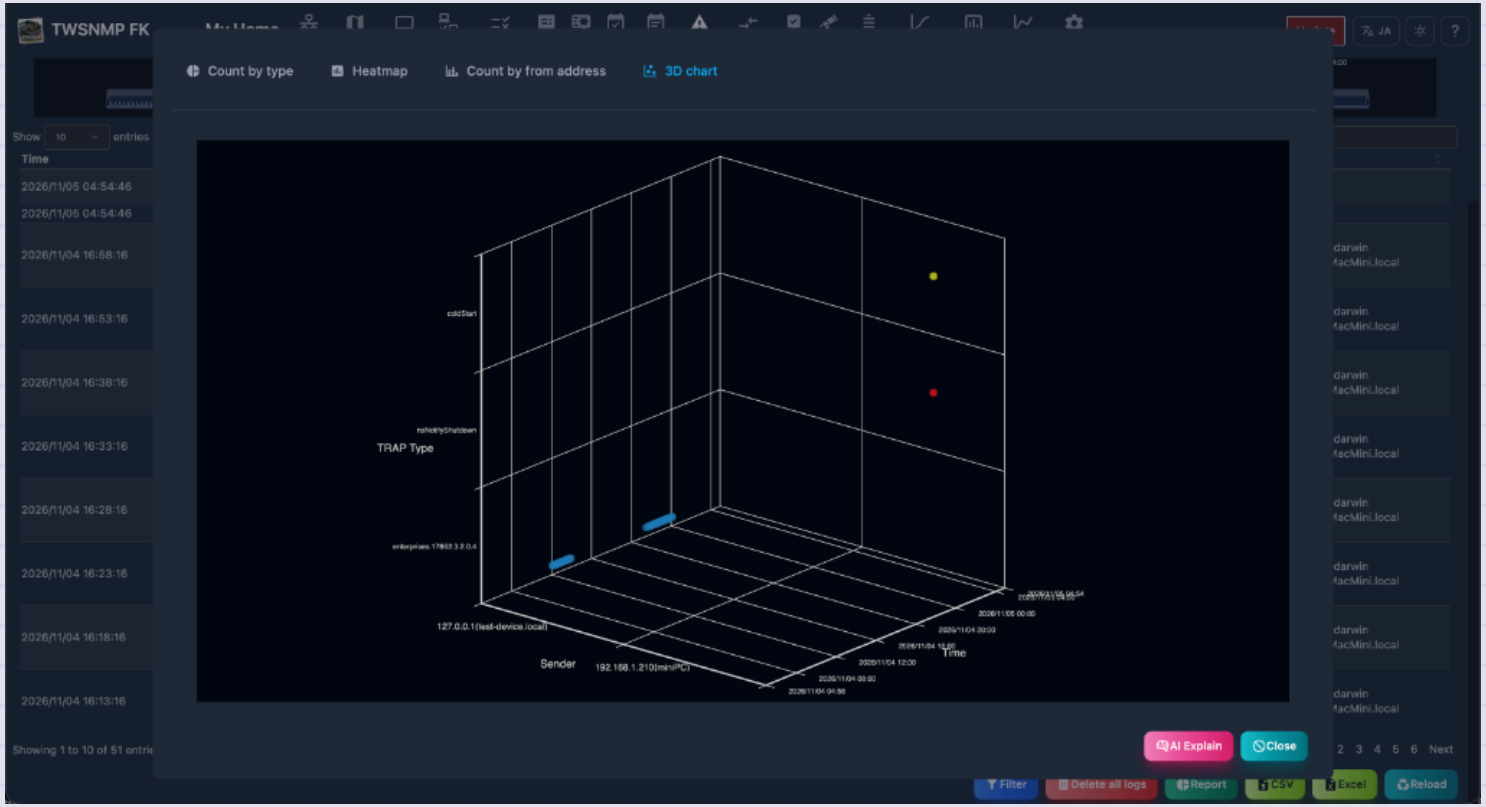
This is a report of the number of SNMP Trap receiving cases by source host.



Copyright (c) 2023-2026 Masayuki Yamai

SNMP TRAP send source and type (3D)

This is a report displayed in the source host, type, and three -dimensional graph of the SNMP Trap receiving log.



Copyright (c) 2023-2026 Masayuki Yamai

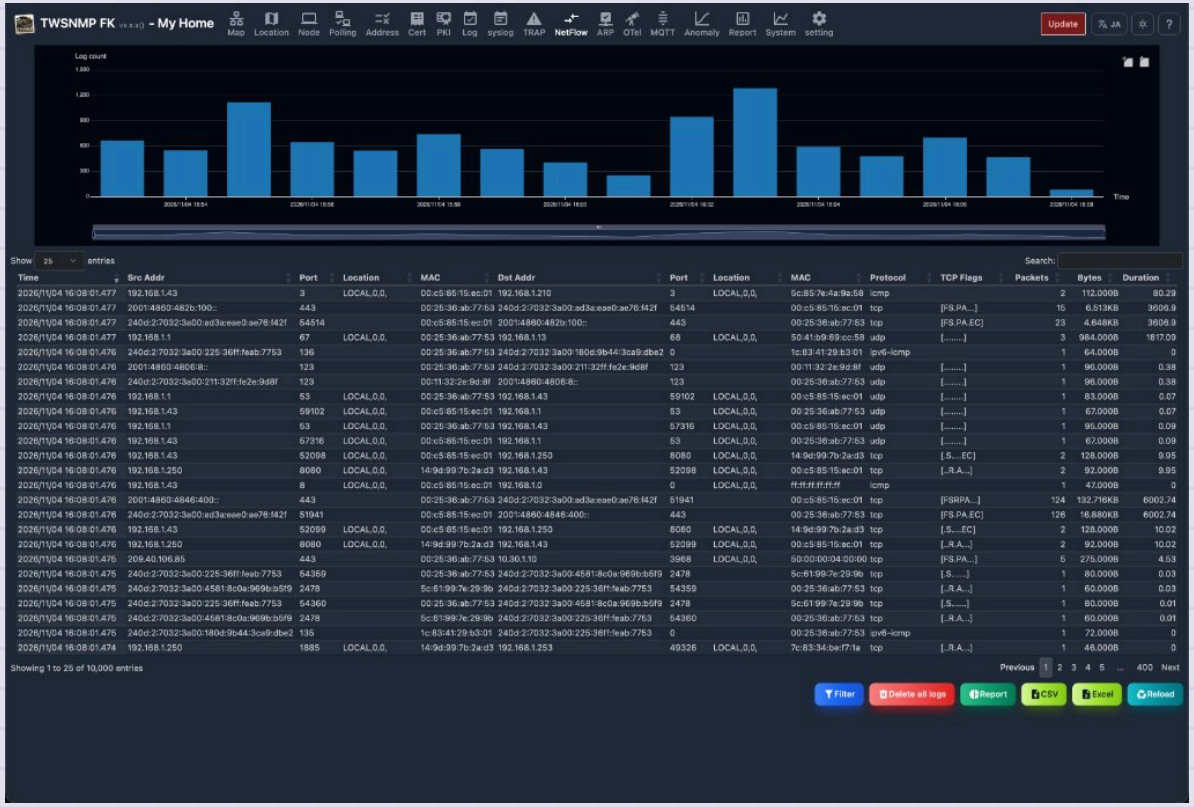
AI Explanation

Sends current SNMP Trap report entries to AI (LLM) for trap frequency analysis, affected equipment identification, and remediation guidance (only visible when AI integration is enabled).

Copyright (c) 2023-2026 Masayuki Yamai

NetFlow

This is the analysis screen of Netflow.



Copyright (c) 2023-2026 Masayuki Yamai

Explanation of table items

Items	Contents
Date and time	The date and time of receiving Netflow.
Sending source	It is the source IP.
Port	It is the port number of the source.
Location	It is the source of the source.GEOIP DB is required.
Address	IP for the destination.
Port	The destination port number.
Location	It is the destination position.GEOIP DB is required.

Copyright (c) 2023-2026 Masayuki Yamai

Items	Contents
Protocol	Protocol such as TCP/UDP/ICMP.
TCP flag	TCP flag.
Packet	The number of sending packets.
Byte	The number of sending bytes.
Period	Flow communication time.

Copyright (c) 2023-2026 Masayuki Yamai

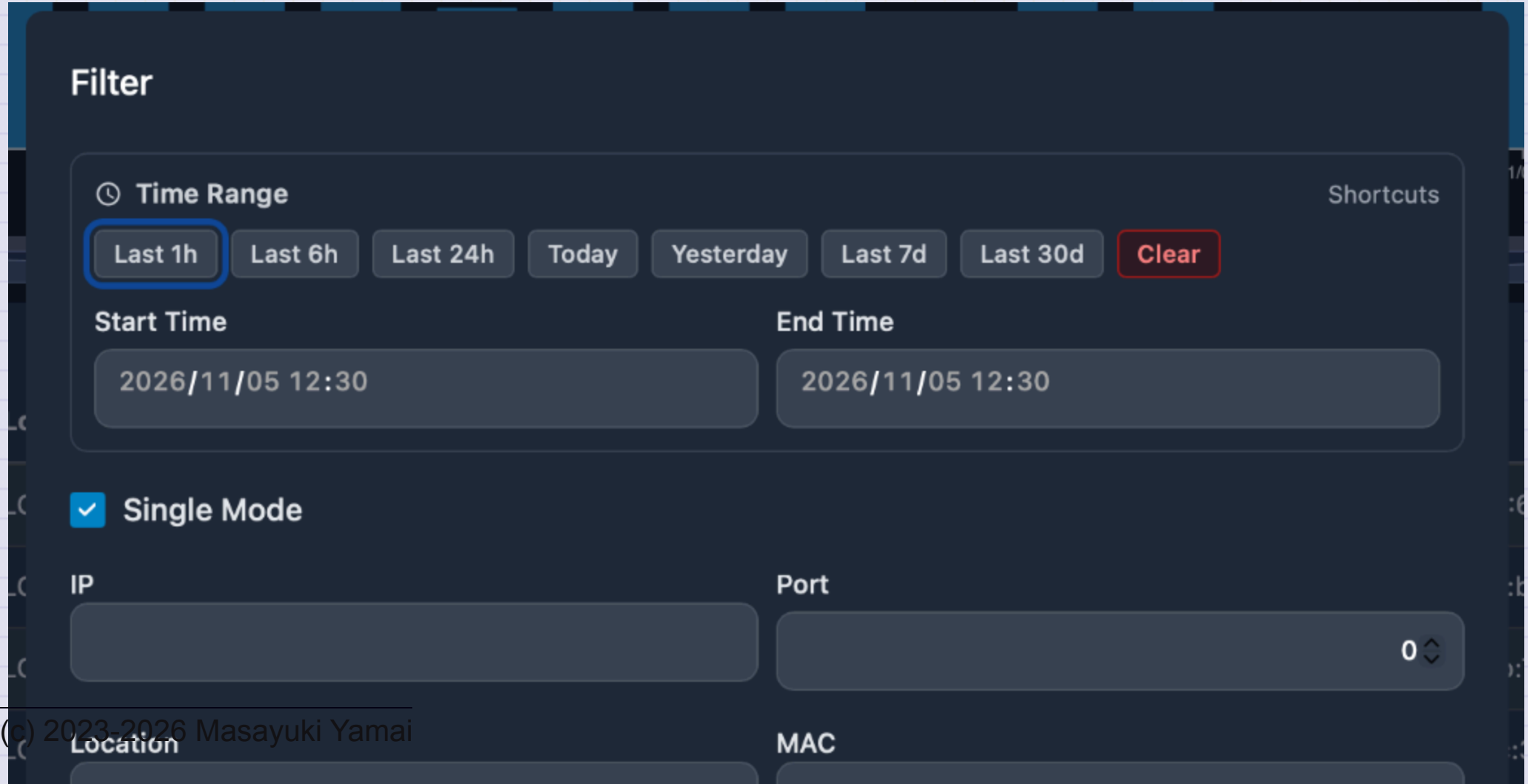
Button

Items	Contents
Filter	Specify the search conditions and display Netflow.
Delete all logs	Delete all Netflow.
Copy	Copy the selected log.
Report	Displays Netflow analysis reports.
CSV	Export Netflow to CSV file.
Excel	Export Netflow to Excel file.
Update	Update the Netflow list to the latest state.

Copyright (c) 2023-2026 Masayuki Yamai

Filter item

It is a filter for netflow search.



Copyright (c) 2023-2026 Masayuki Yamai

Items	Contents
Start date and time	Specify the date and time of the search start.
End date and time	Specify the date and time of the search termination.
Simple mode	Mode to apply IP, port, and position in both directions.
IP	In the case of simple mode, specify the source and destination IP.
Port	In the case of simple mode, specify the source and destination port.
Location	Specify the source and destination position in the case of simple mode.

Copyright (c) 2023-2026 Masayuki Yamai

Items	Contents
Sending source IP	Specify the source IP.
Port	Specify the source port.
Location	Specify the source position.
Destination IP	Specify the destination IP.
Port	Specify the destination port.
Location	Specify the destination position.
Protocol	Specify the protocol name.
TCP flag	Specify the TCP flag.

Copyright (c) 2023-2026 Masayuki Yamai

Netflow report

Report name	Contents
Heat Map	Heat map by time zone of Netflow's receiving number.
Histogram	Histgram of numerical data.
Traffic	This is a time -series graph of traffic.
TOP List	This is a ranking report for each item.
TOP list (3D)	Ranking reports for each item are displayed in 3D graphs.
IP Pair Flow	The combination of communication is displayed in a graph.

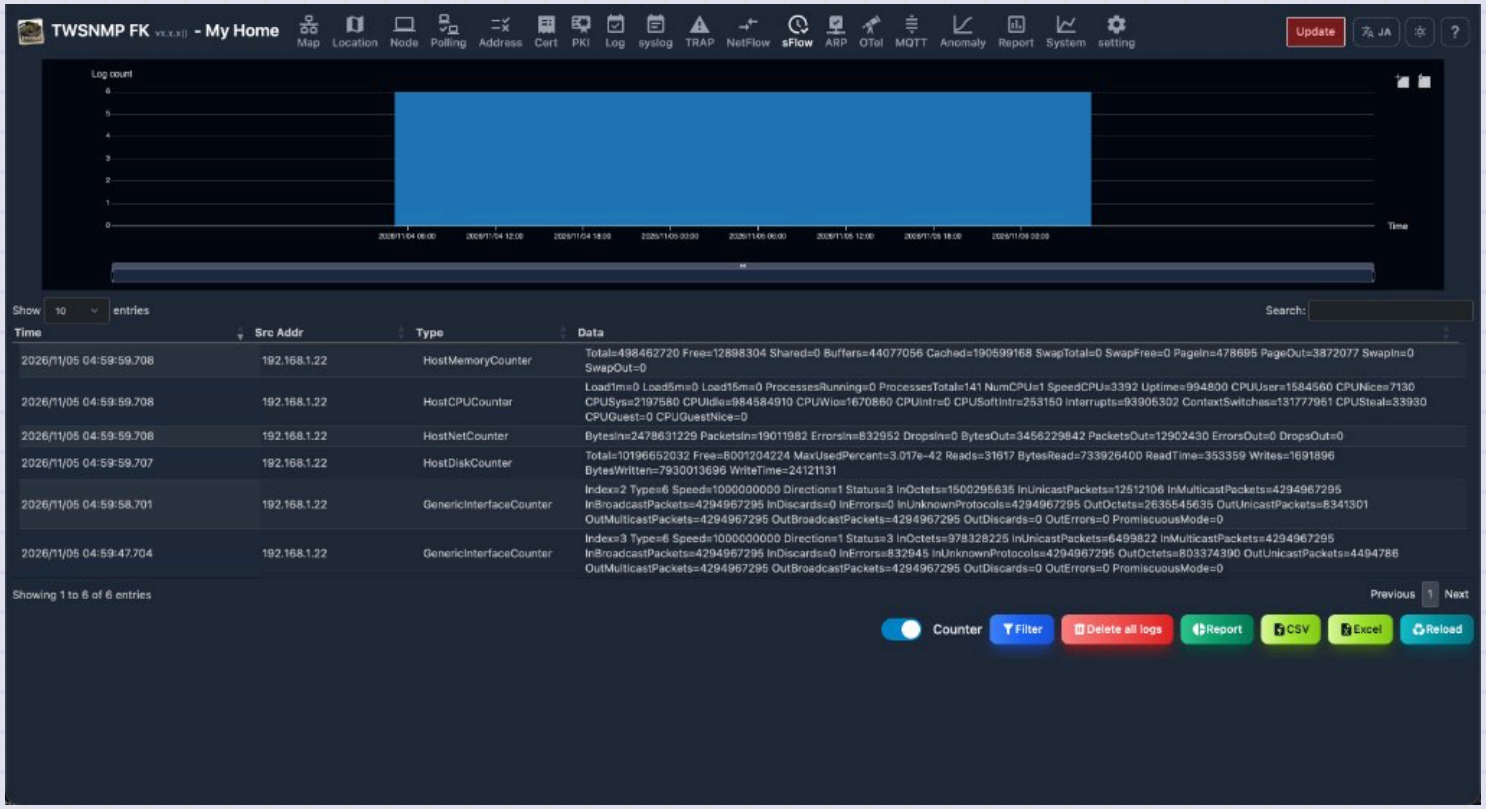
Copyright (c) 2023-2026 Masayuki Yamai

Report name	Contents
FFT Analysis	Analyzing the communication cycle with FFT.
FFT Analysis (3D)	Analyze the communication cycle with FFT and display it on the 3D graph.
Map	Display the position of the IP address on the map.
AI Explanation	Sends current NetFlow data to AI (LLM) for traffic volume, top communicating pairs, bandwidth analysis, and traffic anomaly explanations (only visible when AI integration is enabled).

Copyright (c) 2023-2026 Masayuki Yamai

sFlow

This is the analysis screen of sFLOW.



Flow sample

Copyright (c) 2023-2026 Masayuki Yamai

Items	Contents
Address	IP for the destination.
Port	The destination port number.
Location	It is the destination position.GEOIP DB is required.
Mac	This is the destination MAC address.
Protocol	Protocol such as TCP/UDP/ICMP.
TCP flag	TCP flag.
Byte	The number of sending bytes.

Copyright (c) 2023-2026 Masayuki Yamai

Description of button

Items	Contents
Counter	Switch between flow samples and counter samples.
Filter	Specify the search conditions and display sFlow.
Delete all logs	Delete all sFlow.
Copy	Copy the selected log.

Copyright (c) 2023-2026 Masayuki Yamai

Items	Contents
Report	Displays SFLOW analysis reports.
CSV	Export sFlow to CSV file.
Excel	Export SFLOW to Excel file.
Update	Update the SFLOW log to the latest state.

Copyright (c) 2023-2026 Masayuki Yamai

Flow sample filter item

Items	Contents
Start date and time	Specify the date and time of the search start.
End date and time	Specify the date and time of the search termination.
Simple mode	Mode to apply IP, port, and position in both directions.
IP	In the case of simple mode, specify the source and destination IP.
Port	In the case of simple mode, specify the source and destination port.
Location	Specify the source and destination position in the case of simple mode.

Copyright (c) 2023-2026 Masayuki Yamai

Items	Contents
Port	Specify the source port.
Location	Specify the source position.
Destination IP	Specify the destination IP.
Port	Specify the destination port.
Location	Specify the destination position.
Protocol	Specify the protocol name.
TCP flag	Specify the TCP flag.

- Strings can be searched using regular expressions.

Copyright (c) 2023-2026 Masayuki Yamai

Counter sample

Items	Contents
Date and time	It is the date and time when the SFLOW sample received.
Sending source	It is the source IP.
Type	The type of counter sample.(I/F, CPU, Memory, Disk, Netowk)
Data	Data of counter sample.

Copyright (c) 2023-2026 Masayuki Yamai

Counter sample filter item

Items	Contents
Start date and time	Specify the date and time of the search start.
End date and time	Specify the date and time of the search termination.
Submit	Specify the source.
Type	Specify the type of counter sample.

- The character string can be searched by regular expression.

Copyright (c) 2023-2026 Masayuki Yamai

Flow sample report

Report name	Contents
Heat map	Heat map showing the number of log time.
Communications	This is a time -series graph of communication volume.
TOP List	This is a ranking report for each item.
TOP list (3D)	Ranking reports for each item are displayed in 3D graphs.
IP Pair Flow	The combination of communication is displayed in a graph.
FFT Analysis	Analyzing the communication cycle with FFT.
FFT Analysis (3D)	Analyze the communication cycle with FFT and display it on the 3D graph.
Map	Display the position of the IP address on the map.
	Sends current sFlow sample data to AI (LLM) for flow traffic

Copyright (c) 2023-2026 Masayuki Yamai

Counter sample report

Report name	Contents
Heat map	Heat map showing the number of log time.
I/F BPS	This is a time -series graph of the communication volume (Bytes/Sec) obtained from I/F sample.
I/F PPS	This is a time -series graph of the communication volume (Packets/Sec) obtained from I/F sample.

Copyright (c) 2023-2026 Masayuki Yamai

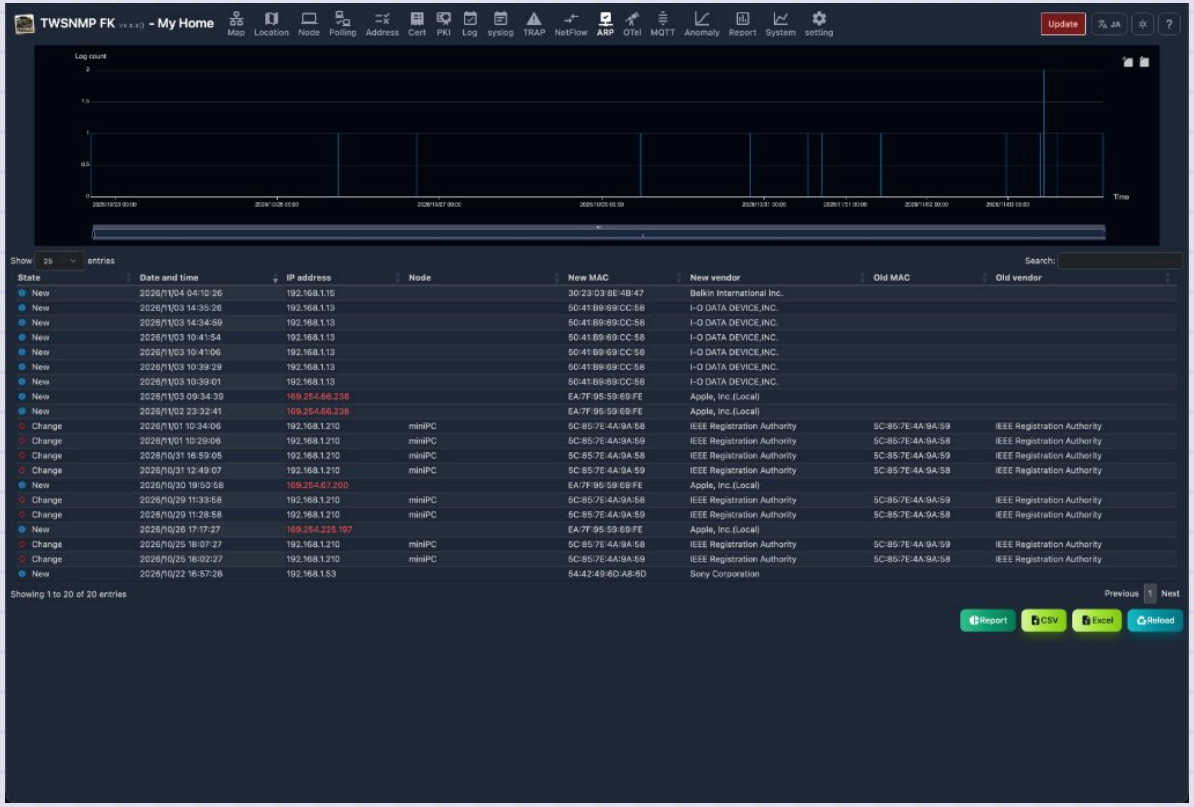
Report name	Contents
CPU	CPU usage rate and load time series graph acquired from CPU sample.
Memory	Memory is a graph of memory usage and available capacity obtained from a sample.
DISK	This is a graph of disk usage and access amount acquired from Disk sample.
Network	Network is a graph of network usage obtained from sample.
AI Explanation	Sends current sFlow counter data to AI (LLM) for port traffic load analysis, interface error diagnostics, and bottleneck detection (only visible when AI integration is enabled).

Copyright (c) 2023-2026 Masayuki Yamai

ARP warch log

ARP watch log screen.

At the top, there is a graph showing the number of logs in chronological order.



Copyright (c) 2023-2026 Masayuki Yamai

Items	Contents
State	Log status.Either new or change.
Date and time	The date and time of the log.
IP address	IP address to log.
Node	The name of the node registered on the map.
New MAC	New discovery or MAC address after change.
New vendor	The vendor name of the new MAC address.
Old MAC	MAC address before change.
Old vendor	This vendor name of the old MAC address.

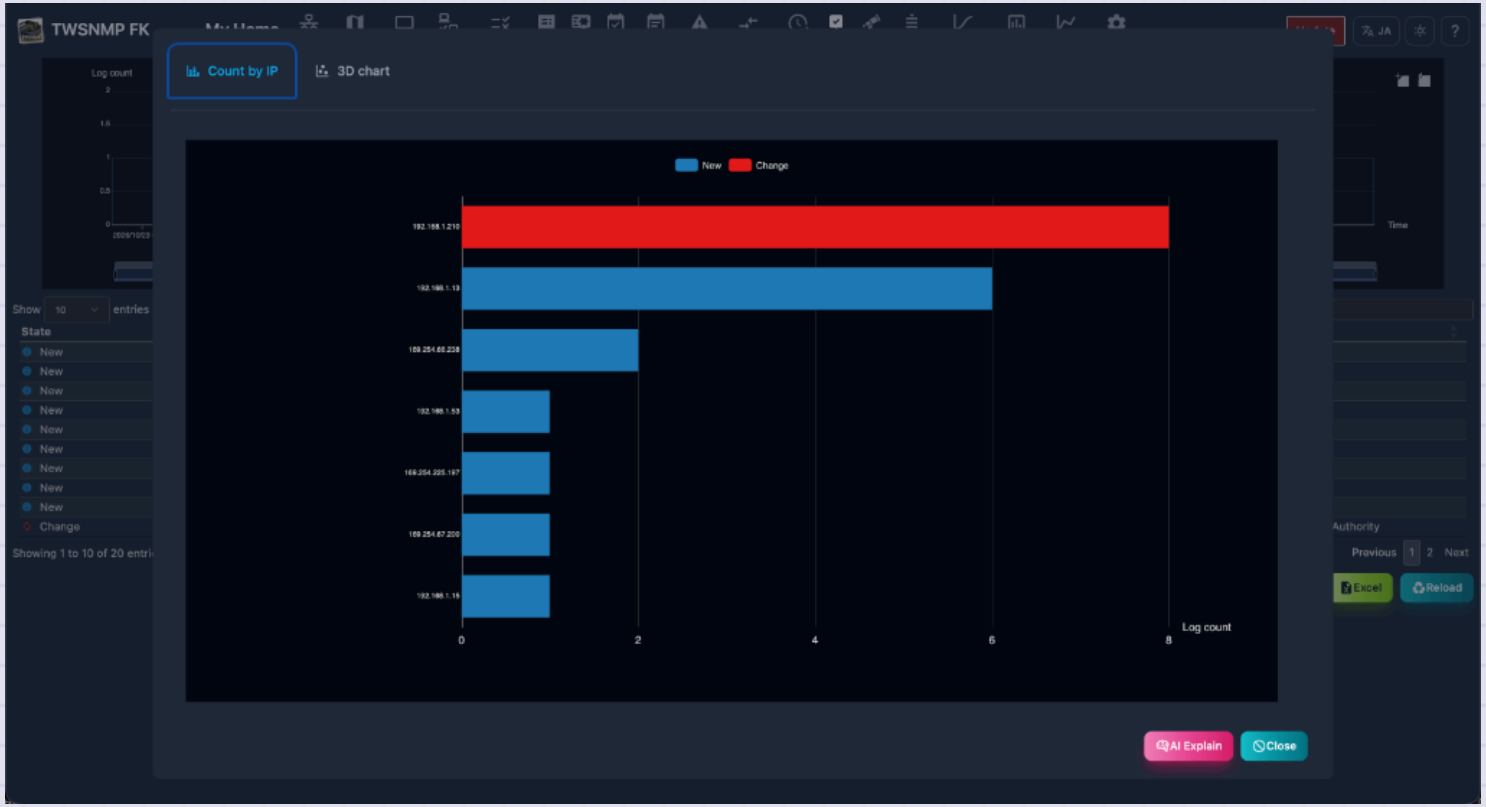
Copyright (c) 2023-2026 Masayuki Yamai

Items	Contents
Report	Displays the ARP watch log analysis report.
CSV	Export the ARP watch log to the CSV file.
Excel	Export the ARP watch log to the Excel file.
Reload	Update the list of ARP watch logs to the latest state.

Copyright (c) 2023-2026 Masayuki Yamai

ARP watch log count by IP address

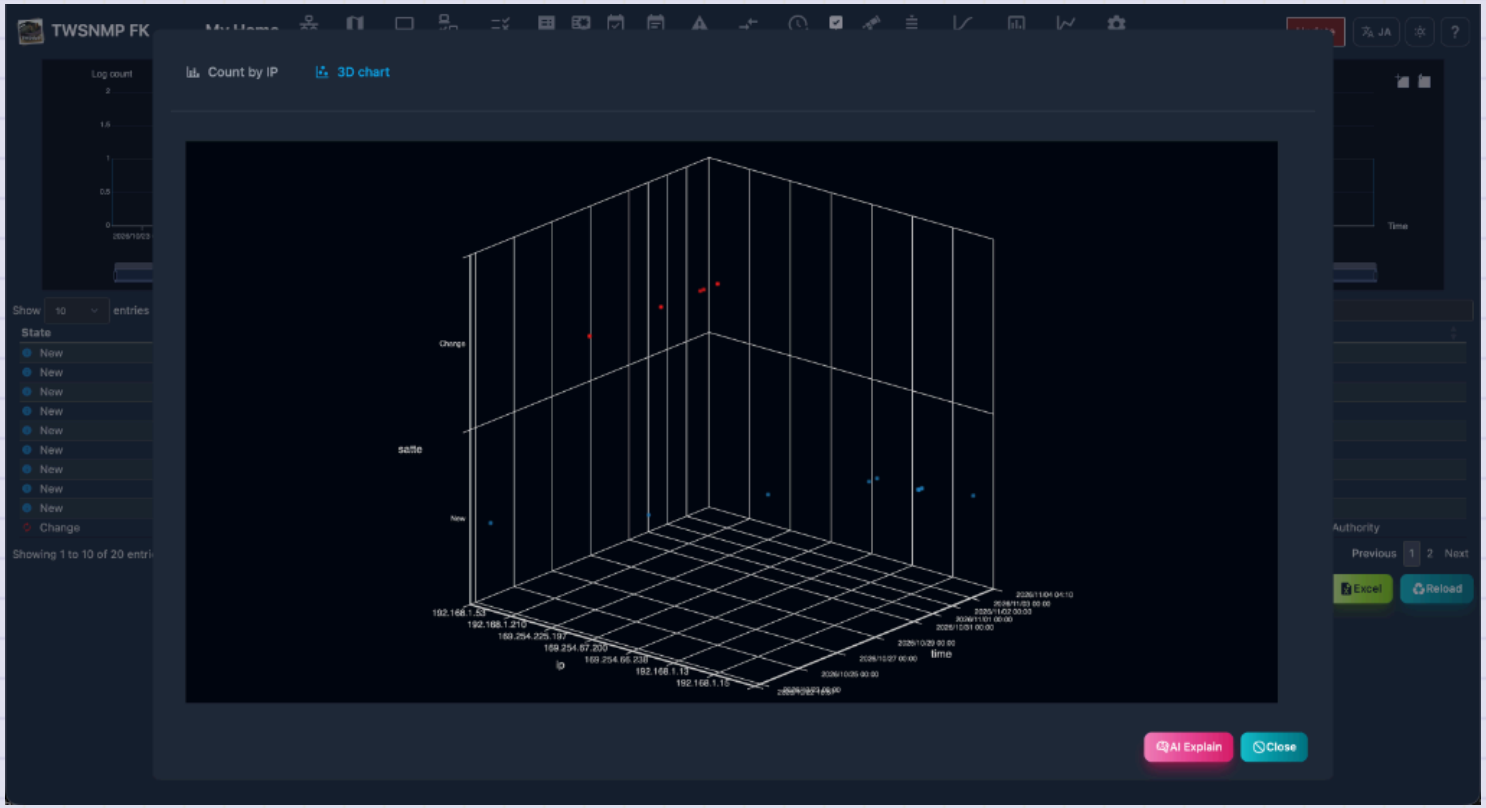
This is a report of the number of logs by IP address.The IP address with many changes is obvious at a glance.



Copyright (c) 2023-2026 Masayuki Yamai

ARP watch log count by IP address (3D)

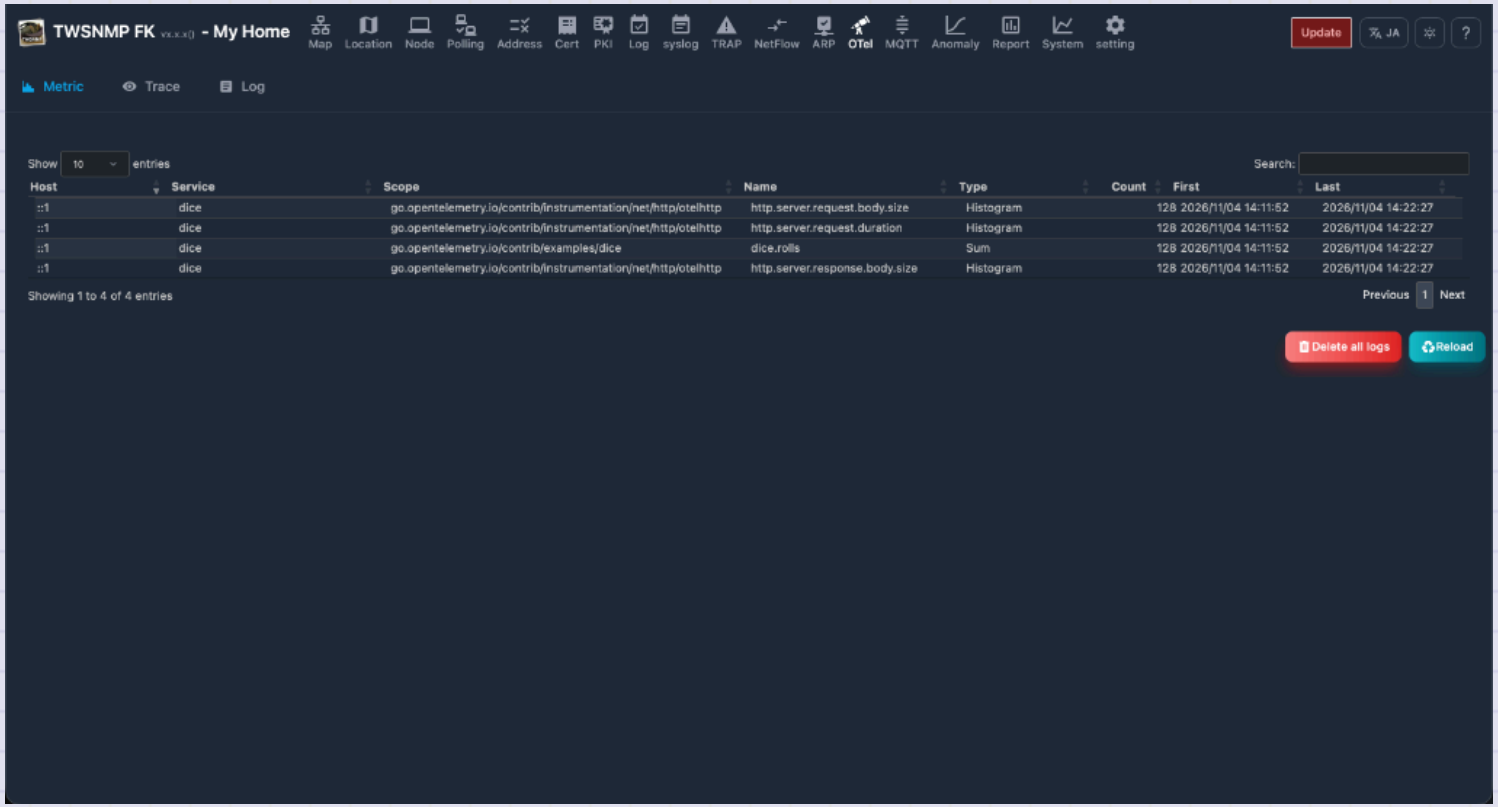
This is a report of ARP watch logs from both IP addresses and time series.The time of new discoveries and changes is obvious at a glance.



Copyright (c) 2023-2026 Masayuki Yamai

OpenTelemetry

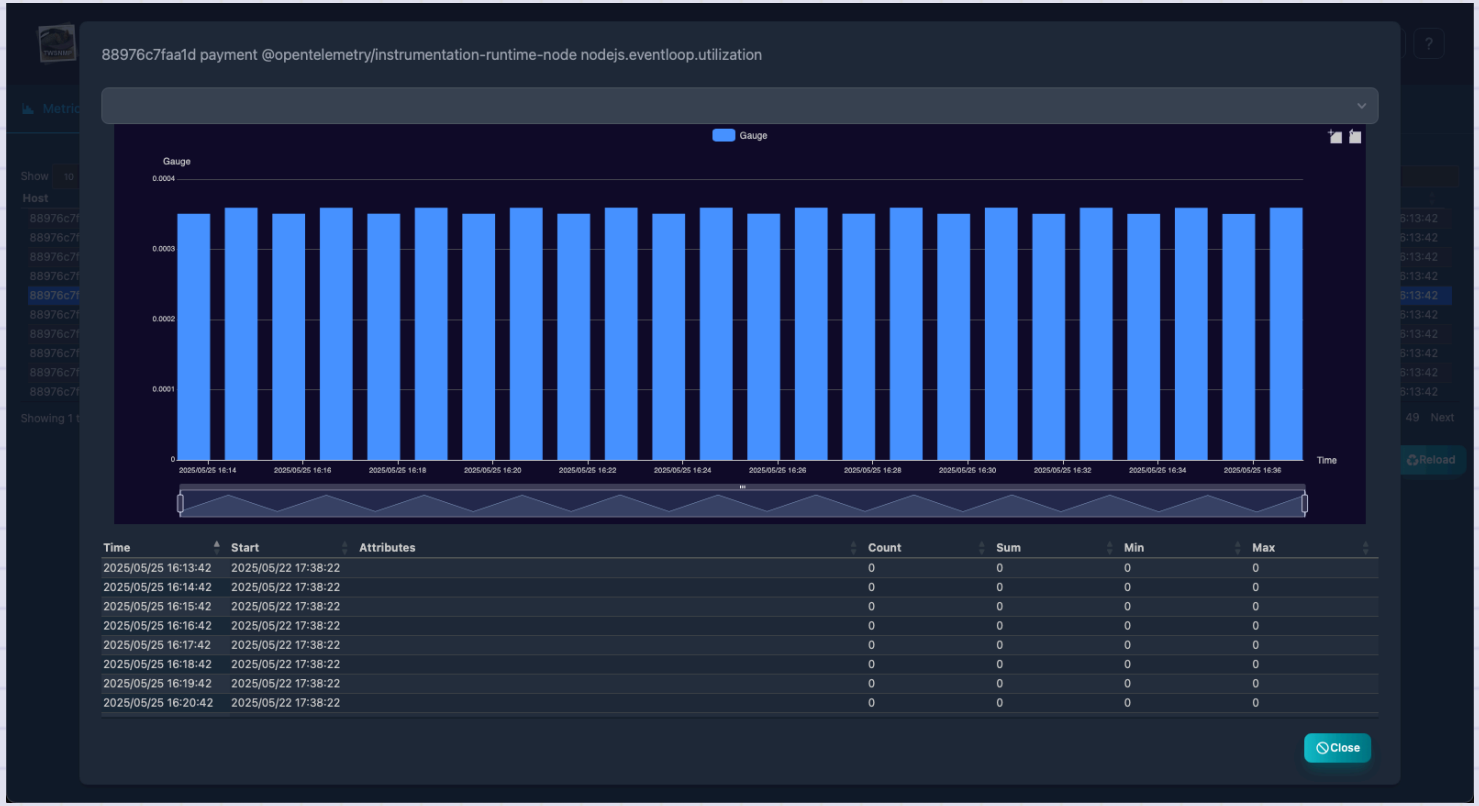
OpenTelemetry Collector screen. Toggle between Metrics, Traces, and Logs.



Copyright (c) 2023-2026 Masayuki Yamai

Metrics

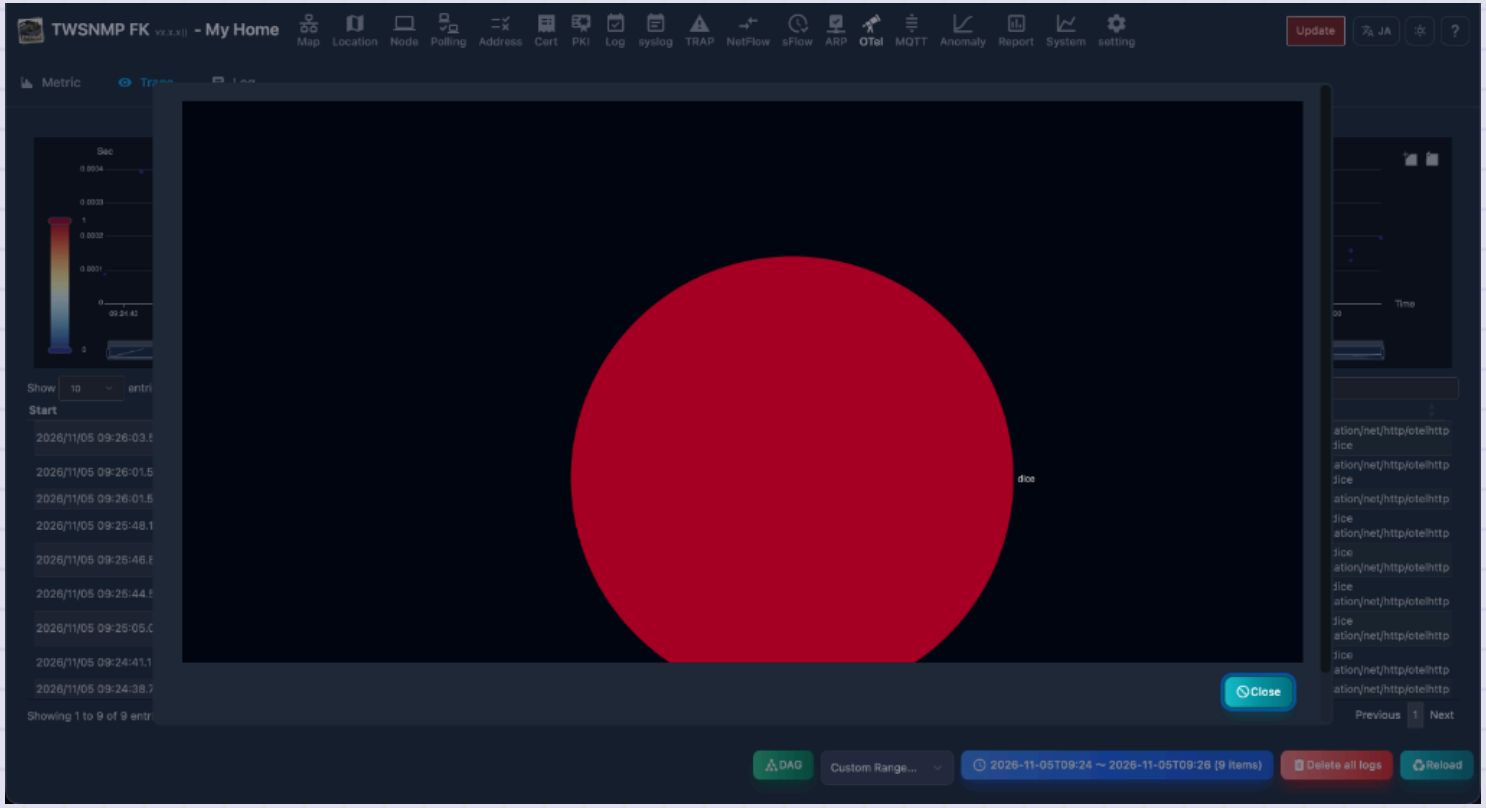
Time charts and histograms are available for visualization.



Copyright (c) 2023-2026 Masayuki Yamai

Traces

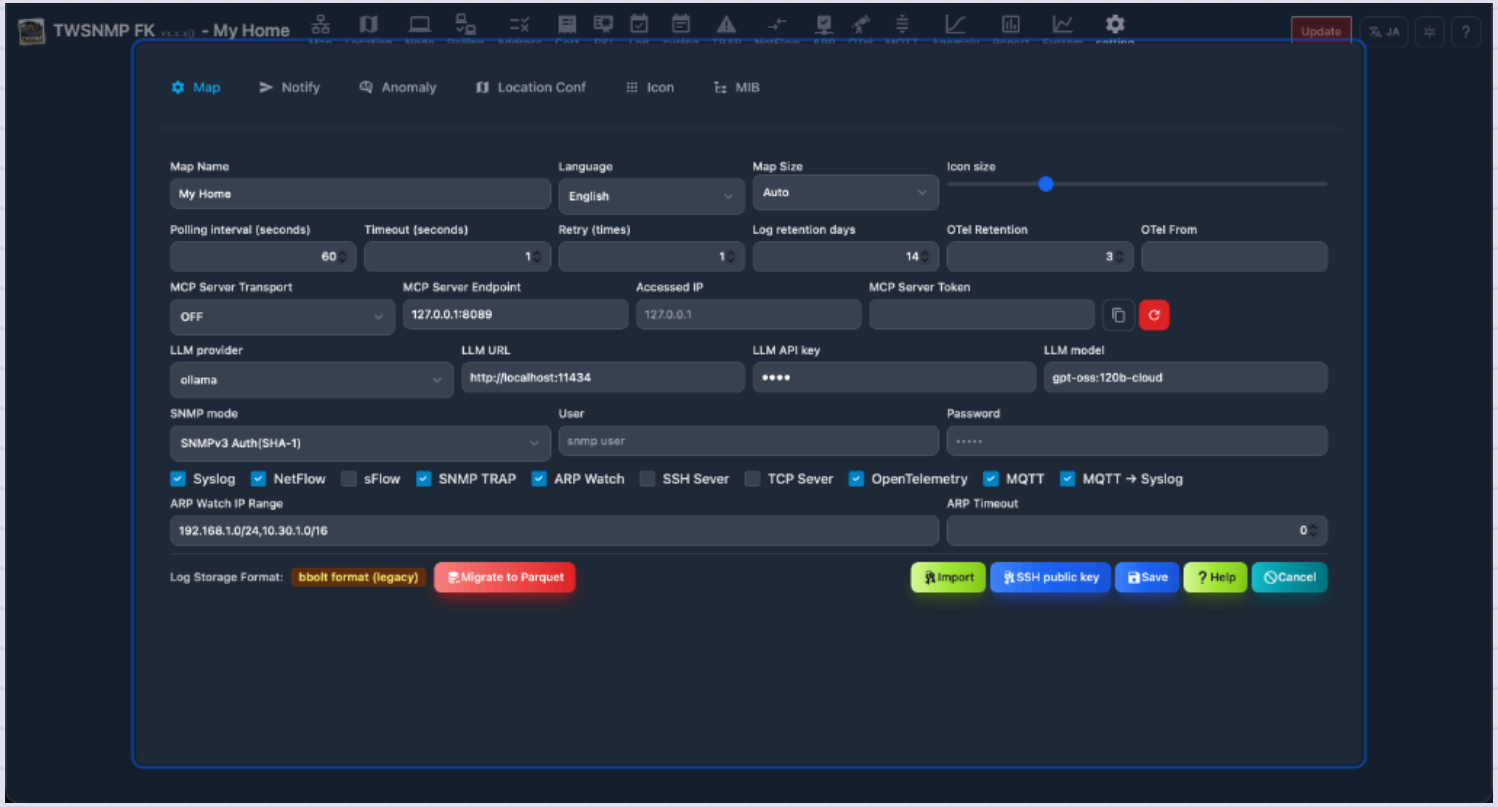
Displays DAG (relationships between services) and timeline graphs.



Copyright (c) 2023-2026 Masayuki Yamai

Settings

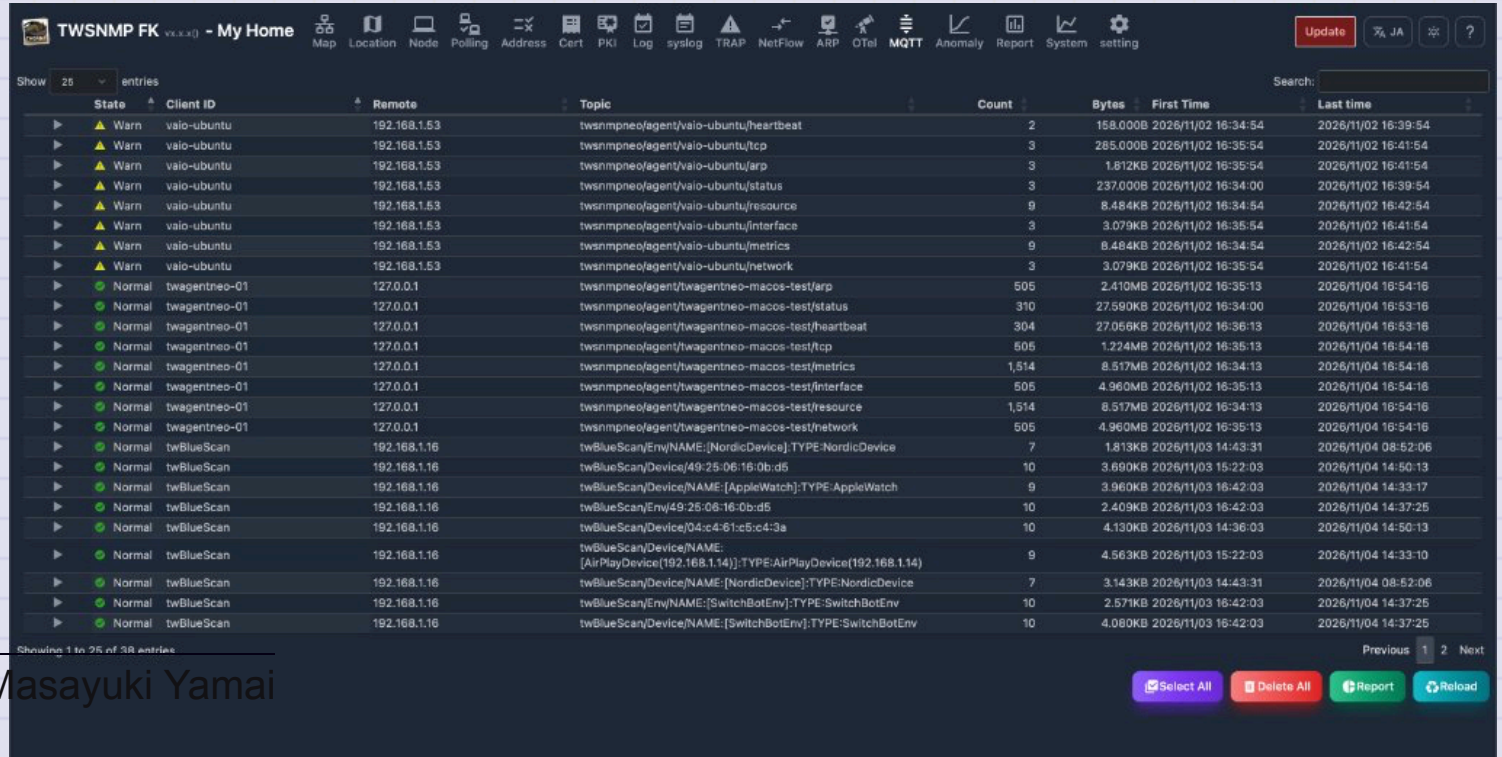
Retention Time, Source IP limits, and reception ON/OFF.



Copyright (c) 2023-2026 Masayuki Yamai

MQTT

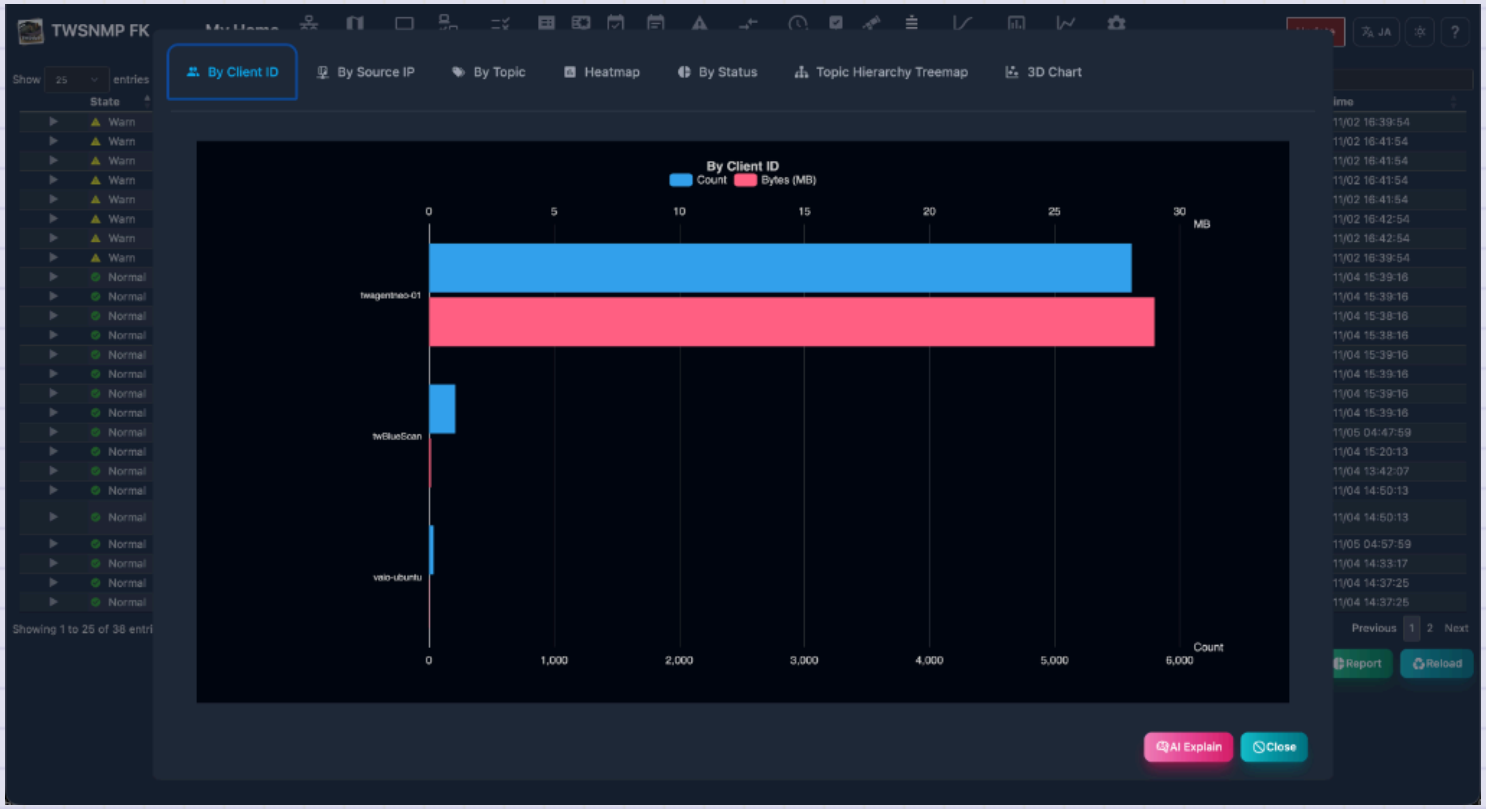
Displays received MQTT topics, status, client ID, and counts. You can expand any topic row to view and inspect the latest received payload data (formatted as JSON, TEXT, or HEX) directly on the screen. Also supports viewing statistical report charts, copying topic names, and creating polling tasks.



Copyright (c) 2023-2026 Masayuki Yamai

MQTT Report

A report screen that provides multi-faceted analysis of received MQTT messages, including statistical charts, topic hierarchy treemaps, and 3D correlation graphs.



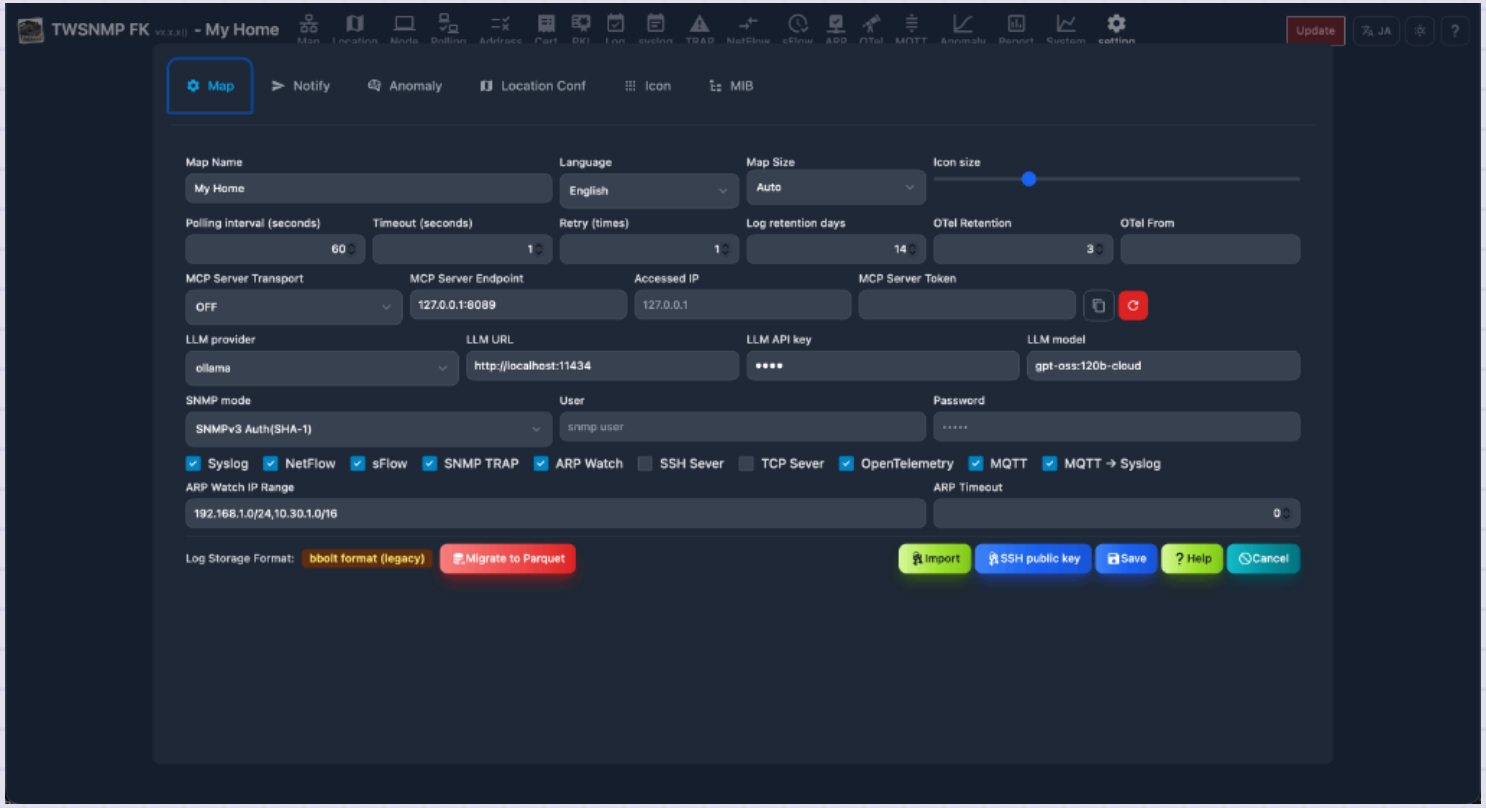
Copyright (c) 2023-2026 Masayuki Yamai

Tab	Contents
By Client ID	Aggregates and displays message counts and data volume per Client ID.
By Source IP	Aggregates message counts and data volume per sender IP address.
By Topic	Aggregates message counts and data volume ratios per topic.
Heatmap	Visualizes message reception frequency over time using a heatmap.
By Status	Displays the distribution and ratio of topics by status level.
Topic Hierarchy Treemap	Visualizes the hierarchical structure of MQTT topics using a Treemap.
	Plots correlations between topics, senders, and data volume

Copyright (c) 2023-2026 Masayuki Yamai

MQTT settings

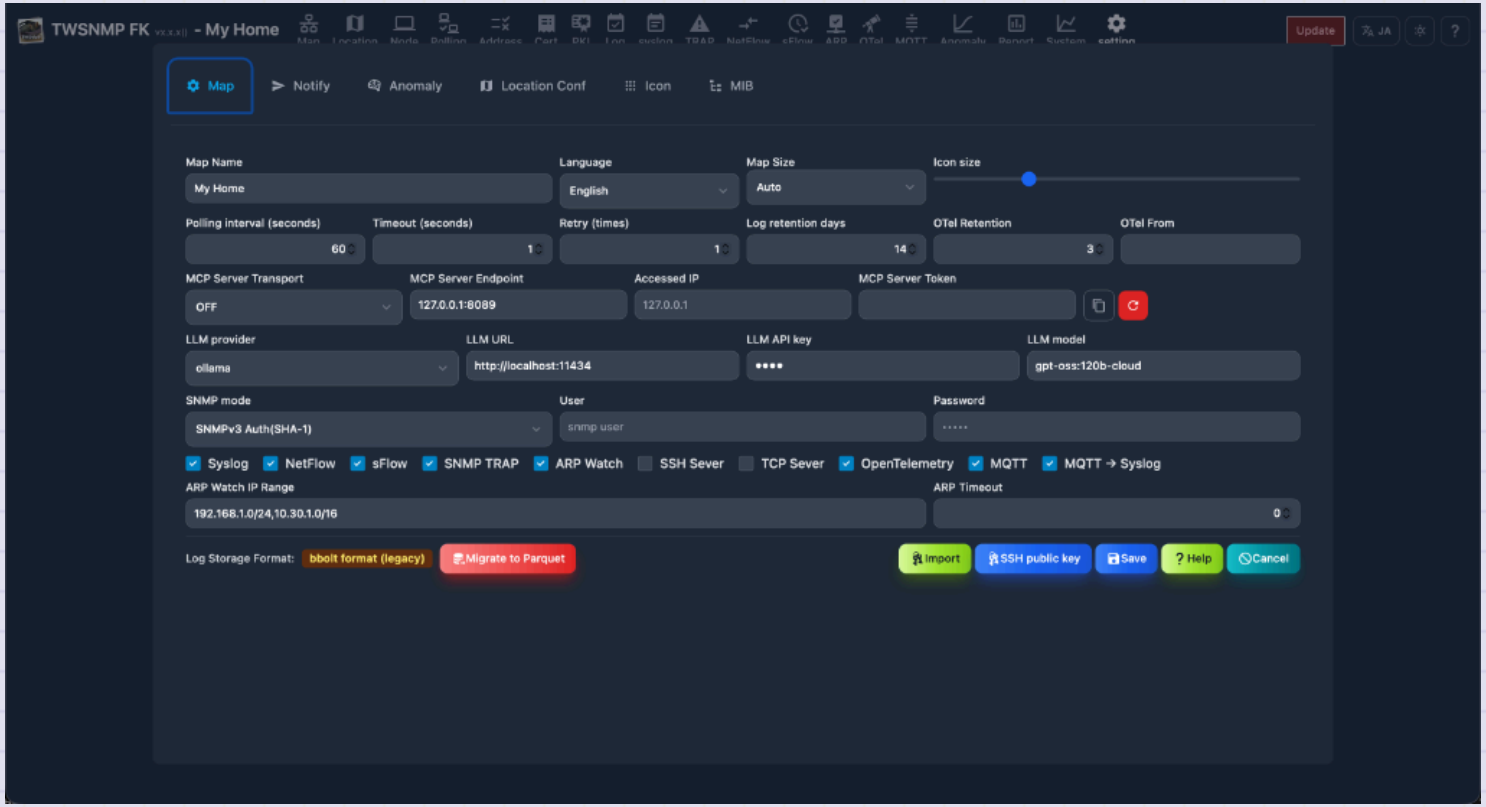
Toggle enable/disable and record MQTT data to syslog.



Copyright (c) 2023-2026 Masayuki Yamai

MCP Server

MCP server function for AI to use TWSNMP FK.



Copyright (c) 2023-2026 Masayuki Yamai

MCP Server Features

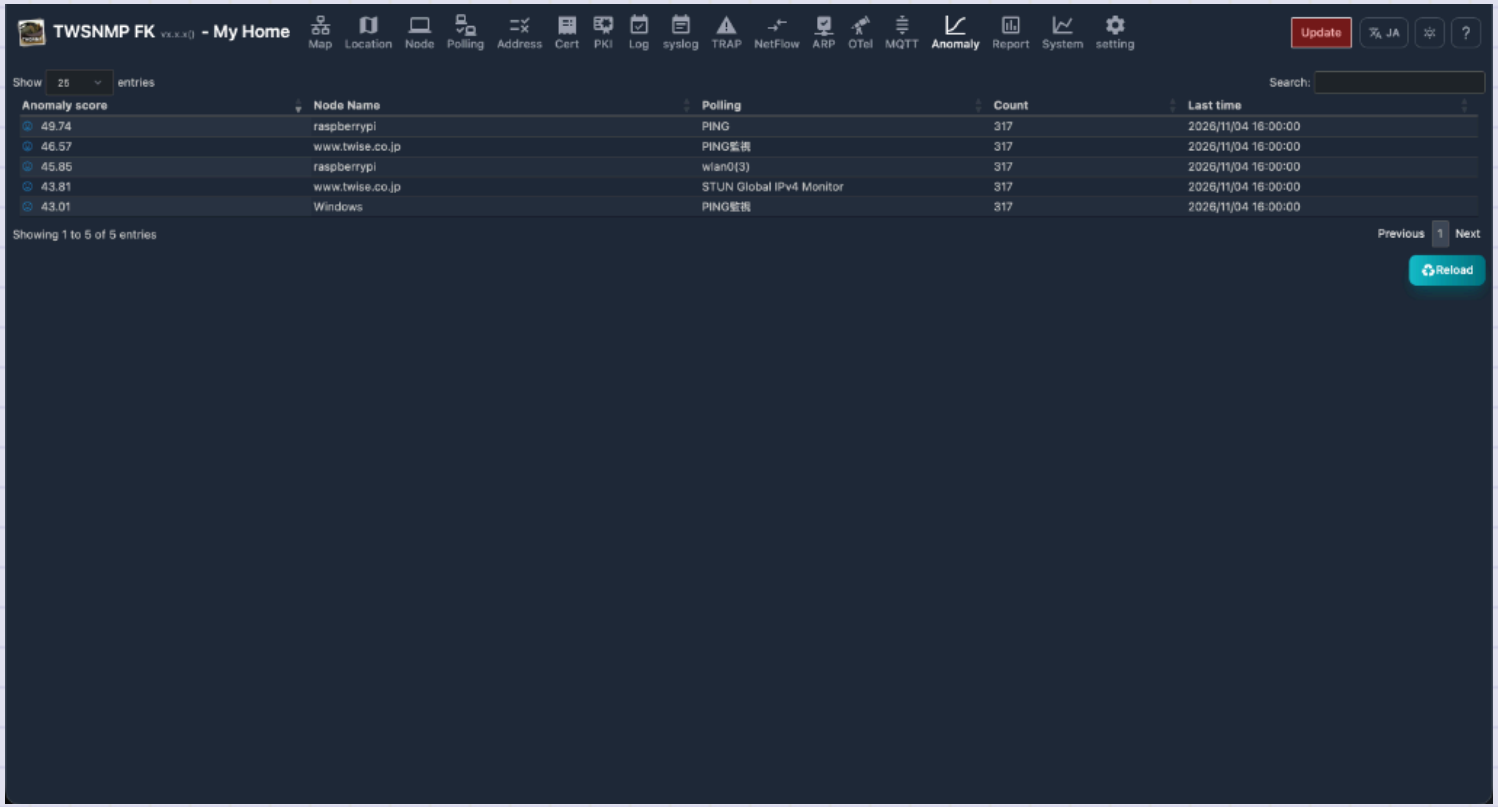
Provides tools for network monitoring and management.

- MCP server specifications
- MCP server prompt specifications

Copyright (c) 2023-2026 Masayuki Yamai

Anomaly

The screen of the anomaly detection list (Menu: "Anomaly"). Displays items for which analysis is performed in the polling log settings.



Copyright (c) 2023-2026 Masayuki Yamai

Items	Contents
anomaly score	A deviation value that indicates the degree of anomaly of analysis results. 50 is average. Large values indicate a high degree of anomaly.
Node name	The name of the node to be analyzed.
Polling	Polling for anomaly detection.
Data count	The number of data points analyzed. If small, the accuracy will be lower.
Last time	The last date and time of analysis.

Copyright (c) 2023-2026 Masayuki Yamai

Items	Contents
Report	Displays reports on the selected anomaly detection results.
clear	Clear the selected anomaly detection results.
Reload	Update the anomaly list to the latest state.

Copyright (c) 2023-2026 Masayuki Yamai

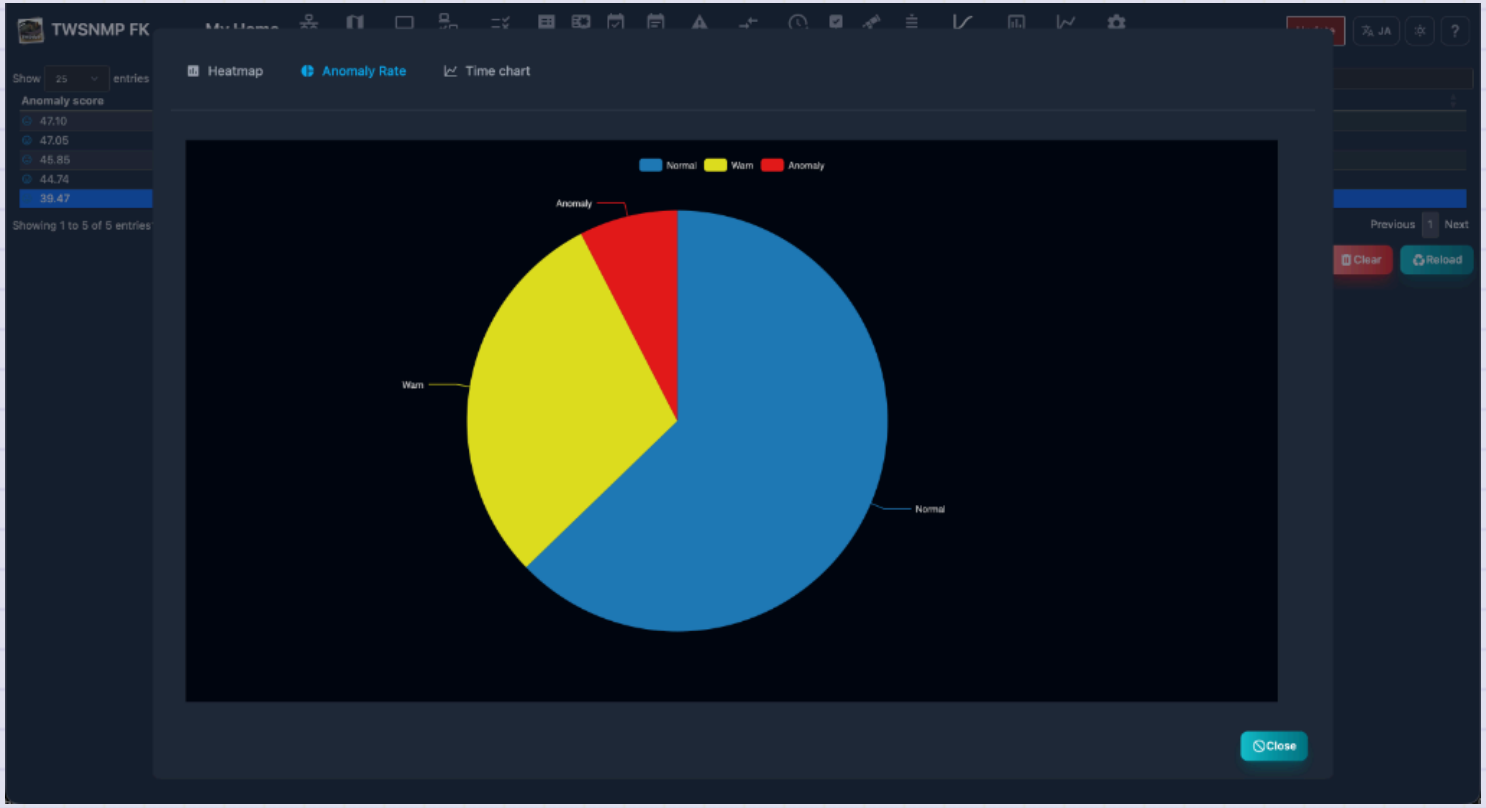
AI anomaly score heatmap

This is a report showing an anomaly score on a daily heat map.It indicates that the red color is the time when the anomaly has occurred.

Copyright (c) 2023-2026 Masayuki Yamai

AI anomaly score percentage

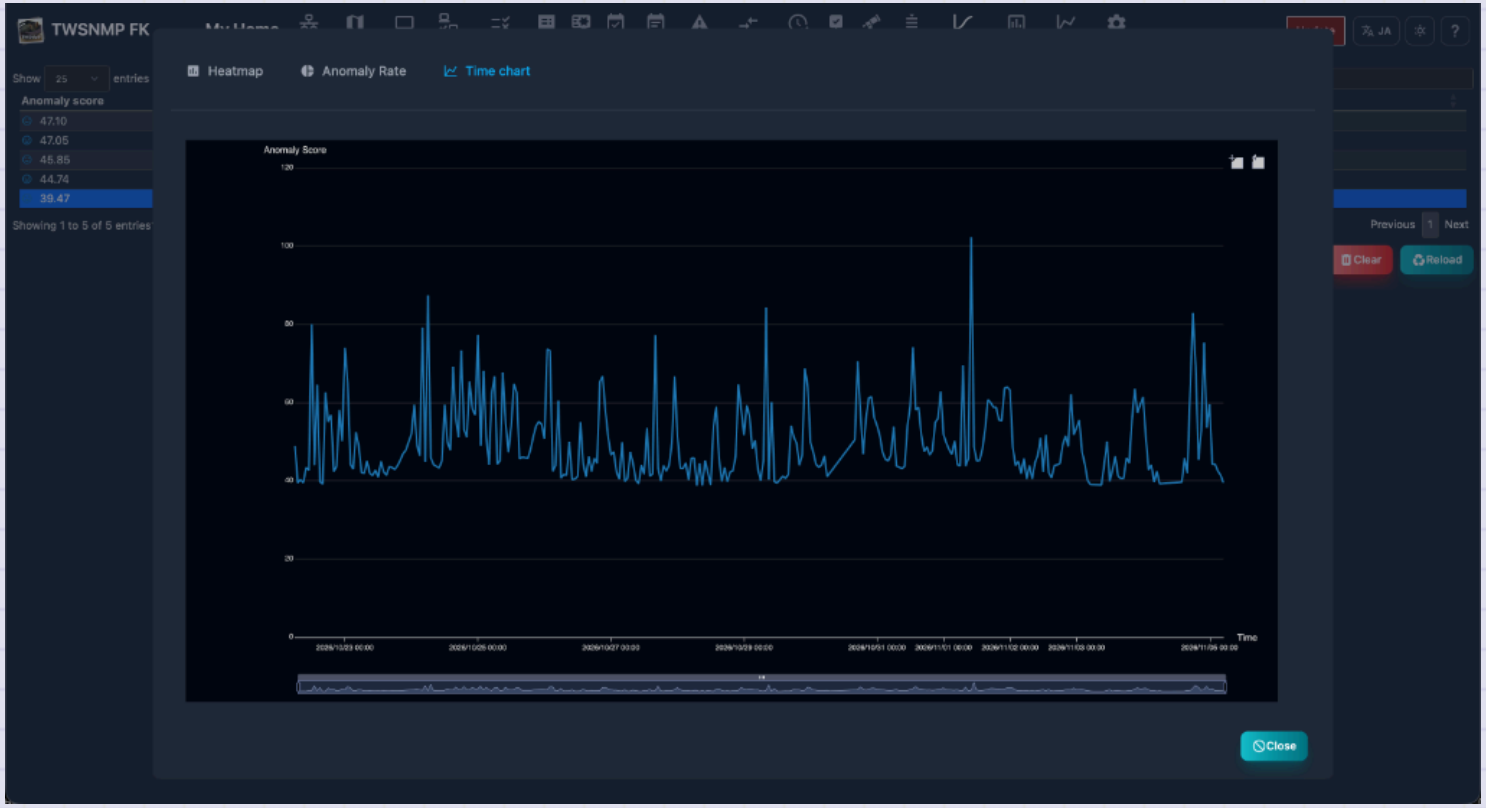
The percentage of the anomaly score in the entire period is shown in a circular graph.



Copyright (c) 2023-2026 Masayuki Yamai

AI anomaly score time chart

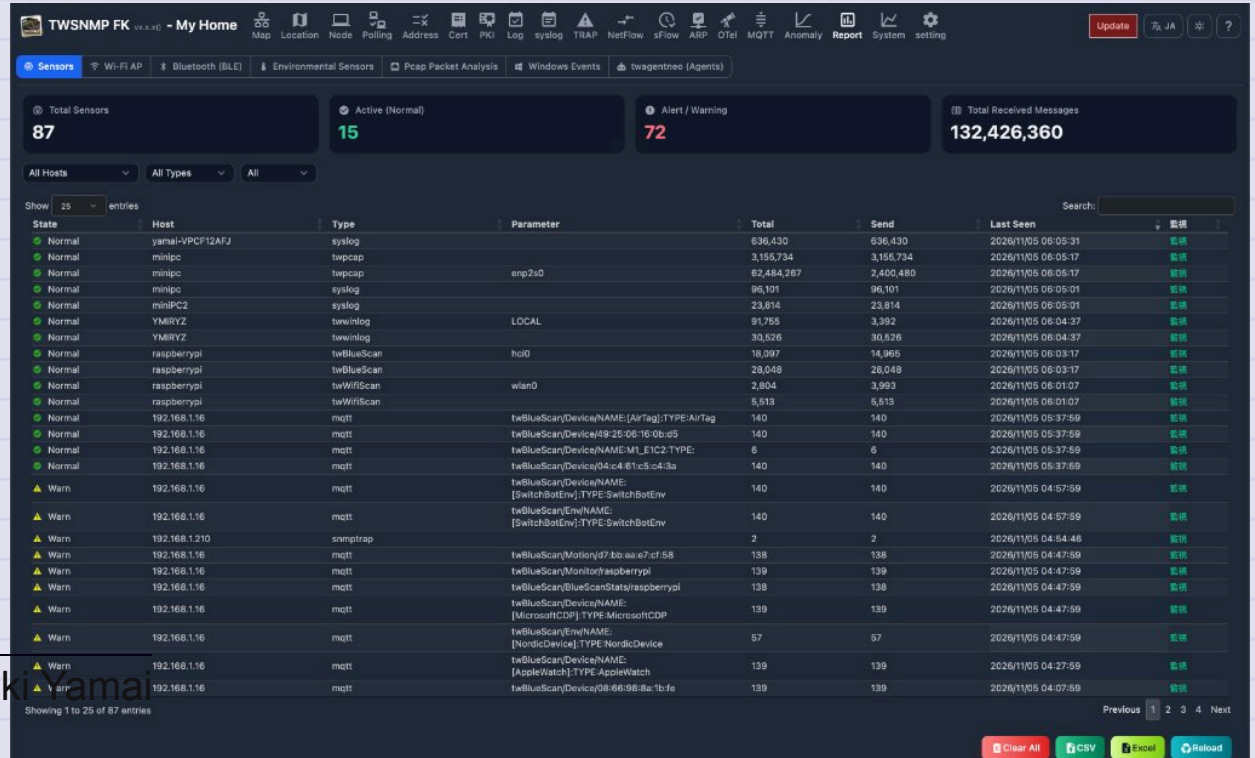
This is a report that displays an anomaly score in chronological order.



Copyright (c) 2023-2026 Masayuki Yamai

Report

Comprehensive sensor and agent report screen (menu "Report").
Aggregates and visualizes messages from syslog, twpcap, twBlueScan, twWifiScan, twwinlog, MQTT, and SNMP TRAP, with dedicated views for Wi-Fi, Bluetooth, environmental data, packet analysis, Windows event logs, and twagentneo agents.



Copyright (c) 2023-2026 Masayuki Yamai

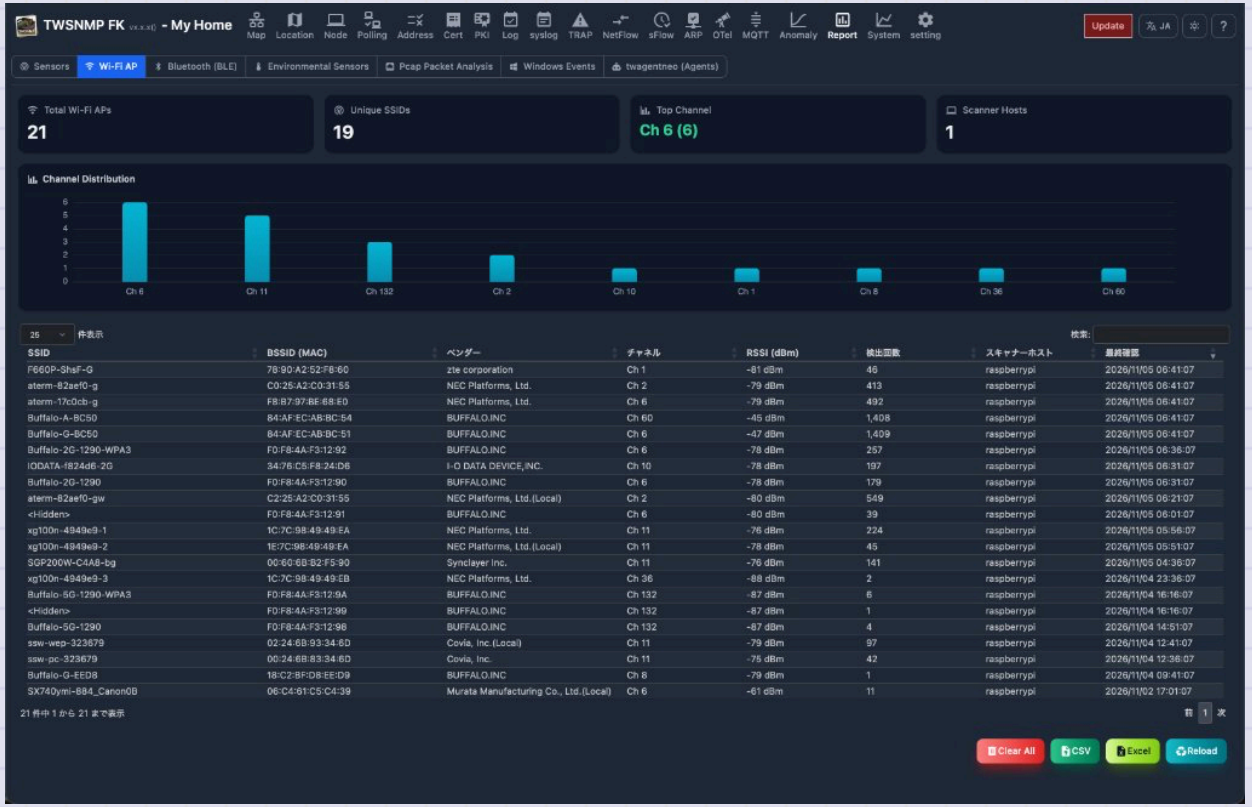
Report Tabs Breakdown

Tab	Contents
Sensors	Registered sensors state, type, message statistics, last seen timestamp, and monitoring controls.
Wi-Fi AP	Wi-Fi access points detected by twWifiScan (SSID, BSSID, RSSI, channel, encryption).
Bluetooth (BLE)	Proximity detection report for BLE beacons and wearables detected by twBlueScan.
Environmental Sensors	Environmental sensor values (temperature, humidity, atmospheric pressure, CO2, etc.).
Pcap Packet Analysis	Traffic flow and protocol statistics from packet captures via twpcap.
	Windows security log aggregation (logon/logoff, user actions

Copyright (c) 2023-2026 Masayuki Yamai

Wi-Fi AP Report

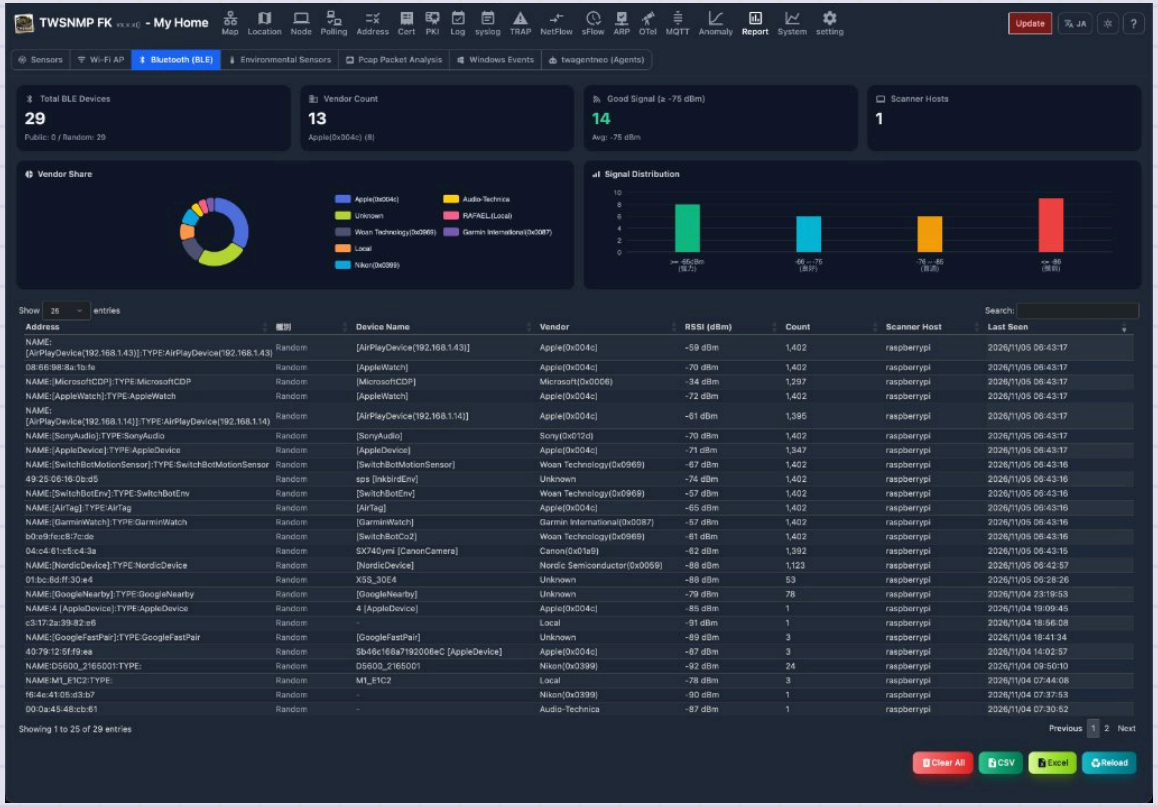
Displays detected Wi-Fi access points and channel distribution graph.



Copyright (c) 2023-2026 Masayuki Yamai

Bluetooth (BLE) Report

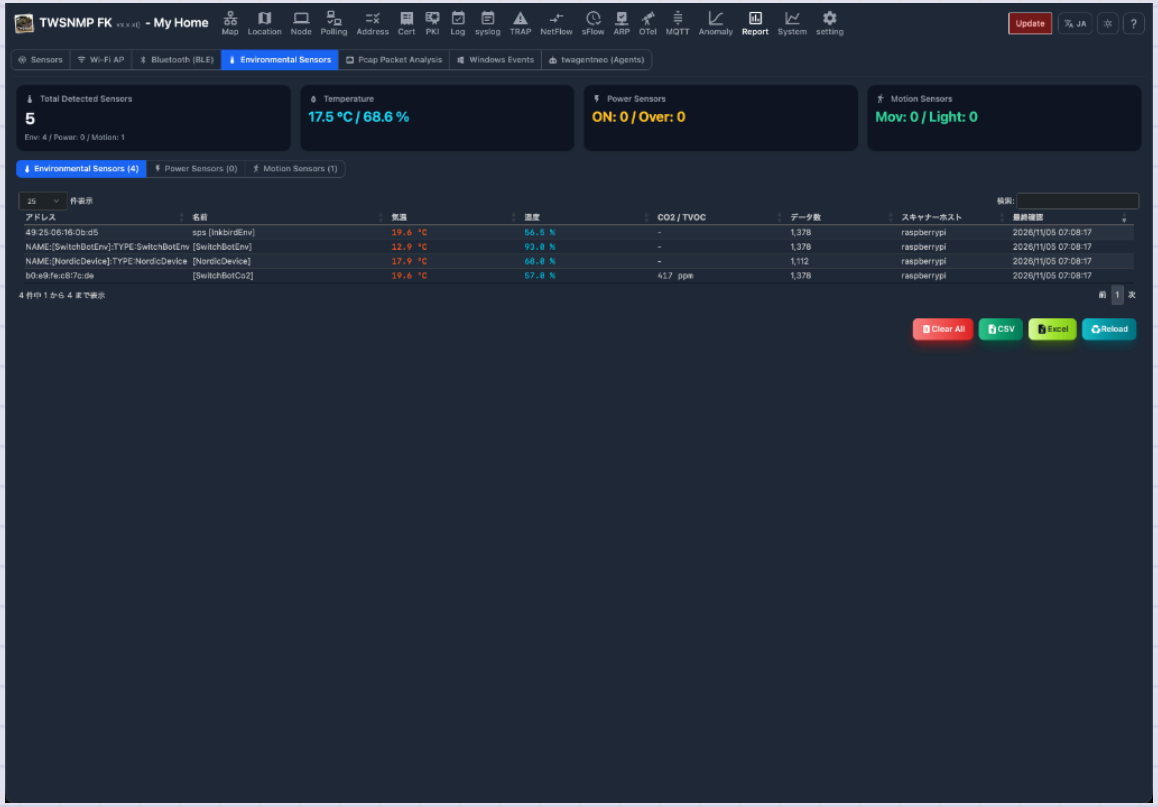
Displays detected Bluetooth / BLE devices, vendor breakdown, and signal strength distribution.



Copyright (c) 2023-2026 Masayuki Yamai

Environmental Sensors Report

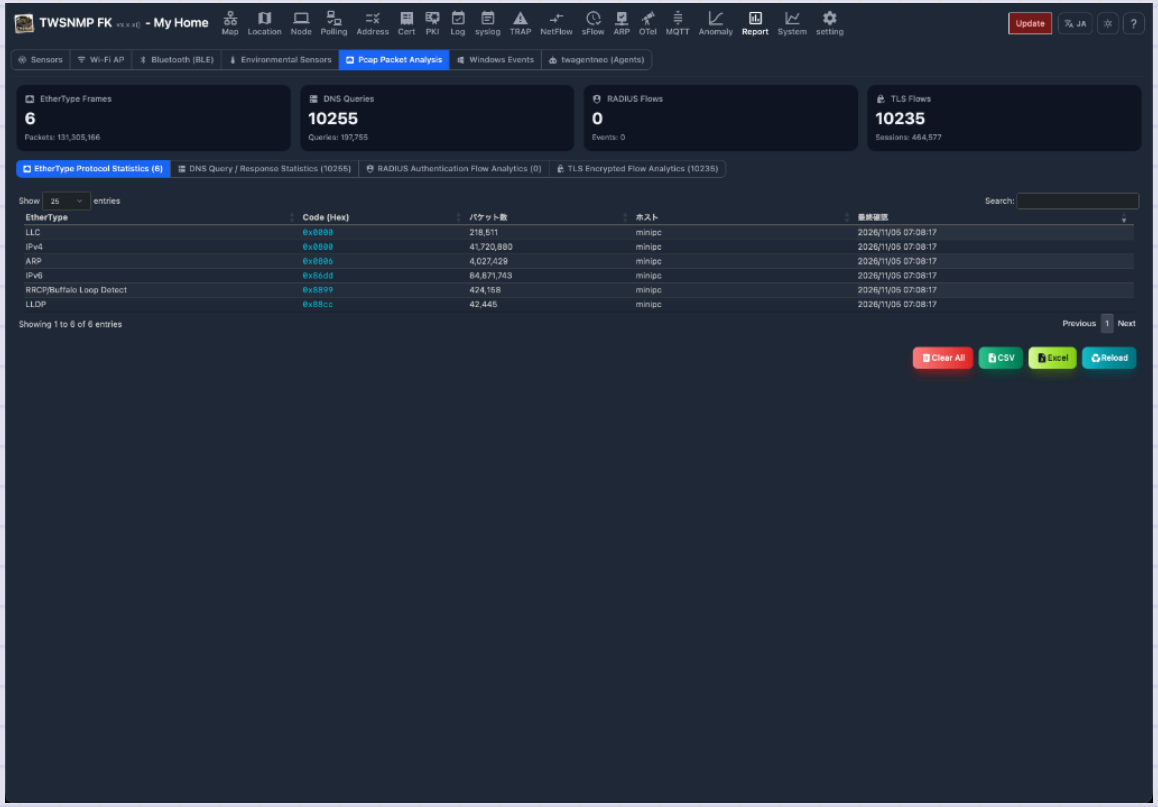
Displays environmental sensor telemetry (temperature, humidity, CO2), power monitors, and motion sensors.



Copyright (c) 2023-2026 Masayuki Yamai

Pcap Packet Analysis Report

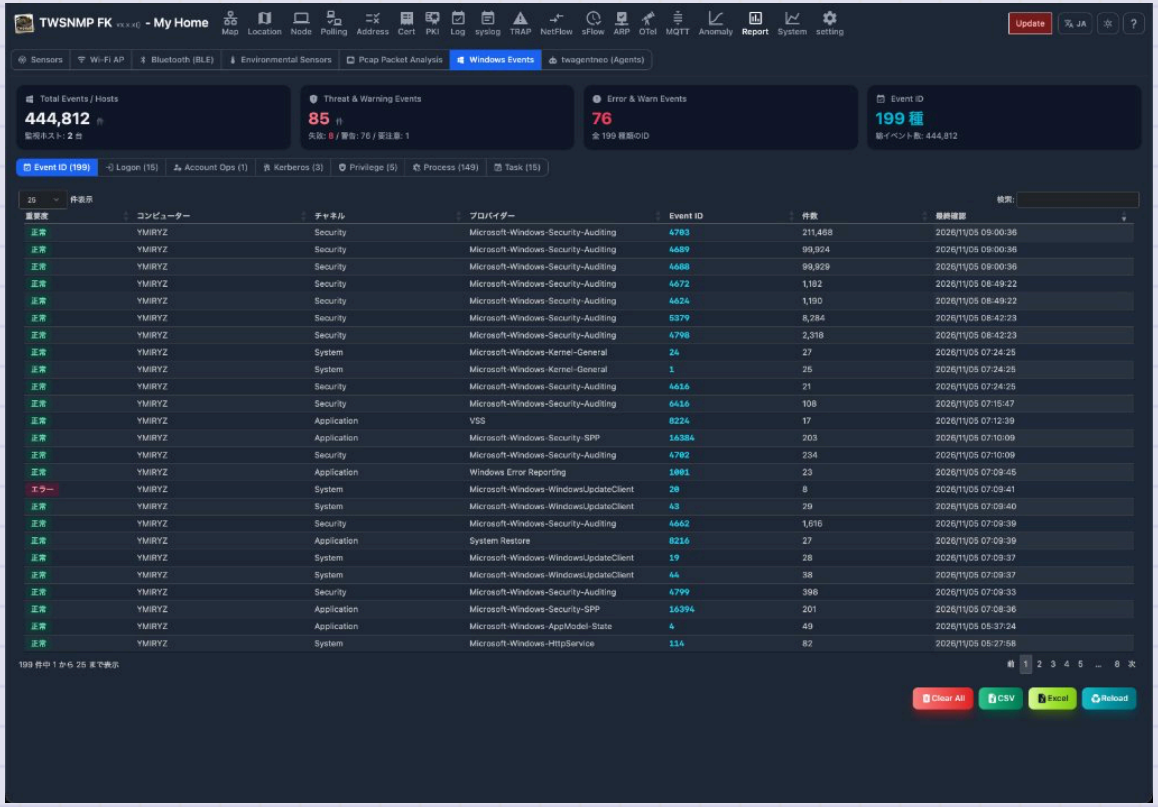
Provides detailed breakdown of EtherTypes, DNS queries, RADIUS flows, and TLS encrypted flows.



Copyright (c) 2023-2026 Masayuki Yamai

Windows Events Report

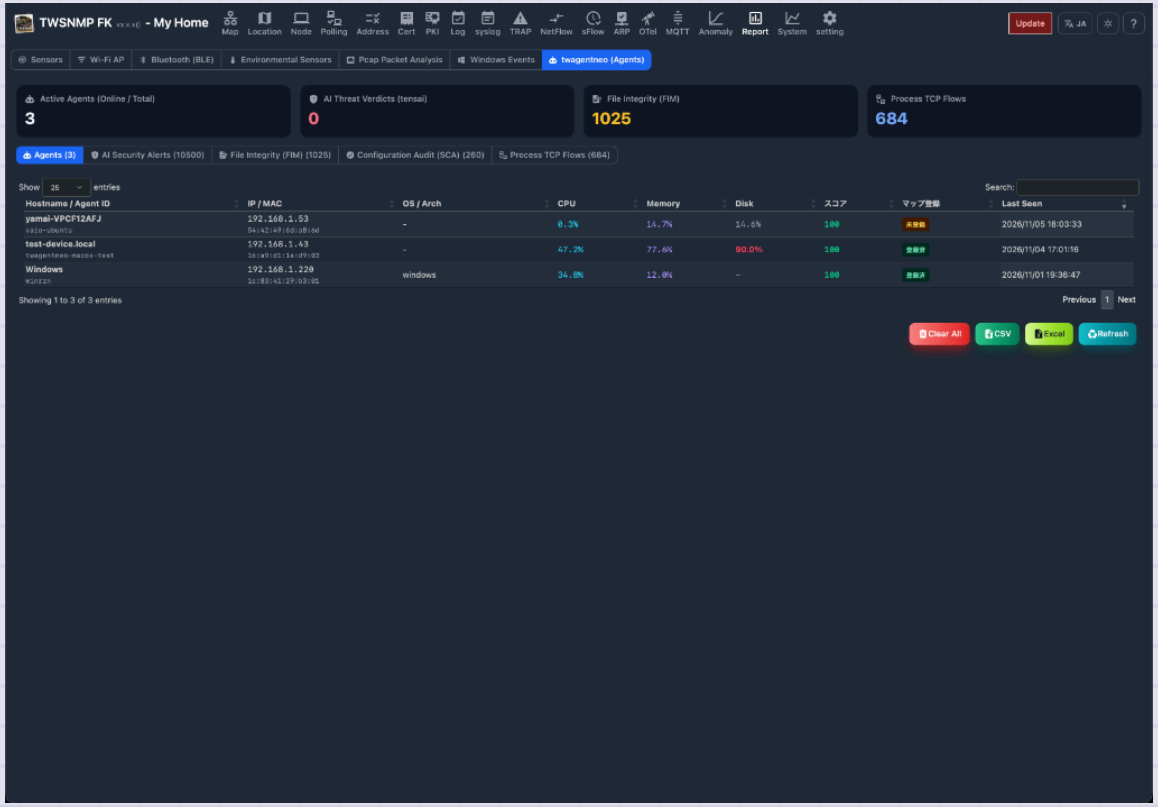
Aggregates and audits Windows security logs (Event ID, logon/logoff, account operations, process activity).



Copyright (c) 2023-2026 Masayuki Yamai

twagentneo Report

Monitors agent operational status, AI threat classification, file integrity (FIM), security configuration assessment (SCA), and process TCP flows.



Copyright (c) 2023-2026 Masayuki Yamai

System

System information screen.At the top, there is a graph showing log resources and communication information in a chronological order.

Time	CPU	Memory	My CPU	MyMemory	Swap	Disk	Load	Network	Connections	Processes	Goroutines	Heap	Sys	DB Size
2026/11/04 16:53:38	47.48%	78.48%	8.06%	1.56%	41.64%	15.60%	5.59	2MBPS	102	583	45	65.394MB	224.978MB	10.879GB
2026/11/04 16:48:38	34.86%	79.09%	10.80%	2.26%	12.46%	15.60%	4.53	2MBPS	105	579	45	87.745MB	224.978MB	10.879GB
2026/11/04 16:43:38	38.27%	81.40%	79.02%	1.39%	12.46%	15.60%	4.33	0BPS	102	553	34	41.250MB	80.823MB	10.879GB

Copyright (c) 2023-2026 Masayuki Yamai

Items	Contents
Date and time	It is the date and time when System information is recorded.
CPU	CPU usage rate.
Memory	Memory usage rate.
Disk	Data folder is the usage rate of disks.
Load	load.
Communication amount	LAN port communication amount.
Connection number	TCP connection number.
Process	Total number of processes.
DB size	Database size.

Copyright (c) 2023-2026 Masayuki Yamai

Items	Contents
Size prediction	Database size and disk usage rate are forecast for one year.
Backup	Get backup.
Reload	Update System information to the latest state.

Copyright (c) 2023-2026 Masayuki Yamai

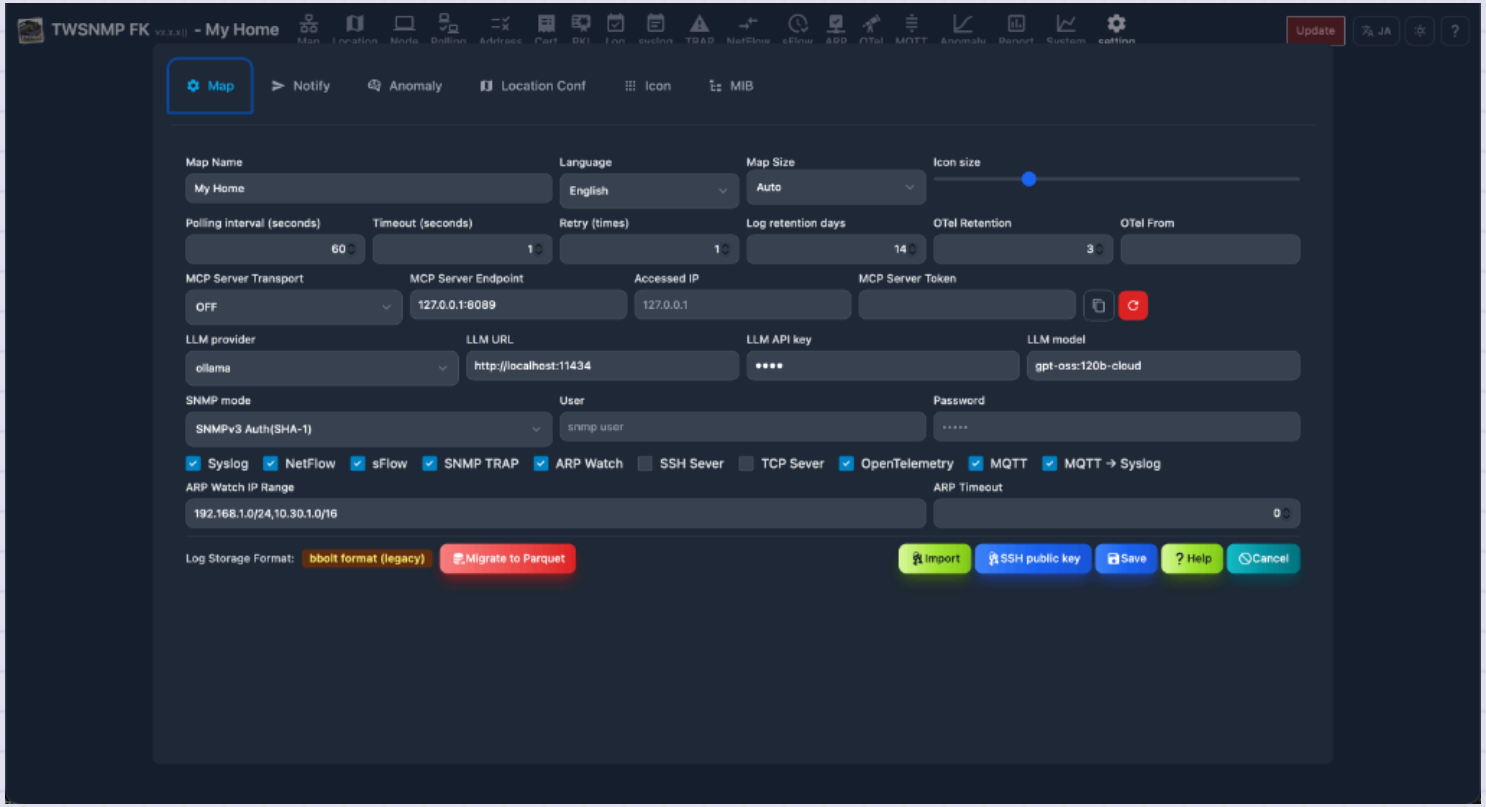
Size prediction

This is a year forecast for the database size and disk usage rate.

Copyright (c) 2023-2026 Masayuki Yamai

Map settings

This is the screen to set the management map.



Copyright (c) 2023-2026 Masayuki Yamai

Items	Contents
Map name	Map name.It will be displayed in the upper left of the screen. Please give your favorite name.
Icon size	It is the size of the icon to be displayed on the map.
Polling interval	Default polling interval.
Timeout	Default timeout.
Retry	Default number of retry times.
Log saving days	It is the number of days to save the log.The log will be deleted automatically after passing.

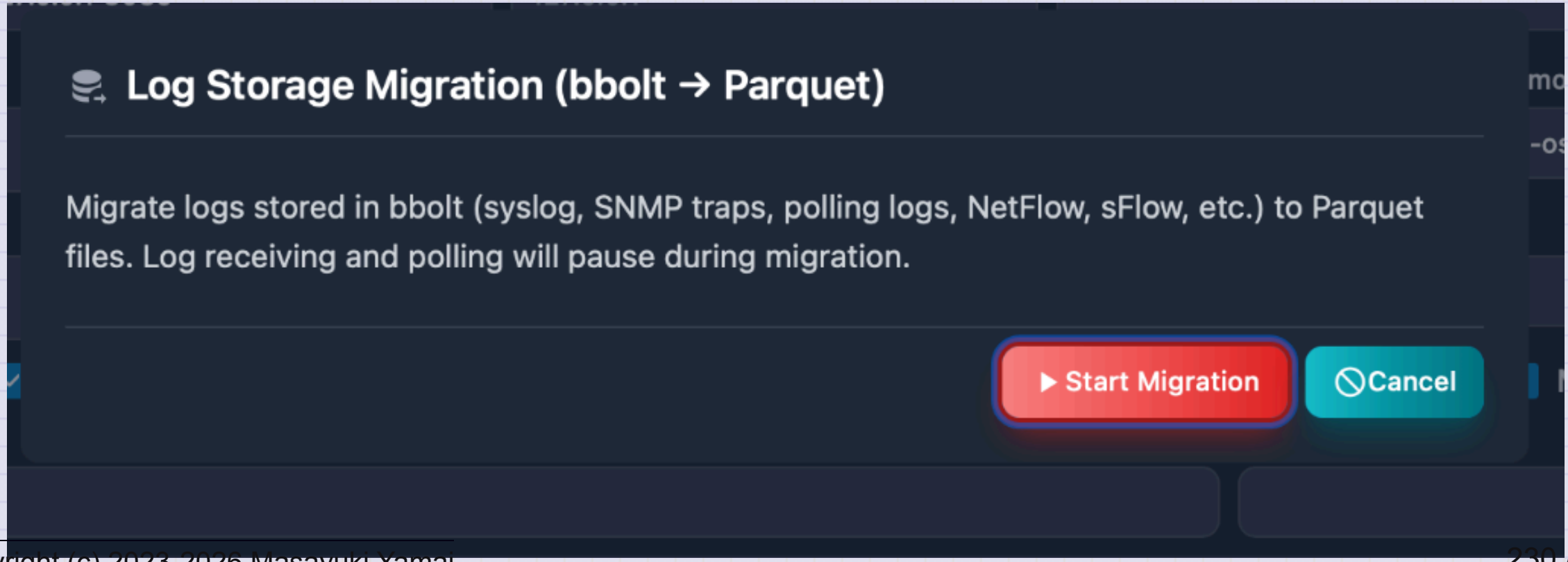
Copyright (c) 2023-2026 Masayuki Yamai

Items	Contents
SNMP mode	SNMP version and type of encryption.(SNMPV1, SNMPv2C, SNMPv3)
SNMP Community	Community name for SNMPV1, V2C.
SNMP user	User name at SNMPv3.
SNMP password	Password name for SNMPv3.
Syslog	Receive syslog.
SNMP Trap	Receive SNMP Trap.
SSH Seerver	SSH Server
ARP Watch	Enable ARP monitoring function.

Copyright (c) 2023-2026 Masayuki Yamai

Log Storage Migration (bbolt → Parquet)

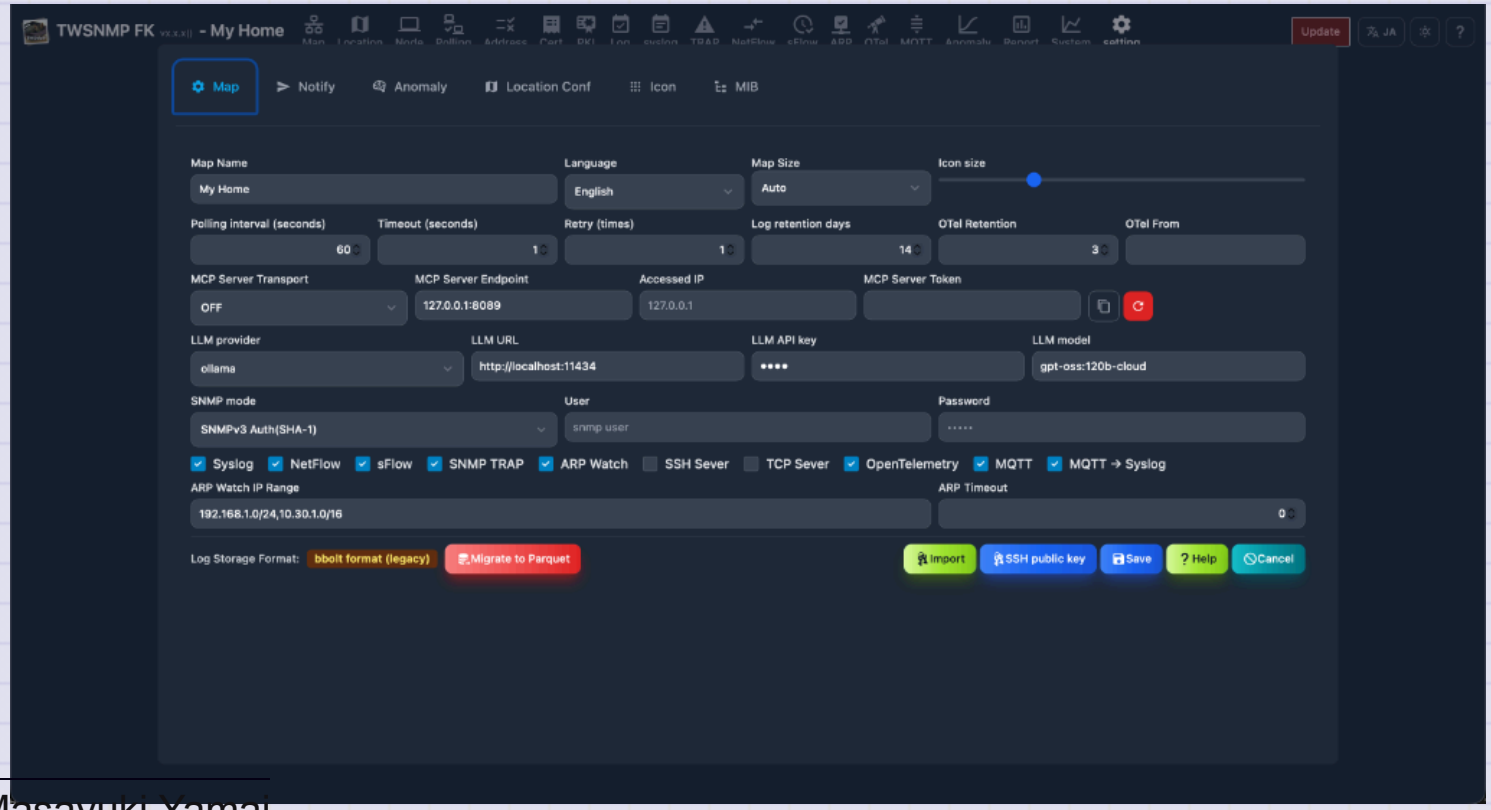
Safely batch migrates legacy logs (Syslog, TRAP, Polling logs, NetFlow, sFlow, etc.) stored in bbolt databases to high-speed, highly compressed Apache Parquet columnar files.



Copyright (c) 2023-2026 Masayuki Yamai

LLM Settings (AI Provider Configuration)

Configuration for setting up the LLM provider used by TWSNMP FK's AI integration features (Node AI Diagnosis, Log Explanation, AI Polling Assistant, and Report AI Summaries).



Copyright (c) 2023-2026 Masayuki Yamai

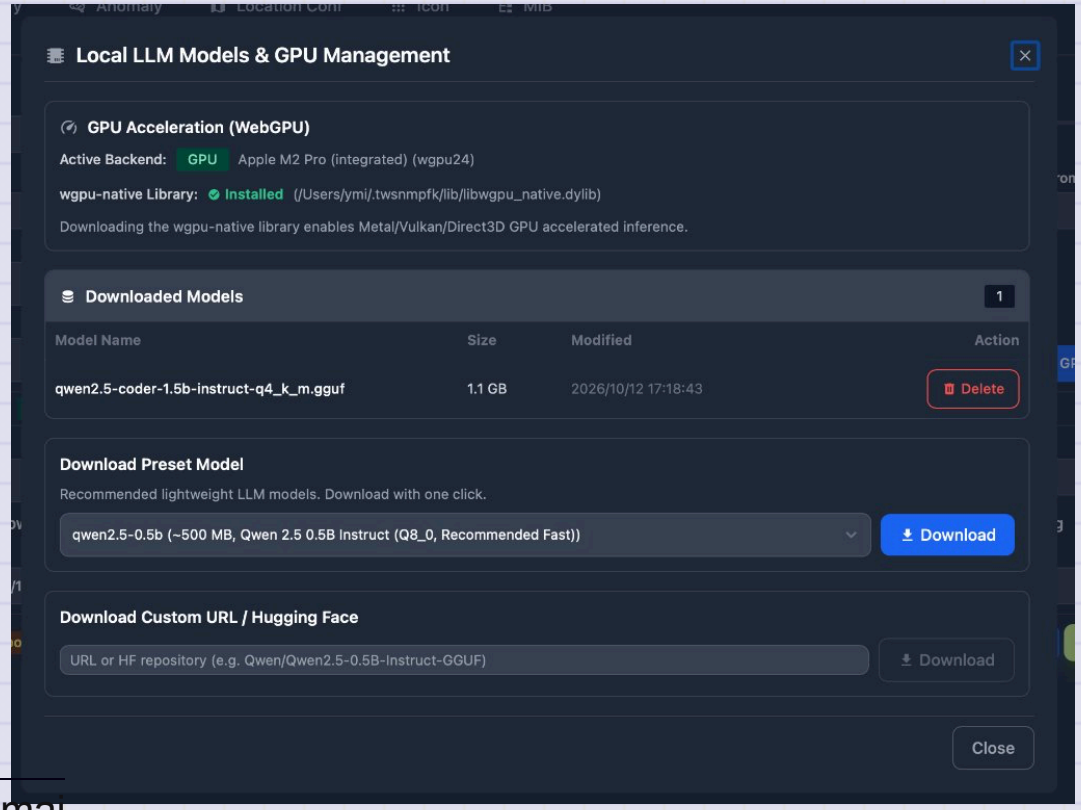
Items	Contents
LLM provider	Select the AI provider to use (Ollama, OpenAI, Gemini, Claude, DeepSeek, etc.).
LLM URL	Specify the provider API endpoint URL (Default for Ollama: `http://localhost:11434`).
LLM API key	Enter the authentication API key for cloud providers (leave blank for local LLM).
LLM model	Enter the model name to use (e.g., `gpt-oss`, `llama3`, `gpt-4o`, etc.).

> [!TIP]
> **Recommended: Local LLM (Ollama)**
> When using cloud LLMs (such as OpenAI or Gemini), network topology, hostnames, IP addresses, and Syslog log contents may be sent to external cloud

Copyright (c) 2023-2026 Masayuki Yamai

Local LLM Models & GPU Management (tensai)

Integrates a fully local LLM engine (tensai) that runs directly inside the application without requiring an external Ollama server. Supports GPU hardware acceleration via WebGPU and one-click downloading/managing of GGUF models.



Copyright (c) 2023-2026 Masayuki Yamai

When you want to change the receiving port of syslog, SNMP Trap

The port number is specified by the startup parameter of the program.

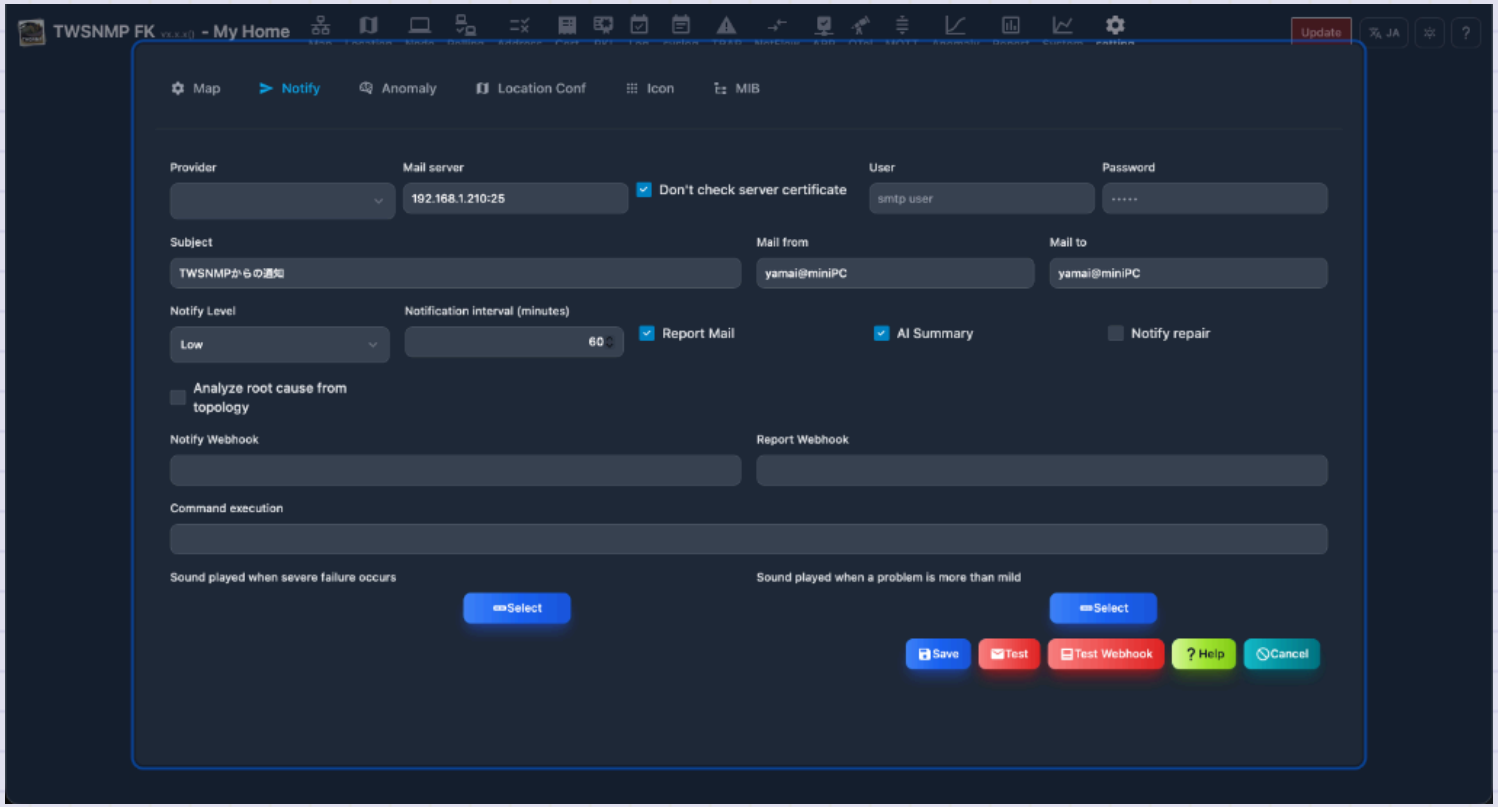
```
-syslogPort int
      Syslog port (default 514)
-trapPort int
    SNMP TRAP port (default 162)
-sshdPort int
    SSH Server port (default 2022)
```

*If SYSLOG or SNMP Trap cannot be received, check the OS and security software firewall settings.

Copyright (c) 2023-2026 Masayuki Yamai

Notification settings

This is the screen to set the notification.



Copyright (c) 2023-2026 Masayuki Yamai

Items	Contents
Provider	Select from smtp/Google/Microsoft/Microsoft (Custom Server).
OAuth2	Client ID, Secret, Tenant Name (Microsoft).
Mail server	Host name or IP address: port number
Certificate	Check when the mail server is self-signed.
Auth	User ID and Password.
From/To	Sending source and destination emails.
Subject	Notification email subject.

Copyright (c) 2023-2026 Masayuki Yamai

Items	Contents
Notification level	Monitoring level to send notifications.
Notification interval	Interval to check notifications.
Regular report	Send a daily report.
Repair notification	Send an email when repaired.
Webhook	Webhook URL for failures and reports.
Command	Run command when state changes.
Audio	Audio file to play for severe/mild status.

Copyright (c) 2023-2026 Masayuki Yamai

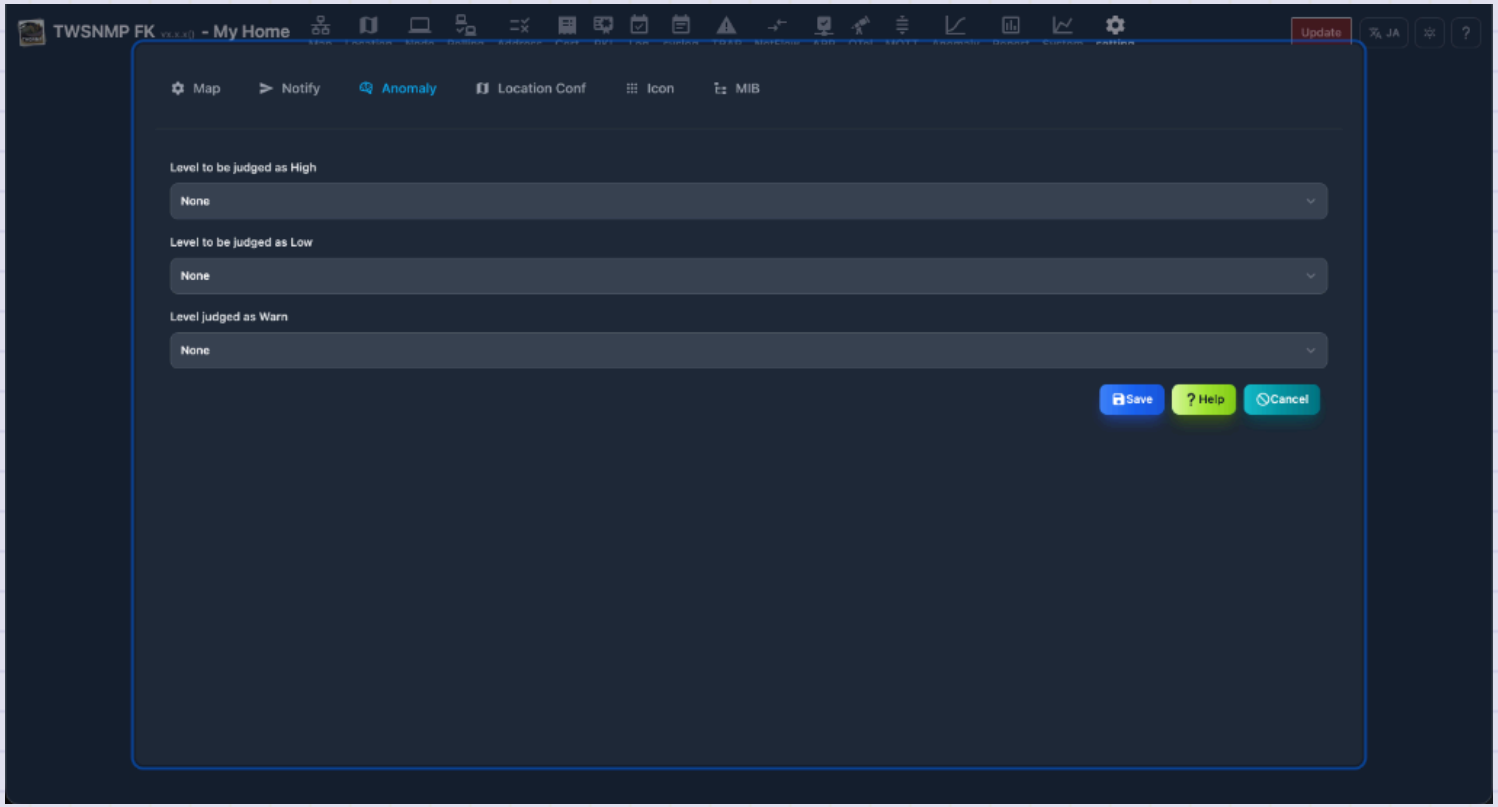
Email send test

Click the <Test> button to send the test email with the configured content.

Copyright (c) 2023-2026 Masayuki Yamai

AI analysis setting

This is the screen to set AI analysis.



Copyright (c) 2023-2026 Masayuki Yamai

Items	Contents
Level to be high	Specify the deviation level of AI analysis determined as severe disorder.
Level to be low	Specify the deviation level of AI analysis determined as mild disorder.
Level to be warn	Specify the deviation level of AI analysis determined as a disorder.

Copyright (c) 2023-2026 Masayuki Yamai

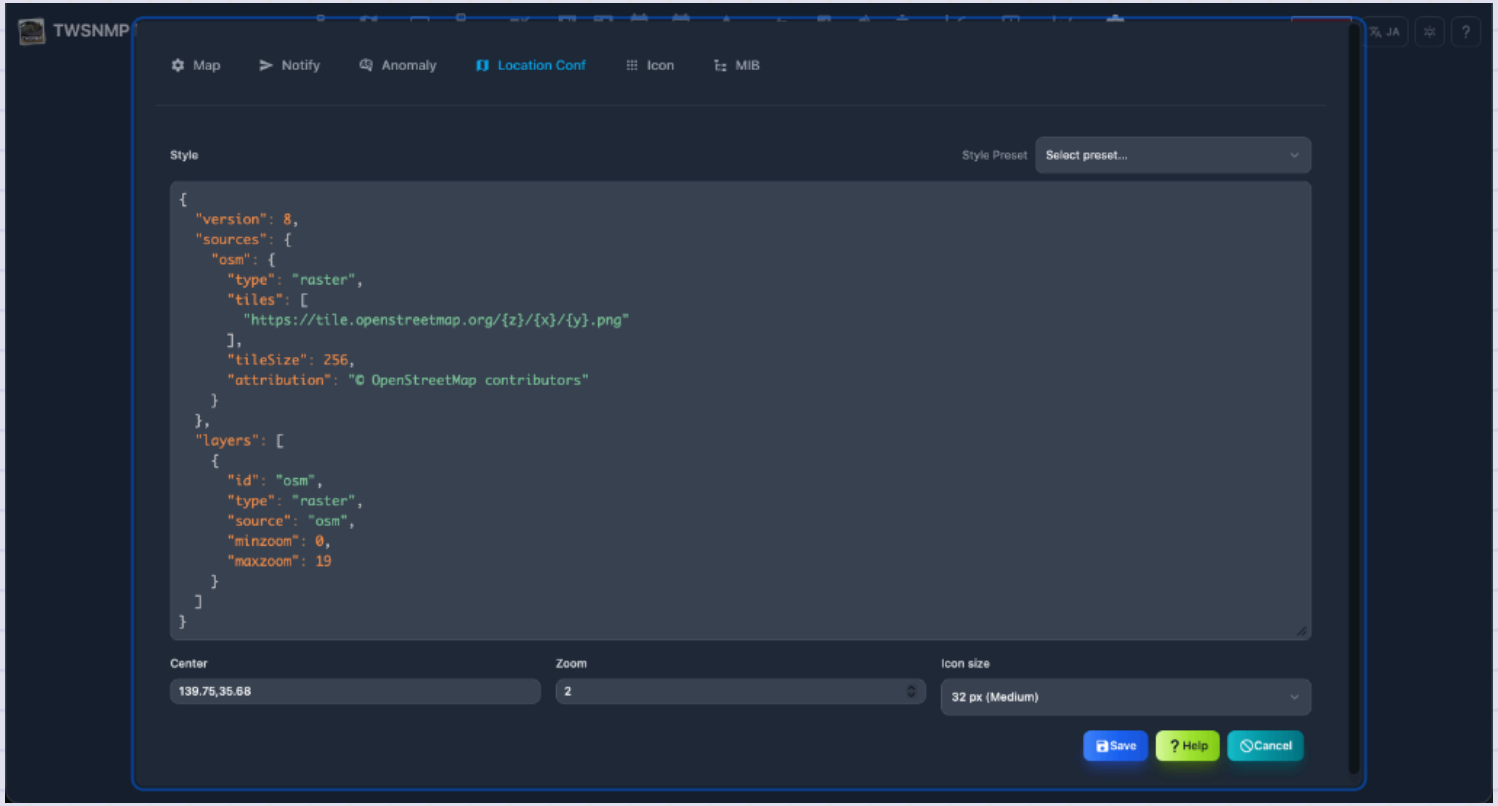
About AI analysis

- The AI analysis is implemented by setting the log mode to "AI analysis" in the polling settings.
- An anomaly detection of the numerical data of the polling result in isolation forest.
- The results are set to deviation values.
- The deviation value is familiar to school results.It shows how rare it is.
- So, the disability level setting is an expression of once every 10,000 times.

Copyright (c) 2023-2026 Masayuki Yamai

Location map settings

This is the screen to set the map.



Copyright (c) 2023-2026 Masayuki Yamai

Items	Contents
Style	Specify the map style.Specify in URL or JSON.
Central coordinates	The central coordinates on the map are in the order of longitude and latitude. Example: 135.3338576281734, 39.614306840830096
Zoom	Specify the enlargement level of the map.
Icon size	Specify the size of the icon to be displayed.

Copyright (c) 2023-2026 Masayuki Yamai

About map style

The map is displayed using Maplibre GL JS.The map to be displayed is specified in the style.
You can specify it with URL or JSON.Search for MAPLIBRE GL JS and find something suitable.

URL example

```
https://tile.openstreetmap.jp/styles/osm-bright-ja/style.json
```

Copyright (c) 2023-2026 Masayuki Yamai

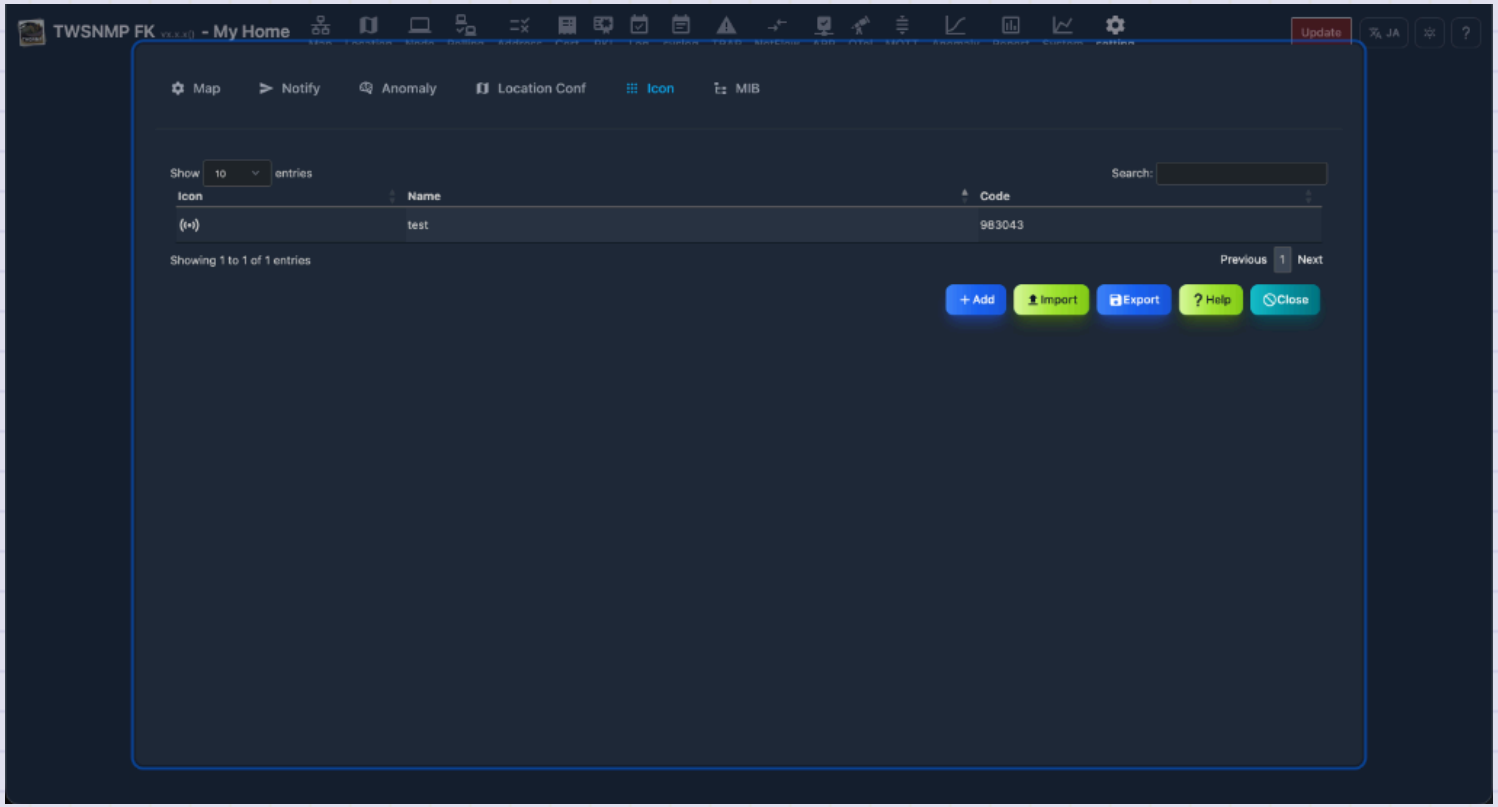
JSON example

```
{
                        "version": 8,
                        "sources": {
                                "MIERUNEMAP": {
                                        "type": "raster",
                                        "tiles": ["https://tile.mierune.co.jp/mierune_mono/{z}/{x}/{y}.png"],
                                        "tileSize": 256,
                                        "attribution":
                                                "Maptiles by <a href='https://mierune.co.jp/' target='_blank'>MIERUNE</a>, under CC BY. Data by <a href='https://osm.org/copyright' target='_blank'>OpenStreetMap</a> contributors, under ODbL."
                                }
                        },
                        "layers": [
                                {
                                        "id": "MIERUNEMAP",
                                        "type": "raster",
                                        "source": "MIERUNEMAP",
                                        "minzoom": 0,
                                        "maxzoom": 18
                                }
                        ]
}
```

Copyright (c) 2023-2026 Masayuki Yamai

Icon management

This is a screen that manages the icon.



Copyright (c) 2023-2026 Masayuki Yamai

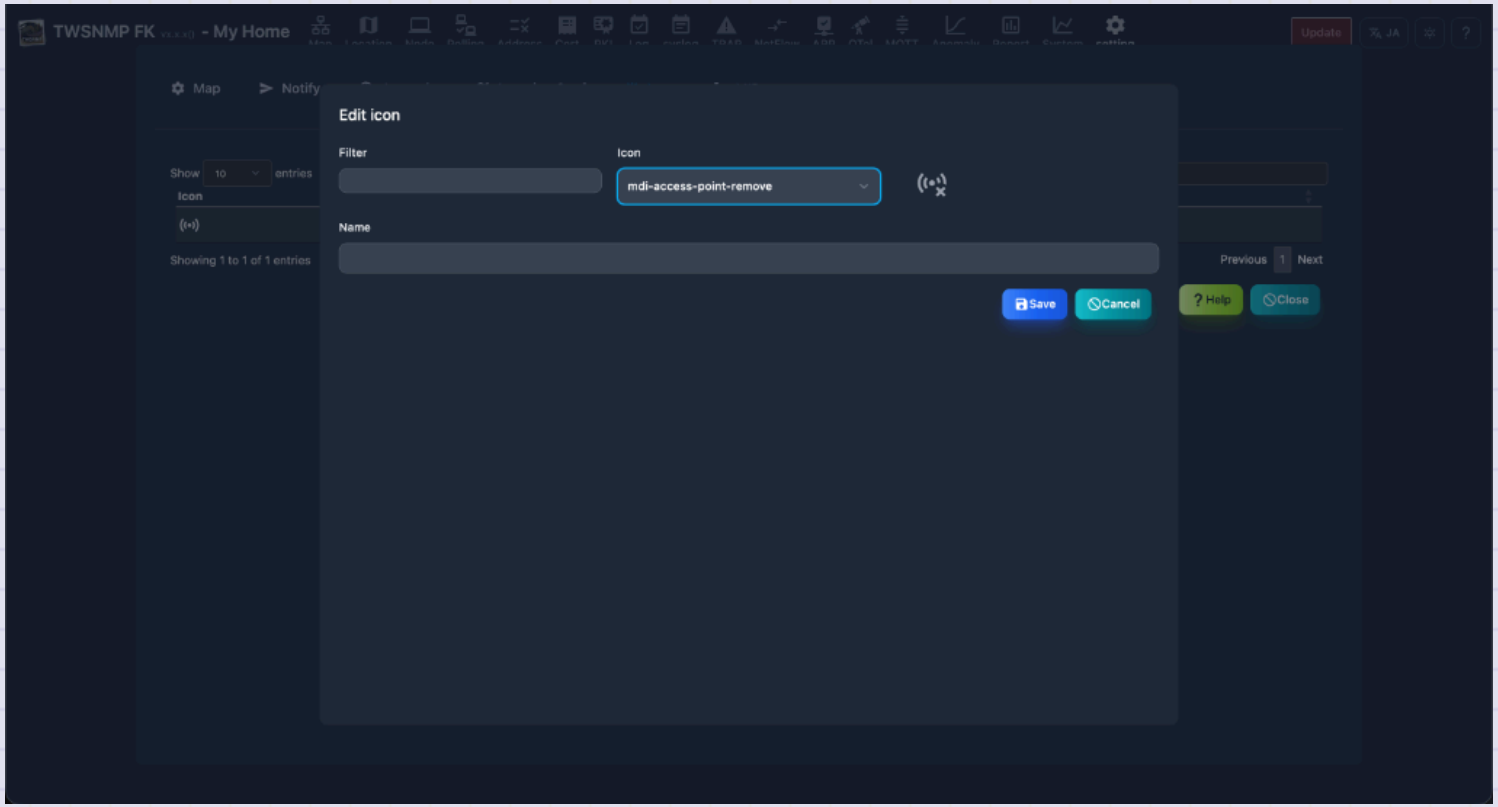
Items	Contents
Icon	It is an image of an icon.
Name	Name when choosing.You can attach it freely.
Code	icon code.

Copyright (c) 2023-2026 Masayuki Yamai

Button	Contents
Added	Add a new icon.
Edit	Edit the name of the selected icon.
Delete	Delete the selected icon.
Help	Display this help.
Close	Close the setting screen.

Copyright (c) 2023-2026 Masayuki Yamai

Icon editing screen



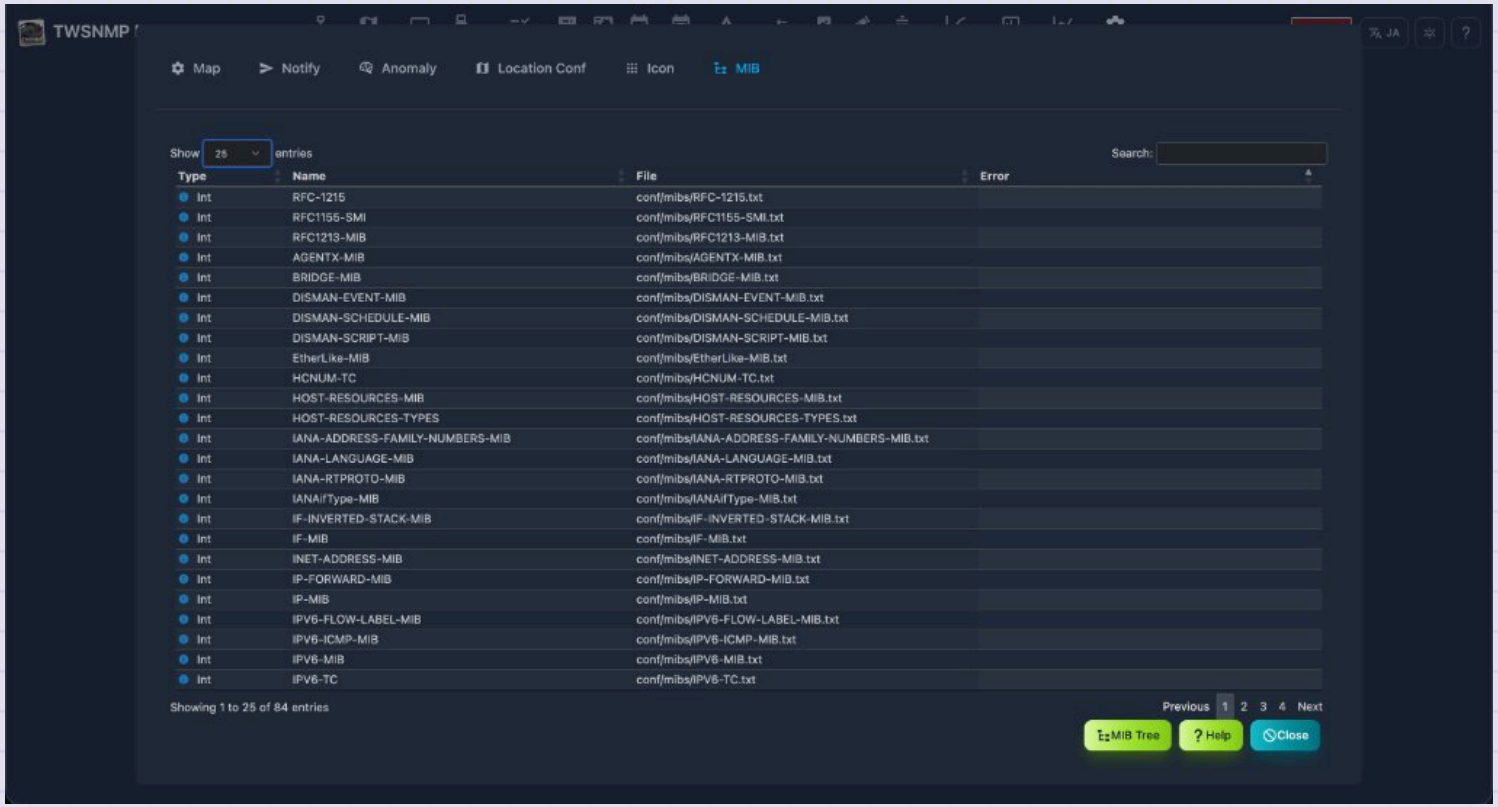
Copyright (c) 2023-2026 Masayuki Yamai

Items	Contents
Icon	Select an icon.The name of the web font of the MDI icon.
Name	Give the icon your favorite name.

Copyright (c) 2023-2026 Masayuki Yamai

MIB management

This is a screen that manages SNMP MIB.



Copyright (c) 2023-2026 Masayuki Yamai

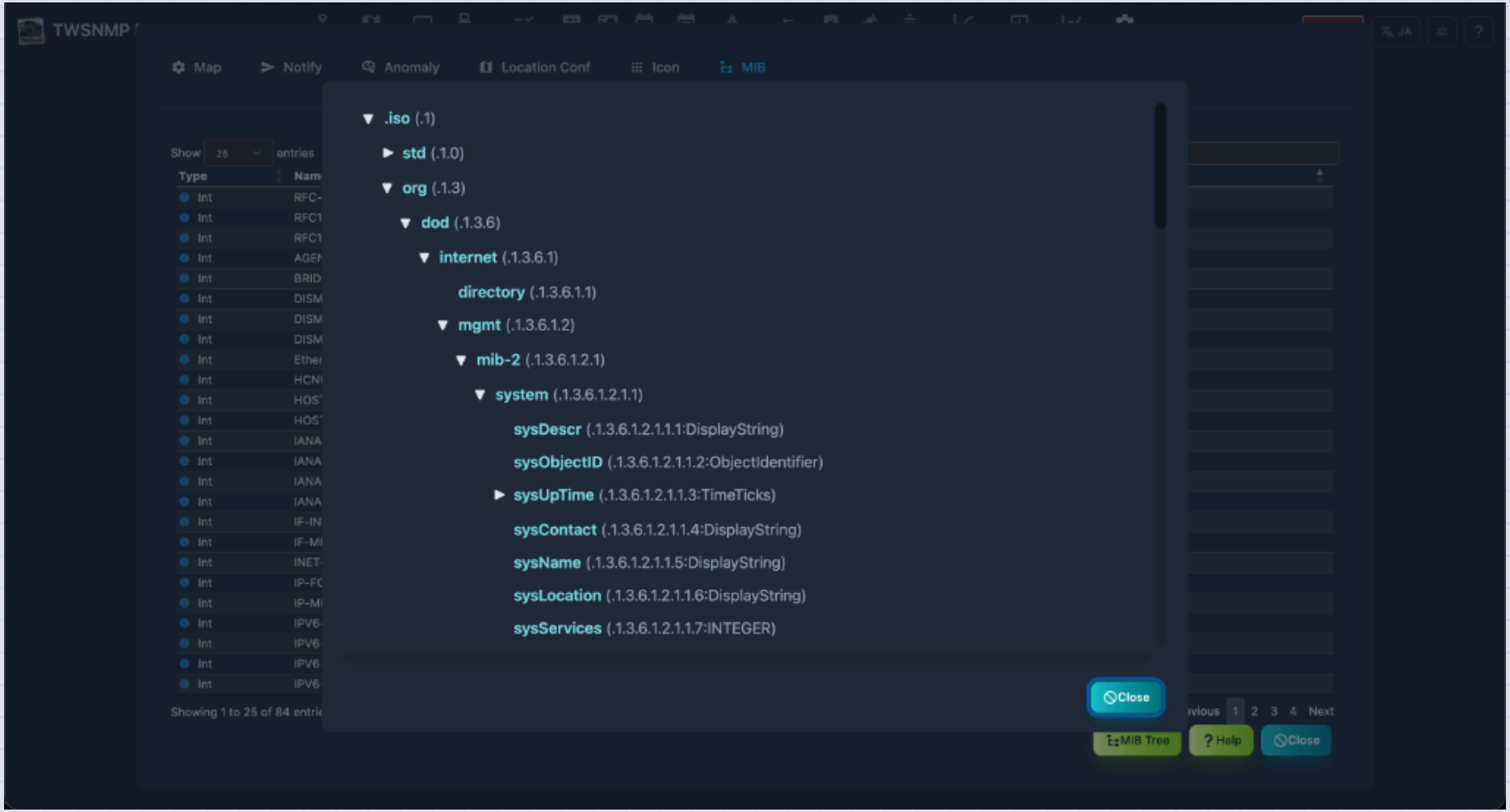
Items	Contents
Type	It is a type of built -in or reading.
Name	MIB module name.
File	It is a read file name.
Error	An error when you read it.

Copyright (c) 2023-2026 Masayuki Yamai

Button	Contents
MIB Tree	Displays MIB tree.
Help	Display this help.
Close	Close the setting screen.

Copyright (c) 2023-2026 Masayuki Yamai

MIB tree screen



Copyright (c) 2023-2026 Masayuki Yamai

File in the datastore

You can customize it by saving the following files in the data folder.

File	Contents
TWSNMPFK.db	Database file.If it does not exist, it will be created automatically.
Services.txt	This is a file to use the service name conversion.(Optional)
Mac-vendors-export.csv	Mac A database that indicates the relationship between the MAC address and the vendor name.(Optional)
Polling.json	Polling settings (optional)

Copyright (c) 2023-2026 Masayuki Yamai

File	Contents
mail_test.html	Notification test mail template (optional)
mail_notify.html	Notification mail template (optional)
Mail_repot.html	Template of report mail (optional)
EXTMIBS/*	Additional reading extended MIB (optional)

Copyright (c) 2023-2026 Masayuki Yamai

Usage

```
Usage of twsnmpfk:
  -datastore string  Path to data store directory
  -kiosk             Kisok mode(frameless and full screen)
  -lang string       Language(en|jp)
  -lock string       Disable edit map and lock page(map or loc)
  -maxDispLog int    Max log size to diplay (default 10000)
  -ping string       ping mode icmp or udp
  -syslogPort int    Syslog port (default 514)
  -trapPort int      SNMP TRAP port (default 162)
  -sshdPort int      SSH server port (default 2022)
  -netflowPort int   Netflow port (default 2055)
  -sFlowPort int     sFlow port (default 6343)
  -tcpdPort int      tcp server port (default 8086)
```

Copyright (c) 2023-2026 Masayuki Yamai

```
-caCert, -clientCert, -clientKey   TLS/mTLS for TWLogEye
-otelGRPCPort, -otelHTTPPort       OTel receiver ports
-otelCert, -otelKey, -otelCA       OTel TLS/mTLS settings
-mqttTCPPort, -mqttWSPort          MQTT server ports
-mqttCert, -mqttKey, -mqttFrom     MQTT TLS and client limits
-mqttUsers                         MQTT user/password list
-mcpCert, -mcpKey                  MCP server TLS settings
-notifyOAuth2Port                  OAuth2 redirect port
```

Copyright (c) 2023-2026 Masayuki Yamai

Parameters	Description
dataStore	Datastore Path
kiosk	Kiosk mode (frameless, full screen)
lock <page>	disable edit map and show fixed page
Maxdisplog <number>	Maximum number of logs (default 10000)
ping <Mode>	Ping operation mode (ICMP or UDP)
syslogPort <PORT>	Syslog receiving port (default 514)
trapPort <Port>	SNMP TRAP Reception port (Default 162)

Copyright (c) 2023-2026 Masayuki Yamai

Parameters	Description
sshdPort <port>	SSH Server Receive Port (Default 2022)
netflowPort <port>	NetFlow/IPFIX receive port (default 2055)
sFlowPort <port>	sFlow receiving port (default 6343)
tcpdPort <port>	TCP log receiving port (default 8086)
otelCert <file>	OpenTelemetry server certificate
otelKey <file>	OpenTelemetry server private key
otelGRPCPPort <number>	GRPC port number for OpenTelemetry
otelHTTPPort <file>	OpenTelemetry HTTP port number

Copyright (c) 2023-2026 Masayuki Yamai

Parameters	Description
mqttTCPPort	MQTT server TCP port (default 1883)
mqttWSPort	MQTT server Websock port (default 1884)
mqttCert	MQTT server cert path
mqttKey	MQTT server key path
mqttFrom	MQTT server Client IP limit
mqttUsers	MQTT server User ID and password list
mcpCert	MCP server cert path
mcpKey	MCP server key path
notifyOAuth2Port	OAuth2 redirect port (default 8180)

Copyright (c) 2023-2026 Masayuki Yamai

Configuration File

Place `.twsnmpfk.ini` in the datastore folder to load boot parameters.

```
#lang=ja
#maxDispLog=10000
#lock=map

[logger]
syslogPort=8514
trapPort=8162
netflowPort=2056
sshdPort=2023

[OTel]
#otelGRPCPort=4317
#otelHTTPPort=4318

[MCP]
#mcpCert=
#mcpKey=

[MQTT]
#mqttTCPPort=1883
```

Copyright (c) 2023-2026 Masayuki Yamai